

HEALTHCARE ANALYTICS MARKET

GLOBAL FORECAST TO 2026

BY TYPE (PREDICTIVE, PRESCRIPTIVE, COGNITIVE) APPLICATION (CLINICAL, RCM,
CLAIM, FRAUD, WASTE, SUPPLY CHAIN, PHM) COMPONENT (SERVICE, SOFTWARE)
DEPLOYMENT (ON-DEMAND, CLOUD) END USER (PAYER, HOSPITAL)

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LIST OF ABBREVIATIONS

ABBREVIATIONS	FULL FORM
ACA	Affordable Care Act
ACOs	Accountable Care Organizations
A-EMRAM	Ambulatory EMR Adoption Model
AHRQ	Agency for Healthcare Research and Quality
AI	Artificial Intelligence
APAC	Asia Pacific
ARRA	American Recovery and Reinvestment Act
BITKOM	German Federal Association for Information Technology
CAGR	Compound Annual Growth Rate
CDC	Centers for Disease Control and Prevention
CHCAA	Canadian Healthcare Anti-Fraud Association
CHOP	Children's Hospital of Philadelphia
CMC	Catholic Medical Centre
CMS	Centers of Medicare & Medicaid
conhIT	Connecting Healthcare IT
CQM	Clinical Quality Measures
CTL	Cohort Term License
DCH	Department of Community Health
EHR	Electronic Health Record
EMA	European Medicines Agency
EMRAM	Electronic Medical Record Adoption Model
ETG	Emerging Technology Group
FCCs	French Cancer Centers
FDA	Food and Drug Administration
FFS	Fee-For-Service
FWA	Fraud, Waste, and Abuse
GBS	Global Business Services
HCAHPS	Hospital Consumer Assessment of Healthcare Providers and Systems

ABBREVIATIONS	FULL FORM
HDAA	Healthcare Data & Analytics Association
HHS	Health and Human Services
HIE	Health Information Exchange
HIM	Healthcare Information Management
HIMSS	Healthcare Information Management Systems Society
HIPAA	Health Insurance Portability and Accountability Act
HITECH	Health Information Technology for Economic and Clinical Health Act
HIX	Health Insurance Exchanges
IaaS	Insights-as-a-Service
ICTs	Information and Communications Technologies
IDNs	Integrated Delivery Networks
IMIA	International Medical Informatics Association
IT	Information Technology
LATAM	Latin America
LHCH	Liverpool Heart and Chest Hospital
MACSS	Medical Allround-Care Service Solutions
ML	Machine Learning
MCOs	Managed Care Organizations
NAHDO	National Association of Health Data Organizations
NCQA	National Committee for Quality Assurance
NCT	National Center for Tumor Diseases
NHCAA	National Health Care Anti-Fraud Association
NIH	National Institutes of Health
NIMHD	National Institute on Minority Health and Health Disparities
NLP	Natural Language Processing
ODDT	Open Data and Digital Transformation
OPPE	Ongoing Professional Practice Evaluation
PACOs	Post-Acute Care Organizations
PAS	Patient Administration System
PDPH	Philadelphia Department of Public Health
PHM	Population Health Management

ABBREVIATIONS	FULL FORM
PPACA	Patient Protection and Affordable Care Act
PQRS	Physician Quality Reporting System
R&D	Research and Development
RCM	Revenue Cycle Management
RISE	Resource Initiative and Society for Education
RoA	Rest of Asia
RoE	Rest of Europe
RoW	Rest of the World
RUHS	Riverside University Health System
RWE	Real-World Evidence
TCC	Transdisciplinary Collaborative Centers
TCCs	Transdisciplinary Collaborative Centers
TPAs	Third-Party Administrators
UK	United Kingdom
US	United States
USD	United States Dollar
UTs	Union Territories
VC	Venture Capital
WAA	World Analytics Association

1 INTRODUCTION

1.1 OBJECTIVES OF THE STUDY

- To define, describe, and forecast the healthcare analytics market based on type, application, component, deployment model, end user, and region
- To provide detailed information regarding the major factors influencing the market growth (drivers, restraints, opportunities, and challenges)
- To strategically analyze the micromarkets¹ with respect to individual growth trends, prospects, and contributions to the overall market
- To analyze the opportunities in the market for stakeholders and provide details of the competitive landscape for market leaders
- To forecast the size of the market segments with respect to six main regions—North America, Europe, Asia, Pacific Countries, Latin America, and the Middle East & Africa
- To strategically profile the key players and comprehensively analyze their product portfolios, market positions, and core competencies²
- To track and analyze competitive developments in the healthcare analytics market, such as joint ventures, mergers and acquisitions, new product developments, expansions, and research and development (R&D) activities

1.2 MARKET DEFINITION

Healthcare analytics is the efficient use of data and related business insights developed through statistical, quantitative, predictive, comparative, cognitive, and other emerging applied analytical models. Healthcare analytics tools help reveal and understand historical data patterns, predict future outcomes, and provide actionable insights to drive fact-based decision-making for improved clinical, financial, and operational performance of healthcare organizations. The use of analytics helps healthcare systems reduce healthcare costs and improve the quality of care.

This report studies the healthcare analytics market by type, component, deployment model, application, end user, and region to arrive at the revenues generated in the global market from 2019 to 2026.

1. Micromarkets are segments and subsegments of the healthcare analytics market.

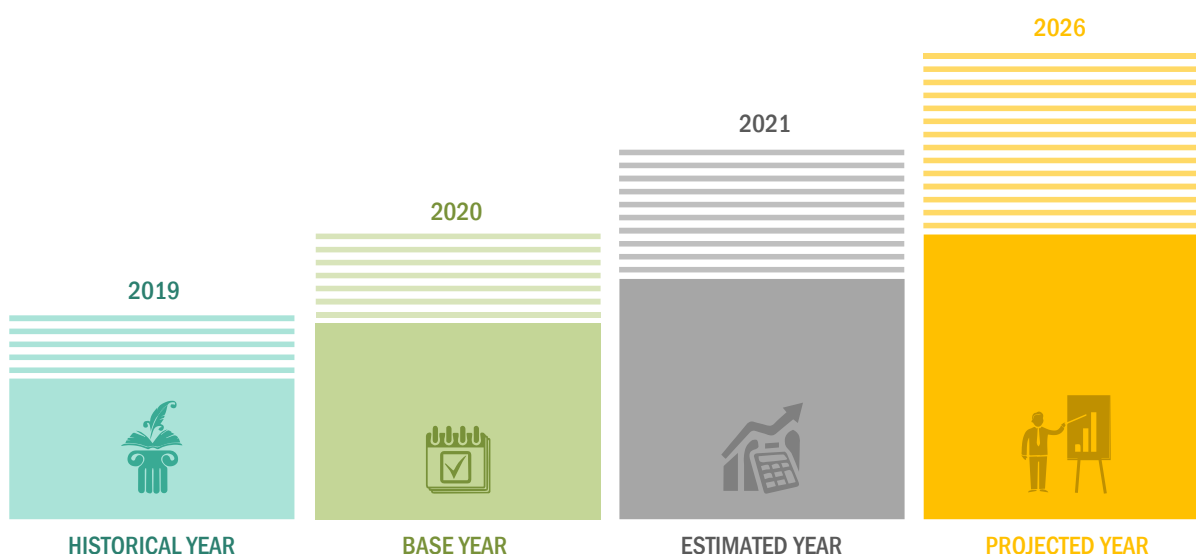
2. Core competencies of companies are captured in terms of their key developments and strategies adopted to sustain their market positions.

1.2.1 INCLUSIONS & EXCLUSIONS OF THE STUDY

CATEGORY	INCLUSION	EXCLUSION
Type	Predictive, cognitive, prescriptive, and descriptive analytics solutions	Life sciences and veterinary analytics products and solutions
Component	Software, hardware associated with software installation and storage hardware, and product support services	Hardware such as medical devices
Application	Clinical, financial, operational & administrative, and population health analytics products and services. The market analysis also includes medical imaging analytics, claim processing analytics, and supply chain analytics	Life sciences and veterinary applications
Deployment Model	On-demand and on-premise deployment models	NA
End User	Healthcare providers (hospitals, ASCs, diagnostic & imaging centers, home health agencies & assisted living facilities, and pharmacies), payers, ACOs, HIEs, MCOs, and TPAs	Pharmaceutical & biotechnology and medical device companies

FIGURE 1 HEALTHCARE ANALYTICS MARKET SEGMENTATION

1.2.2 YEARS CONSIDERED FOR THE STUDY



1.3 CURRENCY

The currency used in this report is the United States Dollar (USD), with the market size indicated in USD million/billion.

- For companies reporting their revenues in USD, the revenues were taken from their annual reports
- For companies reporting their revenues in other currencies, the average annual currency conversion rate for the particular year was used to convert the value to USD

1.4 LIMITATIONS

- This report mainly focuses on the analytics solutions used by healthcare organizations. The application of analytics in the life sciences, pharmaceutical, and medical device industries is not considered under the scope of this study.
- Some companies in the market are privately owned, and their revenues are not available in the public domain. Hence, revenues for those companies are not included in the report.
- Company developments not reported in the public domain are not included in the report.
- For public limited companies that have not provided revenues for a particular product segment in their annual reports, analyst assumptions have been made based on certain parameters to attain revenue contributions from particular product segments of interest.
- Market values have been rounded off till the second decimal, owing to which there may be a slight difference in numbers.

1.5 STAKEHOLDERS

- Healthcare IT firms
- Healthcare analytics vendors
- Health insurance exchanges
- Healthcare payers
- Healthcare providers
- Venture capitalists
- Research and consulting firms
- Accountable care organizations (ACOs)

1.6 SUMMARY OF CHANGES

- Refinements in the segments of the healthcare analytics market:
 - The current edition of the report consists of the impact of COVID-19 on the market.
 - Additional data points in the market overview section have been added, keeping in mind the impact of COVID-19 on the overall market.
 - The competitive landscape chapter has been updated with the market evaluation matrix. The competitive landscape also includes the market ranking analysis of the top players, product and geographic footprint of the major companies, and competitive scenario of the global market.
- Addition of market players in the healthcare IT market:
 - The new edition of the report includes profiles for Flatiron (US), Aplixio (US), Sema4 (US), Komodo Health (US), Evidation Health (US), Aetion, Inc. (US), Enlitic, Inc. (US), Roam Analytics (US), CVS Health (US), Insilico Medicine (US), and HealthEC (US).
- Updated financial information/product portfolios of players:
 - The new edition of the report provides updated financial information until 2020 for each listed company in a single diagram (instead of multiple tables). This will easily help analyze the present status of profiled companies in terms of their financial strength, profitability, key revenue-generating regions/countries, business segment focus in terms of the highest revenue-generating segments, and R&D investments.
- Updated market developments of profiled players:
 - Recent developments are important to understand the market trends and growth strategies adopted by players in the market. The current edition of this report provides updated developments of profiled players from January 2018 to July 2021, indicating a continuation of the previous version. Acquisitions, partnerships, and product launches were the principal growth strategies adopted by market players during this period.

2 RESEARCH METHODOLOGY

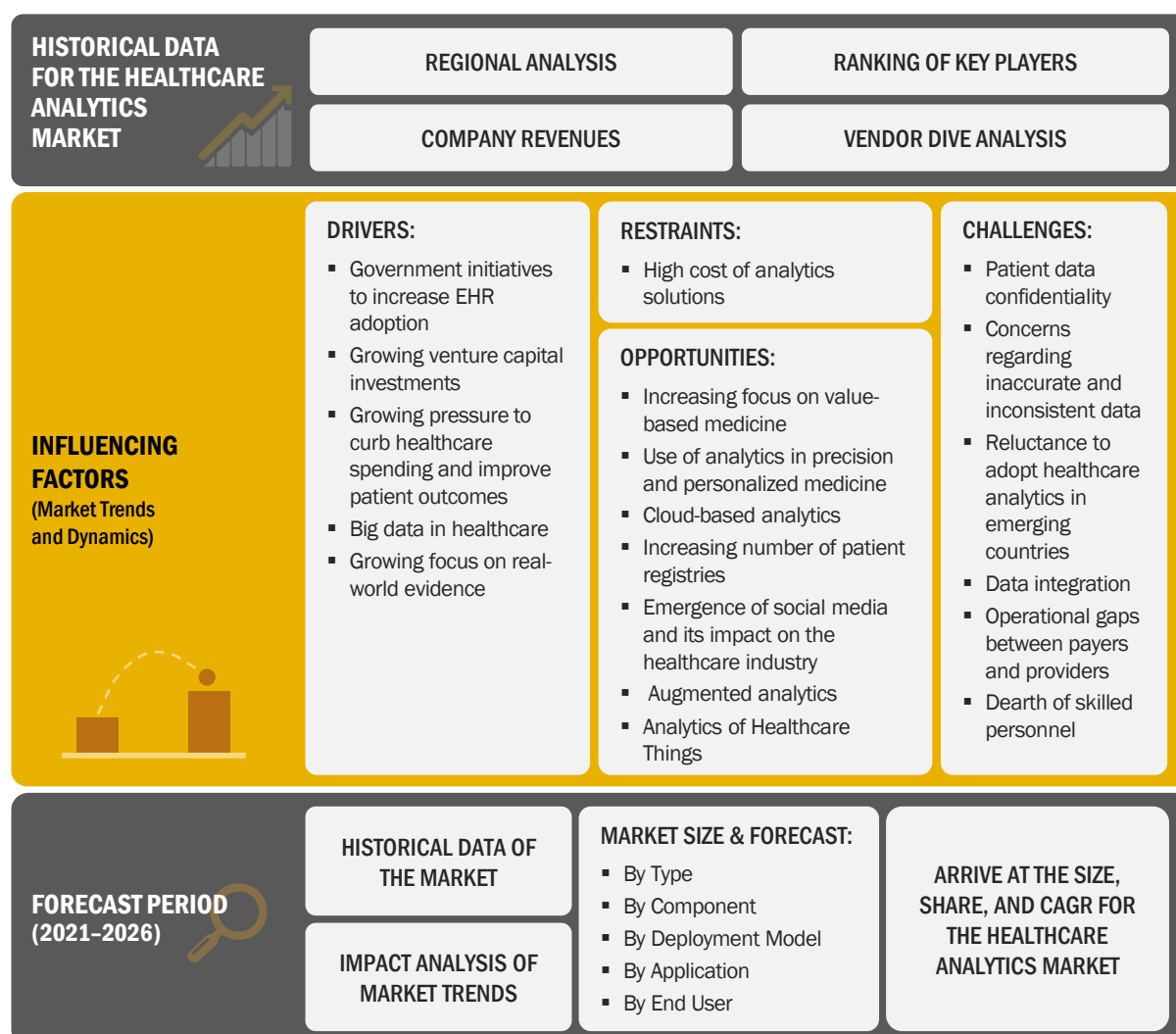
2.1 RESEARCH APPROACH

This report provides a detailed assessment of the healthcare analytics market, along with qualitative inputs and insights from MarketsandMarkets. This research study involved the extensive use of both primary and secondary sources. Various factors affecting the industry were studied to identify segmentation types, industry trends, key players, the competitive landscape of the market, key player strategies, and key market dynamics, such as drivers, restraints, opportunities, and challenges.

The following illustrative figure shows the market research methodology applied in making this report on the healthcare analytics market.

2.2 RESEARCH METHODOLOGY DESIGN

FIGURE 1 HEALTHCARE ANALYTICS MARKET: RESEARCH DESIGN



Source: Secondary Literature, Industry Experts, and MarketsandMarkets Analysis

2.2.1 SECONDARY RESEARCH

This research study involved the usage of widespread secondary sources; directories; databases such as Bloomberg Business, Factiva, and Wall Street Journal; white papers; and annual reports, company house documents, investor presentations, and SEC filings of companies. Secondary research was used to identify and collect information useful for the extensive, technical, market-oriented, and commercial study of the healthcare analytics market. It was also used to obtain important information about key market players, market classification and segmentation according to industry trends to the bottom-most level, and key developments related to market and technology perspectives. A database of the key industry leaders was also prepared using secondary research.

Some of the key secondary sources referred to for this study include:

- World Health Organization (WHO)
- Organisation for Economic Co-operation and Development (OECD)
- World Analytics Association
- Healthcare Data & Analytics Association (HDAA)
- Health Information and Management Systems Society (HIMSS)
- International Medical Informatics Association (IMIA)
- Centers for Medicare & Medicaid Services (CMS)
- National Association of Health Data Organizations (NAHDO)
- American Health Information Management Association (AHIMA)
- Association of Healthcare Providers (AHPI)
- Canadian Institute for Health Information (CIHI)
- Canadian Health Care Anti-Fraud Association (CHCAA)

2.2.1.1 Key data from secondary sources

TYPE	PARAMETER	KEY DATA
REGIONAL ANALYSIS	- Overall market and subsegments in 2020 - CAGR of each region in the forecast period (2021–2026)	Healthcare analytics market, by region—North America, Europe, Asia, Pacific Countries, Latin America, and the Middle East & Africa
MARKET SIZE	- The global market size for 2020 - CAGR for the forecast period (2021–2026)	- Healthcare analytics market - Healthcare analytics market, by type, component, deployment model, application, and end user
MARKET SPLIT	- Descriptive analytics, predictive analytics, prescriptive analytics, and cognitive analytics - Services (support services and business analytics services), software, and hardware - On-premise and on-demand delivery models - Financial analytics, clinical analytics, operational & administrative analytics, and population health analytics - Payers, providers, and ACOS, HIES, MCOS, and TPAS	- Healthcare analytics market by: - Type - Component - Deployment model - Application - End user

2.2.2 PRIMARY DATA

In the primary research process, various sources from both the supply and demand sides were interviewed to obtain qualitative and quantitative information for this report.

The primary sources from the supply side include CEOs, vice presidents, marketing and sales directors, business development managers, and technology and innovation directors of healthcare analytics companies. On the other hand, primary sources from the demand side include industry experts such as directors of hospitals, hospital purchase managers, insurance payers, physicians, and related key opinion leaders.

Some of the major objectives of primary research are:

- To validate market segmentation defined through an assessment of the product portfolios of leading players
- To understand key industry trends and technologies defining the strategic growth objectives of market players
- To gather both demand and supply side validation of the key factors affecting market growth, i.e., market drivers, restraints, opportunities, and challenges
- To validate assumptions for the market sizing and forecasting model used for this study
- To understand the market positions of the leading players in the healthcare analytics market and their market shares/ranking
- To understand ongoing trends in the market and future expectations

FIGURE 2 PRIMARY SOURCES

 SUPPLY SIDE (Healthcare Analytics solutions and services providers)	 DEMAND SIDE (Healthcare Providers, Healthcare Payers, ACOs, HIEs, MCOs, and TPAs)	 OTHERS (Industry Experts)
- CEOs - VPs & Managing Directors - Marketing Heads/Directors and Sales Directors - Marketing Managers - Regional/Area Sales Managers - Product Managers/Technology Experts - Dietitians and Nutrition Experts - Fitness Experts - Counselors and Lifestyle Coaches	- Hospital IT Managers - Physicians and Specialists - HIM Professionals - Medical Consultants - Chief Technology Officers	- Consultants - Government Officials/Regulatory Authorities

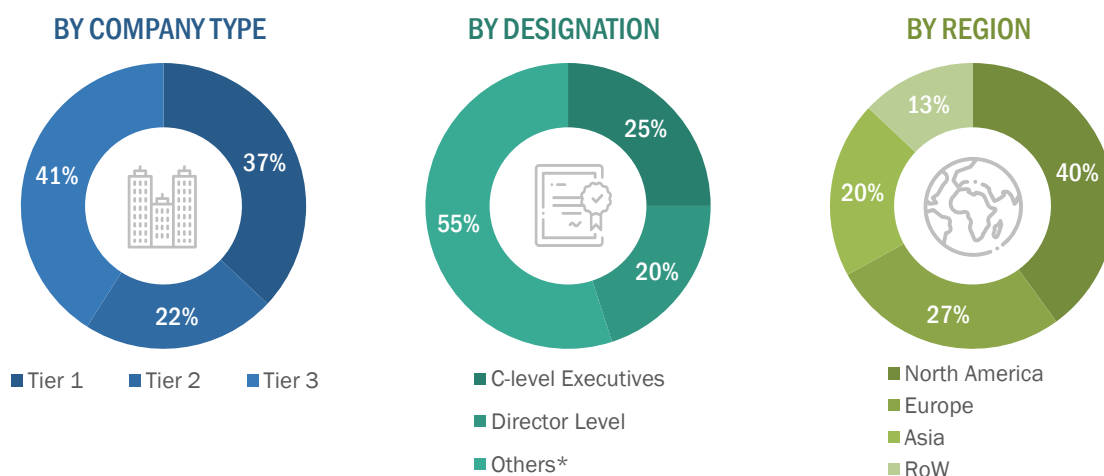
2.2.2.1 Insights from primary experts

FACTORS AFFECTING MARKET GROWTH											
 Healthcare Analytics Consultant, Leading Healthcare Analytics Company	 The adoption of clinical analytics is higher than that of financial analytics due to the rising pressure to curtail healthcare costs and the large volume of healthcare data generated, which contains valuable insights.										
 Vice President, Leading BI Company	 The North American market is estimated to grow at the highest CAGR during the forecast period.										
 Senior BI Consultant, Business Intelligence Company	 Due to the rising focus of payers on the early detection of fraud and reducing preventable costs, the market for fraud analytics is expected to register significant growth during the forecast period.										
MARKET TRENDS	TOP 4 MARKET PLAYERS										
- Virtual care/consultation - Technological innovations, wearable devices, and wellness platforms - Online on-demand nutrition, fitness, and lifestyle coaching services	<table> <tr> <th>COMPANY NAME</th><th>MARKET RANK (2020)</th></tr> <tr> <td>IBM</td><td>1</td></tr> <tr> <td>SAS Institute</td><td>2</td></tr> <tr> <td>Optum</td><td>3</td></tr> <tr> <td>Cerner</td><td>4</td></tr> </table>	COMPANY NAME	MARKET RANK (2020)	IBM	1	SAS Institute	2	Optum	3	Cerner	4
COMPANY NAME	MARKET RANK (2020)										
IBM	1										
SAS Institute	2										
Optum	3										
Cerner	4										

FIGURE 3 BREAKDOWN OF PRIMARY INTERVIEWS: SUPPLY-SIDE AND DEMAND-SIDE PARTICIPANTS

 SUPPLY-SIDE PARTICIPANTS		 DEMAND-SIDE PARTICIPANTS
Cognizant (US)	Xcode Life Sciences (India)	St. Francis Hospital (US)
OpttraHEALTH (US)	Optum (US)	Capital Medical Center (US)
Wipro (India)	Health Catalyst (US)	
Saama (US)	Enlitic (UAE)	

FIGURE 4 BREAKDOWN OF PRIMARY INTERVIEWS: BY COMPANY TYPE, DESIGNATION, AND REGION



Note 1: C-level executives include CEOs, COOs, CTOs, and VPs.

Note 2: Other primaries include sales managers, marketing managers, and product managers.

Note 3: Companies are classified into tiers based on their total revenue. As of 2020: Tier 1 = >USD 1 billion, Tier 2 = USD 500 million to USD 1 billion, and Tier 3 = <USD 500 million.

2.3 MARKET SIZE ESTIMATION: HEALTHCARE ANALYTICS MARKET

TOTAL MARKET SIZE: HEALTHCARE ANALYTICS MARKET

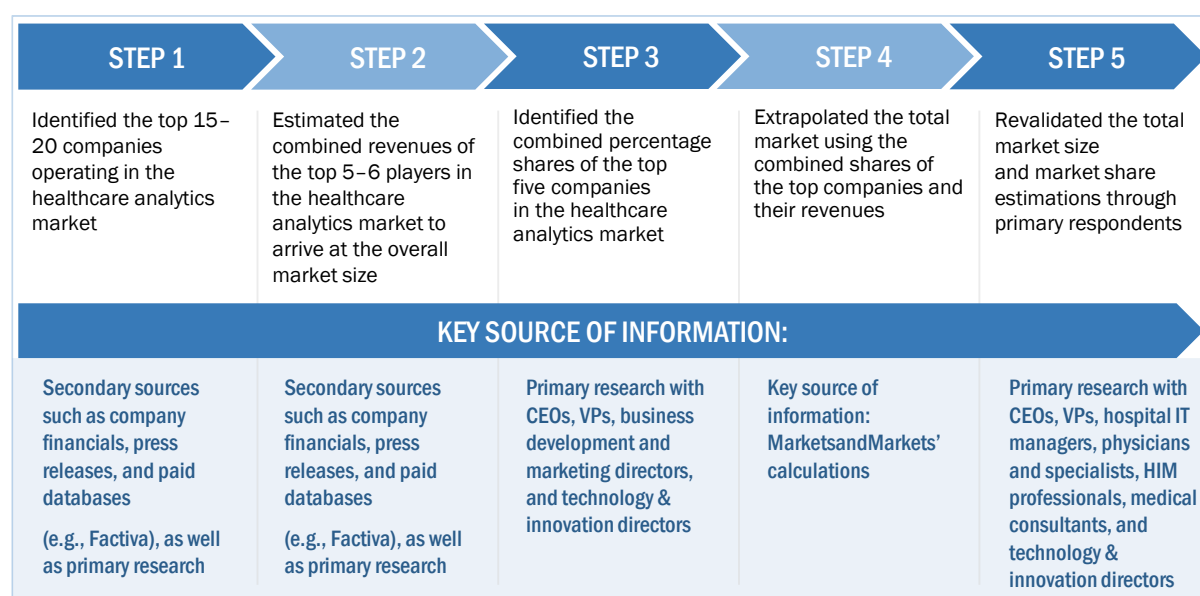
The market size estimates and forecasts provided in this study are derived through a mix of the bottom-up approach (revenue share analysis of the leading players) and top-down approach (assessment of utilization/adoption/penetration trends, by type, component, deployment model, end user, and region).

APPROACH 1: SUPPLY-SIDE ANALYSIS (REVENUE SHARE ANALYSIS) – HEALTHCARE ANALYTICS MARKET

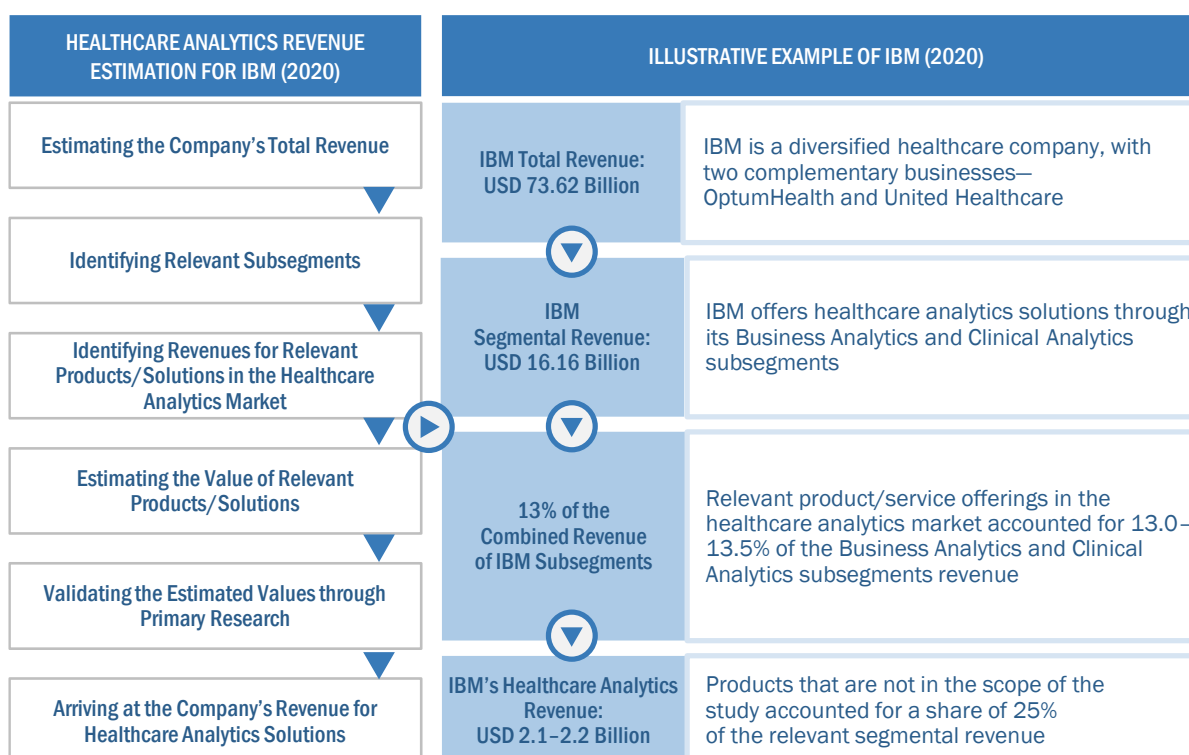
- Revenues of individual companies were gathered from public sources and databases.
- Shares of the healthcare analytics businesses of leading players were gathered from secondary sources to the extent available. In certain cases, shares of the healthcare analytics businesses were ascertained after a detailed analysis of various parameters, including service portfolios, market positioning, average selling price, and geographic reach and strength.
- Individual shares or revenue estimates were validated through expert interviews.

The following figure shows the overall market size estimation process employed for this study.

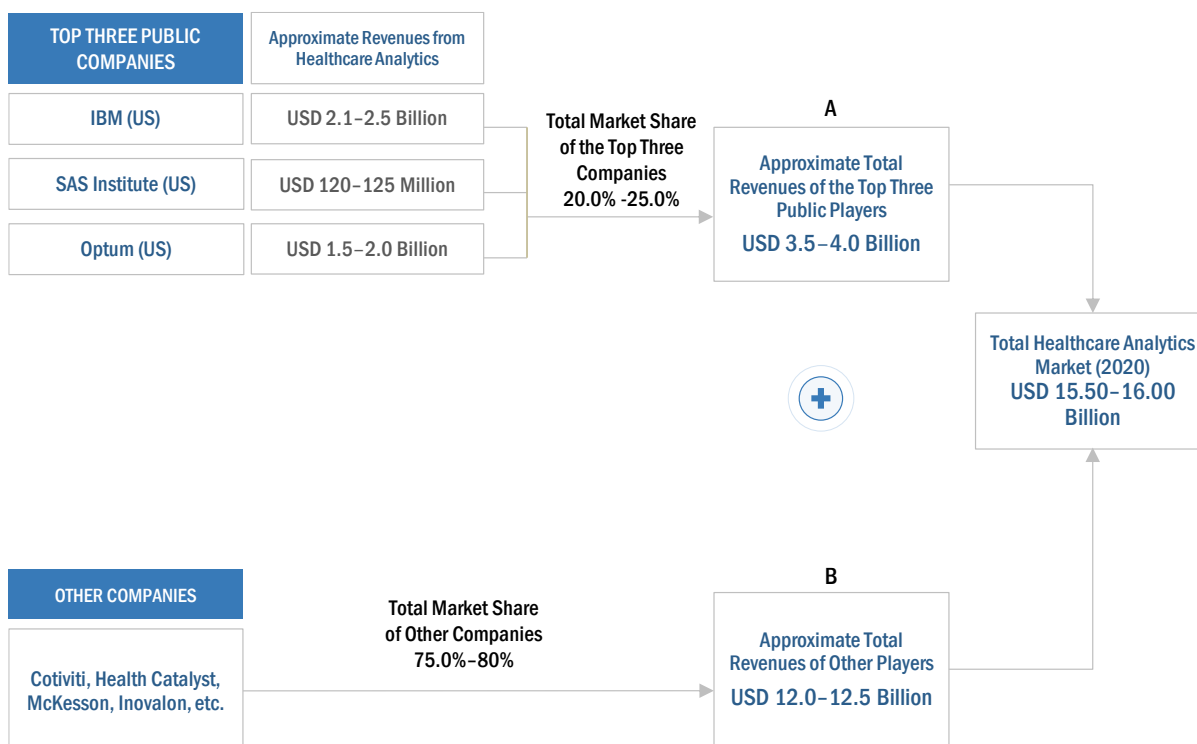
FIGURE 5 SUPPLY-SIDE MARKET SIZE ESTIMATION: REVENUE SHARE ANALYSIS



Source: Annual Reports, SEC Filings, Investor Presentations, Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

FIGURE 6 REVENUE SHARE ANALYSIS ILLUSTRATION

Source: Annual Reports, SEC Filings, Investor Presentations, Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

FIGURE 7 REVENUE ANALYSIS OF THE TOP THREE PUBLIC COMPANIES: HEALTHCARE ANALYTICS MARKET (2020)

Source: Annual Reports, SEC Filings, Investor Presentations, Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

Approach 2: Bottom-up Approach – End-user Adoption (US)

- The size of the healthcare analytics market was estimated through end-user adoption/consumption of these services and solutions in the US.
- The number of small, medium-sized, and large organizations in the US was obtained through public databases.
- The adoption/consumption of healthcare analytics by each organizational category in the US was obtained through secondary research and validated by primary participants.
- The market share of the US in the healthcare analytics market (validated by primaries) was further applied to extrapolate the global market size of the healthcare analytics market.

FIGURE 8 BOTTOM-UP APPROACH: END-USER ADOPTION/CONSUMPTION OF HEALTHCARE ANALYTICS (US)



Source: Annual Reports, SEC Filings, Investor Presentations, Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

APPROACH 3: PARENT MARKET ASSESSMENT

- The healthcare IT (HCIT) market and HCIT ecosystem-related reports from the MarketsandMarkets repository were referred to obtain the market size of the healthcare analytics market.
- The market size of individual segments in the above reports was validated through expert interviews.
- A weighted average of the numbers obtained from the HCIT market and HCIT ecosystem-related reports from the MarketsandMarkets repository was taken to derive the market size of the healthcare analytics market.

APPROACH 4: COMPANY PRESENTATIONS AND PRIMARY INTERVIEWS

- The healthcare analytics market size was obtained from the annual presentations and investor presentations of leading players.
- Shares of different service offerings in the overall healthcare analytics market were obtained from secondary data and validated by primary participants to arrive at the total market size.
- As part of the primary research process, individual respondents' insights on the market size and market growth were taken during the interview (regional and global, as applicable).
- All responses were collated, and a weighted average was taken to derive a probabilistic estimate of the market size and growth rate.

GROWTH FORECAST

The market forecast in this study was arrived at after a detailed assessment of various qualitative and quantitative factors, such as historical revenue growth trends of leading players, major market growth drivers and restraints and their impact over the forecast period, innovation trends, and relevant macro and micro-economic indicators. The growth forecast model was defined based on an assessment of qualitative and quantitative factors affecting the market growth. These mainly included:

- Historic revenue growth trends of leading players in the market
- Growth trends of the parent and peer markets
- Demand-side factor assessment, including growth in the number of small, medium-sized, and large organizations, rising uptake of corporate wellness programs, etc.
- Impact assessment of historic competitive trends driving the market growth. These included the number of service offerings, strategic growth initiatives, innovation trends, and R&D initiatives.
- Impact assessment of regulatory guidelines/mandates on market growth over the forecast period. For this, only the defined regulatory guidelines at the point of the study period have been considered. The impact of any future regulatory uncertainties on market growth is not accounted for.
- Impact assessment of the COVID-19 pandemic has been considered for the short-term forecast for 2020 and 2021.

FIGURE 9 CAGR PROJECTIONS FROM THE ANALYSIS OF DRIVERS, RESTRAINTS, OPPORTUNITIES, AND CHALLENGES OF THE HEALTHCARE ANALYTICS MARKET (2021–2026)

DRIVERS	IMPACT LEVEL
Government initiatives to increase EHR adoption	●
Big data in healthcare	●
Growing pressure to curb healthcare spending and improve patient outcomes	●
RESTRAINTS	RATING
High cost of analytics solutions	●
OPPORTUNITIES	RATING
Increasing focus on value-based medicine	●
Use of analytics in precision and personalized medicine	●
CHALLENGES	RATING
Patient data confidentiality	●
Concerns regarding inaccurate and inconsistent data	●
Reluctance to adopt healthcare analytics in emerging countries	●
CUMULATIVE GROWTH	10.0–20.0%
IMPACT LEVEL : ● 10-15% ● 15-20% ● 20-25%	

Source: Annual Reports, SEC Filings, Investor Presentations, Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

FIGURE 10 CAGR PROJECTIONS: SUPPLY-SIDE ANALYSIS

HEALTHCARE ANALYTICS MARKET CAGR (2021–2026)			
Companies' Revenue Growth	MnM KNOW (Repository Analysis of Adjacent Markets)	Primary Insights	CAGR Projections from the Analysis of DROC's
~15%	~20%	~20-25%	~15-25%
Average CAGR: ~20-25%			
Final CAGR: 25%			

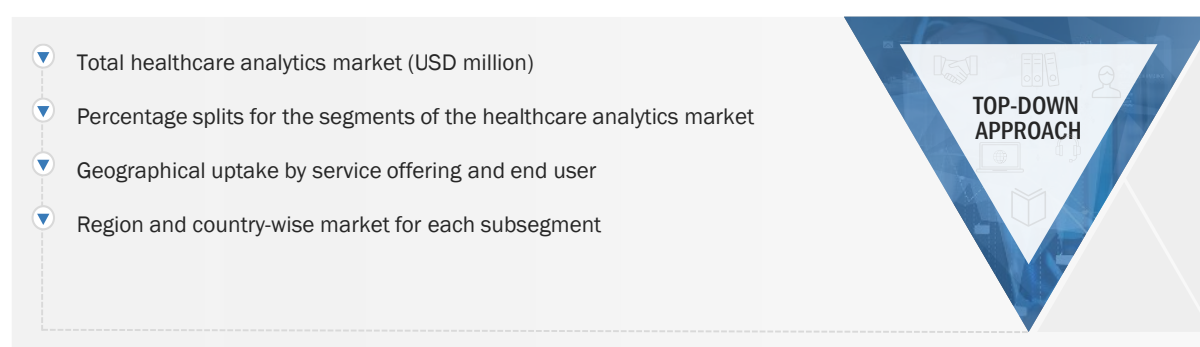
Source: Annual Reports, SEC Filings, Investor Presentations, Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

MARKET SEGMENT ASSESSMENT (BY SERVICE OFFERING AND END USER)

The segmental assessment was conducted using the following approaches:

- Service Offering: Splits for segments were derived by studying the types of service offerings considered in the study and their adoption by end-user organizations. This information was validated through primary interviews.
- End User: Splits for segments were derived by studying end-user adoption trends. This information was validated through primary interviews. Similarly, the adoption of healthcare analytics for each end user was assessed to arrive at the percentage splits.

FIGURE 11 TOP-DOWN APPROACH



Note: Analogy/benchmarking-based estimations have been used to arrive at percentage splits for the product & service segments in countries with limited availability of reliable and recent data

GEOGRAPHIC MARKET ASSESSMENT (BY REGION & COUNTRY)

The geographic assessment was undertaken using the following approaches:

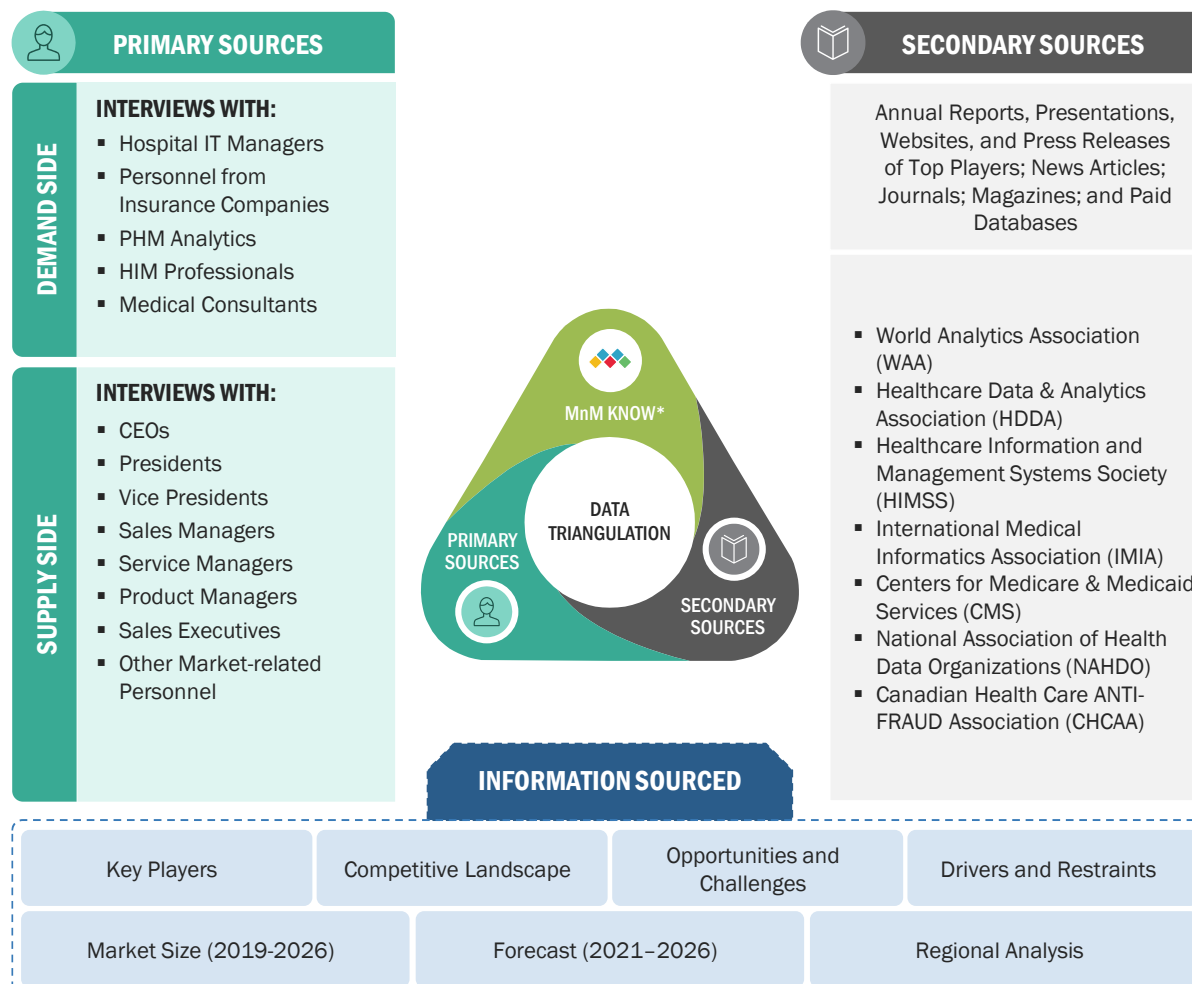
- Approach 1: Geographic revenue contributions/splits of leading players in the market (wherever available), and respective growth trends
- Approach 2: Geographic adoption trends for individual service offering segments and growth prospects for each of the segments (assumptions and indicative estimates validated from primary interviews)

At each point, the assumptions and approaches were validated through industry experts contacted during primary research. Considering the limitations of data available from secondary research, revenue estimates for individual companies (for the overall healthcare analytics market and for geographic market assessment) were ascertained based on detailed analysis of their respective product offerings, geographic reach/strength (direct or through distributors or suppliers), and share of leading players in a particular region or country.

2.4 MARKET BREAKDOWN & DATA TRIANGULATION

After arriving at the market size, the total market was divided into several segments and subsegments. To complete the overall market engineering process and arrive at the exact statistics for all segments and subsegments, data triangulation and market breakdown procedures were employed, wherever applicable. The following figure shows the market validation, source structure, and data triangulation methodology implemented in this report's market engineering process.

FIGURE 12 DATA TRIANGULATION METHODOLOGY



MnM KNOW* stands for MarketsandMarkets' 'Knowledge Asset Management' framework. In this context, it stands for the existing market research knowledge repository of over 5,000 granular markets, our flagship competitive intelligence and market research platform "KnowledgeStore", subject-matter experts, and independent consultants. MnM KNOW acts as an independent source that helps us validate information gathered from primary and secondary sources.

2.5 MARKET SHARE ESTIMATION

Shares of leading market players in the healthcare analytics market were arrived at after a detailed assessment of their revenues using secondary data available through paid and unpaid sources. Owing to data limitations, in certain cases, revenue shares were arrived at after a detailed assessment of the product portfolios of major companies and their respective sales performance. At each point, this data was validated through primary interviews with various industry experts.

2.6 ASSUMPTIONS FOR THE STUDY

INDICATOR/FACTOR	ASSUMPTION	IMPACT
Companies' Revenue Growth Rate	The yearly revenue growth of companies was considered to estimate the growth rate of the overall market	Low
Classification of Segmentation	All market segments and subsegments listed in this report through various forms of classification have been considered to be mutually exclusive of each other	Low
Segmental Splits and Projections	Wherever exact segmental market splits were not available, the historical trends and inputs from primary correspondents have been considered, and accordingly, the splits and projection rates have been assumed	Moderate
Global Economic and Regulatory Factors	The CAGR for the forecast period was assumed to be normalized, and unforeseen regulatory or policy changes during the forecast period were not considered. No downturns and economic collapses, apart from COVID-19, have been considered while estimating the market size in this research study	High
Global Political Outlook	The political environment was assumed to remain stable between 2021 and 2026. In general, these scenarios are not considered for the parent market analysis and are not expected to have a great impact on the market	Low
COVID-19 Pandemic	The current COVID-19 pandemic has been considered in the market assessment during 2020–2021	Medium

2.7 LIMITATIONS

2.7.1 METHODOLOGY-RELATED LIMITATIONS

- End-user adoption (demand-side) is used to cross-verify the market size determined from the supply side.
- All forecast values are based on historical data. Dynamic price variations may change the projected market size.
- The impact of COVID-19 has been considered. This may have a long-term impact on the healthcare analytics market as the pandemic scenario unfolds.

2.7.2 SCOPE-RELATED LIMITATIONS

Market revenues generated from healthcare analytics not provided by healthcare analytics market players are beyond the scope of the study and have been excluded.

2.8 RISK ASSESSMENT

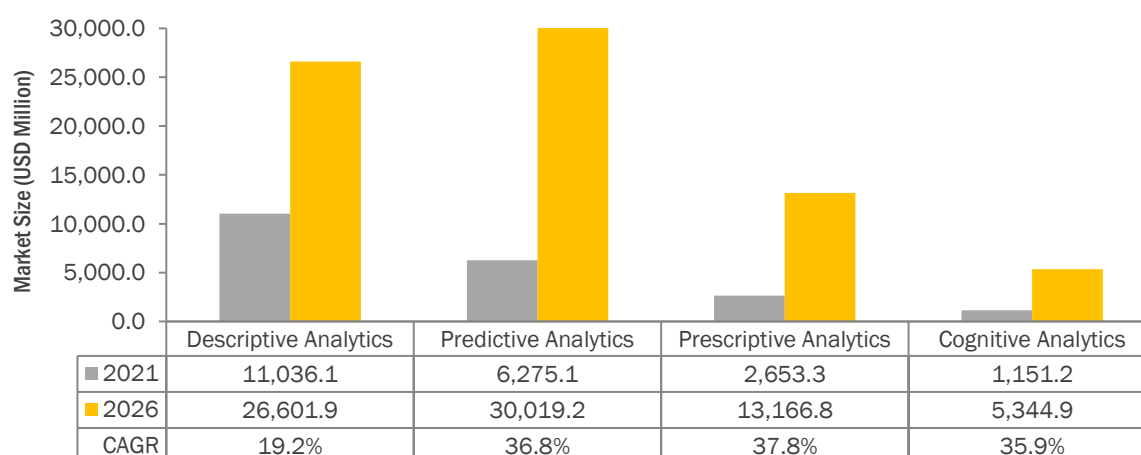
TABLE 1 RISK ASSESSMENT: HEALTHCARE ANALYTICS MARKET

DESCRIPTION	RISK ANALYSIS
The revenue generated for healthcare analytics market players was assumed for a few market players based on the product portfolio analysis and their concerned segmental revenues	High
End-user adoption in the healthcare analytics market has been considered based on the data available in the public domain regarding the number of registered organizations in the small, medium-sized, and large categories	Low
Segmental splits for service offerings in the healthcare analytics market have been provided after considering the adoption rates in the pre-COVID-19 and post-COVID-19 pandemic scenarios, which could vary region-wise	High
Segmental splits for end users in the healthcare analytics market have been provided after considering the adoption rates in the pre-COVID-19 and post-COVID-19 pandemic scenarios, which could vary region-wise.	Moderate
Revenues from the sales of products, such as wearable devices, have not been considered in the market estimation. Only revenues from services/solutions offered through these devices in the healthcare analytics market have been considered	Moderate

3 EXECUTIVE SUMMARY

The global healthcare analytics market is projected to reach USD 75.13 billion by 2026 from USD 21.11 billion in 2021, at a CAGR of 28.9% during the forecast period. The growth of this market is mainly driven by factors such as government initiatives to increase EHR adoption, growing venture capital investments, growing pressure to curb healthcare spending and improve patient outcomes, growing focus on real-world evidence, and availability of big data in healthcare. However, the high cost of analytics solutions is expected to restrain the growth of this market to a certain extent.

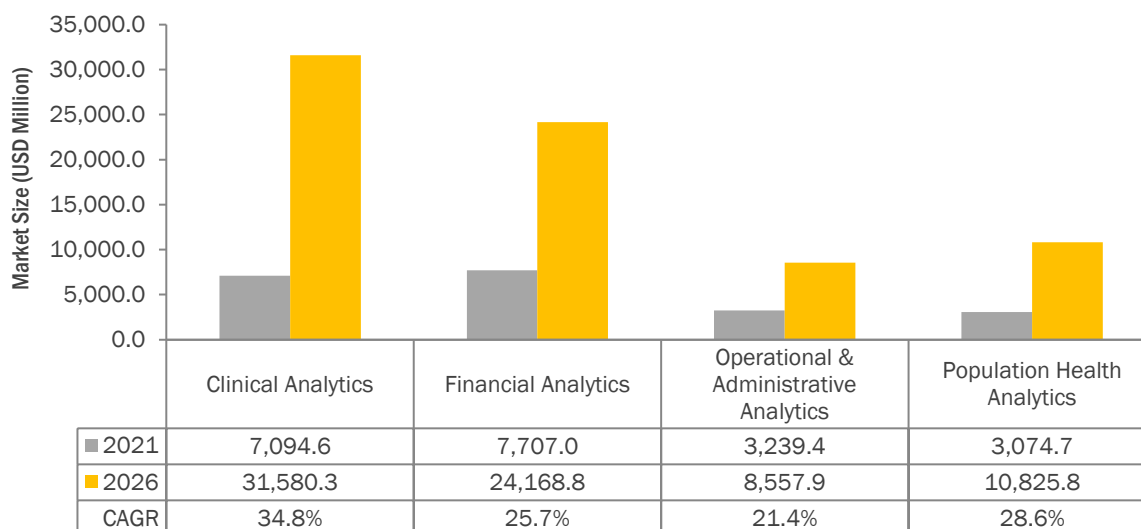
FIGURE 13 HEALTHCARE ANALYTICS MARKET, BY TYPE, 2021 VS. 2026 (USD MILLION)



Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association, American Health Information Management Association, National Health Care Anti-fraud Association, American Medical Informatics Association, Certification Commission for Health Information Technology, Healthcare Information Technology Standards Panel, Health Level Seven International, Information Technology Association of Canada, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network, Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

Based on type, the healthcare analytics market is segmented into descriptive, predictive, prescriptive, and cognitive analytics. The descriptive analytics segment is expected to account for the largest share of 52.3% of the healthcare analytics market in 2021, while the prescriptive analytics segment is projected to grow at the highest CAGR of 37.8% during the forecast period. Descriptive analytics forms the base for the effective application of predictive or prescriptive analytics. Therefore, descriptive analytics is expected to dominate the healthcare analytics market during the forecast period.

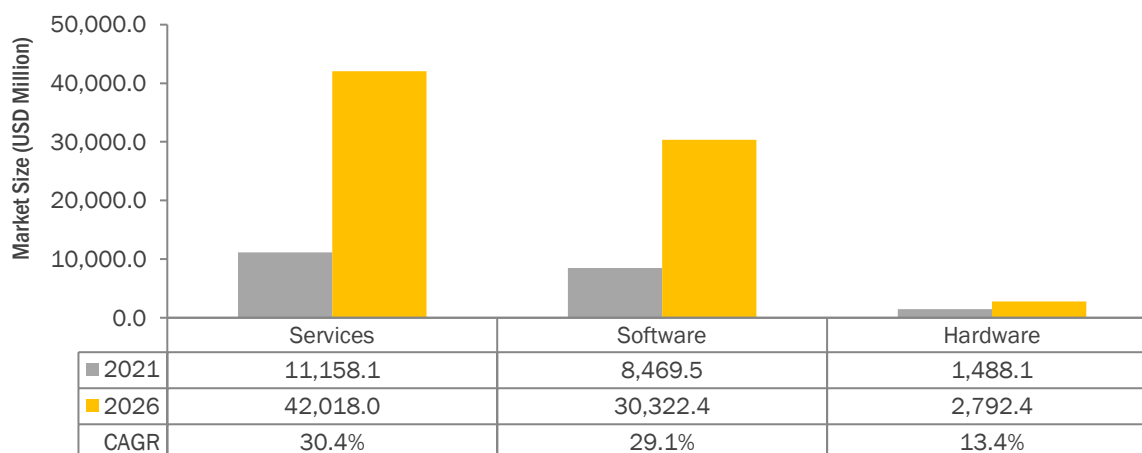
FIGURE 14 HEALTHCARE ANALYTICS MARKET, BY APPLICATION, 2021 VS. 2026 (USD MILLION)



Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association, American Health Information Management Association, National Health Care Anti-fraud Association, American Medical Informatics Association, Certification Commission for Health Information Technology, Healthcare Information Technology Standards Panel, Health Level Seven International, Information Technology Association of Canada, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network, Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

By application, the healthcare analytics market is segmented into financial analytics, clinical analytics, operational & administrative analytics, and population health analytics. The financial analytics segment is estimated to command the largest share of 36.5% of the healthcare analytics market in 2021, while the clinical analytics segment is projected to grow at the highest CAGR of 34.8% during the forecast period. The high growth of this segment is attributed to the growing pressure to curb healthcare costs, federal mandates such as the implementation of ICD-10 code sets, increasing adoption of electronic healthcare records, the need to improve patient outcomes and reduce hospital readmission rates, and rising focus on personalized medicine-based analytics.

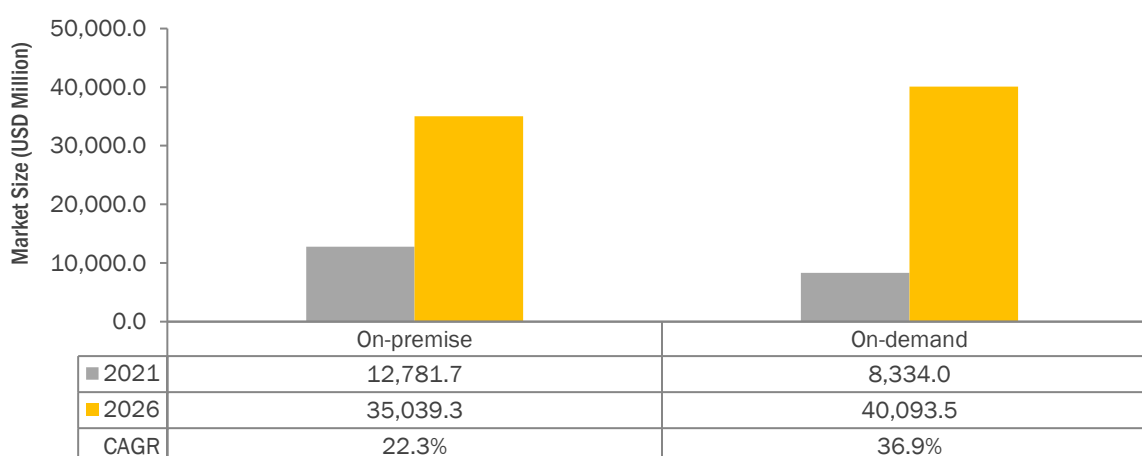
FIGURE 15 HEALTHCARE ANALYTICS MARKET, BY COMPONENT, 2021 VS. 2026 (USD MILLION)



Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association, American Health Information Management Association, National Health Care Anti-fraud Association, American Medical Informatics Association, Certification Commission for Health Information Technology, Healthcare Information Technology Standards Panel, Health Level Seven International, Information Technology Association of Canada, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network, Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

Based on component, the healthcare analytics market is segmented into services, software, and hardware. In 2021, the services segment is expected to account for the largest share of 52.8% of the healthcare analytics market. This segment is also expected to grow at the highest CAGR during the forecast period, primarily due to the recurring need for services such as software upgrades and maintenance.

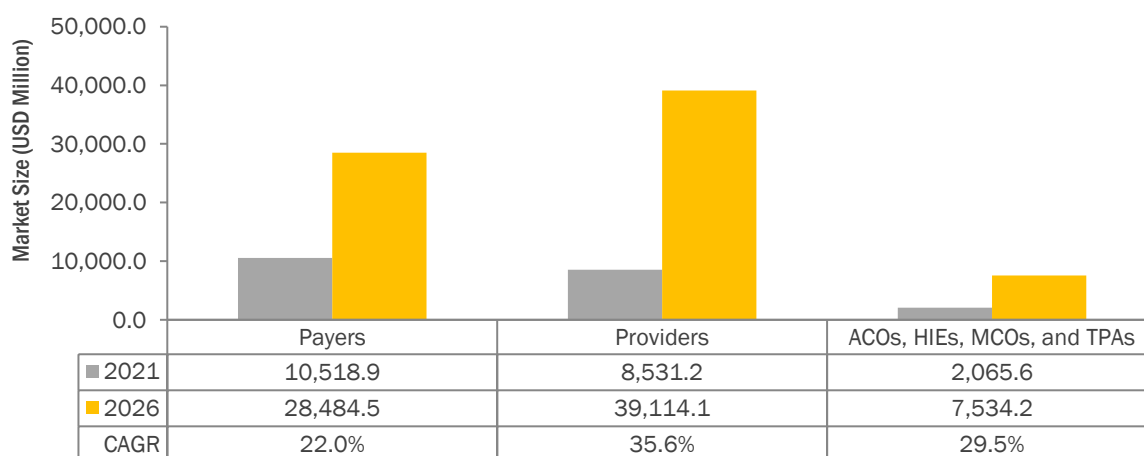
FIGURE 16 HEALTHCARE ANALYTICS MARKET, BY DEPLOYMENT MODEL, 2021 VS. 2026 (USD MILLION)



Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association, American Health Information Management Association, National Health Care Anti-fraud Association, American Medical Informatics Association, Certification Commission for Health Information Technology, Healthcare Information Technology Standards Panel, Health Level Seven International, Information Technology Association of Canada, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network, Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

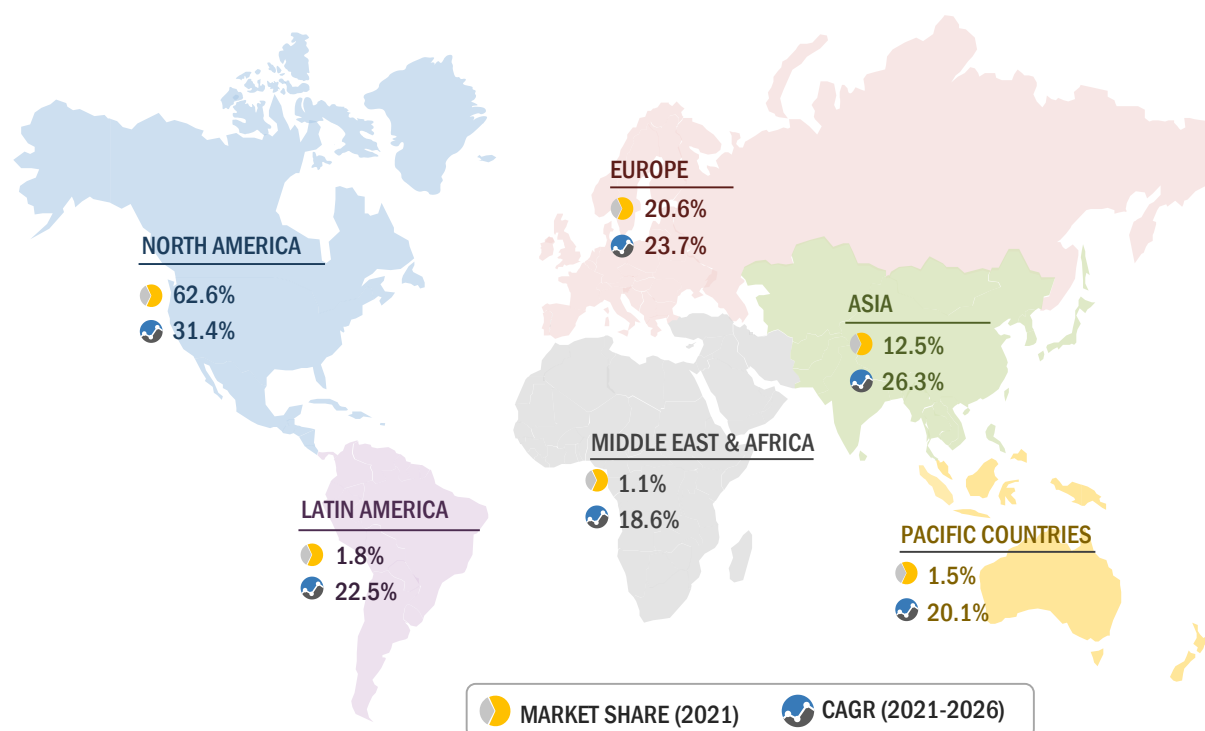
By deployment model, the healthcare analytics market is segmented into on-premise and on-demand models. The on-premise deployment models segment is expected to account for the larger market share of 60.5% in 2021, while the on-demand segment is projected to register the highest growth rate during the forecast period. The on-demand deployment model is the fastest and easiest way of deploying healthcare analytics solutions. On-demand healthcare analytics solutions are sold based on subscription, ranging from one month to multiple years. As compared to on-premise solutions, on-demand solutions do not require significant setup charges. Moreover, the maintenance and support service charges are included in the subscription fee, making these solutions more affordable for customers.

FIGURE 17 HEALTHCARE ANALYTICS MARKET, BY END USER, 2021 VS. 2026 (USD MILLION)



Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association, American Health Information Management Association, National Health Care Anti-fraud Association, American Medical Informatics Association, Certification Commission for Health Information Technology, Healthcare Information Technology Standards Panel, Health Level Seven International, Information Technology Association of Canada, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network, Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

Based on end users, the healthcare analytics market is segmented into payers, providers, and ACOs, HIEs, MCOs, and TPAs. In 2021, the payers segment is estimated to account for the largest share of 49.8% of the healthcare analytics market. The growing need to curb fraudulent claims, optimize provider networks, and reduce operational costs is the key factor driving the growth of this segment.

FIGURE 18 GEOGRAPHICAL SNAPSHOT OF THE HEALTHCARE ANALYTICS MARKET

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association, American Health Information Management Association, National Health Care Anti-fraud Association, American Medical Informatics Association, Certification Commission for Health Information Technology, Healthcare Information Technology Standards Panel, Health Level Seven International, Information Technology Association of Canada, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network, Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

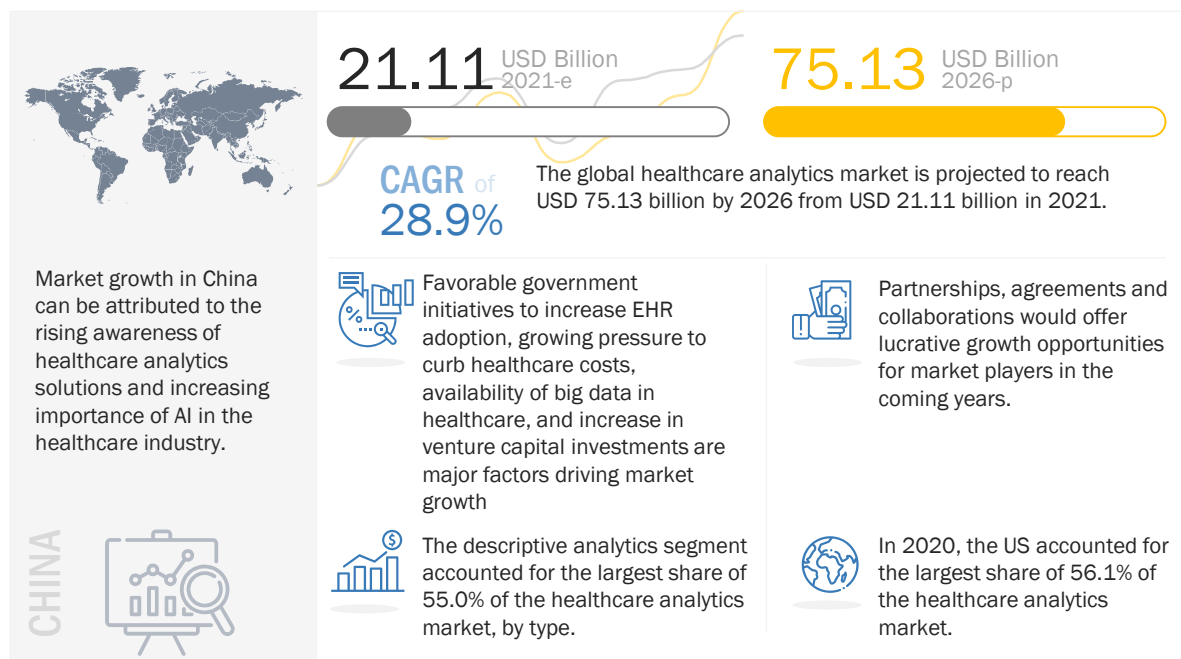
North America dominates the healthcare analytics market in 2021, and this trend is expected to continue during the forecast period. This regional segment is also expected to grow at the highest CAGR during the forecast period. Factors such as growing federal healthcare mandates to curb rising healthcare costs, increasing regulatory requirements, growing EHR adoption, and rising government initiatives focusing on personalized medicine, population health management, and value-based reimbursements are expected to drive market growth in North America.

The healthcare analytics market is consolidated and competitive in nature. Major players in this market include IBM Corporation (US), Optum, Inc. (US), Cerner Corporation (US), SAS Institute, Inc. (US), Allscripts Health Solutions (US), McKesson Corporation (US), MedeAnalytics, Inc. (US), Inovalon (US), Oracle Corporation (US), Health Catalyst (US), SCIO Health Analytics (US), Verscend Technologies, Inc. (US), CitiusTech, Inc. (US), Wipro Limited (India), and VitreosHealth (US). Partnerships, agreements, collaborations, and alliances are the key strategies adopted by players to enhance their presence in the healthcare analytics market.

4 PREMIUM INSIGHTS

4.1 HEALTHCARE ANALYTICS MARKET OVERVIEW

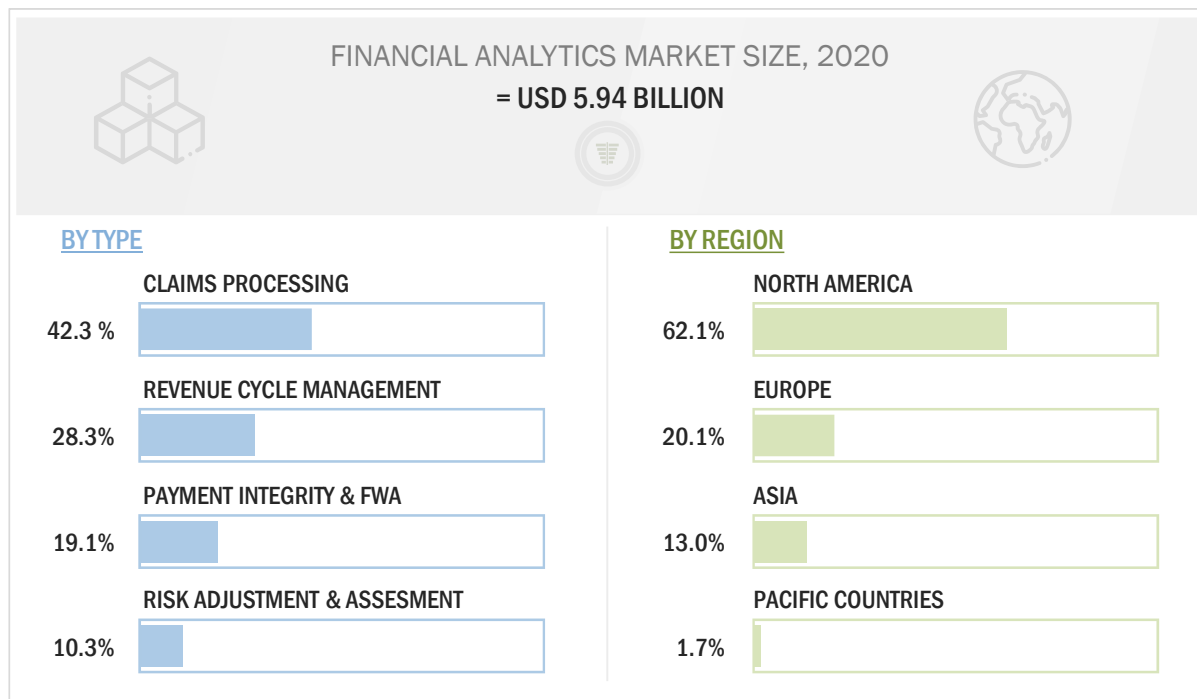
FIGURE 19 GOVERNMENT INITIATIVES TO INCREASE EHR ADOPTION AND GROWING PRESSURE TO CURB HEALTHCARE COSTS ARE DRIVING MARKET GROWTH



Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association, American Health Information Management Association, National Health Care Anti-fraud Association, American Medical Informatics Association, Certification Commission for Health Information Technology, Healthcare Information Technology Standards Panel, Health Level Seven International, Information Technology Association of Canada, Canadian Anti-Fraud Centre, European Healthcare Fraud & Corruption Network, Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

4.2 FINANCIAL ANALYTICS MARKET SHARE, BY TYPE & REGION (2020)

FIGURE 20 NORTH AMERICA TO HOLD THE LARGEST SHARE OF THE FINANCIAL ANALYTICS MARKET IN 2020

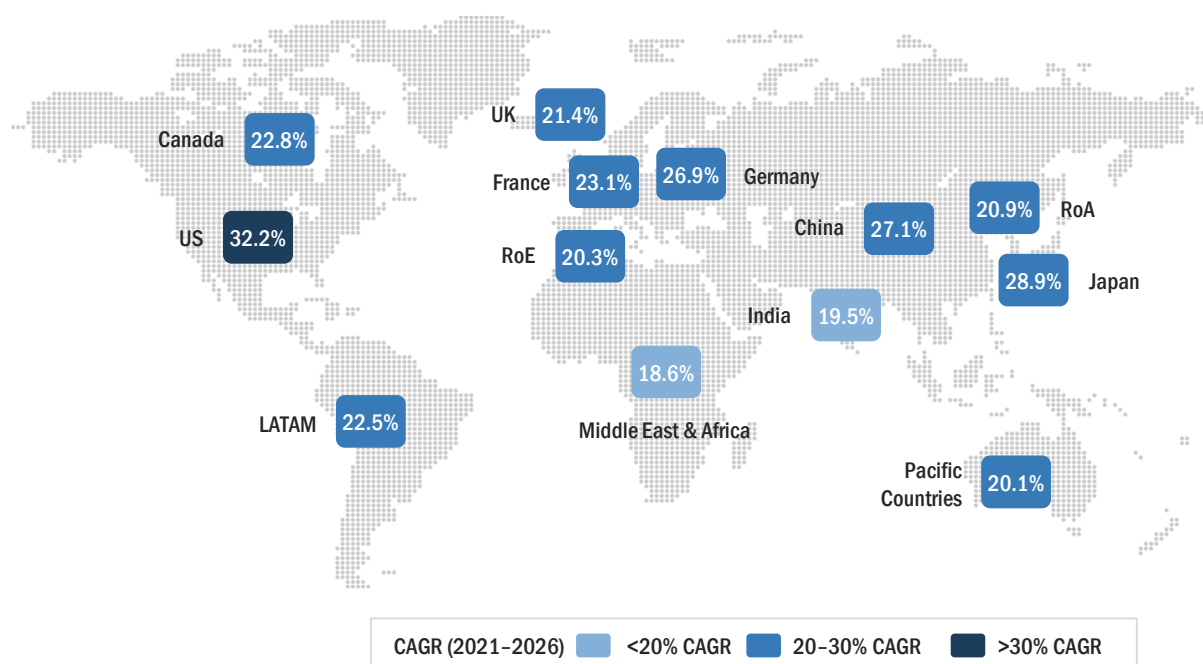


*Note: The regional shares might not add up to 100% as only major regions are considered in this section.

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association, American Health Information Management Association, National Health Care Anti-fraud Association, American Medical Informatics Association, Certification Commission for Health Information Technology, Healthcare Information Technology Standards Panel, Health Level Seven International, Information Technology Association of Canada, Canadian Anti-Fraud Centre, European Healthcare Fraud & Corruption Network, Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

4.3 HEALTHCARE ANALYTICS MARKET: GEOGRAPHIC MIX

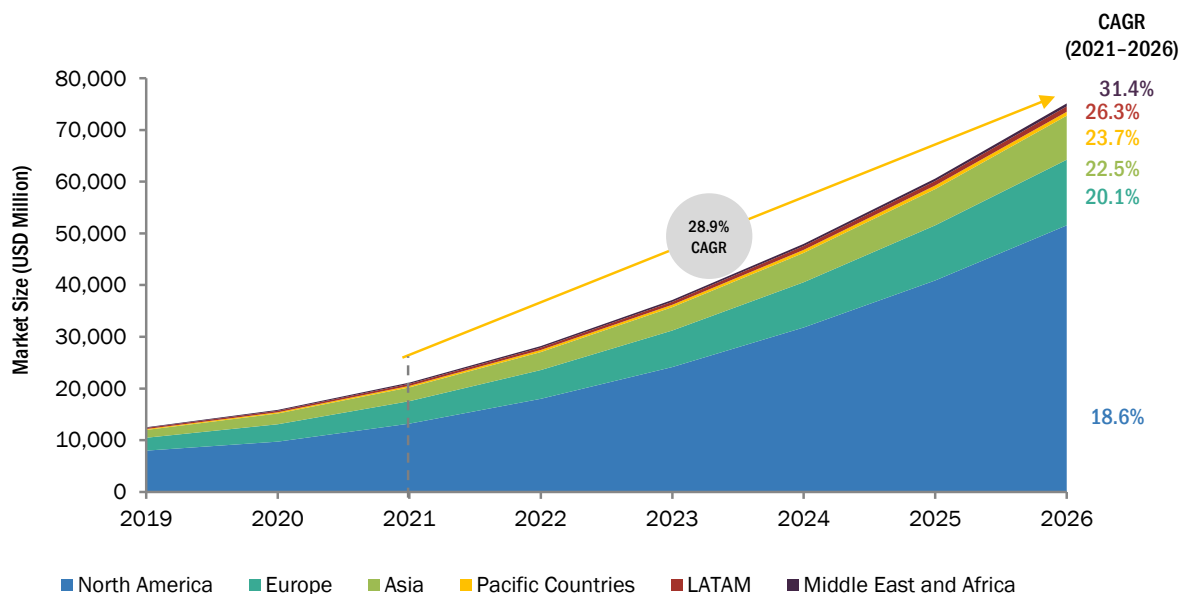
FIGURE 21 THE US TO WITNESS THE HIGHEST GROWTH DURING THE FORECAST PERIOD



Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association, American Health Information Management Association, National Health Care Anti-fraud Association, American Medical Informatics Association, Certification Commission for Health Information Technology, Healthcare Information Technology Standards Panel, Health Level Seven International, Information Technology Association of Canada, Canadian Anti-Fraud Centre, European Healthcare Fraud & Corruption Network, Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

4.4 REGIONAL MIX: HEALTHCARE ANALYTICS MARKET

FIGURE 22 NORTH AMERICA DOMINATED THE MARKET IN 2020



Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association, American Health Information Management Association, National Health Care Anti-fraud Association, American Medical Informatics Association, Certification Commission for Health Information Technology, Healthcare Information Technology Standards Panel, Health Level Seven International, Information Technology Association of Canada, Canadian Anti-Fraud Centre, European Healthcare Fraud & Corruption Network, Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

5 MARKET OVERVIEW

5.1 INTRODUCTION

The growth of the global healthcare analytics market is primarily driven by the implementation of government initiatives to increase EHR adoption. Growing venture capital investments, the emergence of big data in healthcare, increasing focus on curbing the rising healthcare costs, and the growing focus on real-world evidence are some of the other key factors driving the growth of this market. However, market growth is expected to be restrained because of the high cost of analytics solutions.

5.2 MARKET DYNAMICS

FIGURE 23 HEALTHCARE ANALYTICS MARKET: DRIVERS, RESTRAINTS, OPPORTUNITIES, AND CHALLENGES

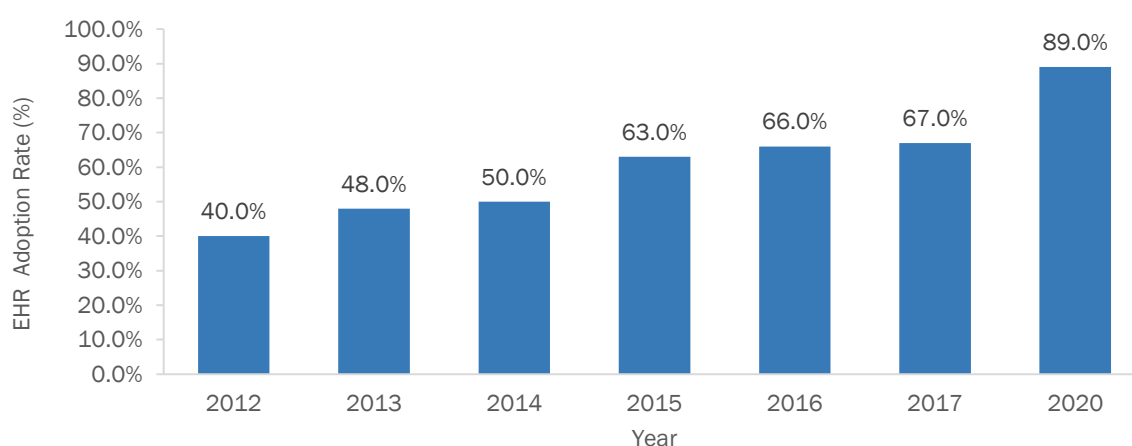
 DRIVERS	▪ Government initiatives to increase EHR adoption ▪ Growing venture capital investments ▪ Growing pressure to curb healthcare spending and improve patient outcomes ▪ Big data in healthcare ▪ Growing focus on real-world evidence
 RESTRAINTS	▪ High cost of analytics solutions
 OPPORTUNITIES	▪ Increasing focus on value-based medicine ▪ Use of analytics in precision and personalized medicine ▪ Cloud-based analytics ▪ Increasing number of patient registries ▪ Emergence of social media and its impact on the healthcare industry ▪ Augmented analytics ▪ Analytics of Healthcare Things
 CHALLENGES	▪ Patient data confidentiality ▪ Concerns regarding inaccurate and inconsistent data ▪ Reluctance to adopt healthcare analytics solutions in emerging countries ▪ Data integration ▪ Operational gaps between payers and providers ▪ Dearth of skilled personnel

5.2.1 DRIVERS

5.2.1.1 Government initiatives to increase EHR adoption

In the US, various federal mandates, such as the American Recovery and Reinvestment Act of 2009 (ARRA) and the implementation of ICD-10 code sets are encouraging healthcare organizations to adopt EHRs, focus on the quality of care to comply with pay-for-performance initiatives, and enhance information exchange between various health systems. The implementation of such initiatives is helping providers gather patient data electronically and supporting data integration and chronic disease management. This ultimately helps prevent hospital readmissions and reduces hospital stays, minimizes claims fraud and abuse, improves patient health, and reduces healthcare costs.

FIGURE 24 EHR ADOPTION RATE AMONG HEALTHCARE PROVIDERS IN THE US (2012–2020)



Source: Practice Fusion, Inc.

As of July 2019, more than 500,000 healthcare providers received payment for participating in the Medicare and Medicaid EHR Incentive Program.

Several other federal regulations in the US compel healthcare organizations to focus on digitization and care quality; these include the Health Insurance Portability and Accountability Act (HIPAA) and the Patient Protection and Affordable Care Act (PPACA). The growing need for hospitals in the US to comply with federal healthcare mandates is encouraging them to use data analytics tools for decision-making, improving care quality, and preventing avoidable costs. Since the US is a major contributor to the healthcare analytics market, the implementation of these initiatives is considered a positive indicator of the growth of the overall market.

In the UK, NHS England is developing a program for collecting data and linking electronic records from general practices with those from hospitals and other care settings (for research purposes and planning in the healthcare and social care services sectors) in the UK. The full implementation of this program has been delayed due to concerns about confidentiality. However, improvements in EHR facilities and improved data protection protocols have enabled the steady growth in the adoption of these systems.

In Asia, governments are increasingly focusing on adopting EHRs and utilizing digital technologies. In China, as of 2018, around 85.3% of hospitals have implemented EHRs in their facilities. It was expected that by 2020, 80% of tertiary hospitals, 50% of urban community health centers, and 50% of rural hospitals would have their own EHR system (Source: Clinical Leader). The government of Hong Kong intends to increase EHR adoption in its healthcare system from 30% in 2015 to 60% by 2019–2020. Similarly, according to an article published in January 2017, the Government of India intends to introduce a uniform system for the maintenance of EHRs/EMRs by hospitals and healthcare providers in the country.

Innovative solutions, such as analytics solutions, extract, aggregate, and interpret the massive volume of data stored within EHR systems into actionable, useful information that can be used by physicians to improve the quality of care. As a result, the growth of the healthcare analytics market can be expected to follow that of the EHRs market.

5.2.1.2 Growing venture capital investments

The potential of healthcare analytics to boost efficiency and ensure better care quality overall has attracted significant venture capital to this sector. Several venture capitalists are currently investing in healthcare analytics startups, with investments majorly focused on enabling startup firms to develop advanced healthcare analytics solutions, such as predictive analytics and prescriptive analytics. Venture capital funding for health tech innovators increased to USD 14 billion in 2019, double that of the previous year (Source: HealthITAnalytics). Healthcare analytics companies secured over USD 1.8 billion in venture capital investments in 2020 (Source: Mercom Capital Group).

Many significant deals have been carried out in recent years to support healthcare analytics market vendors in the successful implementation of their strategic plans. Some of these developments have been mentioned below:

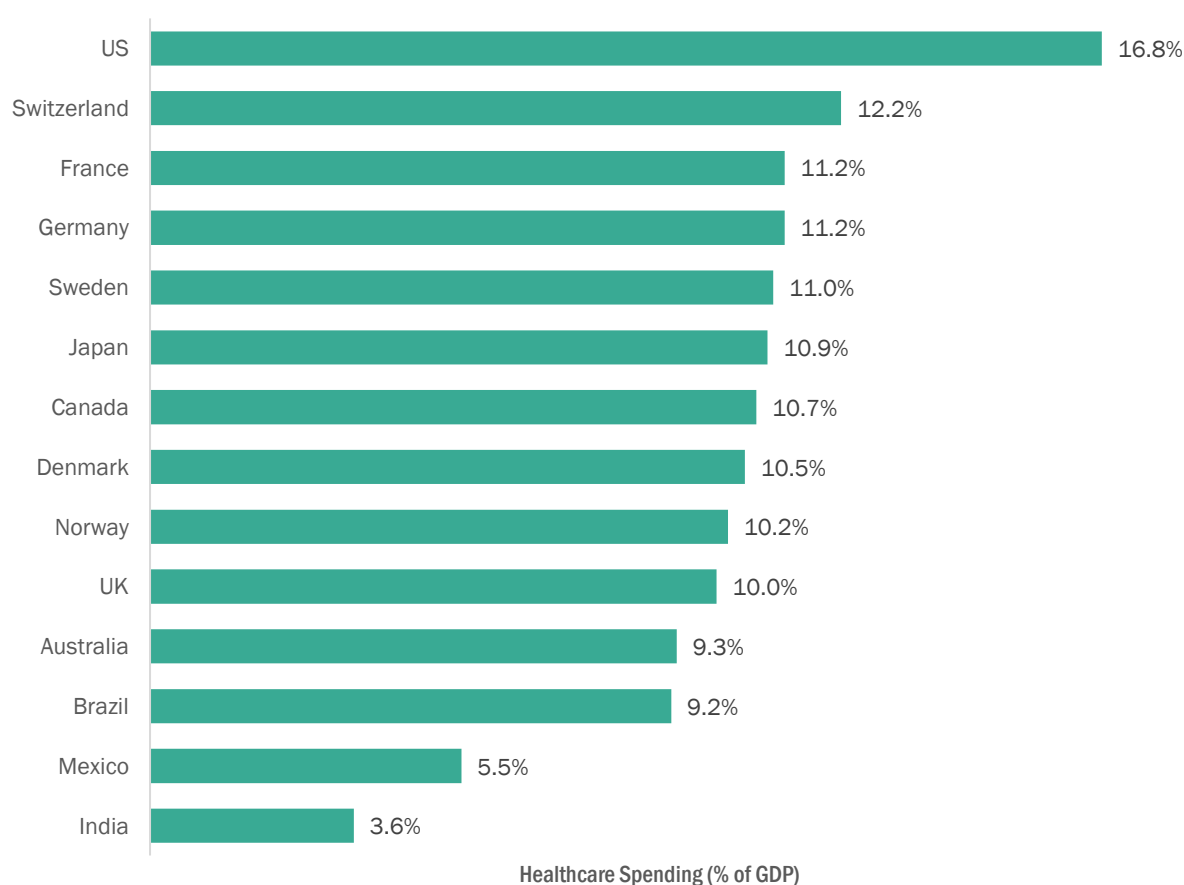
- In March 2021, Komodo Health received USD 220 million Series E funding led by Tiger Global Management. The investors were SVB Capital, Andreessen Horowitz, and ICONIQ Growth. The investment will help improve Komodo Health's product offerings and accelerate further investments in the future.
- In May 2020, Flatiron raised over USD 8 million in Series A funding led by Google Ventures. The investment will help the company upgrade its pilot software platforms into beta testing versions for public use.
- In November 2019, Enlitic, Inc. secured USD 25 million in the Series B-1 round. The company will use the funding in the next phase build-out of its AI software platform and further grow its global footprint with major regional partners in the healthcare industry, paving the way for market deployment in 2020.
- In May 2018, ThoughtSpot (US) received USD 145 million in Series D funding from Lightspeed Ventures, Sapphire Ventures, Khosla Ventures, and General Catalyst to develop AI-driven analytics solutions for optimizing operational applications.
- In June 2018, the NHS received USD 36 million (GBP 20 million) from the UK government for the development of AI-driven analytics tools.
- In February 2018, analytics start-up THB (India) received INR 14 crore (USD 2.1 million) from Blume Venture and HealthQuad.

The availability of funding will enable startups and existing firms to develop innovative products and expand their product portfolios.

5.2.1.3 Growing pressure to curb healthcare spending and improve patient outcomes

Globally, the cost of healthcare is on the rise due to the growing demand for healthcare services, the development of new and expensive prescription drugs and medical technologies, the rising prevalence of chronic diseases, and lack of transparency of information on prices and quality. Operational inefficiencies, hospital readmissions, medical errors, and adverse drug events are the other major factors adding to the increasing healthcare costs. The US, for example, spent USD 3.81 trillion on healthcare in 2019; this figure is estimated to reach USD 6.19 trillion by 2028 and will account for 19.7% of the country's GDP (Source: CMS Data).

FIGURE 25 HEALTHCARE SPENDING AS A PERCENTAGE OF GDP, BY COUNTRY, 2018



Source: OECD

There is increasing pressure on healthcare systems across the globe to reduce healthcare costs, and at the same time, provide better-quality patient outcomes. The need to curb the rising healthcare costs is one of the major growth drivers for the healthcare business intelligence solutions market, which in turn would result in the greater adoption of preventive care approaches, better chronic disease management, optimum operational efficiency, better care coordination, minimal hospital readmissions, and reduced medical errors and adverse drug events. On the other hand, BI helps healthcare payers improve their claims administration process by reducing errors and eliminating rework; it can also help them in identifying and preventing fraudulent claims, thereby significantly reducing the time and cost spent on processing claims.

5.2.1.4 Big data in healthcare

Big data mainly refers to the extremely large datasets that are difficult to analyze with traditional tools. This data includes both structured and unstructured data; however, 90% of big data is unstructured. The amount of data being processed in the healthcare industry is expanding rapidly as data arrays are growing at a high rate.

In the healthcare industry, big data comprises information from clickstream and interactions on the web and social media; readings from medical devices such as sensors, ECGs, X-rays, and pulse oximeters; healthcare claims and other billing records; biometric data; EMRs; and prescriptions. With the increasing digitization and adoption of information systems in healthcare, a large volume of data is being generated, which is measured and analyzed with the help of machine-learning algorithms. According to an article published in April 2016, the volume of big data in healthcare stood roughly at 153 exabytes in 2013 and may increase to 2,314 exabytes by 2020 (Source: International Journal of Advanced Research in Computer Science and Software Engineering).

It has become increasingly important that such multiple streams of data be leveraged with data aggregation and analytics techniques to advance the delivery of healthcare for individual patients as well as disease-specific and condition-specific populations.

5.2.1.5 Growing focus on real-world evidence

Real-world evidence (RWE) is clinical evidence regarding the use and potential benefits or risks of a medical product. Data derived from real-world sources can be used to support regulatory decisions. Depending on the characteristics of data, real-world data (RWD) and associated RWE may constitute valid scientific evidence.

Factors such as the rising need to personalize drug therapy and treatment, limited coverage of randomized clinical trials, and growing patient volumes have made RWE important for healthcare providers. Randomized control trials (RCTs), the gold standard for measuring the safety and efficacy of drugs, are increasingly being challenged by healthcare providers. Players in the wellness ecosystem are asking for RWD to validate whether new drugs provide similar safety and efficacy levels, as indicated by RCT results. With RCTs, results are obtained in a highly controlled environment, over a small patient population base, over a short period of time, and across a handful of highly dispersed clinical trial centers. The lack of real-life supporting evidence on the efficacy and effectiveness of new drug treatments is resulting in an increase in the number of drug recalls, thus raising the need for RWE studies.

Healthcare providers need quick access to evidence-based research to ensure that drug decisions reflect current evidence. As financial risks shift from payers to providers, new investments are required to achieve clinical and business goals. To gain access to real-world data sets that can generate powerful insights, providers are opting for real-world outcomes to build patient support and provide the appropriate treatments. According to a survey by Omnicom Health Group in 2017 with a total of 866 US physicians, respondents ranked RWE as the most important data informing of treatment decisions, followed by clinical trial data. The growing acceptance of RWE has boosted the growth of the overall healthcare analytics market.

5.2.2 RESTRAINTS

5.2.2.1 High cost of analytics solutions

Despite the known benefits of analytics in healthcare organizations, its adoption is low among price-sensitive end users owing to the high implementation and subscription/licensing fees of advanced analytics solutions. Only top-tier hospitals have the capability and funding to utilize big data in order to provide actionable insights effectively.

Advanced on-premise healthcare analytics solutions can cost over USD 200,000 for implementation and about the same for annual subscription/licensing fees. This makes it challenging for hospitals to justify their spending on analytics solutions, at least in the short term when they are focused on implementing EHRs and upgrading RCM systems. The impact of this factor is greater in emerging markets where several end users struggle to secure budgets for IT over medical equipment.

5.2.3 OPPORTUNITIES

5.2.3.1 Increasing focus on value-based medicine

According to the American Journal of Managed Care, value-based medicine is the practice of medicine based on the objective value (improvement in length of life and/quality of life) conferred by healthcare interventions as determined by utilizing a set of standardized parameters (commonly used to evaluate value and cost-effectiveness). Future healthcare trends are more quality-oriented, focusing on the quality of services provided as well as on generating novel approaches to value-based medicine and care management. Gradually, most hospitals are recognizing the inefficiencies of traditional approaches in patient care, quality assurance, and cost management. Therefore, healthcare BI solutions will be among the top priorities on the investment list of hospitals for logistics management, HIE, and predictive analysis of diseases.

Value-based medicine is redefining healthcare and capturing results-oriented data, and making it usable. BI helps healthcare organizations determine what data they need and further helps them to analyze, evaluate, and utilize this data to understand the overall costs versus benefits and gain a competitive advantage. It also enables transparency in all processes to examine every purchase from the standpoint of value, utility, and outcomes.

Healthcare systems, especially in the US, are gradually shifting from volume-based models of reimbursement to models based on measures of value. In December 2020, the Centers for Medicare & Medicaid Services (CMS) announced that the organization made changes to the payment system related to telehealth as well as ACO participation in the Medicare Shared Savings Program. According to the amendments, the payment will now be extended to cover 60 telehealth-delivered services. This has increased the payments to physicians and other practitioners for additional time spent with patients. In March 2015, the Department of Health and Human Services (HHS) announced the next-generation Medicare ACO model with new payment structures, improved performance benchmarks, increased beneficiary engagement authority, and enhanced care settings. Such initiatives will lead to an increase in the number of ACOs.

The rapid transition from fee-for-service (FFS) payments to value-based reimbursement models and the key role of analytics in new pay-for-performance models like ACOs will provide growth opportunities for players operating in the healthcare analytics market.

5.2.3.2 Use of analytics in precision and personalized medicine

Precision medicine involves the use of an individual's genomic data to develop personalized medicine. Advanced analytics and predictive modeling applications would enable healthcare organizations to treat each patient in a personalized way to produce the best possible outcomes. Over the years, analytics has increasingly been used to develop personalized medicine, primarily for cancer and diabetes. For instance, advanced analytics helps physicians to predict which treatment would be most effective and least toxic for patients suffering from cancer.

Various organizations are receiving grants to develop personalized medicine-based big data solutions.

- In July 2020, Genome Medical (US) raised over USD 14 million in Series B extension funding, which will enable the company to accelerate the development of its Genome Care Delivery technology platform for cancer, reproductive health, and pharmacogenomics.
- In February 2020, Karius secured USD 165 million in Series B funding to expand its proprietary microbial cell-free DNA technology to help doctors diagnose infectious diseases, broaden clinical development, and accelerate technology innovation.
- In March 2020, BillionToOne, Inc. raised USD 15 million in Series A+ funding. The funding will be used to support the commercial expansion of UNITY to all US states as well as the research and development of future diagnostic tests, including liquid biopsy products.
- In May 2018, MediSapiens Ltd. was awarded USD 0.5 million funding for its next-generation clinical data curation solution.
- In November 2017, Syapse (US) raised USD 30 million in Series D financing, which enabled healthcare providers to use precision medicine for improving care for their cancer patients.

Several healthcare organizations and analytics vendors are collaboratively undertaking efforts to make optimum use of analytics in personalized medicine. In August 2017, Biovista and Hewlett Packard Enterprise collaborated on next-generation big data analytics and AI to advance personalized medicine at community hospitals. Similarly, in January 2016, The Children's Hospital of Philadelphia (CHOP) joined a group of academic institutions and medical centers that focus on precision medicine and big data analytics. CHOP provides precision medicine in pediatrics through its Center for Data-Driven Discovery in Biomedicine, its precision medicine and big data analytics research center. Such efforts toward precision and personalized medicine are creating an array of opportunities for healthcare analytics vendors.

5.2.3.3 Cloud-based analytics

Cloud-based models have emerged as an important means of delivering healthcare analytics software for enhanced data accessibility and real-time analysis. Cloud-based analytics helps deliver analytical insights from a multi-terabyte database, anywhere and anytime. Every change of data is captured and automatically updated on the dashboard on a real-time basis. Factors such as scalable data storage and scalable computing power, and machine-learning capabilities make cloud platforms ideal for analytics in healthcare.

A cloud-based analytics solution does not require any upfront capital investments for hardware and reduces the burden on healthcare systems, as there is less involvement of IT staff. It also provides faster data transfer between organizations. These advantages will create significant growth opportunities for the vendors of cloud-based healthcare analytics solutions during the forecast period.

5.2.3.4 Increasing number of patient registries

The availability of many patient registries is driving the demand for implementing healthcare analytics solutions for gaining actionable insights from the large volume of data contained in these registries. Analytics solutions also assist in point-of-care healthcare delivery by providing a central point of access for case management to identify operational gaps, study disease patterns, know cost-effective treatment alternatives, notify users about tests or procedures, and track the progress of each patient to better manage patients with chronic diseases.

With the growing demand for population health management and health information exchange, government bodies across the globe are taking initiatives to digitize and integrate healthcare systems, including the development of computerized disease registries. In the US, federal agencies such as the NIH, CDC, CMS, AHRQ, and FDA, as well as several state agencies, are focusing on developing patient registries to determine the long-term outcomes of agents, devices, groups of drugs, or procedures. Rising government support and funding and increasing focus on population health management are resulting in the creation of patient registries, which will provide growth opportunities for healthcare analytics vendors in the coming years.

5.2.3.5 Emergence of social media and its impact on the healthcare industry

The healthcare industry is changing with incredible speed, and one of the major contributors to this change is the increasing popularity of healthcare communication through social media. Not only has social media become a place where people seek health information, but social media channels also allow for two-way public communication between patients, providers, and other third parties. This helps create a large forum for health discussions globally. The vast network of healthcare influencers, leaders, patients, providers, organizations, and governmental entities creates a massive amount of healthcare data on a regular basis. This data, if segregated, segmented, and analyzed in a meaningful way, can offer incredible value for improving treatment efficiencies and health outcomes. This has created a demand for data aggregation and analytical tools and thus acts as an opportunity for the healthcare analytics market.

5.2.3.6 Augmented analytics

The combination of AI and business intelligence together has given rise to a new era of intelligence, known as augmented analytics. Augmented analytics automates data insights by utilizing machine learning and natural language processing technologies to automate data preparation and enable data sharing. Thus, augmented analytics uses machine-learning automation to supplement human intelligence across the entire analytics lifecycle.

Critical areas of healthcare like patient care, training, operations, and drug research present immense potential for the application of augmented analytics tools. For instance, in a hospital, regular analytics tasks involve hospital staffing, operational costs, patient turnover, wait times, and the overall patient experience. Augmented analytics platforms provide instant answers to searched questions, along with answers to additional and important questions that are not asked by employees. Thus, for healthcare, augmented analytics solutions are ideal for common use cases such as prioritizing patients, detecting patient churn, predicting hospital readmissions, or identifying insurance fraud. As a result, augmented analytics can be expected to provide an immense opportunity for players in this market.

5.2.3.7 Analytics of Healthcare Things

Since IoT devices provide valuable patient data that is collected outside clinical settings, they can significantly help in transforming data collection for clinical trials and accelerate the role of technology in clinical development. According to the NHS Long-term Plan published in 2019, the government body has envisioned a healthcare service in which digital devices worn by patients play a crucial role. Currently, wearables and other digital technologies are increasingly being deployed in the process of data collection and evidence generation. Continuous developments in technologies are helping pharmaceutical companies dive deeper into digital health. A measured, practical approach for adopting new technologies has been proven to yield better ROI, efficiency, and valuable learning opportunities.

With the increased focus on leveraging new digital health technologies, companies outside of the healthcare space are focusing on trial setting processes and trying to help provide the tools necessary for using wearables in trials. Apple, Inc. (US), for instance, is continuing to invest in ResearchKit, an open-source framework for the creation of mobile applications that support medical researchers by gathering robust and meaningful data. Similarly, in March 2021, Cievert (UK) announced a strategic collaboration with Novartis AG (Switzerland). The collaboration was created via Novartis BIOME, a project that encourages impactful collaboration in science tech, with a selection procedure involving NHS clinicians and innovation leads. Novartis and Cievert are working with the NHS to launch pilot programs at various sites across the UK to combat the COVID-19 pandemic.

Integrating IoT data and conducting robust, advanced analytics on the data can provide valuable insights far beyond what traditional and more streamlined means of data collection and analysis can achieve. As a result, the Analytics of Healthcare Things is expected to be an important area of opportunity in the healthcare analytics arena.

5.2.4 CHALLENGES

5.2.4.1 Patient data confidentiality

Patient data contains personal, private, or confidential information that requires strict safeguards to prevent its misuse. The growing use of automated technologies (such as EHRs and medical claims processing), as well as the shifting trend toward health information exchanges (HIEs), have altered the healthcare privacy and security landscape. The electronic exchange of patient data offers greater reach and efficiency in healthcare delivery but involves greater data risk and liability due to broader access and can result in data breaches.

According to a report published by Protenus Breach Barometer, more than 3.14 million patient records were breached in 142 disclosed health data breach incidents during a three-month span from April to June 2018 in the US. These breaches were mainly due to the lack of internal control over patient information, lack of top management support for streamlining data management, obsolete policies and procedures or non-adherence to existing ones, and insufficient personnel training. Thus, patient confidentiality is a major challenge faced by healthcare analytics providers.

5.2.4.2 Concerns regarding inaccurate and inconsistent data

The importance of EHRs in healthcare analytics projects is greater than any other data source. Most patient data enter the health IT ecosystem through EHRs, but as patient data is moved, coded, translated, exchanged, analyzed, and reported, there are dozens of opportunities for errors that may seriously impact patient care. Inaccurate data has the potential to become a major health risk for patients and a huge liability to providers. Without robust, accurate, timely, clean, and complete data, healthcare organizations will not be able to move beyond the basics of recordkeeping and develop analytics competencies. This directly affects the use of analytical tools and thus hinders the growth of the healthcare analytics market.

5.2.4.3 Reluctance to adopt healthcare analytics solutions in emerging countries

The continued use of existing basic analytics tools is restricting the implementation of newer solutions among end users in emerging countries. Many hospitals and physicians in these countries continue to rely on existing basic business analytics tools. These tools are used at the departmental level and can use only retrospective data. Reports are seen as 'good enough for now' until they become too inefficient to continue. Since such regions lack established processes to capture data, analyzing unstructured data is also a challenge. Moreover, many hospitals and other end users in these countries are opposing the implementation of advanced EHRs and analytics solutions due to budgetary constraints, lack of assurance for effective outcomes, and lack of operating skills.

This reluctance to change existing systems is one of the major challenges for healthcare analytics vendors operating in emerging markets. In addition, in some emerging economies, the adoption rate of healthcare analytics solutions is low due to the lack of language support in solutions.

5.2.4.4 Data integration

The integration of data from different sources enables coordinated care and plays an important role in the effective communication of patient data between various stakeholders. However, data integration is a major challenge faced by healthcare organizations due to the incompatibility of multiple HCIT infrastructure systems. As the data collected is not unified, the ability to derive analytical value from the data becomes difficult for organizations. This, in turn, is limiting the adoption of healthcare analytics solutions among potential end users.

5.2.4.5 Operational gaps between payers and providers

The lack of maintenance of medical records, either by payers or providers, creates operational gaps in clinical and financial analysis. Moreover, the informal exchange of information and clinical data, both within hospitals as well as between payers and providers, results in errors, poor quality of patient care, and significant information loss. This operational gap results in a highly fragmented and incomplete data set (clinical, financial, and operational data), which can significantly affect the overall efficiency of data analytics solutions in healthcare organizations.

5.2.4.6 Dearth of skilled personnel

Currently, there are only a limited number of healthcare organizations that have in-house experts who can translate large volumes of data into valuable insights. The lack of data management capabilities and skilled analysts is one of the major barriers to the growth of the healthcare analytics market.

The dearth of IT staff in the healthcare industry creates inefficiencies and hinders technological advances like interoperability. According to an article published in December 2019, the need for big data talent in the healthcare industry is expected to increase by 30% in the coming five years. However, changing regulations in the US and various countries across the globe are limiting the hiring of immigrant talent; this, in turn, is expected to weaken the healthcare IT industry in these countries. According to a report published by the US Bureau of Labor Statistics, hospitals and physician practices in the US will need an additional 35,000 HIT workers by 2018. The bureau also states that an additional 41,000 professionals will be required in the healthcare information management (HIM) field by 2022. Other major markets such as Germany, the Netherlands, England, and Australia are also witnessing a significant shortage of HCIT professionals. The lack of in-house IT experts, as well as a trained workforce with cross-functional skills, may hinder the growth of the healthcare analytics market during the forecast period.

5.3 COVID-19 IMPACT ANALYSIS

5.3.1 COVID-19 HEALTH ASSESSMENT

The COVID-19 pandemic is an unprecedented crisis that has placed a considerable strain on healthcare providers, facilities, and systems worldwide. According to the WHO, within 67 days of the outbreak, the first 100,000 confirmed cases were reported worldwide. It took a further 11 days to reach 200,000; four days to reach 300,000; and 12 days to reach 1 million confirmed cases. This highlights the extreme rate of spread of the disease.

As of August 31, 2021, there are 226,686,571 reported cases of COVID-19 (Source: Worldometer) across Africa, Asia Pacific, the Americas, Europe, and Oceania. The virus has resulted in 4,663,332 deaths. The true number of infected people would be much higher than the official total, as a large number of cases are asymptomatic and do not undergo testing.

Disease progression patterns are similar in most countries, but the flattening of the curve and disease containment is largely dependent on the various measures taken by each country and the respective governments. Countries have implemented various measures to control the spread of the disease, such as strict travel restrictions; imposing city-wide quarantines, curfews, and lockdowns; constructing specialty hospitals; increasing medical aid; and ramping up COVID-19 testing. Disturbingly, countries that were initially thought to have succeeded in controlling the spread of the virus are now experiencing a second and third wave of the pandemic. However, vaccination has been initiated in many countries worldwide and is found to effectively control the severity of the effect of the COVID-19 virus.

5.4 COVID-19 ECONOMIC ASSESSMENT

The COVID-19 pandemic affected the healthcare sector and significantly impacted the global economy, severely impacting various financial and industrial sectors, such as the travel and tourism, manufacturing, and aviation industries.

The economic downturn is slowly improving across regions, with better recovery expected by the fourth quarter of 2021 as most developed countries and many developing nations have already initiated the vaccination process. However, growth in exports is expected to reduce drastically. Although economies have opened up in major countries, recovery is predicted to be slow and gradual and is not expected to be complete till the spread of the virus is successfully controlled across all major trade regions.

5.5 ASSESSMENT OF THE IMPACT OF COVID-19 ON THE ECONOMIC SCENARIO

In the recent past, the global economy has become substantially more interconnected. Adverse consequences of various steps related to containment are evident from global supply chain disruptions, weaker demand for imported products and services, and the increasing unemployment rates. Risk aversion has increased in the financial markets, with all-time-low interest rates and sharp declines in equity and commodity prices. Consumer and business confidence have also reduced significantly.

FIGURE 26 CRITERIA IMPACTING THE GLOBAL ECONOMY

Source: WHO and ECDC

This decline in the economy has magnified over time, and the spillover is seen across the globe. In the US, the GDP is expected to increase by 3.5% in 2021, after an estimated 3.6% reduction in 2020. In Europe, the GDP is anticipated to grow by 3.6% in 2021, following a 7.4% decline in 2020. The GDP in Japan reduced by 5.3% during 2020 and is projected to grow by 2.5% in 2021. In emerging markets and developing economies, including China, the aggregate GDP is expected to grow by 5% in 2021, after a contraction of 2.6% in 2020 (Source: World Bank Prospects).

Based on these forecasts, the world economy is expected to recover in 2021.

However, in this fast-changing environment, the full impact of the pandemic on the global economy may not entirely be known. MarketsandMarkets uses three scenario-based approaches (epidemiology and health response as well as an economic response) to assess the economic impact and recovery period on the global level. Countries and regions are expected to experience different impacts and recovery periods.

To provide a perspective on various scenarios, we assume factors that would impact economic recovery from the global recession.

Realistic Scenario: The economic slowdown was seen until the second quarter of 2020, and a slow recovery was observed from the third and fourth quarters of 2020. This recovery was expected at a slow rate going forward in 2021, with uncertainties persisting with newer strains of COVID-19 resulting in lockdowns in major economies in the world. The emergence of various vaccines has positively spurred the economy, but their rollout is still a big challenge, especially in emerging economies. Hence, there will be a recovery in a realistic scenario but at a much slower pace.

Optimistic Scenario: In an optimistic scenario, the rollout of vaccines will be successful, with very few challenges, and vaccines will be available to most of the population by the second quarter of 2021, resulting in a V-shaped economic recovery with a strong and fast improvement in the global economy.

Pessimistic Scenario: In a pessimistic scenario, we assume that all health and economic measures are neither expected to help control the spread of the virus nor lead to economic recovery. New strains of the virus could pose newer and more significant challenges, resulting in additional lockdowns, which could result in unprecedented economic loss, along with social and political turmoil. In this scenario, most economies are expected to face recession till the first quarter of 2021, and a modest recovery is expected

in 2021 Q2–2021 Q3. This is expected to be an L-shaped economic recovery, with countries taking longer to return to pre-crisis levels.

5.6 ASSESSMENT OF THE IMPACT OF COVID-19 ON THE HEALTHCARE ANALYTICS MARKET

The COVID-19 pandemic has positively impacted the healthcare analytics market. As most of the global formal workforce transitioned quickly into working from home between January to April 2020, players in the healthcare analytics market differentiated and enhanced their services to suit the changed requirements of end users.

The predictive and prescriptive analytics has been used to develop prediction models to estimate the emergence of subsequent waves of COVID-19 infection spread in different regions of the world. The increase in the adoption of EHR in developing and developed countries during the pandemic has further influenced the growth of the market in 2020. Operational analytics has proven beneficial in the management of patient data and COVID-19 bed allocation in ICU units during the pandemic. Claim management and risk assessment analytics were widely used in healthcare reimbursement for COVID-19 patients. The healthcare analytics has also witnessed various innovations and developments in the market space owed to the outbreak of the pandemic. All these aforementioned factors have aided the growth of the market in the year 2020.

COVID-19 data and analytics have significantly aided efforts of emergency and department, frontline workers and hospital capacity planning during the outbreak. According to the survey conducted by Health Catalyst, one of the key players in the market, 88% of their respondent said that COVID-19 pandemic has increased the need for analytics. Individuals associated with leadership and financial roles in provider organizations are the most interested in pandemic-response data and analytics.

5.7 ECOSYSTEM/MARKET MAP

The healthcare analytics ecosystem consists of companies providing analytics solutions and services to payer organizations, healthcare providers (hospitals, clinics, diagnostics centers, ambulatory centers and others). Certain players in the healthcare analytics market provide end-to-end services are SAS Institute (US), Accenture (US), IBM Corporation (US), Cerner (US), Optum (US), Allscripts (US), and others provide final software deliverables Cerner (US), McKesson (US), EPIC (US), LexisNexi (US) and Health Catalyst (US).

The key roles of major stakeholders in the healthcare analytics ecosystem and their benefits are listed below.

INFRASTRUCTURE SERVICE PROVIDERS

- Help in managing data and structuring workflows.
- Create high-value innovative products, software, and platforms that provide patients and healthcare professionals with increasingly advanced functionalities

PLATFORM, SERVICE, AND HEALTHCARE ANALYTICS COMPANIES

- Product ability to incorporate advanced technology and include sensors, hardware, and connectivity to EHR so that consumers can consult the data collected
- Data collection hard-drive storage and hardware required to develop healthcare analytics softwares
- Continuous upgrades in products as connectivity with EHR

HOSPITALS/HEALTH SYSTEMS

- Reduce avoidable readmissions
- Keep patients within a health system's network
- Triage ER demand more efficiently
- Manage/track chronically ill patients more effectively
- Improve care transition handoffs
- Differentiate brands in a competitive marketplace
- Improve management of psych patients
- Align with rapidly evolving shared risk models more effectively
- ACOs and PCMHs promote care delivery in the most appropriate setting

SENIOR LIVING/POST-ACUTE/HOSPICES

- Utilize scarce clinical resources more efficiently
- Avoid unnecessary hospitalizations/ER visits
- Manage/track chronically ill patients more effectively
- Increase medication and therapy adherence and compliance
- Creates brand differentiation in a competitive marketplace
- Enhance caregiver/family member communication
- Gain the ability to measure changes in ADLs and vitals 24/7
- See early warning signs of health issues
- Safety monitoring solutions to increase patient security and family
- Member/caregiver comfort
- Slow transition into more acute settings—reduce vacancy rates

PHYSICIANS

- Prioritize patients more efficiently and reduce in-office wait times
- Manage/track the chronically ill more effectively
- Diversify/supplement income streams
- Provide physicians with a complete view of patient health status
- Address growing digital appetite/demands
- Align quickly with rapidly evolving shared risk models
- Increase complexity levels of in-office visits
- Improve productivity of nurses and mid-levels
- Engage patients and drive healthy behavior change

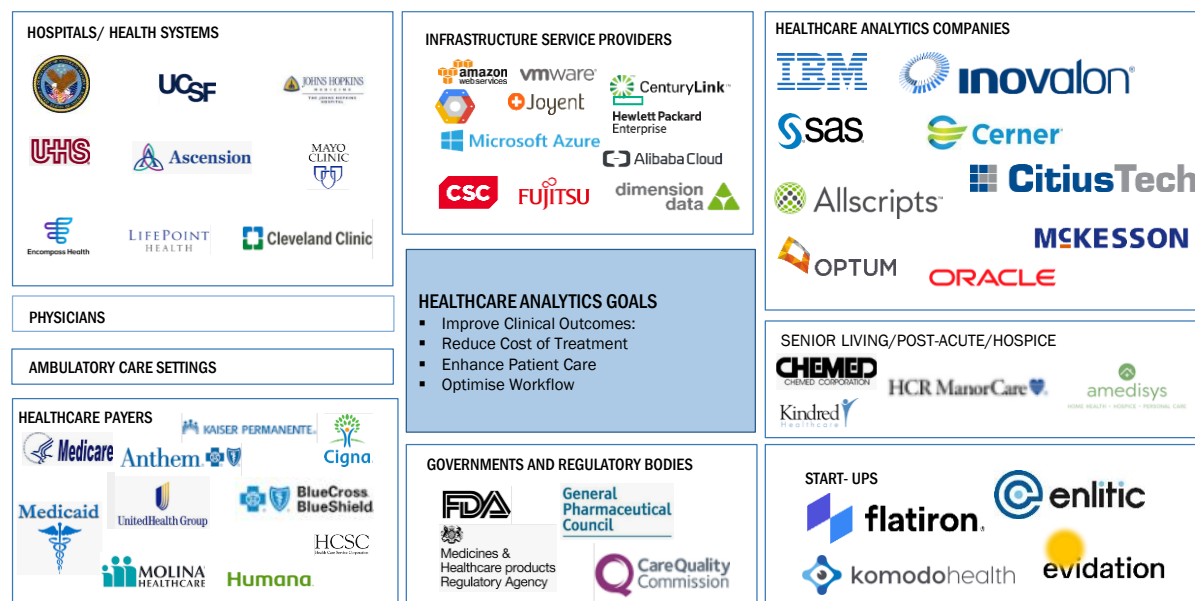
PAYERS

- Provide appropriate care in the most effective cost settings
- Enhance adherence and compliance with evidence-based pathways
- Modify patient behavior – promote healthier living
- Share risk with employees more effectively by increasing alternatives for benefit plan designs
- Reduce unnecessary ER visits
- Solutions effectively align with rapidly evolving shared risk payment models
- Significantly reduce readmissions
- Drive better compliance for annual screenings
- Reduce overall healthcare costs with more timely, consistent delivery of care

GOVERNMENTS/REGULATORY BODIES

- Favorable initiatives to encourage healthcare analytics adoption, improve the quality of care, and bring down healthcare spending
- Softening regulatory approval procedures for healthcare analytics platforms to expedite product development and commercialization, especially after the COVID-19 outbreak in 2019

FIGURE 27 HEALTHCARE ANALYTICS MARKET: ECOSYSTEM



Source: Company Websites and MarketsandMarkets Analysis

5.8 CASE STUDY ANALYSIS

5.8.1 CASE STUDY 1

In November 2019, the University of Chicago Medical Center faced a problem with surgical room delays. Operating room usage cost ranges from \$22 to \$133 per minute and remains one of the largest expenses for medical institutions. The delays are caused because of the number of staff involved in surgical operations and preparations held between each operation. Utilizing real-time data and machine learning algorithms, the hospital managed to schedule operating room activity to pass them between the teams with minimum time waste. The system is based around real-time alerts by consuming real-time data to optimize performance

Result: Turnover time decreased by 15-20 percent per room with an approximately \$600 thousand increase in annual revenue for the organization.

5.8.2 CASE STUDY 2:

Operating across 900 clinical facilities, Envision Physician Services implemented analytics solutions to predict daily emergency room visits. A cloud-based system utilized Azure SQL Database and Power BI interface for reporting features. The analytical system sources data from Envision's billing and patient scheduling databases. The machine learning model applied to this data makes predictions on expected daily emergency room visits with 80 percent accuracy. The provided information helps emergency room operation teams to organize staff and required equipment in advance, optimizing patient serving.

Result: optimized performance of emergency rooms. The same analytical systems installed in approximately 100 clinical facilities supported by Envision Physician Services.

5.8.3 CASE STUDY 3:

In August 2020, Agilon Health used DOS platform by Health Catalyst to gather population symptomatology and mortality data across Italy, China, and the US. The data included prior researches, studies that covered factors like demographics, comorbidities, population density across certain regions, etc. Combining these factors, Agilon Health created a model that considered individual risk scores for the US population. During a one-week period, the Agilon team managed to contact nearly 50 percent of the covered patient base giving precautions and mitigating the risk for those who were in older age groups. Appointments with the medical staff shifted to telehealth channels (consultancy and diagnostics on the web) to minimize clinician exposure to the virus.

Result: As stated in Health Catalyst case study, Agilon Health managed to create a COVID-19 model for approximately 125,000 individuals that were assigned with risk scores. It also increased one partner location's telehealth appointments from none in the first week to 2,200 in weeks 12 and 13, aligning with social distancing and overall pandemic policies.

5.9 REGULATORY ANALYSIS

5.9.1 NORTH AMERICA

REGULATION/LAW	DESCRIPTION
Health Insurance Portability and Accountability Act (HIPAA), US	- The HIPAA Act of 1996 was created primarily to modernize the flow of healthcare information, stipulate how personally identifiable information is maintained by healthcare and healthcare insurance industries to protect them from frauds and theft, and address the limitations of healthcare insurance coverage. - Healthcare applications that process Protected Health Information (PHI) are subject to HIPAA's privacy rules and security rules. HIPAA enforces cloud providers to provide healthcare businesses with written assurances that they would safeguard any PHI received or created.
Security in a Cloud Computing Environment, US	- The US Federal Financial Institutions Examination Council (FFIEC), an inter-agency group comprising federal and state banking regulators, issued guidance on 'Security in a Private Cloud Computing Environment' in April 2020. - FFIEC's guidance has provided several examples related to risk management practices. Some of these examples are listed below: - Governance, including how the use of cloud computing fits with financial institutions' strategic plan - Private cloud security management, including adequate due diligence, service provider contracts that clearly state responsibilities of each party, proper network controls, ongoing oversight, and effective training - Change management, including the effective controls for transitioning the financial systems of financial institutions to cloud computing environments
Control Objectives for Information and Related Technology (COBIT), US	COBIT is an IT industry regulation, which is a standard to provide IT governance and control. It ensures information integrity by providing technical guidelines on various IT-related topics.
The Federal Information Security and Management Act (FISMA), US	FISMA is a federal regulation. It provides guidelines on security controls, user access, identity management, risk assessment, auditing, and monitoring for government agencies and affiliated companies collecting and processing data on behalf of government agencies.
SOC 2, US	SOC 2 is developed by the American Institute of CPA, which defines criteria for managing customer data based on five trust service principles—security, availability, processing integrity, confidentiality, and privacy.

5.9.2 EUROPE

REGULATION/LAW	DESCRIPTION
General Data Protection Regulation (GDPR), European Union (EU)	GDPR 2016/679 is a data protection and privacy regulation in the EU and European Economic Area (EEA). It also addresses the transfer of personal data outside the EU and EEA.
EU Private Cloud Code of Conduct (EU Private Cloud COC)	To secure the trust of cloud customers in CSPs, the EU Private Cloud COC aims at helping cloud providers in complying with GDPR. The EU Private Cloud COC consists of requirements for CSPs that aim to adhere to the Code and a governance section designed to support the effective and transparent implementation, management, and evolution of the Code.
EU Cybersecurity Act	The EU Cybersecurity Act has revamped and strengthened the EU Agency for Cybersecurity (ENISA) and established the EU-wide Cybersecurity Certification Framework for digital products, services, and processes.

5.9.3 ASIA PACIFIC

REGULATION/LAW	DESCRIPTION
Multi-Tier Cloud Security (MTCS), Singapore	▪ The MTCS standard was prepared under the direction of the Information Technology Standards Committee (ITSC) of the Infocomm Media Development Authority (IMDA) of Singapore. ▪ The purpose of MTCS is to provide: • A common standard that CSPs can apply to address customer concerns related to the security and confidentiality of data in the cloud and the impact of using cloud services on businesses • Verifiable operational transparency and visibility into risks to customers when they use cloud services
Private Cloud Computing Service Security Assessment, China	▪ To improve the security and controllability of cloud computing services purchased by parties and government agencies, key information infrastructure operators, the National Internet Information Office, National Development and Reform Commission, Ministry of Industry and Information Technology, and the Ministry of Finance have formulated the Private Cloud Computing Service Security Assessment.
Multi-Level Protection Scheme (MLPS) 2.0, China	▪ MLPS 2.0 is a regulation that outlines the security requirements for different data types and requires all network operators wishing to provide cloud services in China to store data in infrastructure within the country's borders.
China's Cyber Security Law (CSL)	▪ China's CSL is a broad piece of legislation that dictates how companies should approach security and privacy within the country. It has mandatory requirements around breach notifications, the appointment of cybersecurity heads, and incident response plans. It governs almost every aspect of online and network activities and gives law enforcement agencies the authority to inspect and monitor businesses, including in-person inspections on the company premises.
Personal Information Protection Act (PIPA), Japan	▪ PIPA by the Personal Information Protection Commission has established guidelines for CSPs to regulate the collection, use, and processing of personal data.

5.9.4 MIDDLE EAST & AFRICA

REGULATION/LAW	DESCRIPTION
Protection of Personal Information Act, South Africa	The Protection of Personal Information (POPI) Act was formed in 2013. It is South Africa's equivalent regulation of the EU GDPR. It sets some prior conditions for responsible parties to lawfully process the personal information of data subjects.
Private Cloud Computing Regulatory Framework, Saudi Arabia	The Communications and Information Technology Commission (CITC) considers that adopting the Cloud Computing Regulatory Framework (CCRF) would provide increased regulatory clarity and encourage the adoption of cloud computing services in the country. The Framework is based on best international practices and public consultation analysis, including the rights and obligations of service providers, individual customers, government entities, and enterprises.

5.9.5 LATIN AMERICA

REGULATION/LAW	DESCRIPTION
Lei Geral de Proteção de Dados General Data Protection Law, Brazil	Brazil's Lei Geral de Proteção de Dados (LGPD) is a data protection law formed to regulate the personal data of its citizens. LGPD attempts to unify over 40 statutes that currently govern personal online and offline data by replacing certain regulations.

5.10 PORTER'S FIVE FORCES ANALYSIS

The competitive environment in the healthcare analytics market can be analyzed through the Porter's five forces analytical framework.

TABLE 2 HEALTHCARE ANALYTICS MARKET: PORTER'S FIVE FORCES ANALYSIS

FORCE	PARAMETER	IMPACT
Threat from New Entrants	Presence of Established Players	High
	Specialist Knowledge	High
	Technological Advancements	Moderate-to-high
	Entry Barriers	Moderate
	Capital Requirements	Moderate
Intensity of Competitive Rivalry	Number of Competitors	High
	Quality Difference	Moderate
	Customer Loyalty	Moderate-to-high
Bargaining Power of Buyers	Price Sensitivity	Low-to-moderate
	Ability to Substitute	Moderate-to-high
	Cost of Changing	Moderate
Bargaining Power of Suppliers	Number of Suppliers	High

Threat from Substitutes	Ability to Substitute	Low-to-moderate
	Supplier Switching Cost	Moderate
	Ability to Substitute	Low-to-moderate
	Cost of Change	Low

Source: Annual Reports, SEC Filings, Press Releases, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

5.10.1 THREAT FROM NEW ENTRANTS

The threat from new entrants in the healthcare analytics market is moderate, based on the following factors:

There are many established players operating in the market. These players account for a considerable share of the global market. Established industry operators have a competitive advantage over new entrants, as they operate efficiently due to the benefits of economies of scale. Although new entrants in the market can develop various healthcare analytics solutions efficiently and charge low prices for their solutions, already established players operating in the market with clients can pose a hindrance. It becomes difficult for start-ups to get acceptance from end users and incur the high costs associated with them. These high entry barriers pose a competitive disadvantage for new entrants, limiting competition from smaller companies in the global healthcare analytics market. Start-up companies provide innovative solutions, new ways of doing things, and are expected to put pressure on key players through lower pricing strategy, reducing costs, and providing new value propositions to the customers. Since established players such as Cerner and Allscripts keep defining the standards regularly, it becomes difficult for start-up companies to gain extraordinary profits, discouraging new players in the healthcare analytics industry.

5.10.2 THREAT FROM SUBSTITUTES

The threat from substitutes is low-to-moderate in the healthcare analytics market. There are currently limited substitutes for healthcare analytics solutions. The conventional model of analysis of patient data is cheaper than healthcare analytics solutions. However, the conventional model can lead to time-consuming tasks of viewing and analyzing patient data. Thus, the threat from substitutes in terms of using conventional methods is low. When a new product or service meets customer needs in different ways, industry profitability suffers. Services like cognitive analytics are enabling efficient data analysis and diagnosis of disease indications. The threat from a substitute product or service is high if it offers a value proposition that is uniquely different from the present offerings in the industry.

5.10.3 BARGAINING POWER OF SUPPLIERS

The overall bargaining power of suppliers in the healthcare analytics market is low-to-moderate. There are a large number of suppliers and healthcare analytics developers in the market. However, there is very little variation in the services offered by these suppliers. The purchasing decision is largely based on the price offered and the reliability of the supplier. Thus, suppliers and software developing companies can decrease the margins that can be earned by companies in the market. Powerful suppliers use their negotiating power to extract higher prices from healthcare analytics companies. The overall impact of higher supplier bargaining power is that it lowers the overall profitability of healthcare analytics companies. Developing dedicated suppliers and experimenting with product designs using different materials can help companies tackle this problem.

5.10.4 BARGAINING POWER OF BUYERS

In this market, the impact of the bargaining power of buyers is moderate to high. Many large and small players operate in the healthcare analytics market. Buyers want to buy the best offerings available by paying minimal prices. This puts pressure on the company's profitability in the long run. The smaller and more powerful the customer base, the higher the bargaining power of the customers and higher their ability to seek increasing discounts and offers. Building a large base of customers and rapidly innovating new products will help key players reduce the deflection of existing customers to their competitors. Due to possible variations in quality and reliability in this market, buyers (clinics, hospitals, retail pharmacies, laboratories, and research institutes) prefer companies that provide consistent, reliable, and repeatable results. Hence, the presence of a limited supplier pool decreases the bargaining power of buyers in the market.

5.10.5 INTENSITY OF COMPETITIVE RIVALRY

The competitive rivalry within the industry refers to the extent to which market players exert pressure on each other. If the competition among existing players in an industry is intense, it drives down prices and decreases the industry's overall profitability. The healthcare analytics market is marked by moderate-to-high competitive intensity. Many companies in the healthcare analytics market pursue different approaches, such as technological innovation and broadening of their product portfolios, to ensure high market shares in this competitive market. Due to many players operating in the market, there is intense rivalry among firms. Key players pursue different approaches such as product launches and R&D investments to ensure a greater share in this competitive market. Since there are multiple market players, the competitive aspect exists at the regional and global levels in terms of pricing and product quality. The emergence of smaller healthcare analytics companies in low-cost manufacturing locations across the APAC further intensifies the competitiveness in the global healthcare analytics market.

5.11 TECHNOLOGY ANALYSIS

5.11.1 MACHINE LEARNING

ML enables users to dive deep into data and produce actionable insights. When paired with AI and IoT, ML can help employees gain an unprecedented level of insights. These insights are helpful in intelligent maintenance recommendations. Machine learning algorithms quickly process huge datasets and provide useful insights that allow superior healthcare services. ML can also be useful in benchmarking, which helps in determining performances.

5.11.2 ARTIFICIAL INTELLIGENCE

AI is a branch of computer science that deals with building smart machines that can perform tasks related to human intelligence, such as planning, learning, reasoning, and problem-solving. AI in cloud services can help organizations build mission-critical solutions, comprehend speech, make predictions using data, and imitate other intelligent human behaviors. AI can help in mitigating biases and interpret behaviors with privacy and accountability.

5.11.3 INTERNET OF THINGS

IoT is digitally transforming business processes. It enables businesses to automate processes, increase efficiency, effectively manage maintenance tasks, and save costs. IoT can be extremely effective in remote locations, and it can reduce capital expenses. IoT also provides business agility with end-user IoT application development. IoT, when used with ML, can produce actionable insights helpful in analytics and reporting.

5.11.4 BLOCKCHAIN

Blockchain is referred to as a system of recording information in groups that are shared and recorded in a ledger and are not mutable. It is used to build, govern, and expand a network at a scale. Blockchain helps create end-to-end solutions using blockchain data managers with flexible, reliable, and scalable data strings and application integration. It helps organizations monitor contracts, react to transactions and events, and strings on-chain data to off-chain data stores.

5.11.5 BIG DATA

Currently, stakeholders in the healthcare sector are focusing on building integrated healthcare systems for improving care delivery. This has led to the high adoption of solutions, such as EMRs, hospital information systems, and various remote monitoring and diagnostic solutions, which capture health information. These healthcare solutions generate large volumes of high-velocity, complex, and variable data, requiring advanced analysis techniques.

In the healthcare industry, big data comprises information from clickstream and interactions from the web and social media; readings from medical devices such as sensors, ECGs, X-rays, and pulse oximeters; healthcare claims and other billing records; biometric data; EMRs; and prescriptions, among other sources. According to an article published by the International Journal of Advanced Research in Computer Science and Software Engineering, the volume of big data in healthcare stood roughly at 153 exabytes in 2013 and is expected to increase to 2,314 exabytes by 2020. As a result, it has become increasingly important that such multiple streams of data can be leveraged with data aggregation and analytics techniques to advance the delivery of healthcare for individual patients as well as disease-specific and condition-specific populations. The application of big data analytics with healthcare IT solutions provides tools to accumulate, manage, analyze, and assimilate large volumes of disparate, structured, and unstructured data produced by current healthcare systems.

Owing to the benefits of big data analytics in the effective management of health data, many healthcare IT and healthcare analytics companies and federal bodies are working towards fully deploying this technology in the healthcare industry. For instance:

- Quantiphi (US) is a company that provides applied AI and big data solutions to companies from various sectors such as insurance, healthcare, media, technology, and retail. The company received a series A investment of USD 20 million in December 2019. The company will use this funding to build its vertical expertise and deepen its presence in the North American and European markets.
- Researchers at Johns Hopkins launched a new data analytics tool, Precision Medicine Analytics Platform, in June 2019. This solution will be used to facilitate big data research across the JHU enterprise and help support precision medicine research and enhance care delivery.
- In January 2018, the Agency for Healthcare Research and Quality (AHRQ) planned to grant around USD 5 million to support efforts to use big data analytics to improve patient safety and clinical diagnostics.

5.12 VALUE CHAIN ANALYSIS

FIGURE 28 VALUE CHAIN ANALYSIS: HEALTHCARE ANALYTICS MARKET



RESEARCH AND DEVELOPMENT (R&D)

Typically, the value chain of healthcare analytics products and services starts with technology development and is followed by product design, optimization, and deployment/installation. The R&D phase comprises basic research, cost-benefit analyses, prototype, process development, and regulatory approval. This stage adds nearly 10–35% to the overall price of the final product. R&D activities are divided into in-house and outsourced tasks. In-house activities focus on critical parts that deal with the actual analysis. During this phase, software prototypes are built, systematically tested, and evaluated. The final product is then sent to regulatory authorities for approval, which is later marketed and sold to end users. Companies do not usually disclose their R&D activities; however, they may showcase prototypes at trade shows and conferences before launching a new product.

RAW MATERIAL

The raw material accounts for the major share in the value chain of the market (around 60–70%). This phase involves raw materials for the development of healthcare analytics products and services by integrating components supplied to manufacturers by respective suppliers. The raw materials also include data warehouse/data server for healthcare providers. Such systems have high data storage & data movement capacities, and providers use them to collect, store, and analyze patient data. Raw materials include specialized hard drives, advanced computing systems, and additional software that are used for the development of healthcare analytics software.

DISTRIBUTION AND AFTER-SALES SERVICES

Around 10–20% of the value is added during the marketing and after-sales activities, including post-sales services and access. The price of the product supplied by the software developing company to the distributor is usually twice its production cost; when the product moves from the distributor to the end user, the price again doubles.

The key stakeholders in the value chain for products include:

- Research centers conduct basic and applied research to understand the scope of genome editing/engineering products in various applications. They play an important role in the value chain by developing, optimizing, and inventing new techniques, which later become technological platforms to address the challenges and demands of the industry.
- Large firms in the healthcare analytics market include IBM (US), Optum (US), Cerner (US), SAS Institute (US), Allscripts (US). They have the expertise to develop these products either through technology transfer, licensing agreements from startups and research institutes, or acquisitions. Large firms use their global presence to commercialize products successfully and occupy a major share in the value chain.
- Regulatory bodies are governing agencies that set the standards and guidelines for the manufacturing and marketing of genome-related products. They closely monitor data from R&D to assess the reliability and accuracy of healthcare analytics solutions, and software before approval.
- Healthcare providers and payer are end users who significantly influence the market penetration of each product and service.
- Venture capitalists, federal/private investors, and payers form the most important part of the value chain. They provide funds to other stakeholders, which include research institutes, start-up companies, and small firms. Federal investors are mostly government-based fund providers who encourage technology developments to address the challenges and demands of the industry.

6 HEALTHCARE ANALYTICS MARKET, BY TYPE

KEY FINDINGS

- The descriptive analytics segment is expected to hold the largest share of the healthcare analytics market in 2021. This segment is projected to reach USD 26,601.9 million by 2026 from USD 11,036.1 million in 2021, growing at a CAGR of 19.2% during the forecast period.
- The prescriptive analytics segment is expected to record the highest CAGR of 36.8% during the forecast period.
- In 2021, the prescriptive analytics segment is expected to account for a share of 29.7% of the healthcare analytics market. This segment is projected to reach a value of USD 13,166.8 million by 2026 from USD 2,653.3 million in 2021, at a CAGR of 36.8% during the forecast period.

6.1 INTRODUCTION

Based on type, the healthcare analytics market is segmented into descriptive analytics, predictive analytics, prescriptive analytics, and cognitive analytics. While the descriptive analytics segment is estimated to account for the largest share of 52.3% in 2021, the prescriptive analytics segment is projected to grow at the highest CAGR of 37.8% during the forecast period. The high growth of the prescriptive analytics segment can be attributed to the ability of prescriptive analytics to ensure the synergistic integration of predictions and prescriptions. The capability of prescriptive analytics goes beyond description and predicting outcomes of a healthcare scenario; it also provides insights and actionable solutions.

TABLE 3 HEALTHCARE ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2019–2026)
Descriptive Analytics	7,148.3	8,740.6	11,036.1	26,601.9	19.2%
Predictive Analytics	3,353.2	4,446.0	6,275.1	30,019.2	36.8%
Prescriptive Analytics	1,398.9	1,874.3	2,653.3	13,166.8	37.8%
Cognitive Analytics	613.8	824.9	1,151.2	5,344.9	35.9%
Total	12,514.2	15,885.8	21,115.7	75,132.8	28.9%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association, American Health Information Management Association, National Health Care Anti-fraud Association, American Medical Informatics Association, Certification Commission for Health Information Technology, Healthcare Information Technology Standards Panel, Health Level Seven International, Information Technology Association of Canada, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network, Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

6.2 DESCRIPTIVE ANALYTICS

6.2.1 DESCRIPTIVE ANALYTICS IS USED TO CONVERT PAST DATA INTO USEFUL INFORMATION

Descriptive analytics help healthcare payers and providers to better monitor and manage their business practices. It categorizes, characterizes, consolidates, and classifies the past data to convert it into useful information. Most organizations prefer descriptive/retrospective analytics since it can be implemented with the standard aggregate functions built into most databases. Descriptive analytics also help with PHM tasks such as identifying how many patients live with diabetes, benchmark outcomes against government expectations, or identify areas for improvement. Healthcare providers use descriptive analytics to understand better past trends and events occurring in real-time. Analyzing past trends and using that data to predict the future is important for providing quality care.

Visual analytics is emerging as an important aspect of descriptive analytics. Visual analytics unites human cognition, interactive visual interfaces, and data analytics to facilitate interpreting complex data to gain insights. It is based on rapidly increasing health-related digital, optical, and graphical information. Analytical solutions such as SAS Visual Analytics, JMP, and Tableau offer visual analytics solutions to represent clinical data.

As descriptive analytics forms the base for the effective application of predictive or prescriptive analytics, it is expected to continue dominating the healthcare analytics market during the forecast period. However, the growth of this market segment is expected to be slower compared to other emerging analytics fields that offer advanced analytics tools and are considered to be a priority investment by many healthcare organizations.

Some key vendors offering descriptive analytics solutions are Optum (US), SICO Health (US), SAS Institute (US), CitiusTech (US), PluralSoft, IBM (US), MedAssets (US), and MedeAnalytics (US).

TABLE 4 DESCRIPTIVE ANALYTICS MARKET, BY REGION, 2019–2026 (USD MILLION)

Region	2019	2020	2021	2026	CAGR (2021–2026)
North America	4,279.4	4,898.2	6,244.4	14,635.4	18.6%
Europe	1,517.4	2,049.0	2,540.2	6,135.4	19.3%
Asia	1,001.7	1,331.1	1,692.2	4,636.6	22.3%
LATAM	146.4	180.5	220.9	496.0	17.6%
Pacific Countries	112.6	159.1	191.2	391.8	15.4%
Middle East & Africa	90.6	122.8	147.2	306.6	15.8%
Total	7,148.3	8,740.6	11,036.1	26,601.9	19.2%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association, American Health Information Management Association, National Health Care Anti-fraud Association, American Medical Informatics Association, Certification Commission for Health Information Technology, Healthcare Information Technology Standards Panel, Health Level Seven International, Information Technology Association of Canada, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network, Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

6.3 PREDICTIVE ANALYTICS

6.3.1 THE INCREASING INCIDENCE OF HEALTHCARE FRAUD IS DRIVING THE MARKET GROWTH FOR PREDICTIVE ANALYTICS

Predictive analytics involves using historical performance data combined with several statistical and analytical techniques to develop simulations that predict future events and trends. This helps healthcare organizations avoid preventable events, provide better patient care, and reduce healthcare costs.

Predictive analytics is increasingly being used across the healthcare system to identify high-risk, high-impact patients based on models developed by analyzing the medical records. This enables healthcare providers to analyze future risks better, reduce patient hospital readmission rates, improve patient care, manage chronic disease, and improve hospital administration & supply chain efficiency.

Predictive analytics finds significant applications in the following areas:

TECHNIQUE	DESCRIPTION
Signal processing and time-series analysis techniques	These techniques are applied to extract sensor data, such as physiological data, obtained by monitoring patients regularly and measuring them in real-time at a high frequency.
Natural language processing methods	These techniques can be applied to extract information from clinical texts and detect errors.
Stream analytics	These methods are used to mine sensor data and biomedical signals.
Social media analytics	These techniques detect health trends and adverse drug interactions, improve interventional capabilities for health-related activities, and analyze customer satisfaction.
Sophisticated analytics methods	These techniques are used to detect healthcare fraud, waste, and abuse.

Healthcare fraud prevention is the principal application area of predictive analytics for payers. It helps payers explore hidden patterns in claims and identify undetected fraudulent claims that could otherwise be overlooked due to lack of data and time & resource constraints. In 2019, the Department of Health and Human Services noted that the number of participants in the Healthcare Fraud Prevention Partnership (HFPP) increased to 144 public, state and private organizations by the end of 2019. Moreover, improper payments in Medicare and Medicaid programs totaled USD 610 million in 2019.

The Affordable Care Act (ACA) is slated to provide the CMS with USD 350 million for the next ten years (2015–2025). This amount is expected to help CMS undertake provider screenings and collaborate with various software companies to develop a robust model for detecting improper payments and fix penalty rates for improper claim submissions. Through the effective use of predictive analytics, payers can detect suspicious/fraudulent claims early and prevent FWA cost-effectively. Payers also use predictive analytics to identify members at risk of hospitalization or undergoing other costly procedures and identify coding errors and omissions. This approach helps in reducing healthcare costs. The increasing utilization of analytical tools to identify risks and prevent fraud is expected to boost the healthcare analytics market during the forecast period.

Various market players are focusing on introducing new predictive analytics solutions to leverage growth opportunities in the predictive analytics market. Some of the recent launches in this market are Healthcare.AI (Health Catalyst, March 2021), COTIVITI COVID-19 outbreak tracker (April 2020), Medical Intelligence v7.8 (Cotiviti—formerly Verscend), August 2018).

Various market players focus on acquisitions, collaborations, partnerships, and agreements to expand their product offerings in the predictive analytics market. For instance, in April 2021, Allscripts entered a strategic partnership with Manorama Infosolutions to deliver EHR and Healthcare Management Information Systems in hospital settings in India. In April 2021, Microsoft announced plans to acquire Nuance, a developer of speech-to-text software, for an estimated amount of US\$ 19 Billion. Nuance is one of the leading companies providing speech-to-text software for healthcare professionals and providers.

Some commercially available predictive analytics solutions in the market are:

SOLUTION	COMPANY
IBM Content and Predictive Analytics for Healthcare	IBM
Population Explorer	Health Catalyst
Medictiv	CitiusTech
Predictive Readmission Analyzer	Health Catalyst
Predictive Clinical Insight System	Inovalon
McKesson Performance Management Explorer	McKesson

Source: Company Websites

Other prominent vendors in this market are Optum (US), Oracle (US), Mu Sigma (India), Cotiviti (formerly known as Verscend) (US), SAS Institute (US), Information Builders, Predixion Software (US), InsightOne (US), Accenture (US), Rapid Insight (India), CGI Group (Canada), and Saama Technologies (US).

TABLE 5 PREDICTIVE ANALYTICS MARKET, BY REGION, 2019–2026 (USD MILLION)

Region	2019	2020	2021	2026	CAGR (2021–2026)
North America	2,420.0	3,093.0	4,448.1	22,943.3	38.8%
Europe	568.2	828.9	1,113.2	4,166.7	30.2%
Asia	253.5	365.3	504.8	2,199.5	34.2%
LATAM	52.6	70.6	94.4	344.4	29.6%
Pacific Countries	34.0	52.4	68.8	231.5	27.5%
Middle East & Africa	24.9	35.8	45.8	133.8	23.9%
Total	3,353.2	4,446.0	6,275.1	30,019.2	36.8%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association, American Health Information Management Association, National Health Care Anti-fraud Association, American Medical Informatics Association, Certification Commission for Health Information Technology, Healthcare Information Technology Standards Panel, Health Level Seven International, Information Technology Association of Canada, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network, Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

6.4 PRESCRIPTIVE ANALYTICS

6.4.1 PRESCRIPTIVE ANALYTICS TO REGISTER THE HIGHEST GROWTH RATE DURING THE FORECAST PERIOD

Prescriptive analytics differs from predictive analytics because it predicts an outcome and provides suggested actions. Prescriptive analytics requires a predictive model with two additional components—actionable data and a feedback system—to track the outcome produced by the action taken. It automatically combines big data, mathematical models, business rules, and machine learning to predict outcomes and provide alternatives to leverage predictions. It also shows decision-makers the implications of each decision option.

Prescriptive analytics enables healthcare organizations to understand patients and protocols better and take the necessary actions to improve outcomes. Strategic planning is the prime application area of prescriptive analytics; it helps analyze operational and usage data, coupled with data from external sources (such as economic data, population demographic trends, and population health trends). For instance, providers can plan how many hospital beds to build, when, where, and for what type of care with prescriptive analytics tools. It enables them to better plan for future capital investments such as new equipment and new facilities. However, expert human intervention is crucial for maximizing the value of prescriptive analytics in healthcare.

Prescriptive analytics helps highlight strengths, weaknesses, and opportunities in treatment methods and prescribe better business practices, ultimately reducing costs while mitigating care and financial risks for all stakeholders. Investing in prescriptive analytics enables healthcare providers to understand their patients and their programs better.

Some commercially available prescriptive analytics solutions in the market are:

SOLUTION	COMPANY
SunLeaper	Medalogix
SAS Visual Analytics	SAS Institute
Q-IRI	Mu Sigma

Source: Company Websites

Other prominent vendors offering prescriptive analytics solutions are IBM (US), Health Catalyst (US), NextHealth (US), Predixion (US), ZeOmega (US), and Verisk Analytics (US).

TABLE 6 **PRESCRIPTIVE ANALYTICS MARKET, BY REGION, 2019–2026 (USD MILLION)**

Region	2019	2020	2021	2026	CAGR (2021–2026)
North America	944.0	1,230.9	1,804.8	10,248.9	41.5%
Europe	254.2	362.7	476.0	1,591.6	27.3%
Asia	149.6	209.4	281.0	1,053.8	30.3%
LATAM	22.0	28.7	37.2	117.0	25.8%
Pacific Countries	16.5	24.9	32.1	98.3	25.1%
Middle East and Africa	12.6	17.7	22.1	57.3	21.0%
Total	1,398.9	1,874.3	2,653.3	13,166.8	37.8%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association, American Health Information Management Association, National Health Care Anti-fraud Association, American Medical Informatics Association, Certification Commission for Health Information Technology, Healthcare Information Technology Standards Panel, Health Level Seven International, Information Technology Association of Canada, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network, Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

6.5 COGNITIVE ANALYTICS

6.5.1 VENDORS ARE NOW FOCUSING ON INTRODUCING AI-DRIVEN ANALYTICS SOLUTIONS

Cognitive analytics uses artificial intelligence (AI) technologies such as machine learning, deep learning, natural language processing (NLP), and reasoning or logic systems to provide analysis. It automates the decision-making process and provides insights and suggestions to enhance human decisions. Advanced technologies such as AI are beginning to play a vital role in automated fraud detection. AI-based tools can uncover hidden correlations and patterns that may be difficult to detect. With the support of big data analytics and real-time IoT data streams, AI can efficiently build accurate predictive risk models to support extremely enhanced fraud detection. According to Accenture, 75% of insurance executives believe that AI will provide significant industry changes in the next three years. The benefits of AI in the healthcare analytics industry are expected to drive its adoption and implementation in the coming years. Currently, AI-driven analytics is at a very nascent stage but is a key area of opportunity for the players in the market.

To leverage the growth opportunities in this potentially high-growth market, various analytics vendors aim to introduce AI-driven analytics products and services. Some of the recent launches in this market segment include the following:

- In April 2021, Mayo Clinic launched an AI-focused diagnostic platform to deliver clinical decision support tools, care recommendations, and diagnostic insights to healthcare providers.
- In March 2021, Cleveland Clinic and IBM announced a ten-year partnership to launch the Discovery Accelerator; a center aimed at using artificial intelligence, high performance on hybrid cloud, and quantum computing technologies to speed discovery in healthcare.
- In August 2020, Stanford launched an online program on Artificial Intelligence in Healthcare for healthcare providers, computer scientists, and technologists.

Players in the cognitive analytics market also focus on collaborations, partnerships, and agreements to expand their product offerings. For instance, in November 2018, Virginia Premier (US) and Inovalon (US) entered into an agreement to implement Inovalon's Clinical Data Extraction-as-a-Service (CDEaaS) and Natural Language Processing-as-a-Service (NLPaaS) within the Inovalon ONE Platform.

Such developments increase the availability of advanced analytics solutions for cognitive analytics, thereby increasing its adoption among the payers and providers who are the major end users of this market segment. Some of the commercially available cognitive analytics solutions include Open Source Machine Learning (Health Catalyst), Integrated Analytics Systems with Machine (IBM), and SAS Viya (SAS).

TABLE 7 COGNITIVE ANALYTICS MARKET, BY REGION, 2019–2026 (USD MILLION)

Region	2019	2020	2021	2026	CAGR (2021–2026)
North America	360.8	468.3	682.2	3,757.9	40.7%
Europe	122.9	174.9	228.8	750.5	26.8%
Asia	91.3	127.6	170.8	629.2	29.8%
LATAM	16.8	22.0	28.7	92.0	26.2%
Pacific Countries	11.9	17.8	22.7	66.2	23.9%
Middle East & Africa	10.1	14.3	18.0	49.0	22.1%
Total	613.8	824.9	1,151.2	5,344.9	35.9%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association, American Health Information Management Association, National Health Care Anti-fraud Association, American Medical Informatics Association, Certification Commission for Health Information Technology, Healthcare Information Technology Standards Panel, Health Level Seven International, Information Technology Association of Canada, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network, Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

7 HEALTHCARE ANALYTICS MARKET, BY COMPONENT

KEY FINDINGS

- The services segment is expected to account for the largest share of 52.8% of the healthcare analytics market in 2021. This segment is projected to reach USD 42,018.0 million by 2026 from USD 11,158.1 in 2021 at a CAGR of 30.4% during the forecast period.
- The high growth of the services segment is attributed to the recurring need for services such as software upgrades and maintenance.
- The support services segment is expected to account for 67.5% of the healthcare analytics services market in 2021.
- The software segment is expected to account for 40.1% of the healthcare analytics market in 2021. The healthcare analytics software market is projected to reach USD 30,322.4 million by 2026 from USD 8,469.5 million in 2021, at a CAGR of 29.1% during the forecast period.

7.1 INTRODUCTION

Based on components, the healthcare analytics market is segmented into software, hardware, and services. In 2021, services dominated the market with a share of 52.8%. This segment is also expected to grow at the highest rate during the forecast period, primarily due to the recurring need for services such as software upgrades and maintenance.

TABLE 8 HEALTHCARE ANALYTICS MARKET, BY COMPONENT, 2019–2026 (USD MILLION)

Component	2019	2020	2021	2026	CAGR (2021–2026)
Services	6,359.8	8,249.6	11,158.1	42,018.0	30.4%
Software	5,020.2	6,367.3	8,469.5	30,322.4	29.1%
Hardware	1,134.1	1,269.0	1,488.1	2,792.4	13.4%
Total	12,514.2	15,885.8	21,115.7	75,132.8	28.9%

Source: Healthcare Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), The National Health Care Anti-Fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Health Care Anti-fraud Association (CHCAA), The European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports and SEC Filings, Investors Presentations, Expert Interviews, and MarketsandMarkets Analysis

The software segment is estimated to witness a significant growth rate of 29.1% during the forecast period as it is an integral part of healthcare analytics offerings and works as an interface between the database and the end user.

7.2 SERVICES

Services form a large component of the healthcare analytics market due to the indispensable role of service providers when it comes to the use of data analytics solutions. In terms of IT usage and skills, the healthcare industry is still nascent and relies heavily on service providers for business analytics services and support services such as implementation, integration, training & education, maintenance, and upgrades.

The healthcare analytics services market is expected to witness the highest growth during the forecast period, primarily due to the rapid adoption of cloud services and the introduction of technologically advanced healthcare analytics software that requires regular upgrades and trained users.

Various players in the market focus on providing optimum services to gain an edge over their competitors and expand their existing customer base. McKesson announced the launch of HX Analytics at HIMSS17 in Orlando in February 2017. Ketchum Analytics introduced Vital Health in 2018, a predictive analytics-driven service for its existing healthcare customer base. This cloud-based analytics-as-a-service helps healthcare organizations to provide value-based care.

The key players offering healthcare analytics services are SAS, Accenture, IBM, Cerner, Optum, Allscripts, Verisk Health, ZIETA Technologies, Hexaware, Avanade, SCIO Health, Health Catalyst, Inovalon, CitiusTech, and Wipro.

TABLE 9 HEALTHCARE ANALYTICS SERVICES MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Support Services	4,355.2	5,608.3	7,533.1	27,323.2	29.4%
Business Analytics Services	2,004.7	2,641.3	3,625.0	14,694.8	32.3%
Total	6,359.8	8,249.6	11,158.1	42,018.0	30.4%

Source: Healthcare Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), The National Health Care Anti-Fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Health Care Anti-fraud Association (CHCAA), The European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports and SEC Filings, Investors Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 10 HEALTHCARE ANALYTICS SERVICES MARKET, BY REGION, 2019–2026 (USD MILLION)

Region	2019	2020	2021	2026	CAGR (2021–2026)
North America	4,017.6	4,966.1	6,876.3	28,448.4	32.8%
Europe	1,299.4	1,833.3	2,374.8	7,282.9	25.1%
Asia	753.0	1,051.1	1,402.3	4,913.7	28.5%
Pacific Countries	128.0	165.7	212.8	621.5	23.9%
LATAM	87.3	129.1	162.6	432.4	21.6%
Middle East & Africa	74.5	104.3	129.2	319.1	19.8%
Total	6,359.8	8,249.6	11,158.1	42,018.0	30.4%

Source: Healthcare Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), The National Health Care Anti-Fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Health Care Anti-fraud Association (CHCAA), The European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports and SEC Filings, Investors Presentations, Expert Interviews, and MarketsandMarkets Analysis

7.2.1 SUPPORT SERVICES

7.2.1.1 Support services include implementation, ongoing IT support, system integration, training, and maintenance services

The support services segment accounts for a larger share of the healthcare analytics services market due to the recurring nature of these services. This segment includes implementation, ongoing IT support, system integration, training & education, and maintenance services.

Implementation services include the validation and verification of hardware and software. Healthcare analytics providers offer rapid and smooth implementation to ensure that solutions create sustainable value, usually within 90 days from the start of the contract. Ongoing IT support is extremely crucial to avoid the failure of services or networks. Implementation services and ongoing IT support include continuous monitoring, adjustment and repair, applying patches, restarting services and servers, and backup creation. Aftersales services ensure interoperability, software testing, maintenance, and resolution of software complexity issues.

Onsite user training and education help healthcare organizations upgrade their business and technical analytical skills to make full use of available technologies and apply the results of analytics to the appropriate business issues. These services also help increase awareness among healthcare providers and payers about the latest healthcare analytics solutions available and the benefits of applying analytics

in their business. Most large and medium-sized companies actively provide technical assistance to hospitals and other healthcare providers in healthcare analytics solutions.

TABLE 11 HEALTHCARE ANALYTICS SUPPORT SERVICES MARKET, BY REGION, 2019–2026 (USD MILLION)

Region	2019	2020	2021	2026	CAGR (2021–2026)
North America	2,764.3	3,399.4	4,682.1	18,843.4	32.1%
Europe	875.1	1,218.3	1,555.3	4,326.3	22.7%
Asia	517.9	719.4	954.8	3,253.3	27.8%
Pacific Countries	86.2	111.0	141.8	402.5	23.2%
LATAM	61.5	90.5	113.4	294.3	21.0%
Middle East & Africa	50.1	69.7	85.7	203.3	18.9%
Total	4,355.2	5,608.3	7,533.1	27,323.2	29.4%

Source: Healthcare Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), The National Health Care Anti-Fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Health Care Anti-fraud Association (CHCAA), The European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports and SEC Filings, Investors Presentations, Expert Interviews, and MarketsandMarkets Analysis

7.2.2 BUSINESS ANALYTICS SERVICES

7.2.2.1 Business analytics services help healthcare organizations to optimize the use of analytics and drive performance improvement

Business analytics services help healthcare organizations plan and design analytics strategies and visions, conduct feasibility studies before implementing solutions, provide advisory services to identify care gaps, assess and efficiently deploy analytics solutions, provide technical consulting, and optimize the use of analytics solutions to drive continuous performance improvements.

Companies also provide analytics services for applications such as claims analysis, performance management, risk assessment, predictive analysis, fraud detection, hospital bill review audit, payment integrity validation, and regulatory compliance assistance.

Factors such as the expanding healthcare industry, increase in data generation, increasing awareness and availability of analytics solutions for healthcare, advantages of analytics in improving patient outcomes and reducing costs, and the implementation of regulations to curb rising healthcare costs and enhance the quality of care are driving the market for business analytics services.

The growing demand for business analytics services has prompted players to invest in this segment and offer sophisticated solutions. For instance, in April 2018, Inovalon launched Clinical Data Extraction-as-a-Service (CDEaaS) and Natural Language Processing-as-a-Service (NLPaaS) to the Inovalon ONE Platform. Similarly, Wipro announced in 2017 that its Big Data Analytics-as-a-Service on Microsoft Azure Data Discovery Platform was available for various industries, including healthcare. This platform leverages the company's business analytics service—Power BI—to build analytical applications.

TABLE 12 HEALTHCARE BUSINESS ANALYTICS SERVICES MARKET, BY REGION, 2019–2026 (USD MILLION)

Region	2019	2020	2021	2026	CAGR (2021–2026)
North America	1,253.3	1,566.6	2,194.2	9,604.9	34.4%
Europe	424.3	615.0	819.6	2,956.6	29.3%
Asia	235.1	331.7	447.5	1,660.4	30.0%
LATAM	41.8	54.7	71.0	219.0	25.3%
Pacific Countries	25.8	38.6	49.1	138.1	22.9%
Middle East & Africa	24.4	34.6	43.5	115.8	21.6%
Total	2,004.7	2,641.3	3,625.0	14,694.8	32.3%

Source: Healthcare Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), The National Health Care Anti-Fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Health Care Anti-fraud Association (CHCAA), The European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports and SEC Filings, Investors Presentations, Expert Interviews, and MarketsandMarkets Analysis

7.3 SOFTWARE

7.3.1 HEALTHCARE ANALYTICS SOFTWARE WORKS AS AN INTERFACE BETWEEN THE DATABASE AND END USERS

Healthcare analytics software is a non-tangible component used to run a system or perform primary tasks in any healthcare organization or health plan. It is an integral part of the healthcare analytics market and works as an interface between the database and end users. Platform solutions, desktop solutions, and cloud-based solutions are the broad categories of healthcare analytics software available in the market.

Healthcare analytics software solutions enable hospitals, clinics, and physician practices to improve the quality of patient care and outcomes by using predictive analytics to identify the best course of treatment, provide better insights into claims processing, meet regulatory requirements, optimize performance, and reduce risk. They also help providers increase patient access to services, drive process efficiencies, and enable better revenue cycle and financial performance management.

With increased healthcare expenditure and rising demand for quality patient care, healthcare providers, payers, and other healthcare stakeholders are increasingly focusing on developing data analytics software. Some of the recent launches in the market include the SAS Viya and SAS Platform with AI capabilities; integrated analytics systems with machine learning capabilities launched by IBM; and dbMotion solution through Microsoft Azure, introduced by Allscripts in 2017.

Additionally, the availability of funding for the development of new analytical software has increased in recent years. IBM, Cerner, McKesson, SAS, Optum, EPIC, LexisNexis, Health Catalyst, Tableau Software, MedAnalytics, Inovalon, and Allscripts are some of the major vendors providing healthcare analytics software.

TABLE 13 HEALTHCARE ANALYTICS SOFTWARE MARKET, BY REGION, 2019–2026 (USD MILLION)

Region	2019	2020	2021	2026	CAGR (2021–2026)
North America	3,250.5	3,944.8	5,376.5	21,296.3	31.7%
Europe	943.9	1,306.6	1,665.4	4,804.3	23.6%
Asia	610.7	824.0	1,065.5	3,304.2	25.4%
LATAM	88.5	112.3	141.6	388.0	22.3%
Pacific Countries	75.2	108.8	134.2	329.2	19.7%
Middle East & Africa	51.4	70.7	86.3	200.4	18.3%
Total	5,020.2	6,367.3	8,469.5	30,322.4	29.1%

Source: Healthcare Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), The National Health Care Anti-Fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Health Care Anti-fraud Association (CHCAA), The European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports and SEC Filings, Investors Presentations, Expert Interviews, and MarketsandMarkets Analysis

7.4 HARDWARE

7.4.1 INCREASING ADOPTION OF CLOUD SERVERS IS EXPECTED TO HINDER THE GROWTH OF THE HARDWARE MARKET FOR ANALYTICS

Configurational hardware acts as a data warehouse/data server for healthcare providers. Such systems have high data storage & data movement capacities, and providers use them to collect, store, and analyze patient data. These systems are integrated with analytics solutions that access data and help create powerful analytical combinations to provide valuable insights.

Data accessibility and the availability of centralized information are essential in the healthcare industry. With the growing span and size of networks, the demand for faster and more efficient servers to manage communication within and across various networks has also increased. Furthermore, with the growing adoption of EHRs and the digitization of information, a large amount of data is being generated, which requires storage devices with larger capabilities. This “big data” comprises medical images, medical device readings, patient medical records, emergency correspondence, billing records, radiofrequency identification tags, and barcode tags. This data is used for various analytical needs and must be stored for future reference, which drives the demand for storage devices.

The demand for hardware in the HCIT infrastructure is growing due to the increasing number of hospitals that upgrade or replace the hardware to make optimal use of the latest software available in the market. Emerging technologies such as high-speed storage area networks and high-speed fiber channel switches have helped improve data storage and processing capabilities across networks.

However, the growing adoption of cloud servers is expected to hinder the growth of the hardware market for analytics. Hardware reuse and integration with existing components to optimize costs and implementation times and the rising preference for service-oriented architecture that requires minimal use of hardware are also hindering the market growth.

The players operating in this market include IBM, Oracle, Cisco Systems, and EMC.

TABLE 14 HEALTHCARE ANALYTICS HARDWARE MARKET, BY REGION, 2019–2026 (USD MILLION)

Region	2019	2020	2021	2026	CAGR (2021–2026)
North America	736.2	779.5	926.7	1,840.8	14.7%
Europe	219.5	275.5	318.1	557.2	11.9%
Asia	132.4	158.2	181.0	301.2	10.7%
LATAM	21.2	23.8	26.6	39.8	8.4%
Pacific Countries	12.5	16.3	18.1	26.3	7.8%
Middle East & Africa	12.3	15.6	17.6	27.2	9.1%
Total	1,134.1	1,269.0	1,488.1	2,792.4	13.4%

Source: Healthcare Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), The National Health Care Anti-Fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Health Care Anti-fraud Association (CHCAA), The European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports and SEC Filings, Investors Presentations, Expert Interviews, and MarketsandMarkets Analysis

8 HEALTHCARE ANALYTICS MARKET, BY DEPLOYMENT MODEL

KEY FINDINGS

- The on-premise deployment model segment is expected to account for the larger market share of 60.5% in 2021, while the on-demand segment is projected to register the highest growth rate during the forecast period.
- The on-premise deployment model segment is projected to reach USD 35,039.3 million by 2026 from USD 12,781.7 million in 2021, at a CAGR of 22.3% during the forecast period.
- On-demand deployment models are increasingly being adopted in hospitals due to the increase in data generated across the healthcare provider industry.
- In 2021, the on-demand deployment model segment is expected to account for a share of 39.5% of the healthcare analytics market. This market is projected to reach USD 40,093.5 million by 2026 from USD 8,334.0 million in 2021, at a CAGR of 36.9% during the forecast period.
- In 2021, North America is expected to command the largest share of 59.5% of the healthcare analytics market for the on-premise deployment model.

8.1 INTRODUCTION

Based on the deployment model, the healthcare analytics market is segmented into on-premise and on-demand solutions. The on-premise deployment model segment is expected to account for the larger market share of 60.5% in 2021, while the on-demand segment is projected to register the highest growth rate during the forecast period.

TABLE 15 HEALTHCARE ANALYTICS MARKET, BY DEPLOYMENT MODEL, 2019–2026 (USD MILLION)

Deployment Model	2019	2020	2021	2026	CAGR (2021–2026)
On-premise Deployment	8,091.1	9,975.7	12,781.7	35,039.3	22.3%
On-demand Deployment	4,423.1	5,910.1	8,334.0	40,093.5	36.9%
Total	12,514.2	15,885.8	21,115.7	75,132.8	28.9%

Source: Healthcare Information and Management Systems Society (HIMSS), International Medical Informatics Association, American Health Information Management Association, National Health Care Anti-Fraud Association, American Medical Informatics Association, Certification Commission for Health Information Technology, Healthcare Information Technology Standards Panel, Health Level Seven International, Information Technology Association of Canada, Canadian Health Care Anti-fraud Association, European Healthcare Fraud & Corruption Network, Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

The on-demand deployment model is the fastest and easiest way of deploying healthcare analytics solutions. On-demand healthcare analytics solutions are sold based on subscription—ranging from one month to multiple years. As compared to on-premise solutions, on-demand solutions do not require significant setup charges. Moreover, the maintenance and support service charges are included in the subscription fee, making these solutions more affordable for customers.

8.2 ON-PREMISE DEPLOYMENT

8.2.1 ON-PREMISE DEPLOYMENT MODELS OFFER MULTIVENDOR ARCHITECTURE, WHICH LOWERS THE RISK OF DATA BREACHES & EXTERNAL ATTACKS

On-premise deployment is the traditional method of deploying healthcare analytics solutions. In this delivery model, the buyer must purchase a software license from the vendor and is responsible for purchasing, installing, and maintaining the server. Most healthcare decision-makers prefer to maintain access to healthcare analytics solutions inside the organization due to the control, security, and cost benefits offered by this model.

On-premise costs include the costs of servers, dashboards, data warehouses, initial software licenses, and yearly maintenance fees. In this delivery model, the buyer takes decisions regarding software upgrades and employee training. On-premise models offer multivendor architecture, which lowers the risk of data breaches and external attacks. Buyers mainly opt for these delivery models due to the security benefits associated with them. These factors are driving the growth of the on-premise deployment model segment.

However, lack of data transferability and remote access and the burden on IT staff to maintain system flexibility may limit the growth of this market segment to some extent. Moreover, the increasing volume of data being generated requires systems that can store large amounts of data. These drawbacks are expected to restrain the growth of the on-premise deployment model segment during the forecast period.

TABLE 16 ON-PREMISE SOLUTIONS AVAILABLE IN THE MARKET

SOLUTION	PROVIDER
IBM Cognos	IBM
Oracle Enterprise Healthcare Analytics	Oracle
Allscripts Analytics PHA	Allscripts

Source: Company Websites

TABLE 17 HEALTHCARE ANALYTICS MARKET FOR ON-PREMISE DEPLOYMENT MODEL, BY REGION, 2019–2026 (USD MILLION)

Region	2019	2020	2021	2026	CAGR (2021–2026)
North America	5,018.9	5,842.6	7,600.9	21,605.5	23.2%
Europe	1,569.1	2,130.7	2,658.6	6,769.0	20.6%
Asia	1,125.9	1,501.3	1,915.3	5,369.6	22.9%
LATAM	151.6	185.9	226.0	477.8	16.2%
Pacific Countries	137.0	195.2	236.8	515.1	16.8%
Middle East & Africa	88.6	120.1	144.1	302.3	16.0%
Total	8,091.1	9,975.7	12,781.7	35,039.3	22.3%

Source: Healthcare Information and Management Systems Society (HIMSS), International Medical Informatics Association, American Health Information Management Association, National Health Care Anti-Fraud Association, American Medical Informatics Association, Certification Commission for Health Information Technology, Healthcare Information Technology Standards Panel, Health Level Seven International, Information Technology Association of Canada, Canadian Health Care Anti-fraud Association, European Healthcare Fraud & Corruption Network, Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

8.3 ON-DEMAND DEPLOYMENT

8.3.1 INCREASING VOLUME OF DATA GENERATED ACROSS THE HEALTHCARE PROVIDER INDUSTRY TO DRIVE THE ADOPTION OF ON-DEMAND DEPLOYMENT

In an on-demand deployment model, service providers store customer data in an isolated data center, either as a substitute for or enhancement to a customer's on-premise infrastructure. On-demand analytics is available as a cloud-based solution that can be offered as a web-based or platform/app-based solution. On-demand models are among the most effective delivery models due to the elimination of the purchasing cost of software and hardware. These models are increasingly being adopted in hospitals due to the increasing volume of data generated across the healthcare provider industry.

The cloud was originally considered a feasible option for small organizations due to the lack of devoted IT staff to manage on-premise infrastructure. However, this is no longer the case. Hospital data centers are extremely congested, and advancements in scanning technologies and the increasing adoption of EHRs and other digital health information tools have led to the generation of large volumes of data. This encourages a growing number of healthcare organizations to store their data in the cloud.

Cloud-based healthcare analytics applications help deliver analytic insights from a multi-terabyte database anywhere, anytime, on any platform with ease. Every change of data is captured and automatically updated on the dashboard on a real-time basis. These factors, coupled with the lack of upfront capital investments for hardware, extreme capacity flexibility, and a pay-as-you-go pricing model, are expected to boost the demand for on-demand healthcare analytics solutions.

To capitalize on the growth opportunities in this market, various market players are focusing on offering cloud-based analytics solutions. For example, in February 2018, Infor introduced the Infor CloudSuite, a new analytics solution intended to optimize workforce and operational effectiveness.

TABLE 18 ON-DEMAND SOLUTIONS AVAILABLE IN THE MARKET

SOLUTION	PROVIDER
SAS Business Analytics Platform	SAS
Enterprise Performance Management Cloud	Oracle
Truven Health Unify	Truven Health
RelayAnalytics Pulse	McKesson
Oracle Analytics Cloud	Oracle

Source: Company Websites

On the other hand, complexities related to data integration and data security concerns are expected to hamper the growth of this market during the forecast period.

TABLE 19 HEALTHCARE ANALYTICS MARKET FOR ON-DEMAND DEPLOYMENT MODEL, BY REGION, 2019–2026 (USD MILLION)

Region	2019	2020	2021	2026	CAGR (2021–2026)
North America	2,985.4	3,847.8	5,578.7	29,979.9	40.0%
Europe	893.7	1,284.8	1,699.7	5,875.3	28.2%
Asia	370.3	532.1	733.5	3,149.5	33.8%
LATAM	86.1	115.9	155.1	571.6	29.8%
Middle East & Africa	49.5	70.5	89.0	244.4	22.4%
Pacific Countries	38.0	59.0	78.1	272.8	28.4%
Total	4,423.1	5,910.1	8,334.0	40,093.5	36.9%

Source: Healthcare Information and Management Systems Society (HIMSS), International Medical Informatics Association, American Health Information Management Association, National Health Care Anti-Fraud Association, American Medical Informatics Association, Certification Commission for Health Information Technology, Healthcare Information Technology Standards Panel, Health Level Seven International, Information Technology Association of Canada, Canadian Health Care Anti-fraud Association, European Healthcare Fraud & Corruption Network, Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

9 HEALTHCARE ANALYTICS MARKET, BY APPLICATION

KEY FINDINGS

- The financial analytics segment is expected to account for the largest share of 36.5% of the healthcare analytics market in 2021. This segment is projected to reach USD 24,168.8 million by 2026 from USD 7,707.0 million in 2021, at a CAGR of 25.7% during the forecast period.
- The clinical analytics segment is expected to register the highest CAGR of 34.8% during the forecast period.
- Growth in this segment is mainly due to the growing pressure to curtail healthcare spending, the rising generation of big data, government incentives to improve care quality, and the use of EHRs.
- The quality improvement & clinical benchmarking segment is expected to account for the largest share of the clinical analytics market in 2021.
- In 2021, population health analytics is estimated to account for 14.6% of the healthcare analytics market. This segment is projected to reach USD 10,825.8 million by 2026 from an estimated USD 3,074.7 million in 2021, at a CAGR of 28.6% during the forecast period.

9.1 INTRODUCTION

Based on application, the healthcare analytics market is categorized into clinical analytics, financial analytics, operational & administrative analytics, and population health analytics. The clinical analytics segment is expected to register the highest CAGR of 35.1% during the forecast period. Growth in this segment is mainly due to the growing pressure to curtail healthcare spending, large volumes of healthcare data containing valuable insights, increasing incentives by governments (especially in the US) for improving the quality and reducing the cost of care, growing adoption of electronic health records (EHRs), and rising focus on building patient registries.

TABLE 20 HEALTHCARE ANALYTICS MARKET, BY APPLICATION, 2019–2026 (USD MILLION)

Application	2019	2020	2021	2026	CAGR (2021-2026)
Financial Analytics	4,812.9	5,944.3	7,707.0	24,168.8	25.7%
Clinical Analytics	3,698.3	5,041.4	7,094.6	31,580.3	34.8%
Operational & Administrative Analytics	2,167.2	2,584.7	3,239.4	8,557.9	21.4%
Population Health Analytics	1,835.7	2,315.4	3,074.7	10,825.8	28.6%
Total	12,514.2	15,885.8	21,115.7	75,132.8	28.9%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association, American Health Information Management Association, National Health Care Anti-fraud Association, American Medical Informatics Association, Certification Commission for Health Information Technology, Healthcare Information Technology Standards Panel, Health Level Seven International, Information Technology Association of Canada, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network, Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

The market for financial analytics is expected to hold the largest share during the forecast period. Market growth is mainly driven by provider & payer focus on process efficiency, the need for regulatory compliance & cost-curtailment, and the drive to ensure operational profitability.

9.2 FINANCIAL ANALYTICS

With the growing number of regulations and rising economic burden across the globe, healthcare organizations are increasingly focusing on cutting costs and implementing practices that will help their operations remain profitable. Financial analytics provides healthcare payers and providers with relevant data for effective revenue cycle management (RCM), budgeting, and forecasting. It enables healthcare organizations to understand profitability and make informed investment decisions. Payers mainly use financial analytics for processing claims to prevent payment errors, analyze data, and manage revenue cycles. Providers use financial analytics mainly for future capital investments in equipment, facilities, or human resources; analyzing insurance risks for a defined population; hospital budgeting; and RCM.

The financial analytics market is further segmented into RCM, claims processing, payment integrity & FWA (fraud, waste & abuse), and risk adjustment & assessment.

TABLE 21 **COMMERCIALLY AVAILABLE FINANCIAL ANALYTICS SOLUTIONS**

SOLUTION	COMPANY
Revenue Cycle Explorer: Hospital	Health Catalyst
ClaimsXten	McKesson
Claims Check	McKesson
Optum 360	Optum
Oracle Financial Analytics	Oracle
Claims Accuracy	Cotiviti

Source: Company Websites

TABLE 22 **FINANCIAL ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)**

Type	2019	2020	2021	2026	CAGR (2021–2026)
Claims Processing	2,041.0	2,515.9	3,276.8	10,316.1	25.8%
Revenue Cycle Management	1,388.1	1,685.0	2,119.5	5,723.2	22.0%
Payment Integrity & FWA	887.8	1,133.7	1,520.3	5,650.3	30.0%
Risk Adjustment & Assessment	496.1	609.8	790.5	2,479.2	25.7%
Total	4,812.9	5,944.3	7,707.0	24,168.8	25.7%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association, American Health Information Management Association, National Health Care Anti-fraud Association, American Medical Informatics Association, Certification Commission for Health Information Technology, Healthcare Information Technology Standards Panel, Health Level Seven International, Information Technology Association of Canada, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network, Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 23 **FINANCIAL ANALYTICS MARKET, BY REGION, 2019–2026 (USD MILLION)**

Region	2019	2020	2021	2026	CAGR (2021–2026)
North America	3,130.3	3,693.3	4,896.7	16,851.6	28.0%
Europe	880.4	1,192.2	1,486.0	3,832.4	20.9%
Asia	586.7	774.9	981.0	2,723.2	22.7%
LATAM	93.4	114.7	140.1	326.0	18.4%
Pacific Countries	70.6	99.6	119.9	258.8	16.6%
Middle East & Africa	51.5	69.6	83.4	176.8	16.2%
Total	4,812.9	5,944.3	7,707.0	24,168.8	25.7%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association, American Health Information Management Association, National Health Care Anti-fraud Association, American Medical Informatics Association, Certification Commission for Health Information Technology, Healthcare Information Technology Standards Panel, Health Level Seven International, Information Technology Association of Canada, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network, Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

9.2.1 CLAIMS PROCESSING

9.2.1.1 Need to improve claims response time and avoid fraudulent claims will support the use of analytics tools

Claims processing is a costly function involving both healthcare payers and providers. In claims processing, insurers are obligated to receive, investigate, and act upon a claim filed by an insured individual. Claims processing comprises numerous administrative and customer service layers that include reviewing, investigating, adjusting (if necessary), remitting, or denying the claim.

Healthcare organizations are shifting from paper-based claims processing to electronic claims processing. According to CMS, the US healthcare payments market is expected to reach an estimated USD 5 trillion per year, as a total of both payer and consumer payments, by 2022. However, around 30% of this value is wasted due to inefficient payment processing and antiquated paper-based billing and administrative processes. Claims data comprises billing codes that providers submit to payers—an abundant and standardized source of data. To leverage this data, healthcare providers and payers are turning to analytics tools and services to determine prime risk areas, optimize pricing, facilitate denial management, improve claims response times, avoid fraudulent claims, and improve provider/patient relationships. Claims analytics, therefore, helps reduce false claims and provides vital information.

TABLE 24 COMMERCIALY AVAILABLE CLAIMS PROCESSING SOLUTIONS

SOLUTION	COMPANY
Claims Integrity	Inovalon
Optum Claim	Optum
Connectivity	Optum
SAS Claims Analytics	SAS
Claims Accuracy	Cotiviti
Claims Editing	Cerner
Oracle Health Insurance Analytics	Oracle

Source: Company Websites

TABLE 25 CLAIMS PROCESSING MARKET, BY REGION, 2019–2026 (USD MILLION)

Region	2019	2020	2021	2026	CAGR (2020–2026)
North America	1,379.8	1,625.4	2,159.5	7,385.6	27.9%
Europe	319.8	436.7	549.4	1,450.7	21.4%
Asia	238.2	315.5	400.0	1,103.5	22.5%
Pacific Countries	44.4	62.8	75.7	164.3	16.8%
LATAM	37.5	46.5	57.2	135.7	18.9%
Middle East & Africa	21.3	29.0	35.0	76.3	16.8%
Total	2,041.0	2,515.9	3,276.8	10,316.1	25.8%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association, American Health Information Management Association, National Health Care Anti-fraud Association, American Medical Informatics Association, Certification Commission for Health Information Technology, Healthcare Information Technology Standards Panel, Health Level Seven International, Information Technology Association of Canada, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network, Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

9.2.2 REVENUE CYCLE MANAGEMENT

9.2.2.1 Healthcare organizations are rapidly adopting predictive analytics solutions for proactive decision-making

Revenue cycle analytics provides business intelligence across the revenue cycle to healthcare providers and payers, transforming data points into self-explanatory, accessible analytics and predictive information to help improve financial performance. These solutions help improve the overall efficiency of the RCM process, including scheduling, registration, bill payment, claims denials, revenue adjustments, payer contracts, and billing and collection activities. These solutions provide healthcare organizations with a complete picture of critical revenue cycle performance data, lower operational costs, accelerate cash flows, and reduce accounts receivables.

TABLE 26 COMMERCIALY AVAILABLE RCM SOLUTIONS

SOLUTION	COMPANY
Revenue Integrity	MedeAnalytics
Optum 360	Optum
Revenue Cycle	Cotiviti
Revenue Cycle Explorer: Hospital	Health Catalyst
McKesson Performance Analytics	McKesson Corporation

Source: Company Websites

TABLE 27 REVENUE CYCLE MANAGEMENT MARKET, BY REGION, 2019–2026 (USD MILLION)

Region	2019	2020	2021	2026	CAGR (2021–2026)
North America	854.8	990.3	1,275.5	3,805.3	24.4%
Europe	301.4	397.1	480.3	1,067.3	17.3%
Asia	176.0	226.6	280.0	686.8	19.7%
LATAM	28.6	34.0	40.3	80.1	14.7%
Middle East & Africa	15.3	20.3	23.8	45.6	13.9%
Pacific Countries	12.0	16.6	19.6	38.1	14.2%
Total	1,388.1	1,685.0	2,119.5	5,723.2	22.0%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association, American Health Information Management Association, National Health Care Anti-fraud Association, American Medical Informatics Association, Certification Commission for Health Information Technology, Healthcare Information Technology Standards Panel, Health Level Seven International, Information Technology Association of Canada, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network, Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

9.2.3 PAYMENT INTEGRITY & FRAUD, WASTE, AND ABUSE

9.2.3.1 Implementation of ICD-10 code sets and adoption of fee-for-value systems will drive market growth

The rising demand for healthcare has also increased the number of people covered under healthcare schemes. However, with the rising number of insured people, fraudulent billings and medically unnecessary services billed to healthcare insurers are also likely to rise. Fraud and abuse are persistent problems faced by health plans across the globe.

To overcome the losses incurred due to fraud, the CMS decided to implement the prepayment review model in 2010 to detect fraudulent activities. According to this model, claims are reviewed using predictive analytical tools before paying the claim amount. The fraud analytics tool is an integral component of this model. This tool helps detect fraudulent activities and subjects them to an investigation. This helps reduce improper payments and the cost and time involved with the traditional pay-and-chase model. In 2019, the Massachusetts Office of Information Technology launched the prepayment predictive approach and observed a 27% drop in claims for duplicative services. The US government recovered USD 2.4 billion in 2015 and USD 3.3 billion in 2016 because of healthcare fraud judgments, settlements, and additional administrative impositions in healthcare fraud cases and proceedings.

To control fraud and abuse in healthcare insurance exchanges, the ACA will provide CMS with USD 350 million for the next ten years from 2011. This development aims to facilitate provider screening and collaborations with software companies for developing software to detect improper payments and fix penalty rates for improper claim submissions. While fraud and abuse cannot completely be eradicated from HIXs, predictive analytics enables payers to detect fraud early.

Payment integrity uses data mining and analytics to help insurers detect fraudulent payments and recover claims payments, shorten investigation cycles, and control medical spending, thus providing accuracy and consistency in claims reimbursement, reducing administrative costs, and preventing inappropriate claims payments.

This market segment is expected to grow faster during the forecast period due to the implementation of a prepayment review model by the CMS, technological advancements, increasing awareness of various fraud analytics tools, and high returns on investment in these solutions. Federal mandates such as the implementation of ICD-10 code sets, alternative payment models, and the adoption of fee-for-value systems will also drive the market segment. In the US, the ACA has imposed regulations on healthcare organizations to curb rising healthcare costs. For instance, the Act requires insurers to pay about 80–85% of the premium on claims, leaving only 15–20% for administration and profit.

However, the lack of skilled professionals and the time-consuming deployment of these solutions may hinder the growth of this market segment.

TABLE 28 **COMMERCIALLY AVAILABLE PAYMENT INTEGRITY & FWA SOLUTIONS**

SOLUTION	COMPANY
Emdeon Edge	Emdeon
ClaimsXten	McKesson
Prepayment and Post-Payment Solutions	Optum
DataProbe	Truven Health
Claims Integrity	Inovalon
Claims Accuracy	Verisk Analytics
IBM Cognos	IBM
Program Integrity: Prepay Review and Prevention	Truven Health
Investicclaim	McKesson
Fico Insurance Fraud Manager 3, Healthcare Edition	FICO
Claim Focus (SM)	LexisNexis
SAS Fraud Framework for Healthcare	SAS
SICOMine COB	SICO Health

Source: Company Websites

TABLE 29 **PAYMENT INTEGRITY & FWA MARKET, BY REGION, 2019–2026 (USD MILLION)**

Region	2019	2020	2021	2026	CAGR (2021–2026)
North America	567.3	691.0	949.2	3,896.8	32.6%
Europe	169.7	237.6	306.3	933.7	25.0%
Asia	114.1	155.5	203.0	657.7	26.5%
LATAM	18.1	22.9	29.0	80.2	22.6%
Middle East & Africa	9.9	13.7	16.9	40.9	19.4%
Pacific Countries	8.8	12.9	16.0	41.0	20.7%
Total	887.8	1,133.7	1,520.3	5,650.3	30.0%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association, American Health Information Management Association, National Health Care Anti-fraud Association, American Medical Informatics Association, Certification Commission for Health Information Technology, Healthcare Information Technology Standards Panel, Health Level Seven International, Information Technology Association of Canada, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network, Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

9.2.4 RISK ADJUSTMENT & ASSESSMENT

9.2.4.1 Risk adjustment & assessment allow for optimization of cash flows and improved financial performance

Risk adjustment includes using patient data to explain variations in healthcare spending, resource utilization, and health outcomes over a fixed period. An ideal risk adjustment system aligns incentives, predicts future risks, and finds ways to optimize cash flows and improve financial performance. It aims to pay health plans according to health risk enrollees and reduce the risk of negative financial consequences of insuring individuals at high risk. The health risks of enrollees can be quantified by using risk assessment tools.

Risk adjustment payments are adopted by both public and private health insurance companies. Risk assessment tools have been adopted by the Medicaid managed care program, the Medicare+Choice program, state employee programs, and a small number of private purchase cooperatives and large employers. These tools are also used to set initial managed care rates.

TABLE 30 COMMERCIALY AVAILABLE RISK ADJUSTMENT & ASSESSMENT SOLUTIONS

SOLUTION	COMPANY
Risk Model Analyzer	Health Catalyst
Optum Risk Reporting and Population Management Tools	Optum
Population Risk Assessment	Optum
Medicare Risk Adjustment	Cotiviti
DxCG Risk Analytics - Risk Analytics Solution for Providers	Cotiviti

Source: Company Websites

TABLE 31 RISK ADJUSTMENT & ASSESSMENT MARKET, BY REGION, 2019–2026 (USD MILLION)

Region	2019	2020	2021	2026	CAGR (2021–2026)
North America	328.5	386.5	512.6	1,763.9	28.0%
Europe	89.5	120.8	150.0	380.7	20.5%
Asia	58.4	77.3	98.1	275.1	22.9%
LATAM	9.2	11.2	13.6	30.1	17.2%
Pacific Countries	5.4	7.4	8.6	15.5	12.6%
Middle East & Africa	5.0	6.6	7.7	14.0	12.7%
Total	496.1	609.8	790.5	2,479.2	25.7%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association, American Health Information Management Association, National Health Care Anti-fraud Association, American Medical Informatics Association, Certification Commission for Health Information Technology, Healthcare Information Technology Standards Panel, Health Level Seven International, Information Technology Association of Canada, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network, Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

9.3 CLINICAL ANALYTICS

In the past few years, healthcare providers have figured out ways to manage their operational and financial data; however, the concept of managing and utilizing clinical data is relatively new. Clinical data contains patient information such as diagnoses, treatments administered, prescribed medicines, procedures and lab tests conducted, and hospitalizations. This data is collected from various sources such as EHRs, disease registries, patient surveys, and information exchanges among care providers. Effective analytics makes this exhaustive data more usable.

Clinical analytics encourages preventive care over reactive care and helps provide evidence-based treatment and personalized care to patients, enhancing healthcare quality and making it cost-effective. Clinical analytics tools capture and use clinical data to provide critical data insights for retrospective performance measures and predictive analytics, comparative effectiveness, quality improvement & clinical benchmarking, clinical decision support (CDS), and regulatory reporting & compliance.

The adoption of EHRs in healthcare organizations is increasing the volume of data available for performance measurement and quality improvement. According to data from the Office of the National Coordinator for Health Information Technology, in January 2019, about eight in ten (83%) office-based physicians in the US had adopted EHRs. In August 2019, the Riverside University Health System (RUHS) spent USD 53 million implementing Epic's EHR in its 439-bed medical center, ten health centers, and several primary and specialty clinics throughout Riverside County, California. The significant focus on the implementation of EHRs is influencing healthcare organizations to adopt clinical data analytics that helps them transition from data and information to knowledge and actionable insights.

Under the current ICD-9 system, only 17,000 codes are available for identifying diagnoses and procedures in the US. This transition is required for everyone (providers, including physicians) covered by the HIPAA. The implementation of ICD-10 has increased the amount of clinical data, the management of which requires advanced analytics. Thus, the growing adoption of EHRs, the implementation of ICD-10, and increased focus on delivering quality care while reducing healthcare costs will drive the growth of the clinical analytics market during the forecast period.

Despite the advantages offered by analytics, its penetration is limited to large hospitals on the provider front. A major barrier is the lack of IT knowledge among medical professionals and data security issues; companies are, therefore, increasing their efforts to develop solutions that are easy to understand and interpret.

TABLE 32 **COMMERCIALLY AVAILABLE CLINICAL ANALYTICS SOLUTIONS**

SOLUTION	COMPANY
DxCG Intelligence	Cotiviti
Clinical Quality Metrics	IntrinsiQ
InfoSphere Clinical Analytics Solution	IBM
Optum Compliance Suite	Optum

Source: Company Websites

TABLE 33 CLINICAL ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Quality Improvement & Clinical Benchmarking	1,209.4	1,744.5	2,586.5	14,142.2	40.5%
Clinical Decision Support	897.3	1,190.7	1,626.7	6,245.6	30.9%
Precision Health	594.4	778.5	1,064.4	4,070.5	30.8%
Reporting & Compliance	496.1	644.1	864.4	3,025.6	28.5%
Medical Imaging Analytics	255.0	346.1	475.7	1,926.6	32.3%
Comparative Analytics/Effectiveness	246.1	337.6	476.9	2,169.8	35.4%
Total	3,698.3	5,041.4	7,094.6	31,580.3	34.8%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association, American Health Information Management Association, National Health Care Anti-fraud Association, American Medical Informatics Association, Certification Commission for Health Information Technology, Healthcare Information Technology Standards Panel, Health Level Seven International, Information Technology Association of Canada, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network, Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 34 CLINICAL ANALYTICS MARKET, BY REGION, 2019–2026 (USD MILLION)

Region	2019	2020	2021	2026	CAGR (2021–2026)
North America	2,217.2	2,892.8	4,203.9	21,166.2	38.2%
Europe	803.2	1,169.0	1,558.1	5,375.4	28.1%
Asia	498.6	721.0	993.1	3,965.8	31.9%
LATAM	61.4	94.3	122.9	373.6	24.9%
Pacific Countries	75.2	102.3	137.4	476.9	28.3%
Middle East & Africa	42.8	62.0	79.2	222.4	22.9%
Total	3,698.3	5,041.4	7,094.6	31,580.3	34.8%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association, American Health Information Management Association, National Health Care Anti-fraud Association, American Medical Informatics Association, Certification Commission for Health Information Technology, Healthcare Information Technology Standards Panel, Health Level Seven International, Information Technology Association of Canada, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network, Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

9.3.1 QUALITY IMPROVEMENT & CLINICAL BENCHMARKING

9.3.1.1 Benchmarking allows for the identification of issues that are negatively impacting healthcare systems

Quality improvement and clinical benchmarking enable healthcare providers to identify and manage issues negatively impacting healthcare systems. They also aid in designing solutions for those problems and measuring the effects of changes, enabling healthcare organizations to improve quality and reduce costs.

Clinical benchmarking relies on several standard measures used by healthcare providers to benchmark performance while gathering information from various organizations to identify and quantify the highest performance levels from a sample of providers. It helps providers identify and quantify opportunities for improvements that may not be identified by internal data alone. For instance, clinical benchmarking could show potential cost savings for a hospital that has reduced its surgical complication rates to the level of the best group surgeons in the comparative database. However, the standardization of measurement is crucial for meaningful benchmarking. Patient registries focused on narrow populations of patients with a specific condition or procedure implement unique data definitions that allow deeper clinical analysis with

the help of analytics tools. This increases the use of analytics solutions and drives the healthcare analytics market for quality improvement and clinical benchmarking.

TABLE 35 **COMMERCIALLY AVAILABLE QUALITY IMPROVEMENT & CLINICAL BENCHMARKING SOLUTIONS**

SOLUTION	COMPANY
Care Discovery	Truven Health
Healthcare Performance Benchmarks	McKesson
Quality Advisor	Premium
Caradigm Quality Improvement	Caradigm Health
HEIDS Quality Management	MedeAnalytics
SAS Data Quality	SAS
Quality Reporting	Verisk Analytics
Clinical Quality	Allscripts

Source: Company Websites

TABLE 36 **QUALITY IMPROVEMENT & CLINICAL BENCHMARKING MARKET, BY REGION, 2019–2026 (USD MILLION)**

Region	2019	2020	2021	2026	CAGR (2021–2026)
North America	706.3	975.2	1,500.0	9,376.0	44.3%
Europe	270.4	416.8	585.6	2,491.9	33.6%
Asia	170.7	258.3	371.4	1,781.1	36.8%
LATAM	24.7	36.0	51.5	225.3	34.3%
Pacific Countries	23.9	38.0	51.2	178.8	28.4%
Middle East & Africa	13.3	20.2	26.9	89.1	27.1%
Total	1,209.4	1,744.5	2,586.5	14,142.2	40.5%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association, American Health Information Management Association, National Health Care Anti-fraud Association, American Medical Informatics Association, Certification Commission for Health Information Technology, Healthcare Information Technology Standards Panel, Health Level Seven International, Information Technology Association of Canada, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network, Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

9.3.2 CLINICAL DECISION SUPPORT

9.3.2.1 Predictive clinical analytics reduce medical errors and ensure that appropriate tests are performed, a key factor for market growth

CDS helps healthcare providers by giving them specific and filtered information to make the necessary choices to improve the quality of care. CDS also involves providing the right information at the right time to help a clinician provide the best care to an individual.

CDS was earlier limited to providing physicians with access to documentation of standard care paths or details of drug dosage interactions. However, with technological advancements, smart CDS solutions use predictive analytics to tailor specific care guidance based on details of the individual patient's condition. For instance, some organizations use predictive clinical analytics to predict a patient's risk of readmission using clinical data elements from an EMR. These solutions can help expedite diagnosis, reduce medical errors, and ensure that the appropriate tests are performed, resulting in optimum patient outcomes.

Players in the healthcare analytics market are focusing on developing and providing new solutions for CDS. For instance, in October 2017, the Indianapolis-based Community Health Network announced the deployment of Stanson Health's CDS and analytics tools. The Community Health Network is a non-profit health system with more than 200 sites and integrates hundreds of physicians, specialty and acute care hospitals, surgery centers, home care services, MedChecks, and behavioral health and employer health services.

Such developments increase the availability of advanced analytics solutions for CDS, thereby increasing their adoption.

TABLE 37 **COMMERCIALLY AVAILABLE SOLUTIONS FOR CDS**

SOLUTION	COMPANY
ZynxOrder	Zynx Health
UpToDate	Wolters Kluwer Health
Clinical Decision Support System	IBM
InterQual Evidence-based Clinical Solutions	McKesson

Source: Company Websites

TABLE 38 CLINICAL DECISION SUPPORT MARKET, BY REGION, 2019–2026 (USD MILLION)

Region	2019	2020	2021	2026	CAGR (2021–2026)
North America	555.0	707.1	993.4	4,258.2	33.8%
Europe	184.3	262.0	340.7	1,036.4	24.9%
Asia	115.7	163.2	218.9	760.5	28.3%
LATAM	18.5	23.6	29.8	76.0	20.6%
Pacific Countries	13.3	20.2	25.9	73.0	23.0%
Middle East & Africa	10.5	14.6	18.0	41.6	18.3%
Total	897.3	1,190.7	1,626.7	6,245.6	30.9%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association, American Health Information Management Association, National Health Care Anti-fraud Association, American Medical Informatics Association, Certification Commission for Health Information Technology, Healthcare Information Technology Standards Panel, Health Level Seven International, Information Technology Association of Canada, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network, Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

9.3.3 REPORTING & COMPLIANCE

9.3.3.1 Stringent regulatory requirements have led to the need for reporting & compliance solutions

Clinical analytics tools in the regulatory reporting and compliance segment help track regulatory changes, implement policies and procedures, define & assess controls, identify and assess the risk of non-compliance, and perform internal audits.

Physician performance evaluation is gaining importance among clinical and financial decision-makers, as physician performance directly influences the overall performance of hospitals. The ongoing Professional Practice Evaluation (OPPE) accreditation requirement in the US (which mandates an ongoing process by hospitals for physician performance reviews) will also drive the integration of data analytics for physician performance evaluation. Other regulatory measures such as the Hospital Consumer Assessment of Healthcare Providers and Systems (HCAHPS), Physician Quality Reporting System (PQRS), Meaningful Use, and Core Measures also demand timely reporting and compliance, which can be easily achieved with the use of analytics. Analytics solutions help providers evaluate a physician's performance pattern against benchmarks and adhere to set guidelines and best practices, optimizing physician performance and patient treatment. Analytics enables payers to select physicians for inclusion in their provider networks and pays quality bonuses to well-performing physicians.

TABLE 39 REPORTING & COMPLIANCE MARKET, BY REGION, 2019–2026 (USD MILLION)

Region	2019	2020	2021	2026	CAGR (2021–2026)
North America	306.5	380.5	524.2	2,026.8	31.1%
Europe	103.6	143.6	182.2	490.6	21.9%
Asia	65.0	90.8	120.7	406.7	27.5%
LATAM	9.9	13.1	17.1	50.9	24.4%
Pacific Countries	5.7	8.3	10.3	24.2	18.7%
Middle East & Africa	5.4	7.8	9.9	26.5	21.8%
Total	496.1	644.1	864.4	3,025.6	28.5%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association, American Health Information Management Association, National Health Care Anti-fraud Association, American Medical Informatics Association, Certification Commission for Health Information Technology, Healthcare Information Technology Standards Panel, Health Level Seven International, Information Technology Association of Canada, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network, Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

9.3.4 COMPARATIVE ANALYTICS/EFFECTIVENESS

9.3.4.1 Growing adoption of EHRs to provide significant growth opportunities

Comparative analytics is used in the healthcare industry to measure performance, ensure timely reimbursements, and avoid financial penalties. Comparative analytics/effectiveness brings together big data from healthcare payers, healthcare providers, the supply chain, and patient data in real-time, enabling healthcare stakeholders to compare their performance with their peers. The rapid adoption of EHRs results in the aggregation of much broader and deeper clinical data that makes comparing the effectiveness and costs of alternative patterns of care more feasible. Therefore, the growing adoption of EHRs is expected to provide significant growth opportunities for this market segment.

TABLE 40 COMMERCIALLY AVAILABLE COMPARATIVE ANALYTICS/EFFECTIVENESS SOLUTIONS

SOLUTION	COMPANY
RelayAnalytics Pulse	McKesson Corporation
Truven Health ActionOI Practice Insights	Truven Health Analytics
TITAN	RemitDATA Inc.

Source: Company Websites

TABLE 41 COMPARATIVE ANALYTICS/EFFECTIVENESS MARKET, BY REGION, 2019–2026 (USD MILLION)

Region	2019	2020	2021	2026	CAGR (2021–2026)
North America	152.5	201.6	293.5	1,505.2	38.7%
Europe	54.0	78.8	105.5	370.4	28.5%
Asia	29.7	43.0	59.2	236.3	31.9%
LATAM	3.9	5.3	7.2	26.2	29.4%
Pacific Countries	3.3	5.0	6.4	17.5	22.4%
Middle East & Africa	2.7	4.0	5.1	14.2	22.8%
Total	246.1	337.6	476.9	2,169.8	35.4%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association, American Health Information Management Association, National Health Care Anti-fraud Association, American Medical Informatics Association, Certification Commission for Health Information Technology, Healthcare Information Technology Standards Panel, Health Level Seven International, Information Technology Association of Canada, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network, Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

9.3.5 PRECISION HEALTH

9.3.5.1 Strong government support for personalized medicine is likely to drive market growth

Precision health uses a patient's genomic data to develop personalized medicine. Precision health uses predictive analytics to provide personalized treatment for diseases such as cancer and diabetes by selecting the most effective and least toxic treatments for a particular patient.

The US government supports the growth of precision health by providing financial aid for the R&D of personalized medicine. For instance, in September 2018, The *All of Us* Research Program granted USD 28.6 million to establish three genome centers across the country to accelerate research in precision medicine [Source: National Institutes of Health (NIH)]. Such government initiatives will create a demand for analytical solutions for personalized medicine, thereby driving the growth of this market segment.

Other contributing factors include the increased adoption of EHRs, strong government support for personalized medicine (especially in the US), rising prevalence of chronic diseases, collaborations between market players and universities to develop efficient analytics solutions for personalized health, and frequent product launches.

TABLE 42 COMMERCIALY AVAILABLE PRECISION HEALTH SOLUTIONS

SOLUTION	COMPANY
- SAP Foundation for Health - SAP Medical Research Insights Solution	SAP
Allscripts Precision Medicine	Allscripts

Source: Company Websites

TABLE 43 PRECISION HEALTH MARKET, BY REGION, 2019–2026 (USD MILLION)

Region	2019	2020	2021	2026	CAGR (2021–2026)
North America	373.9	468.5	660.6	2,834.2	33.8%
Europe	123.1	172.7	222.1	638.4	23.5%
Asia	71.1	100.2	134.3	465.7	28.2%
LATAM	10.7	14.1	18.4	55.5	24.7%
Pacific Countries	9.1	13.7	17.4	46.5	21.8%
Middle East & Africa	6.4	9.2	11.6	30.1	21.1%
Total	594.4	778.5	1,064.4	4,070.5	30.8%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association, American Health Information Management Association, National Health Care Anti-fraud Association, American Medical Informatics Association, Certification Commission for Health Information Technology, Healthcare Information Technology Standards Panel, Health Level Seven International, Information Technology Association of Canada, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network, Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

9.3.6 MEDICAL IMAGING ANALYTICS

9.3.6.1 Imaging analytics aided by artificial intelligence and machine learning is rapidly expanding within the healthcare IT industry

Medical imaging analytics involves the use of data mining and machine learning techniques to analyze medical images. Medical imaging is one of the fastest-growing areas of discovery that offers radiologists, pathologists, ophthalmologists, and other medical practitioners an opportunity to enhance their workflows with advanced systems and solutions.

Imaging analytics offers various benefits in the healthcare industry, such as:

- Real-time insights to healthcare providers during diagnosis and treatment
- Diagnosis of certain genetic disorders
- Reduces varying subjective interpretations and human errors
- Improves the standard of care and speed of diagnosis, treatment, and recovery
- Offers the potential to make faster and more informed decisions, streamlines costs, and broadly improves the quality and economics of healthcare

Imaging analytics aided by artificial intelligence and machine learning is rapidly expanding within the healthcare IT industry. From reducing administrative burdens to supporting decisions, AI and machine learning are showing various prospects across financial, clinical, and operational domains; thus, representing significant growth opportunities for the medical imaging analytics market.

TABLE 44 MEDICAL IMAGING ANALYTICS MARKET, BY REGION, 2019–2026 (USD MILLION)

Region	2019	2020	2021	2026	CAGR (2021–2026)
North America	123.0	160.0	232.2	1,165.8	38.1%
Europe	67.7	95.0	122.0	347.8	23.3%
Asia	46.3	65.6	88.5	315.5	29.0%
LATAM	7.6	10.1	13.4	43.1	26.3%
Pacific Countries	6.0	9.1	11.8	33.5	23.3%
Middle East & Africa	4.3	6.2	7.8	20.9	21.7%
Total	255.0	346.1	475.7	1,926.6	32.3%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association, American Health Information Management Association, National Health Care Anti-fraud Association, American Medical Informatics Association, Certification Commission for Health Information Technology, Healthcare Information Technology Standards Panel, Health Level Seven International, Information Technology Association of Canada, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network, Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

9.4 OPERATIONAL & ADMINISTRATIVE ANALYTICS

Operational analytics solutions help healthcare providers, insurance providers, and ACOs to achieve better results through the efficient and effective utilization of resources, supply chains, and strategic planning. This market is further segmented into supply chain analytics, workforce analytics, and strategic analytics.

Operational analytics is an emerging application area in the healthcare analytics market. Hospitals and other healthcare providers can utilize operational analytics to reduce treatment and outcome analysis costs and help improve results, resource utilization, and rare event detection. The emerging applications and increasing awareness of paperless technology are factors driving the operational analytics market. However, a lack of data analysis expertise is one of the major challenges for the growth of this market segment.

TABLE 45 **COMMERCIALLY AVAILABLE SOLUTIONS FOR OPERATIONAL & ADMINISTRATIVE ANALYTICS**

SOLUTION	COMPANY
SmartLinx IQ	CapsuleTech
Oracle Enterprise Healthcare Analytics	Oracle
Allscripts Patient Flow	Allscripts

Source: Company Websites

TABLE 46 **OPERATIONAL & ADMINISTRATIVE ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)**

Type	2019	2020	2021	2026	CAGR (2021–2026)
Supply Chain Analytics	976.0	1,156.0	1,433.7	3,574.1	20.0%
Workforce Analytics	848.3	1,019.0	1,288.6	3,557.0	22.5%
Strategic Analytics	342.9	409.7	517.0	1,426.8	22.5%
Total	2,167.2	2,584.7	3,239.4	8,557.9	21.4%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association, American Health Information Management Association, National Health Care Anti-fraud Association, American Medical Informatics Association, Certification Commission for Health Information Technology, Healthcare Information Technology Standards Panel, Health Level Seven International, Information Technology Association of Canada, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network, Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 47 OPERATIONAL & ADMINISTRATIVE ANALYTICS MARKET, BY REGION, 2019–2026 (USD MILLION)

Region	2019	2020	2021	2026	CAGR (2021–2026)
North America	1,426.3	1,624.8	2,080.0	6,001.8	23.6%
Europe	419.3	551.1	666.9	1,481.0	17.3%
Asia	235.6	299.2	364.4	833.5	18.0%
LATAM	25.3	34.6	40.2	73.4	12.8%
Pacific Countries	37.1	44.1	52.1	103.0	14.6%
Middle East & Africa	23.6	30.9	35.9	65.1	12.7%
Total	2,167.2	2,584.7	3,239.4	8,557.9	21.4%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association, American Health Information Management Association, National Health Care Anti-fraud Association, American Medical Informatics Association, Certification Commission for Health Information Technology, Healthcare Information Technology Standards Panel, Health Level Seven International, Information Technology Association of Canada, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network, Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

9.4.1 SUPPLY CHAIN ANALYTICS

9.4.1.1 The growing requirement to reduce healthcare costs will increase the adoption of supply chain analytics solutions

Supply chain analytics enables healthcare organizations to make informed decisions that enhance operational efficiencies and outcomes for care delivery and result in cost savings. It helps healthcare organizations to evaluate the performance of the inventory and central supply processes that address recalls, expired products, and consignment items and make day-to-day improvements. By capturing and analyzing patient care, inventory levels, shipping costs, and more, healthcare organizations can accurately determine the right amount of medication and equipment for purchase. Such insights provided by supply chain analytics allow hospitals to streamline their supply chain to increase efficiency and reduce expenses.

Supply chain expenses can be as high as 35–45% of a health system's total operating costs, consequently putting pressure on the margins and profitability in an already competitive and resource-constrained environment. The timely availability of the right product and service and cost-effective procurement are important for patient health outcomes.

Supply chain analytics also assists organizations in reducing manual efforts so that more time can be devoted to value-added projects. The growing need to reduce healthcare costs, implement better inventory management practices in hospitals, and comply with regulatory requirements is a major factor driving market growth.

TABLE 48 COMMERCIALY AVAILABLE SUPPLY CHAIN ANALYTICS SOLUTIONS

SOLUTION	COMPANY
Oracle Healthcare Provider Supply Chain Analytics	Oracle
CareAware Critical Care	Cerner
Vizient Savings Actualizer	Vizient

Source: Company Websites

TABLE 49 SUPPLY CHAIN ANALYTICS MARKET, BY REGION, 2019–2026 (USD MILLION)

Region	2019	2020	2021	2026	CAGR (2021–2026)
North America	630.0	709.8	897.0	2,417.4	21.9%
Europe	192.0	249.9	299.3	629.2	16.0%
Asia	114.5	146.5	179.7	424.1	18.7%
LATAM	17.2	20.3	23.8	45.2	13.7%
Pacific Countries	11.4	15.3	17.7	30.5	11.5%
Middle East & Africa	10.9	14.2	16.3	27.9	11.4%
Total	976.0	1,156.0	1,433.7	3,574.1	20.0%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association, American Health Information Management Association, National Health Care Anti-fraud Association, American Medical Informatics Association, Certification Commission for Health Information Technology, Healthcare Information Technology Standards Panel, Health Level Seven International, Information Technology Association of Canada, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network, Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

9.4.2 WORKFORCE ANALYTICS

9.4.2.1 Need for real-time resource demand and supply data to drive the adoption of workforce analytics solutions

Healthcare organizations need to maximize the productivity and effectiveness of various assets and quantify how they perform against set targets to make the best use of available resources. Workforce analytics enables healthcare organizations to control employee costs by selecting the right person at the right time. It also improves workforce productivity by making rapid adjustments in workforce availability to match fluctuating patient volumes. Workforce analytics also helps in planning effective recruitment and retaining talent. These solutions help healthcare organizations to overcome cost pressures.

Hospitals have begun adopting workforce analytics as a model of choice. Workforce analytics solutions are increasingly being used in healthcare organizations as they provide real-time resource demand and supply data, thus helping executives and managers to manage their workforce.

TABLE 50 COMMERCIALLY AVAILABLE WORKFORCE ANALYTICS SOLUTIONS

SOLUTION	COMPANY
Workforce Planning	SAP
Workforce Analytics Solution	Kronos
Outcomes-driven Tracking	Cerner
IBM Kenexa Talent Suite	IBM
PeopleSoft Workforce Management Solution	Oracle
Care-Aware Tracking	Cerner
Workforce Analytics	SAP

Source: Company Websites

TABLE 51 WORKFORCE ANALYTICS MARKET, BY REGION, 2019–2026 (USD MILLION)

Region	2019	2020	2021	2026	CAGR (2021–2026)
North America	565.6	649.3	837.9	2,517.2	24.6%
Europe	161.5	214.9	263.4	622.8	18.8%
Asia	87.8	111.7	136.2	314.4	18.2%
LATAM	14.2	17.1	20.4	43.1	16.1%
Pacific Countries	10.2	14.1	16.6	32.2	14.2%
Middle East & Africa	9.0	12.0	14.1	27.2	14.1%
Total	848.3	1,019.0	1,288.6	3,557.0	22.5%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association, American Health Information Management Association, National Health Care Anti-fraud Association, American Medical Informatics Association, Certification Commission for Health Information Technology, Healthcare Information Technology Standards Panel, Health Level Seven International, Information Technology Association of Canada, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network, Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

9.4.3 STRATEGIC ANALYTICS

9.4.3.1 Strategic analytics uses advanced analytics tools to leverage operational and usage data, a key advantage

Strategic analytics helps healthcare organizations adhere to their goals and objectives; develop strategic business plans to evaluate market potential; and identify the necessary steps for geographic expansion, new venture initiation, or service line expansion. It also facilitates better planning for future capital investments such as new equipment and new facilities. Strategic analytics uses advanced analytics tools such as predictive and prescriptive analytics to leverage operational and usage data, coupled with data from external sources.

TABLE 52 STRATEGIC ANALYTICS MARKET, BY REGION, 2019–2026 (USD MILLION)

Region	2019	2020	2021	2026	CAGR (2021–2026)
North America	230.7	265.8	345.0	1067.2	25.3%
Europe	65.8	86.3	104.2	229.0	17.1%
Asia	33.3	41.0	48.5	95.1	14.4%
LATAM	5.7	6.7	7.8	14.7	13.5%
Pacific Countries	3.8	5.1	5.9	10.8	12.7%
Middle East & Africa	3.6	4.8	5.5	10.0	12.5%
Total	342.9	409.7	517.0	1426.8	22.5%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association, American Health Information Management Association, National Health Care Anti-fraud Association, American Medical Informatics Association, Certification Commission for Health Information Technology, Healthcare Information Technology Standards Panel, Health Level Seven International, Information Technology Association of Canada, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network, Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

9.5 POPULATION HEALTH ANALYTICS

9.5.1 POPULATION HEALTH ANALYTICS IMPROVES POPULATION HEALTH OUTCOMES AND LOWERS COSTS, A KEY FACTOR FOR MARKET GROWTH

Population health analytics is a part of the overall population health management (PHM) market. Various solutions such as data integration, visualization, and analytics are used for PHM. Population health analytics is considered a separate application segment in this report as it includes both clinical and financial applications.

Population health analytics solutions help improve population health outcomes and lower costs through data aggregation and insight-driven prioritization, risk stratification, coordination, and management of patient-centered care. This segment focuses on analytics tools used for PHM that continuously aggregate and analyze clinical and financial data across enterprises and provide actionable insights to healthcare providers and payers.

These insights help them view population health trends and care patterns, understand high-risk patient populations, and identify clinical gaps in care to take preventive measures before their care becomes costly. Population risk management, population care management, patient engagement, clinical outcome engagement, and activity-based costing are areas where PHM analytics solutions are used frequently.

TABLE 53 COMMERCIALY AVAILABLE POPULATION HEALTH ANALYTICS SOLUTIONS

SOLUTION	COMPANY
DxCG Intelligence	Cotiviti
McKesson Population Health Manager	McKesson
Population Health Intelligence Platform	Conifer Health
Health Intent Platform	Cerner
Allscripts dB Motion Population Health Management	Allscripts Health

Source: Company Websites

TABLE 54 POPULATION HEALTH ANALYTICS MARKET, BY REGION, 2019–2026 (USD MILLION)

Region	2019	2020	2021	2026	CAGR (2021–2026)
North America	1,230.5	1,479.5	1,999.0	7,565.9	30.5%
Europe	359.9	503.1	647.3	1,955.5	24.7%
Asia	175.4	238.2	310.3	996.7	26.3%
LATAM	32.0	40.7	51.6	143.4	22.7%
Middle East & Africa	20.4	28.2	34.6	82.4	18.9%
Pacific Countries	17.6	25.7	32.0	82.0	20.7%
Total	1,835.7	2,315.4	3,074.7	10,825.8	28.6%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association, American Health Information Management Association, National Health Care Anti-fraud Association, American Medical Informatics Association, Certification Commission for Health Information Technology, Healthcare Information Technology Standards Panel, Health Level Seven International, Information Technology Association of Canada, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network, Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

10 HEALTHCARE ANALYTICS MARKET, BY END USER

KEY FINDINGS

- The payers segment accounted for the largest share of 51.8% of the global healthcare analytics market in 2020. This segment is projected to reach USD 28,484.5 million by 2026 from USD 10,518.9 million in 2021, growing at a CAGR of 28.9% during the forecast period.
- Factors such as the rising healthcare costs and the growing need to increase memberships, curb fraudulent claims, optimize provider networks, and reduce operational costs are compelling healthcare payers to adopt analytics solutions.
- The healthcare providers segment is projected to reach USD 39,114.1 million by 2026 from USD 8,531.2 million in 2021, growing at the highest CAGR of 35.6% during the forecast period.

10.1 INTRODUCTION

Based on end users, the healthcare analytics market is segmented into payers; providers; and HIEs (Health Insurance Exchange), ACOs (Accountable Care Organizations), MCOs (Managed Care Organization), and TPAs (Third Party Administrators).

Payers include private healthcare insurance companies, government agencies, employers, and private insurance exchanges. Healthcare providers include hospitals, physician practices, Integrated Delivery Networks (IDNs), post-acute care organizations, and ambulatory care settings.

The payers segment is estimated to account for the largest share of 51.2% of the market during the forecast period. The growing need to curb fraudulent claims, optimize provider networks, and reduce operational costs are key factors compelling healthcare payers to adopt healthcare analytics solutions, thus driving the market growth. On the other hand, the providers' segment is projected to grow at the highest CAGR of 35.9% during the forecast period.

TABLE 55 HEALTHCARE ANALYTICS MARKET, BY END USER, 2019–2026 (USD MILLION)

End User	2019	2020	2021	2026	CAGR (2021–2026)
Payers	6,692.1	8,223.2	10,518.9	28,484.5	22.0%
Providers	4,610.1	6,116.5	8,531.2	39,114.1	35.6%
ACOs, HIEs, MCOs, and TPAs	1,211.9	1,546.2	2,065.6	7,534.2	29.5%
Total	12,514.2	15,885.8	21,115.7	75,132.8	28.9%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association, American Health Information Management Association, National Health Care Anti-fraud Association, American Medical Informatics Association, Certification Commission for Health Information Technology, Healthcare Information Technology Standards Panel, Health Level Seven International, Information Technology Association of Canada, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network, Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

10.2 PAYERS

Healthcare payers include government agencies, private insurance companies, and employers & private insurance exchanges. Payers use healthcare analytics tools for risk management, identifying claims trends, and maximizing payment accuracy. Data analytics also helps payers identify members likely to develop adverse risk conditions and identify members that can be safely redirected from unnecessary treatments to lower-cost alternatives.

With the increasing adoption of predictive analytics, payers can identify patterns in claims and proactively intervene in fraudulent claims that would be otherwise overlooked due to a lack of data, resources, or time constraints. Several payers have begun using analytical tools that make extensive use of data, statistical and quantitative analytics, and predictive models to be more proactive, make better fact-based decisions, and take effective actions. This approach helps optimize various activities, such as billing, care delivery, and fraud reduction. The high utilization of analytics solutions by payers is expected to boost the healthcare analytics market during the forecast period. The market is moving towards an enterprise-wide view of analytics, which helps payers obtain integrated insights to enhance business efficiency and effectiveness.

Some commercially available solutions for healthcare payers are:

SOLUTION	COMPANY
IBM Research	IBM
Payment Integrity	McKesson
Claims Accuracy	Verisk Analytics
Oracle Financial Analytics	Oracle
Optum 360	Optum
Revenue Integrity	MedeAnalytics

Source: Company Websites

Other vendors offering payer healthcare analytics solutions are Inovalon (US), Fuzzy Logix (US), Hexaware Technologies (US), Emdeon (US), Xerox (US), Samma Technologies (US), TriZetto (US), Wipro (India), and PluralSoft (US).

TABLE 56 HEALTHCARE ANALYTICS MARKET FOR PAYERS, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Private Insurance Companies	3,931.9	4,754.0	5,969.2	14,543.2	19.5%
Government Agencies	2,127.1	2,687.5	3,541.9	11,109.8	25.7%
Employers and Private Exchanges	633.1	781.7	1,007.8	2,831.5	23.0%
Total	6,692.1	8,223.2	10,518.9	28,484.5	22.0%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association, American Health Information Management Association, National Health Care Anti-fraud Association, American Medical Informatics Association, Certification Commission for Health Information Technology, Healthcare Information Technology Standards Panel, Health Level Seven International, Information Technology Association of Canada, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network, Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 57 HEALTHCARE ANALYTICS MARKET FOR PAYERS, BY REGION, 2019–2026 (USD MILLION)

Region	2019	2020	2021	2026	CAGR (2021–2026)
North America	4,227.1	4,899.2	6,341.0	17,350.9	22.3%
Europe	1,272.3	1,723.2	2,144.5	5,366.8	20.1%
Asia	880.2	1,183.0	1,524.8	4,623.6	24.8%
LATAM	123.3	152.1	186.4	426.8	18.0%
Pacific Countries	116.4	166.2	202.3	455.2	17.6%
Middle East & Africa	72.9	99.3	119.9	261.3	16.9%
Total	6,692.1	8,223.2	10,518.9	28,484.5	22.0%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association, American Health Information Management Association, National Health Care Anti-fraud Association, American Medical Informatics Association, Certification Commission for Health Information Technology, Healthcare Information Technology Standards Panel, Health Level Seven International, Information Technology Association of Canada, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network, Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

10.2.1 PRIVATE INSURANCE COMPANIES

10.2.1.1 Private insurance companies accounted for the largest share of the payers market in 2020

Private insurance companies worldwide are increasingly relying on analytics solutions for fraud prevention, claims processing, managing costs, improving member and provider satisfaction, and differentiating services in the marketplace. According to the US Census Bureau, 2017, private health insurance coverage (62.7%) is more widespread than government coverage (37.7%) in the US. The high number of enrollees for private insurance creates large volumes of patient data, which is expected to drive the adoption of healthcare analytics solutions among the private insurance companies to transform this data into actionable insights.

However, the lack of infrastructure for capturing data is the biggest challenge private insurance companies face; this could hamper the growth of this segment during the forecast period.

TABLE 58 HEALTHCARE ANALYTICS MARKET FOR PRIVATE INSURANCE COMPANIES, BY REGION, 2019–2026 (USD MILLION)

Region	2019	2020	2021	2026	CAGR (2021–2026)
North America	2,472.6	2,810.9	3,560.0	8,585.6	19.3%
Europe	741.7	987.6	1,206.6	2,701.7	17.5%
Asia	526.6	703.4	900.8	2,641.2	24.0%
Pacific Countries	76.3	107.7	129.4	270.5	15.9%
LATAM	71.7	86.4	103.2	202.6	14.4%
Middle East & Africa	43.0	58.0	69.2	141.6	15.4%
Total	3,931.9	4,754.0	5,969.2	14,543.2	19.5%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association, American Health Information Management Association, National Health Care Anti-fraud Association, American Medical Informatics Association, Certification Commission for Health Information Technology, Healthcare Information Technology Standards Panel, Health Level Seven International, Information Technology Association of Canada, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network, Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

10.2.2 GOVERNMENT AGENCIES

10.2.2.1 Increasing use of analytics solutions by government agencies to boost the market growth of this end-user segment

Government insurers are increasingly adopting analytics solutions to maximize payment accuracy; set reimbursement rates; target preventable and costly inefficiencies; assess, interpret, and prevent population risks; prevent fraud; and improve their Healthcare Effectiveness Data and Information Set (HEDIS) scores, CMS 5-Star programs, and revenue compliance initiatives.

In the US, the Patient Protection and Affordable Care Act (ACA) mandates every state to offer an exchange to its citizens, irrespective of its mode of provision. According to the Hospital for Special Surgery (HSS), as of December 2019, nearly 8.0 million people enrolled for the 2019 health insurance coverage in federal exchanges. The increase in the number of enrollees in such government schemes is expected to create a large volume of patient data, which will drive the adoption of analytics solutions among public exchanges to transform the data into valuable insights.

TABLE 59 HEALTHCARE ANALYTICS MARKET FOR GOVERNMENT AGENCIES, BY REGION, 2019–2026 (USD MILLION)

Region	2019	2020	2021	2026	CAGR (2021–2026)
North America	1,349.0	1,614.0	2,160.3	6,973.6	26.4%
Europe	406.8	567.5	727.6	2,125.8	23.9%
Asia	276.1	375.6	490.1	1,579.1	26.4%
LATAM	40.3	51.7	66.0	184.7	22.9%
Pacific Countries	31.7	46.5	58.3	151.3	21.0%
Middle East & Africa	23.2	32.2	39.6	95.3	19.2%
Total	2,127.1	2,687.5	3,541.9	11,109.8	25.7%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association, American Health Information Management Association, National Health Care Anti-fraud Association, American Medical Informatics Association, Certification Commission for Health Information Technology, Healthcare Information Technology Standards Panel, Health Level Seven International, Information Technology Association of Canada, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network, Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

10.2.3 EMPLOYERS AND PRIVATE EXCHANGES

10.2.3.1 Private exchanges and employers use data analytics for rapid data integration, health plan analysis, and risk modeling

Private exchanges are run by health insurance companies, brokerage firms, or consulting firms. In the case of public exchanges, private exchanges work on a defined contribution model and do not offer subsidies for low-wage workers. Rising healthcare costs are the major factors compelling employers to shift from traditional and defined benefit plans to contribution plans. This helps employers maintain a constant healthcare budget for a year by shifting the costs to their employees and the risks to the insurance companies. As a result, there is growing interest in private health insurance exchanges (HIXs). Approximately 8 million people in the US enrolled in HIXs in 2016, representing a 35% increase over 2015. Moreover, 20% of employees are expected to be insured by private exchanges in the US by the end of 2017. Similarly, 40 million out of the 150 million non-elderly people with employer health insurance will be choosing their plans through private exchanges by 2018 (Source: Accenture).

Private exchanges and employers use data analytics for rapid integration, quality assurance, benchmarking, health plan analysis, and risk modeling. Robust analytics platforms allow employers and private exchanges to turn their data into actionable information and measurable outcomes. This helps them control spending, understand employee disease or wellness trends, and ensure accurate budgeting and forecasting. While the private HIX market is currently nascent, it is expected to register significant growth in the coming five years. The growing private HIX market will provide significant growth opportunities for players operating in the healthcare analytics market during the forecast period.

TABLE 60 HEALTHCARE ANALYTICS MARKET FOR EMPLOYERS AND PRIVATE EXCHANGES, BY REGION, 2019–2026 (USD MILLION)

Region	2019	2020	2021	2026	CAGR (2021–2026)
North America	405.5	474.4	620.7	1,791.7	23.6%
Europe	123.8	168.2	210.3	539.3	20.7%
Asia	77.5	104.0	133.9	403.3	24.7%
LATAM	11.3	14.0	17.2	39.5	18.1%
Pacific Countries	8.4	12.0	14.6	33.4	18.0%
Middle East & Africa	6.7	9.1	11.1	24.3	17.1%
Total	633.1	781.7	1,007.8	2,831.5	23.0%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association, American Health Information Management Association, National Health Care Anti-fraud Association, American Medical Informatics Association, Certification Commission for Health Information Technology, Healthcare Information Technology Standards Panel, Health Level Seven International, Information Technology Association of Canada, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network, Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

10.3 PROVIDERS

Healthcare providers need to analyze a large amount of healthcare data to make critical decisions that impact individual patients and the entire healthcare system. To improve the quality of care and reduce costs, healthcare providers are increasingly turning to IT-enabled business strategies and analytics. Provider organizations need to understand which operational and clinical processes and costs can be reduced to improve financial, administrative, and clinical decision-making. Healthcare analytics is used by providers mainly for the following functions:

- To meet the future business requirements and identify process and cost inefficiencies by determining the true cost of delivering care to patients
- To reduce the risk of hospital-acquired conditions and readmissions by analyzing vital indicator trends in patients
- To increase profitability by managing resource and utilization costs and identifying the best services to invest in or outsource
- To measure quality and outcome improvements of care processes and treatments
- To report on the quality of care delivered by providers and by service lines across the healthcare system

Although provider organizations collect and store most of the requisite data, this data is often not accessible to analytics systems for timely analysis and reporting, affecting day-to-day operations and patient care. Providing analytics solutions that can integrate disparate data sets is a key challenge in this market.

Some commercially available analytics solutions for healthcare providers are:

SOLUTION	COMPANY
Revenue Integrity	MedeAnalytics
ACO Reporting	Actian
Healthcare Analytics and Informatics	Anthelio Healthcare
Population Health for Providers	MedeAnalytics
Provider Intelligence	Cotiviti (formerly known as Verscend)
IBM Enterprise Content Management	IBM
IBM Content Analytics	IBM
Truven Health Custom Solutions	Truven Health
Oracle Enterprise Healthcare Analytics	Oracle

Source: Company Websites

Other key vendors offering analytics solutions for healthcare providers include Health Catalyst (US), McKesson (US), Inovalon (US), Optum (US), Cotiviti (US), SAS (US), and Orion Systems Integrators (US).

TABLE 61 HEALTHCARE ANALYTICS MARKET FOR PROVIDERS, BY TYPE, 2019–2026 (USD MILLION)

Providers	2019	2020	2021	2026	CAGR (2021–2026)
Hospitals, Physician Practices, IDNs	3,695.2	4,946.8	6,957.5	33,062.4	36.6%
Post-acute Care Organizations	552.5	721.8	987.6	4,095.5	32.9%
Ambulatory Settings	362.5	447.9	586.1	1,956.3	27.3%
Total	4,610.1	6,116.5	8,531.2	39,114.1	35.6%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association, American Health Information Management Association, National Health Care Anti-fraud Association, American Medical Informatics Association, Certification Commission for Health Information Technology, Healthcare Information Technology Standards Panel, Health Level Seven International, Information Technology Association of Canada, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network, Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

10.3.1 HOSPITALS, PHYSICIAN PRACTICES, AND IDNS

10.3.1.1 The adoption of healthcare analytics solutions is the highest in large hospitals

In the US, Medicare penalizes hospitals that record excessive patient readmissions within 30 days of discharge. This has influenced hospitals to adopt analytics solutions to keep records and avoid patient readmissions. The adoption of healthcare analytics solutions is the highest in large hospitals, as they need to capture and analyze larger volumes of patient data compared to small hospitals and clinics.

Small hospitals are also less inclined to implement healthcare analytics software due to budgetary constraints. The high implementation and licensing fees, lack of availability of integrated clinical information with financial and administrative data, communication barriers within hospitals, and continued use of existing basic analytics tools (especially in the emerging economies) are other major factors that could restrain the growth of this market segment during the forecast period.

IDNs are organizations that encourage building and integrating healthcare provider networks, exchanging healthcare data, and providing quality care to patients at a lower cost. In the US, federal mandates [such as the Health Information Technology for Economic and Clinical Health Act (HITECH) (2009) and the Patient Protection and Affordable Care Act (2010)] are encouraging the use of analytics for quality care and cost containment. The growing implementation of EHRs/EMRs, modernization of HCIT infrastructure in hospitals, availability of a large amount of clinical data, increasing need to comply with regulatory standards, improve population health, reduce readmission rates & medical errors, and improve physician performance are compelling hospitals and IDNs worldwide to adopt healthcare analytics solutions.

With the healthcare industry moving from the fee-for-service model to an outcome-driven model, medical groups and physician practices are moving towards the adoption of healthcare analytics solutions for practice improvement, identifying areas for improvement, performing periodic reviews, making electronic data submissions to regulatory bodies, monitoring patient outcomes, and assessing and improving financial performance. This growing utilization of analytics solutions by hospitals, medical groups, and physician practices will drive the growth of the healthcare analytics market during the forecast period.

To leverage growth opportunities in this market segment, various market players focus on introducing analytics solutions for hospitals, physician practices, and IDNs. For instance, In June 2018, Innovaccer Inc. (US) introduced a Financial Analytics Solution for Hospitals and Health Networks. Similarly, in June 2017, SSI Group (US) introduced Analytics 2.0 for hospitals.

Some commercially available healthcare analytics solutions for hospitals, physician practices, and IDNs are:

SOLUTION	COMPANY
Allscripts Sunrise	Allscripts
Optum Hospital Claim Solutions	Optum
Discern nCode	Cerner
Micromedex 360 Care Insights Suite	IBM

Source: Company Websites

Other players offering healthcare analytics solutions for this end-user segment are Innovaccer (US), Cotiviti (US) (formerly known as Verscend), Oracle (US), and Health Catalyst (US).

TABLE 62 HEALTHCARE ANALYTICS MARKET FOR HOSPITALS, PHYSICIAN PRACTICES, AND IDNS, BY REGION, 2019–2026 (USD MILLION)

Region	2019	2020	2021	2026	CAGR (2021–2026)
North America	2,418.9	3,130.6	4,550.9	24,668.2	40.2%
Europe	742.4	1,068.3	1,414.2	4,902.5	28.2%
Asia	381.6	534.3	715.6	2,625.9	29.7%
LATAM	71.9	95.5	125.9	426.4	27.6%
Middle East & Africa	40.9	58.1	73.0	195.2	21.7%
Pacific Countries	39.4	60.0	77.8	244.1	25.7%
Total	3,695.2	4,946.8	6,957.5	33,062.4	36.6%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association, American Health Information Management Association, National Health Care Anti-fraud Association, American Medical Informatics Association, Certification Commission for Health Information Technology, Healthcare Information Technology Standards Panel, Health Level Seven International, Information Technology Association of Canada, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network, Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

10.3.2 POST-ACUTE CARE ORGANIZATIONS (PACOS)

10.3.2.1 PACOs focus on reviewing and studying patient data with the help of healthcare analytics

With the introduction of ACOs and the move towards PHM, PACOs have gained momentum in the healthcare industry. To better manage care, PACOs are focusing on reviewing and analyzing patient data. The adoption of analytics solutions helps these organizations maximize accessibility to all data relevant to the treatment history of their patients and the acquisition and retention of data.

Some of the commercially available analytics solutions include Sunrise Financial Manager (Allscripts), Post-Acute Care (PAC) Benefits Management Platform (Inovalon); Post-Acute Care Coordination (Loopback Analytics), and Total Patient Tracker (Post-Acute Analytics)

TABLE 63 HEALTHCARE ANALYTICS MARKET FOR PACOS, BY REGION, 2019–2026 (USD MILLION)

Region	2019	2020	2021	2026	CAGR (2021–2026)
North America	352.4	442.7	624.9	2,941.1	36.3%
Europe	118.9	168.8	220.2	712.1	26.5%
Asia	57.7	78.8	102.9	336.5	26.7%
LATAM	11.6	14.7	18.6	51.8	22.7%
Middle East & Africa	6.8	9.5	11.7	28.8	19.7%
Pacific Countries	5.0	7.3	9.3	25.2	22.2%
Total	552.5	721.8	987.6	4,095.5	32.9%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association, American Health Information Management Association, National Health Care Anti-fraud Association, American Medical Informatics Association, Certification Commission for Health Information Technology, Healthcare Information Technology Standards Panel, Health Level Seven International, Information Technology Association of Canada, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network, Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

10.3.3 AMBULATORY SETTINGS

10.3.3.1 The need to control healthcare costs to support the market growth of ambulatory settings

Ambulatory settings provide health services and primary care to patients from outpatient departments. These settings use healthcare analytics solutions mainly to manage population health and reduce hospitalization, medical errors, and fraud. The rising need to control the cost of healthcare has driven the overall demand for/adoption of healthcare analytics solutions in ambulatory settings.

Government organizations and analytics vendors are encouraging the adoption of analytics in ambulatory settings. In February 2016, Iatric Systems, Inc. launched Analytics on Demand, a new analytics solution that can be deployed quickly and immediately delivers key decision-making data in ambulatory and hospital settings. HIMSS Analytics has developed an Ambulatory EMR Adoption Model (A-EMRAM) for monitoring EMR adoption in ambulatory settings. This model helps ambulatory settings focus on key IT solutions that need to be implemented to achieve higher access, quality, efficiency, and safety.

Some commercially available healthcare analytics solutions for ambulatory settings have been mentioned below:

SOLUTION	COMPANY
Analytics on Demand	Iatric Systems
Ambulatory Care Analytics	Carolinas HealthCare System
Caradigm Population Health Management Ambulatory Solution	GE Healthcare

Source: Company Websites

The other prominent analytics vendors in this segment include Cerner (US), Inovalon (US), Allscripts (US), and Optum (US).

TABLE 64 HEALTHCARE ANALYTICS MARKET FOR AMBULATORY SETTINGS, BY REGION, 2019–2026 (USD MILLION)

Region	2019	2020	2021	2026	CAGR (2021–2026)
North America	244.9	293.5	397.0	1,507.5	30.6%
Europe	72.2	96.5	118.8	287.3	19.3%
Asia	33.1	42.4	52.1	125.2	19.2%
LATAM	6.3	7.5	9.0	19.0	16.1%
Middle East & Africa	3.6	4.7	5.4	9.3	11.5%
Pacific Countries	2.4	3.3	3.9	7.8	14.8%
Total	362.5	447.9	586.2	1,956.1	27.3%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association, American Health Information Management Association, National Health Care Anti-fraud Association, American Medical Informatics Association, Certification Commission for Health Information Technology, Healthcare Information Technology Standards Panel, Health Level Seven International, Information Technology Association of Canada, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network, Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

10.4 ACOS, HIES, MCOS, AND TPAS

Accountable Care Organizations

An ACO is a group of hospitals, doctors, and other healthcare providers who provide high-quality care to patients. While the structure of an ACO is flexible, three major components are defined for all ACOs: (i) ACOs are provider-led organizations that have a strong base of primary care and are collectively held accountable for offering quality and low per capita cost across the continuum of care (ii) ACO payments are subject to quality improvements that lower the overall cost of healthcare (iii) ACOs have a reliable and sophisticated performance measurement system to support improvement and provide confidence that savings are achieved through improvements in care.

The ACO provision was included in the ACA in the US, which provides incentives from the government to physicians who use IT solutions, including analytics solutions. Physicians are increasingly adopting these solutions to transform a traditional organization into an ACO. The increasing number of ACOs primarily drives the growth of this end-user segment. The number of ACOs in the US has grown from a few dozen in 2010 to nearly 782 by December 2015 (Source: American Academy of Neurology). This number reached 1,011 ACOs at the end of the first quarter of 2018 (Source: Health Catalyst).

Analytics solutions help ACOs in patient tracking, risk stratification, quality improvement, and performance evaluation. These solutions also help sustain a successful ACO model by closing gaps in care, preventing unnecessary tests from reducing costs, improving patient experience and clinical outcomes, and enhancing care coordination and population health. Thus, with the growing number of ACOs, the adoption of healthcare analytics solutions is expected to increase in the coming years.

To leverage the growth opportunities in this segment, market players are introducing solutions specifically for ACOs. For example, in April 2017, MTBC launched its ACO analytics module, which is a business intelligence and analytics platform.

Some commercially available solutions for ACOs have been mentioned below:

SOLUTION	COMPANY
ACO Explorer	Health Catalyst
EZ-ANALYTICS	MZI HealthCare
DxCG Risk Analytics	Verisk Analytics
Late-Bing Architecture	Health Catalyst
Truven Health Unify	Truven Health

Source: Company Websites

Health Information Exchanges (HIEs)

HIEs enable the transmission of healthcare-related data among payers, providers, employers, health information organizations, and government agencies by following national standards. The HITECH Act includes private and secure electronic HIEs. Stage 2 and Stage 3 of the Meaningful Use criteria focus on more rigorous HIE and access to widespread patient data through patient-centered HIEs. This is integral in facilitating a 360-degree patient view, enabling coordinated and connected care across the continuum of health delivery and payment settings, thereby improving healthcare delivery quality, efficiency, and safety. Such healthcare reforms are increasing the adoption of HIEs across the US.

HIEs are vital in achieving an interoperable healthcare system. With the growing adoption of HIEs, analytics, predictive modeling, and care management strategies are expected to increase significantly. Public health officials, healthcare intermediaries, and government agencies can access operational and clinical statistics housed within state-wide HIEs. Applying analytics to this data helps them identify insights and opportunities that enable timely care, lower costs, and improved outcomes. The ability to build a 360-degree view of a patient and a population in real-time will allow for cost reductions, improve the quality of care, and facilitate the proactive recognition of at-risk population groups, thus improving the entire healthcare delivery process.

Managed Care Organizations (MCOs) and Third-party Administrators (TPAs)

MCOs and TPAs are bodies that help other organizations to reduce costs and manage their customers through provider network management, utilization management, and quality assurance. MCOs pay for certain or all services used by patients. These organizations utilize healthcare analytics to gain knowledge regarding patients who need their services, manage risks, and reduce readmission rates.

TABLE 65 HEALTHCARE ANALYTICS MARKET FOR ACOS, HIES, MCOS, AND TPAS, BY REGION, 2019–2026 (USD MILLION)

Region	2019	2020	2021	2026	CAGR (2021–2026)
North America	761.0	924.3	1,265.8	5,117.8	32.2%
Europe	256.9	358.6	460.6	1,375.6	24.5%
Asia	143.6	194.9	253.4	807.9	26.1%
LATAM	24.7	32.0	41.2	125.3	24.9%
Middle East & Africa	13.9	19.1	23.1	52.1	17.6%
Pacific Countries	11.8	17.3	21.5	55.5	20.9%
Total	1,211.9	1,546.2	2,065.6	7,534.2	29.5%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association, American Health Information Management Association, National Health Care Anti-fraud Association, American Medical Informatics Association, Certification Commission for Health Information Technology, Healthcare Information Technology Standards Panel, Health Level Seven International, Information Technology Association of Canada, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network, Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

11 HEALTHCARE ANALYTICS MARKET, BY REGION

KEY FINDINGS

- In 2020, North America accounted for the largest share of the global healthcare analytics market. This regional segment is projected to reach USD 51,585.4 million by 2026 from USD 13,179.5 million in 2021, at a CAGR of 31.4% during the forecast period.
- The large share of North America can be attributed to the support for adopting HCIT, increasing focus on PHM, significant amount of VC investments, and greater product and service availability.
- In 2020, the US commanded the largest share of 89.1% of the North American healthcare analytics market.
- Europe accounted for 21.5% of the global healthcare analytics market in 2020.
- The market in Asia is projected to grow at a CAGR of 26.3% during the forecast period.

11.1 INTRODUCTION

The global healthcare analytics market has been segmented based on region into North America, Europe, Asia, the Pacific Countries, Latin America, and the Middle East & Africa.

North America is expected to account for the largest market share in 2021. This regional segment is also slated to grow at the highest CAGR during the forecast period. Growth in this regional segment is largely driven by the increasing adoption of healthcare analytics solutions and services, especially in the US, to provide better quality care and lower healthcare costs.

TABLE 66 HEALTHCARE ANALYTICS MARKET, BY REGION, 2019–2026 (USD MILLION)

Region	2019	2020	2021	2026	CAGR (2021–2026)
North America	8,004.3	9,690.4	13,179.5	51,585.4	31.4%
Europe	2,462.8	3,415.5	4,358.3	12,644.3	23.7%
Asia	1,496.2	2,033.4	2,648.8	8,519.1	26.3%
LATAM	237.8	301.8	381.1	1,049.4	22.5%
Pacific Countries	174.9	254.2	314.9	787.9	20.1%
Middle East & Africa	138.2	190.6	233.1	546.7	18.6%
Total	12,514.2	15,885.8	21,115.7	75,132.8	28.9%

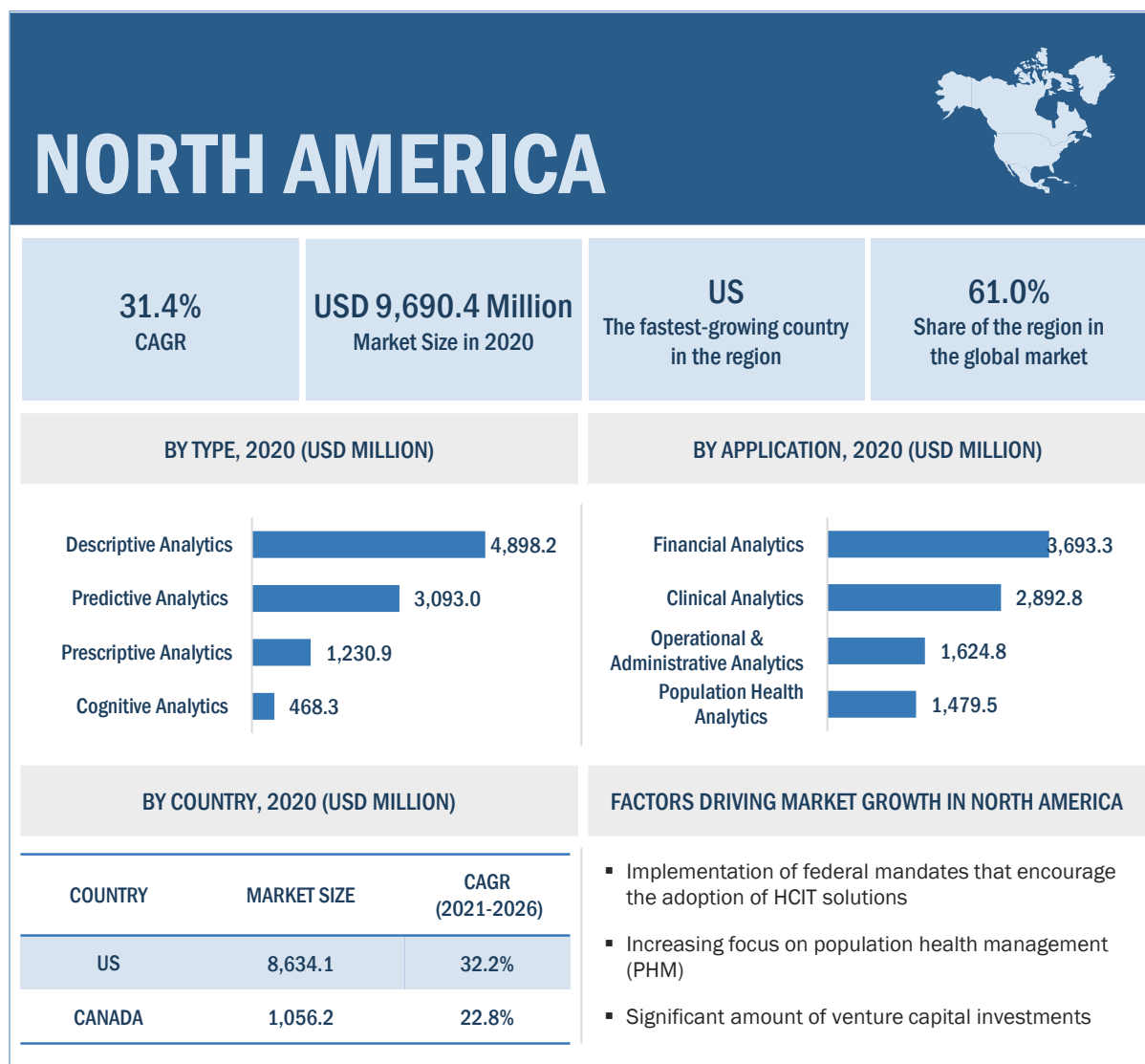
Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

11.2 NORTH AMERICA

North America, comprising the US and Canada, is the largest regional segment in the global healthcare analytics market.

The large share of North America in the global healthcare analytics market can be attributed to the implementation of federal health mandates that encourage the adoption of HCIT solutions, increasing focus on PHM, a significant amount of VC investments, growing pressure to reduce healthcare costs, technological advancements, and high product and service availability in this region. Moreover, most of the leading players in the healthcare analytics market have their headquarters in North America.

Currently, the healthcare industry in North America witnesses heavy investments in HCIT to meet regulatory requirements, comply with government reimbursement programs, and qualify for incentives. With the growing importance of analytics in healthcare and the increasing utilization of technologically advanced tools, North America is expected to register high growth in the field of analytics during the forecast period.

FIGURE 29 NORTH AMERICA: HEALTHCARE ANALYTICS MARKET SNAPSHOT (2020)

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 67 NORTH AMERICA: HEALTHCARE ANALYTICS MARKET, BY COUNTRY, 2019–2026 (USD MILLION)

Country	2019	2020	2021	2026	CAGR (2021–2026)
US	7,188.5	8,634.1	11,836.4	47,828.6	32.2%
Canada	815.8	1,056.2	1,343.2	3,756.8	22.8%
Total	8,004.3	9,690.4	13,179.5	51,585.4	31.4%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 68 NORTH AMERICA: HEALTHCARE ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Descriptive Analytics	4,279.4	4,898.2	6,244.4	14,635.4	18.6%
Predictive Analytics	2,420.0	3,093.0	4,448.1	22,943.3	38.8%
Prescriptive Analytics	944.0	1,230.9	1,804.8	10,248.9	41.5%
Cognitive Analytics	360.8	468.3	682.2	3,757.9	40.7%
Total	8,004.3	9,690.4	13,179.5	51,585.4	31.4%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 69 NORTH AMERICA: HEALTHCARE ANALYTICS MARKET, BY APPLICATION, 2019–2026 (USD MILLION)

Application	2019	2020	2021	2026	CAGR (2021–2026)
Financial Analytics	3,130.3	3,693.3	4,896.7	16,851.6	28.0%
Clinical Analytics	2,217.2	2,892.8	4,203.9	21,166.2	38.2%
Operational & Administrative Analytics	1,426.3	1,624.8	2,080.0	6,001.8	23.6%
Population Health Analytics	1,230.5	1,479.5	1,999.0	7,565.9	30.5%
Total	8,004.3	9,690.4	13,179.5	51,585.4	31.4%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 70 NORTH AMERICA: FINANCIAL ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Claims Processing	1,379.8	1,625.4	2,159.5	7,385.6	27.9%
Revenue Cycle Management	854.8	990.3	1,275.5	3,805.3	24.4%
Payment Integrity & FWA	567.3	691.0	949.2	3,896.8	32.6%
Risk Adjustment & Assessment	328.5	386.5	512.6	1,763.9	28.0%
Total	3,130.3	3,693.3	4,896.7	16,851.6	28.0%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 71 NORTH AMERICA: CLINICAL ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Quality Improvement & Clinical Benchmarking	706.3	975.2	1,500.0	9,376.0	44.3%
Clinical Decision Support	555.0	707.1	993.4	4,258.2	33.8%
Precision Health	373.9	468.5	660.6	2,834.2	33.8%
Reporting & Compliance	306.5	380.5	524.2	2,026.8	31.1%
Comparative Analytics/Effectiveness	152.5	201.6	293.5	1,505.2	38.7%
Medical Imaging Analytics	123.0	160.0	232.2	1,165.8	38.1%
Total	2,217.2	2,892.8	4,203.9	21,166.2	38.2%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 72 NORTH AMERICA: OPERATIONAL & ADMINISTRATIVE ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Supply Chain Analytics	630.0	709.8	897.0	2,417.4	21.9%
Workforce Analytics	565.6	649.3	837.9	2,517.2	24.6%
Strategic Analytics	230.7	265.8	345.0	1,067.2	25.3%
Total	1,426.3	1,624.8	2,080.0	6,001.8	23.6%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 73 NORTH AMERICA: HEALTHCARE ANALYTICS MARKET, BY COMPONENT, 2019–2026 (USD MILLION)

Component	2019	2020	2021	2026	CAGR (2021–2026)
Services	4,017.6	4,966.1	6,876.3	28,448.4	32.8%
Software	3,250.5	3,944.8	5,376.5	21,296.3	31.7%
Hardware	736.2	779.5	926.7	1,840.8	14.7%
Total	8,004.3	9,690.4	13,179.5	51,585.4	31.4%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 74 NORTH AMERICA: HEALTHCARE ANALYTICS MARKET FOR SERVICES, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Support Services	2,764.3	3,399.4	4,682.1	18,843.4	32.1%
Business Analytics Services	1,253.3	1,566.6	2,194.2	9,604.9	34.4%
Total	4,017.6	4,966.1	6,876.3	28,448.4	32.8%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 75 NORTH AMERICA: HEALTHCARE ANALYTICS MARKET, BY DEPLOYMENT MODEL, 2019–2026 (USD MILLION)

Deployment Model	2019	2020	2021	2026	CAGR (2021–2026)
On-premise Deployment	5,018.9	5,842.6	7,600.9	21,605.5	23.2%
On-demand Deployment	2,985.4	3,847.8	5,578.7	29,979.9	40.0%
Total	8,004.3	9,690.4	13,179.5	51,585.4	31.4%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 76 NORTH AMERICA: HEALTHCARE ANALYTICS MARKET, BY END USER, 2019–2026 (USD MILLION)

End User	2019	2020	2021	2026	CAGR (2021–2026)
Payers	4,227.1	4,899.2	6,341.0	17,350.9	22.3%
Providers	3,016.2	3,866.8	5,572.8	29,116.8	39.2%
ACOs, HIEs, MCOs, and TPAs	761.0	924.3	1,265.8	5,117.8	32.2%
Total	8,004.3	9,690.4	13,179.5	51,585.4	31.4%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 77 NORTH AMERICA: HEALTHCARE ANALYTICS MARKET FOR PAYERS, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Private Insurance Companies	2,472.6	2,810.9	3,560.0	8,585.6	19.3%
Government Agencies	1,349.0	1,614.0	2,160.3	6,973.6	26.4%
Employers & Private Exchanges	405.5	474.4	620.7	1,791.7	23.6%
Total	4,227.1	4,899.2	6,341.0	17,350.9	22.3%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 78 NORTH AMERICA: HEALTHCARE ANALYTICS MARKET FOR PROVIDERS, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Hospitals, Physician Practices, and IDNs	2,418.9	3,130.6	4,550.9	24,668.2	40.2%
Post-acute Care Organizations	352.4	442.7	624.9	2,941.1	36.3%
Ambulatory Settings	244.9	293.5	397.0	1,507.5	30.6%
Total	3,016.2	3,866.8	5,572.8	29,116.8	39.2%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

11.2.1 US

11.2.1.1 High adoption of EHRs to drive the healthcare analytics market in the US

The US accounted for 89.1% of the North American healthcare analytics market in 2020. The implementation of federal healthcare mandates that encourage the adoption of HCIT solutions to reduce healthcare costs is a key factor driving market growth in North America. The growing focus on quality care, significant VC investments, increasing focus on PHM, the rising adoption of personalized medicine and value-based medicine, and greater accessibility of analytics solutions and services are the other key factors expected to drive the market in the US.

In 2017, the US spent USD 3.5 trillion (USD 10,739 per person) on healthcare; this figure is expected to grow by 5.5% per year to reach USD 5.7 trillion by 2026 and will account for 17.9% of the country's GDP. Owing to its significant healthcare spending, there is growing pressure to reduce healthcare costs while providing better quality care. As analytics tools help predict and prevent unnecessary costs and facilitate evidence-based decision-making, the adoption of healthcare analytics solutions in the US is expected to increase in the coming years.

The EHR adoption incentive program, launched under the ARRA in 2009, also stimulates the growth of the US healthcare analytics market. According to data presented by the CMS in August 2017, more than USD 24.8 billion in Medicare and more than USD 12 billion in Medicaid EHR incentive program payments have been made to eligible hospitals and professionals participating in the federal EHR incentive program between January 2011 and August 2017. As of August 2017, more than 535,700 healthcare providers received payment for participating in the Medicare and Medicaid EHR Incentive Programs. Growth in the adoption of EHRs will result in the creation of a significant amount of electronic data. In such a scenario, stakeholders in the healthcare industry will be compelled to adopt analytics solutions to derive actionable insights from the high volume of electronic data.

The healthcare system in the US is shifting towards personalized and value-based medicine. This is evident from an announcement by the CMS about implementing the next-generation (NG) Accountable Care Organization (ACO) model in January 2016. By the end of 2016, the HHS planned to offer 30% of traditional Medicare payments to value-based models such as ACOs or bundled payments; the HHS planned to increase this figure to 50% by the end of 2018. The effective use of analytics solutions is essential for the successful transformation to personalized and value-based medicine; therefore, the shift toward personalized and value-based medicine will drive the healthcare analytics market in the US.

The presence of a favorable funding scenario in the US is also expected to support the growth of the healthcare analytics market in the country. Favorable government initiatives help healthcare organizations use advanced IT solutions such as analytics to transform their processes to align with federal motives of population health management, preventive care, and lower overall spending. In July 2016, the HHS announced a grant of USD 35 million to support PHM and big data analytics projects. This funding also focuses on other health IT development initiatives. The HHS also invested USD 87 million later that year to support the continued adoption of health IT tools.

The emergence of newer technologies such as AI is another factor influencing the growth of the market in the US. The benefits of AI in the healthcare analytics industry will further ensure their implementation in the coming years; however, currently, the adoption of AI-driven analytics is in the nascent stage. To tap the growth opportunities in this market space, various healthcare analytics companies are focusing on developing AI solutions; venture capital has also been made available for companies in this market space. For example, an analytics startup, ThoughtSpot (US), received USD 145 million in a Series D funding round from Lightspeed Ventures, Sapphire Ventures, Khosla Ventures, and the General Catalyst to develop AI-driven analytics solutions in 2018. Protenus, a start-up firm providing analytics solutions for healthcare compliance, raised USD 11 million in a Series B funding round to accelerate R&D to expand its analytics

offerings in the same year. Such initiatives, leading to the introduction of better and more powerful solutions, will contribute to market growth in the coming years.

TABLE 79 US: HEALTHCARE ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Descriptive Analytics	3,809.3	4,325.7	5,566.3	13,633.8	19.6%
Predictive Analytics	2,200.9	2,788.8	4,033.4	21,301.2	39.5%
Prescriptive Analytics	855.9	1,105.2	1,628.7	9,448.2	42.1%
Cognitive Analytics	322.5	414.4	607.9	3,445.3	41.5%
Total	7,188.5	8,634.1	11,836.4	47,828.6	32.2%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 80 US: HEALTHCARE ANALYTICS MARKET, BY APPLICATION, 2019–2026 (USD MILLION)

Application	2019	2020	2021	2026	CAGR (2021–2026)
Financial Analytics	2,819.3	3,298.2	4,403.9	15,599.3	28.8%
Clinical Analytics	1,976.9	2,564.3	3,764.9	19,670.3	39.2%
Operational & Administrative Analytics	1,284.8	1,450.5	1,869.2	5,543.2	24.3%
Population Health Analytics	1,107.6	1,321.0	1,798.3	7,015.8	31.3%
Total	7,188.5	8,634.1	11,836.4	47,828.6	32.2%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 81 US: FINANCIAL ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Claims Processing	1,283.2	1,500.9	2,002.1	6,962.9	28.3%
Revenue Cycle Management	711.3	811.4	1,056.3	3,296.6	25.6%
Payment Integrity & FWA	523.1	633.4	875.3	3,682.3	33.3%
Risk Adjustment & Assessment	301.7	352.6	470.3	1,657.5	28.7%
Total	2,819.3	3,298.2	4,403.9	15,599.3	28.8%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 82 US: CLINICAL ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Quality Improvement & Clinical Benchmarking	659.6	904.2	1,396.1	8,883.6	44.8%
Clinical Decision Support	457.1	577.0	824.2	3,755.5	35.4%
Precision Health	358.7	448.5	634.8	2,761.0	34.2%
Reporting & Compliance	266.4	328.2	457.8	1,850.8	32.2%
Comparative Analytics/Effectiveness	124.4	162.8	241.2	1,317.8	40.4%
Medical Imaging Analytics	110.7	143.6	210.8	1,101.5	39.2%
Total	1,976.9	2,564.3	3,764.9	19,670.3	39.2%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 83 US: OPERATIONAL & ADMINISTRATIVE ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Supply Chain Analytics	561.9	626.6	797.4	2,211.2	22.6%
Workforce Analytics	511.1	581.7	755.5	2,331.7	25.3%
Strategic Analytics	211.8	242.2	316.2	1,000.3	25.9%
Total	1,284.8	1,450.5	1,869.2	5,543.2	24.3%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 84 US: HEALTHCARE ANALYTICS MARKET, BY COMPONENT, 2019–2026 (USD MILLION)

Component	2019	2020	2021	2026	CAGR (2021–2026)
Services	3,617.7	4,437.9	6,192.6	26,398.8	33.6%
Software	2,909.8	3,505.4	4,820.0	19,770.6	32.6%
Hardware	661.0	690.7	823.8	1,659.2	15.0%
Total	7,188.5	8,634.1	11,836.4	47,828.6	32.2%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 85 US: HEALTHCARE ANALYTICS MARKET FOR SERVICES, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Support Services	2,487.0	3,035.5	4,214.2	17,490.9	32.9%
Business Analytics Services	1,130.8	1,402.4	1,978.4	8,908.0	35.1%
Total	3,617.7	4,437.9	6,192.6	26,398.8	33.6%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 86 US: HEALTHCARE ANALYTICS MARKET, BY DEPLOYMENT MODEL, 2019–2026 (USD MILLION)

Deployment Model	2019	2020	2021	2026	CAGR (2021–2026)
On-premise Deployment	4,479.3	5,171.8	6,786.2	20,000.6	24.1%
On-demand Deployment	2,709.2	3,462.3	5,050.1	27,828.0	40.7%
Total	7,188.5	8,634.1	11,836.4	47,828.6	32.2%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 87 US: HEALTHCARE ANALYTICS MARKET, BY END USER, 2019–2026 (USD MILLION)

End User	2019	2020	2021	2026	CAGR (2021–2026)
Payers	3,765.6	4,325.7	5,644.2	15,946.5	23.1%
Providers	2,722.0	3,462.3	5,026.4	27,052.5	40.0%
ACOs, HIEs, MCOs, and TPAs	701.0	846.1	1,165.8	4,829.6	32.9%
Total	7,188.5	8,634.1	11,836.4	47,828.6	32.2%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 88 US: HEALTHCARE ANALYTICS MARKET FOR PAYERS, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Private Insurance Companies	2,181.5	2,457.0	3,140.0	7,853.5	20.1%
Government Agencies	1,218.7	1,444.8	1,945.5	6,434.1	27.0%
Employers & Private Exchanges	365.4	423.9	558.7	1,658.9	24.3%
Total	3,765.6	4,325.7	5,644.2	15,946.5	23.1%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 89 US: HEALTHCARE ANALYTICS MARKET FOR PROVIDERS, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Hospitals, Physician Practices, and IDNs	2,177.6	2,796.5	4,096.6	22,902.1	41.1%
Post-acute Care Organizations	317.8	395.7	562.5	2,722.4	37.1%
Ambulatory Settings	226.6	270.1	367.4	1,428.0	31.2%
Total	2,722.0	3,462.3	5,026.4	27,052.5	40.0%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

11.2.2 CANADA

11.2.2.1 Need for managing patient volumes, clinical data, and healthcare costs will favor market growth in Canada

Canada accounted for 10.9% of the North American healthcare analytics market in 2020. The Canadian healthcare analytics market is expected to reach USD 1,713.9 million by 2026 from USD 1,056.2 million in 2020, at a CAGR of 22.8% during the forecast period.

More than half the primary care physicians in Canada use EHRs in their clinical settings. According to the Canadian Medical Association's (CMA) workforce survey, the adoption rate of EHRs/EMRs in Canada increased from about 77% of primary care physicians to 85% as of August 2017. EHRs are increasingly being used at POC clinics, hospitals, and long-term care facilities. Furthermore, in 2001, the Canadian government invested about USD 500 million in the Canada Health Infoway Program, an independent, not-for-profit organization funded by the federal government, which focuses on increasing the adoption of EHRs and related technologies in the country. Infoway established an Emerging Technology Group (ETG) in 2011 to identify and guide the use of ICT in healthcare innovation. This program is also investing in the training and development of new data analytics professionals. In its 2016–2017 budget, the government of Canada allocated USD 50 million to Infoway for digital health for a period of two years to improve the health system's efficiency, accuracy, and patient outcomes.

Digitization enables the deployment of core systems such as registries of patients and providers, electronic prescriptions and laboratory test results, digital diagnostic image capturing and storage, and electronic clinical reports and immunization records. This will improve patient care through faster access to patient information, reduced duplicate tests, and better communication across care teams. As digitized patient data is a base for the use of analytics, increased digitization will boost the demand for analytics solutions in Canada during the forecast period.

The Canadian healthcare system is expected to face challenges in managing patient volumes, patient clinical data, and healthcare costs. The major factors responsible for this include growth in the number of hospitals in the country; the rising geriatric population and increasing burden of chronic diseases; adoption of newer but more expensive diagnostic tests and pharmaceutical treatments; and growing expectations for better health outcomes, quality of care, and active patient engagement. Authorities in the country are focusing on adopting data analytics solutions to counter these challenges. Growing awareness about the benefits of healthcare analytics solutions is also expected to support market growth.

Several events have also been held in Canada to drive awareness of HCIT. This includes the Statistical Society of Canada's 3rd Annual Big Data & Analytics Summit in Toronto in February 2017 and the Institute for Health Technology Transformation (iHT2) Toronto Health IT Summit in September 2016.

TABLE 90 CANADA: HEALTHCARE ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Descriptive Analytics	470.2	572.5	678.0	1,001.5	8.1%
Predictive Analytics	219.2	304.2	414.7	1,642.0	31.7%
Prescriptive Analytics	88.1	125.7	176.1	800.7	35.4%
Cognitive Analytics	38.3	53.9	74.3	312.6	33.3%
Total	815.8	1,056.2	1,343.2	3,756.8	22.8%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 91 CANADA: HEALTHCARE ANALYTICS MARKET, BY APPLICATION, 2019–2026 (USD MILLION)

Application	2019	2020	2021	2026	CAGR (2021–2026)
Financial Analytics	311.0	395.0	492.8	1,252.3	20.5%
Clinical Analytics	240.4	328.5	438.9	1,495.9	27.8%
Operational & Administrative Analytics	141.5	174.3	210.8	458.6	16.8%
Population Health Analytics	122.9	158.4	200.7	550.1	22.3%
Total	815.8	1,056.2	1,343.2	3,756.8	22.8%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 92 CANADA: FINANCIAL ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Revenue Cycle Management	143.5	179.0	219.2	508.7	18.3%
Claims Processing	96.5	124.4	157.4	422.6	21.8%
Payment Integrity & FWA	44.2	57.7	73.9	214.5	23.8%
Risk Adjustment & Assessment	26.8	34.0	42.3	106.4	20.3%
Total	311.0	395.0	492.8	1,252.3	20.5%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 93 CANADA: CLINICAL ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Clinical Decision Support	97.8	130.1	169.1	502.6	24.3%
Quality Improvement & Clinical Benchmarking	46.7	71.0	103.9	492.4	36.5%
Reporting & Compliance	40.2	52.2	66.4	175.9	21.5%
Comparative Analytics/Effectiveness	28.1	38.8	52.3	187.4	29.1%
Precision Health	15.2	20.0	25.8	73.2	23.2%
Medical Imaging Analytics	12.3	16.4	21.4	64.3	24.6%
Total	240.4	328.5	438.9	1,495.9	27.8%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 94 CANADA: OPERATIONAL & ADMINISTRATIVE ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Supply Chain Analytics	68.1	83.1	99.6	206.1	15.7%
Workforce Analytics	54.5	67.6	82.4	185.5	17.6%
Strategic Analytics	18.9	23.5	28.8	66.9	18.3%
Total	141.5	174.3	210.8	458.6	16.8%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 95 CANADA: HEALTHCARE ANALYTICS MARKET, BY COMPONENT, 2019–2026 (USD MILLION)

Component	2019	2020	2021	2026	CAGR (2021–2026)
Services	399.9	528.1	683.8	2,049.6	24.6%
Software	340.7	439.4	556.5	1,525.7	22.3%
Hardware	75.1	88.7	102.9	181.6	12.0%
Total	815.8	1,056.2	1,343.2	3,756.8	22.8%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 96 CANADA: HEALTHCARE ANALYTICS MARKET FOR SERVICES, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Support Services	277.4	363.9	467.9	1,352.6	23.7%
Business Analytics Services	122.5	164.2	215.8	697.0	26.4%
Total	399.9	528.1	683.8	2,049.6	24.6%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 97 CANADA: HEALTHCARE ANALYTICS MARKET, BY DEPLOYMENT MODEL, 2019–2026 (USD MILLION)

Deployment Model	2019	2020	2021	2026	CAGR (2021–2026)
On-premise Deployment	539.6	670.7	814.7	1,604.9	14.5%
On-demand Deployment	276.2	385.5	528.5	2,151.9	32.4%
Total	815.8	1,056.2	1,343.2	3,756.8	22.8%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 98 CANADA: HEALTHCARE ANALYTICS MARKET, BY END USER, 2019–2026 (USD MILLION)

End User	2019	2020	2021	2026	CAGR (2021–2026)
Payers	461.6	573.5	696.9	1,404.4	15.0%
Providers	294.2	404.5	546.3	2,064.3	30.5%
ACOs, HIEs, MCOs, and TPAs	60.0	78.2	100.0	288.2	23.6%
Total	815.8	1,056.2	1,343.2	3,756.8	22.8%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 99 CANADA: HEALTHCARE ANALYTICS MARKET FOR PAYERS, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Private Insurance Companies	291.1	353.9	420.0	732.1	11.8%
Government Agencies	130.3	169.2	214.8	539.5	20.2%
Employers & Private Exchanges	40.1	50.5	62.1	132.8	16.4%
Total	461.6	573.5	696.9	1,404.4	15.0%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 100 CANADA: HEALTHCARE ANALYTICS MARKET FOR PROVIDERS, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Hospitals, Physician Practices, and IDNs	241.3	334.2	454.3	1,766.1	31.2%
Post-acute Care Organizations	34.6	46.9	62.4	218.7	28.5%
Ambulatory Settings	18.3	23.5	29.6	79.5	21.8%
Total	294.2	404.5	546.3	2,064.3	30.5%

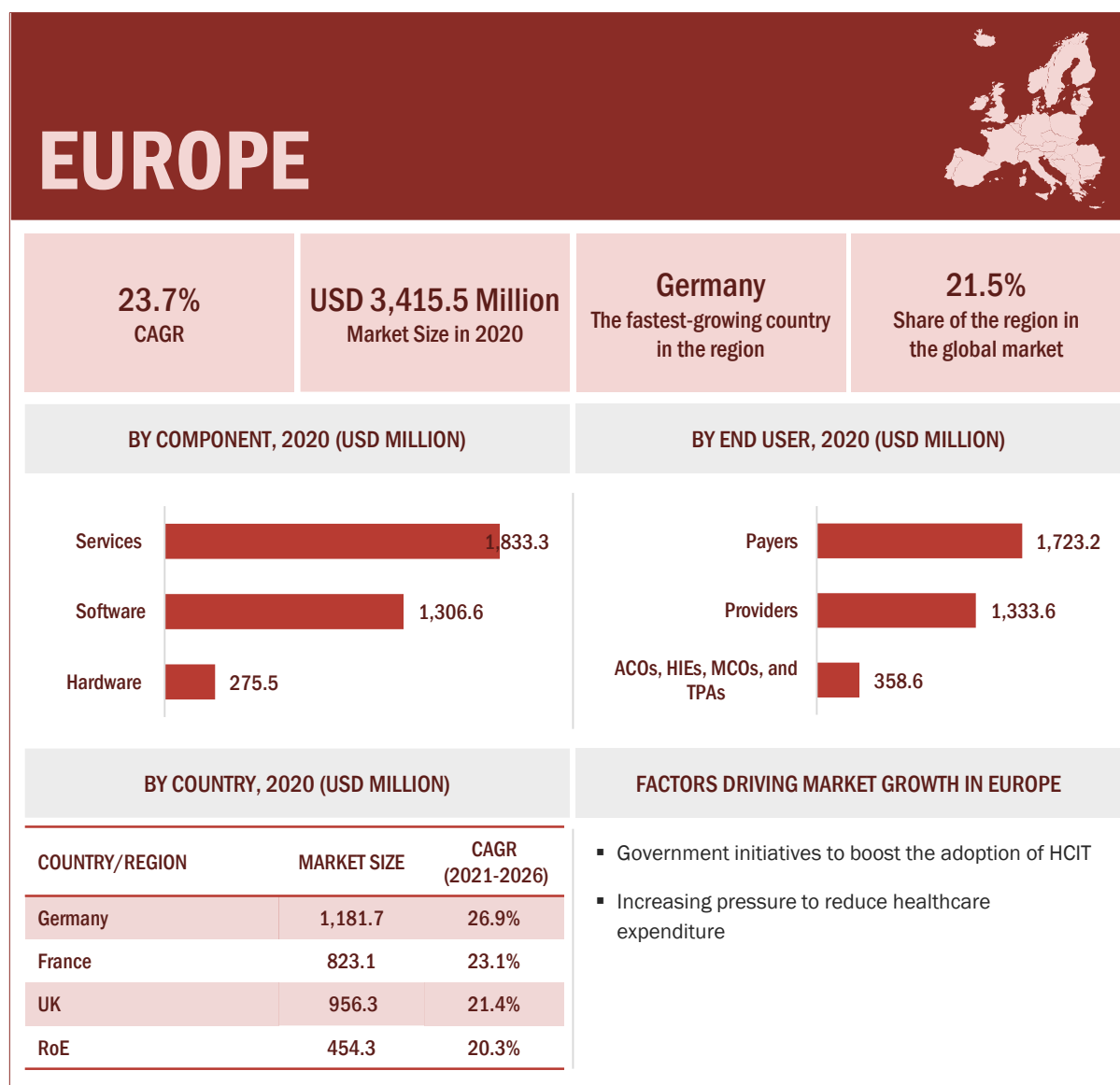
Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

11.3 EUROPE

Europe accounted for 21.5% of the global healthcare analytics market in 2020. This regional segment is projected to reach USD 12,644.3 million by 2026 from USD 4,358.3 million in 2021, at a CAGR of 23.7% during the forecast period.

Despite the aftereffects of the Eurozone debt crisis, governments in various European countries are focusing on increasing the adoption of HCIT solutions to curtail rising healthcare costs and provide quality patient care. In December 2012, the European Commission launched the eHealth Action Plan 2012–2020, aimed at improving population health, increasing healthcare quality and access, and making eHealth tools more effective, user-friendly, and widely accepted by the healthcare system. Such initiatives will increase the adoption of advanced HCIT infrastructure. In 2015, the EC proposed using the Connecting Europe Facility and the European Regional Development Fund from 2013 to 2020 to support healthcare interoperability. Since a strong HCIT infrastructure is a prerequisite for the effective implementation of healthcare analytics solutions, this development is expected to positively impact the growth of the healthcare analytics market in the region.

However, interoperability and data security have emerged as major concerns in European markets. Factors such as a lack of structured data, language barriers, and changing regulations from country to country are expected to restrain the growth of this market.

FIGURE 30 EUROPE: HEALTHCARE ANALYTICS MARKET SNAPSHOT (2020)

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 101 EUROPE: HEALTHCARE ANALYTICS MARKET, BY COUNTRY, 2019–2026 (USD MILLION)

Country/Region	2019	2020	2021	2026	CAGR (2021–2026)
Germany	867.6	1,181.7	1,552.5	5,112.3	26.9%
UK	679.1	956.3	1,197.5	3,161.3	21.4%
France	595.2	823.1	1,044.6	2,948.1	23.1%
RoE	320.9	454.3	563.7	1,422.6	20.3%
Total	2,462.8	3,415.5	4,358.3	12,644.3	23.7%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 102 EUROPE: HEALTHCARE ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Descriptive Analytics	1,517.4	2,049.0	2,540.2	6,135.4	19.3%
Predictive Analytics	568.2	828.9	1,113.2	4,166.7	30.2%
Prescriptive Analytics	254.2	362.7	476.0	1,591.6	27.3%
Cognitive Analytics	122.9	174.9	228.8	750.5	26.8%
Total	2,462.8	3,415.5	4,358.3	12,644.3	23.7%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 103 EUROPE: HEALTHCARE ANALYTICS MARKET, BY APPLICATION, 2019–2026 (USD MILLION)

Application	2019	2020	2021	2026	CAGR (2021–2026)
Financial Analytics	880.4	1,192.2	1,486.0	3,832.4	20.9%
Clinical Analytics	803.2	1,169.0	1,558.1	5,375.4	28.1%
Operational & Administrative Analytics	419.3	551.1	666.9	1,481.0	17.3%
Population Health Analytics	359.9	503.1	647.3	1,955.5	24.7%
Total	2,462.8	3,415.5	4,358.3	12,644.3	23.7%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 104 EUROPE: FINANCIAL ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Claims Processing	319.8	436.7	549.4	1,450.7	21.4%
Revenue Cycle Management	301.4	397.1	480.3	1,067.3	17.3%
Payment Integrity & FWA	169.7	237.6	306.3	933.7	25.0%
Risk Adjustment & Assessment	89.5	120.8	150.0	380.7	20.5%
Total	880.4	1,192.2	1,486.0	3,832.4	20.9%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 105 EUROPE: CLINICAL ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Quality Improvement & Clinical Benchmarking	270.4	416.8	585.6	2,491.9	33.6%
Clinical Decision Support	184.3	262.0	340.7	1,036.4	24.9%
Precision Health	123.1	172.7	222.1	638.4	23.5%
Reporting & Compliance	103.6	143.6	182.2	490.6	21.9%
Medical Imaging Analytics	67.7	95.0	122.0	347.8	23.3%
Comparative Analytics/Effectiveness	54.0	78.8	105.5	370.4	28.5%
Total	803.2	1,169.0	1,558.1	5,375.4	28.1%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 106 EUROPE: OPERATIONAL & ADMINISTRATIVE ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Supply Chain Analytics	192.0	249.9	299.3	629.2	16.0%
Workforce Analytics	161.5	214.9	263.4	622.8	18.8%
Strategic Analytics	65.8	86.3	104.2	229.0	17.1%
Total	419.3	551.1	666.9	1,481.0	17.3%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 107 EUROPE: HEALTHCARE ANALYTICS MARKET, BY COMPONENT, 2019–2026 (USD MILLION)

Component	2019	2020	2021	2026	CAGR (2021–2026)
Services	1,299.4	1,833.3	2,374.8	7,282.9	25.1%
Software	943.9	1,306.6	1,665.4	4,804.3	23.6%
Hardware	219.5	275.5	318.1	557.2	11.9%
Total	2,462.8	3,415.5	4,358.3	12,644.3	23.7%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 108 EUROPE: HEALTHCARE ANALYTICS MARKET FOR SERVICES, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Support Services	875.1	1,218.3	1,555.3	4,326.3	22.7%
Business Analytics Services	424.3	615.0	819.6	2,956.6	29.3%
Total	1,299.4	1,833.3	2,374.8	7,282.9	25.1%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 109 EUROPE: HEALTHCARE ANALYTICS MARKET, BY DEPLOYMENT MODEL, 2019–2026 (USD MILLION)

Deployment Model	2019	2020	2021	2026	CAGR (2021–2026)
On-premise Deployment	1,569.1	2,130.7	2,658.6	6,769.0	20.6%
On-demand Deployment	893.7	1,284.8	1,699.7	5,875.3	28.2%
Total	2,462.8	3,415.5	4,358.3	12,644.3	23.7%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 110 EUROPE: HEALTHCARE ANALYTICS MARKET, BY END USER, 2019–2026 (USD MILLION)

End User	2019	2020	2021	2026	CAGR (2021–2026)
Payers	1,272.3	1,723.2	2,144.5	5,366.8	20.1%
Providers	933.6	1,333.6	1,753.1	5,901.9	27.5%
ACOs, HIEs, MCOs, and TPAs	256.9	358.6	460.6	1,375.6	24.5%
Total	2,462.8	3,415.5	4,358.3	12,644.3	23.7%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 111 EUROPE: HEALTHCARE ANALYTICS MARKET FOR PAYERS, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Private Insurance Companies	741.7	987.6	1,206.6	2,701.7	17.5%
Government Agencies	406.8	567.5	727.6	2,125.8	23.9%
Employers & Private Exchanges	123.8	168.2	210.3	539.3	20.7%
Total	1,272.3	1,723.2	2,144.5	5,366.8	20.1%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 112 EUROPE: HEALTHCARE ANALYTICS MARKET FOR PROVIDERS, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Hospitals, Physician Practices, and IDNs	742.4	1,068.3	1,414.2	4,902.5	28.2%
Post-acute Care Organizations	118.9	168.8	220.2	712.1	26.5%
Ambulatory Settings	72.2	96.5	118.8	287.3	19.3%
Total	933.6	1,333.6	1,753.1	5,901.9	27.5%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

11.3.1 GERMANY

11.3.1.1 Government initiatives to increase the use of analytics in healthcare are expected to contribute to market growth in Germany

Germany accounted for the largest share of 34.6% of the European healthcare analytics market in 2020. The healthcare analytics market in Germany is expected to reach USD 2,046.6 million by 2026 from USD 1,181.7 million in 2020, at a CAGR of 26.9% from 2021 to 2026.

While initiatives have been taken to improve and create awareness on healthcare analytics, the country still faces challenges in terms of poor cross-linking between doctors, health insurance, hospitals, and corporate health services, which makes it difficult to merge medical data. This highlights the need for digital healthcare. As a result, greater emphasis has been laid on healthcare digitization through measures such as the e-Health Law that foresees the creation of an electronic health card for every person in Germany by 2018. This card will include important health information that doctors, hospitals, and pharmacies across the country can access. Another area of concern is the emergence of big data, which has further boosted the drive toward digitization to manage and better analyze the vast amounts of data generated by healthcare processes.

Consequently, support for big data and analytics projects has risen. For instance, the Medical Allround-Care Service Solutions (MACSS) project was launched in January 2016 and is run in Berlin's Charité Hospital. This patient-centered comprehensive electronic health service platform for the integrated care of chronically ill patients was funded by the Federal Ministry of Economics and Technology. Similarly, Germany's National Center for Tumor Diseases (NCT) launched a project with SAP (Germany) in 2015. With NCT's cancer expertise and SAP's real-time analytics platforms, this project aims to gather and analyze information to predict the best-suited therapies for individual cancer patients. Such initiatives by the government are expected to improve the use of analytics in healthcare and thereby contribute to market growth.

TABLE 113 GERMANY: HEALTHCARE ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Descriptive Analytics	535.1	710.2	907.4	2,506.4	22.5%
Predictive Analytics	201.6	288.3	397.8	1,671.6	33.3%
Prescriptive Analytics	88.4	124.1	168.1	644.7	30.9%
Cognitive Analytics	42.5	59.1	79.3	289.6	29.6%
Total	867.6	1,181.7	1,552.5	5,112.3	26.9%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 114 GERMANY: HEALTHCARE ANALYTICS MARKET, BY APPLICATION, 2019–2026 (USD MILLION)

Application	2019	2020	2021	2026	CAGR (2021–2026)
Financial Analytics	310.5	412.4	528.3	1,532.8	23.7%
Clinical Analytics	279.6	401.8	554.2	2,206.5	31.8%
Operational & Administrative Analytics	149.6	192.6	239.1	593.5	19.9%
Population Health Analytics	127.8	174.9	230.9	779.6	27.6%
Total	867.6	1,181.7	1,552.5	5,112.3	26.9%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 115 GERMANY: FINANCIAL ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Claims Processing	128.4	171.2	219.7	635.1	23.6%
Revenue Cycle Management	90.6	117.1	146.1	371.7	20.5%
Payment Integrity & FWA	59.5	81.7	108.2	370.9	28.0%
Risk Adjustment & Assessment	32.1	42.5	54.2	155.1	23.4%
Total	310.5	412.4	528.3	1,532.8	23.7%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 116 GERMANY: CLINICAL ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Quality Improvement & Clinical Benchmarking	96.2	147.9	216.6	1,085.0	38.0%
Clinical Decision Support	61.4	85.6	114.5	391.5	27.9%
Precision Health	43.7	60.3	79.8	259.1	26.6%
Reporting & Compliance	35.4	48.2	63.1	193.4	25.1%
Medical Imaging Analytics	23.5	32.1	42.1	129.8	25.2%
Comparative Analytics/Effectiveness	19.4	27.7	38.0	147.7	31.2%
Total	279.6	401.8	554.2	2,206.5	31.8%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 117 GERMANY: OPERATIONAL & ADMINISTRATIVE ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Supply Chain Analytics	68.0	86.9	107.0	253.9	18.9%
Workforce Analytics	57.9	75.3	94.4	246.3	21.1%
Strategic Analytics	23.7	30.4	37.7	93.2	19.8%
Total	149.6	192.6	239.1	593.5	19.9%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 118 GERMANY: HEALTHCARE ANALYTICS MARKET, BY COMPONENT, 2019–2026 (USD MILLION)

Component	2019	2020	2021	2026	CAGR (2021–2026)
Services	455.0	631.1	842.4	2,935.7	28.4%
Software	337.0	458.5	601.8	1,971.7	26.8%
Hardware	75.6	92.2	108.4	205.0	13.6%
Total	867.6	1,181.7	1,552.5	5,112.3	26.9%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 119 GERMANY: HEALTHCARE ANALYTICS MARKET FOR SERVICES, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Support Services	307.1	414.0	535.8	1,519.3	23.2%
Business Analytics Services	147.9	217.1	306.6	1,416.4	35.8%
Total	455.0	631.1	842.4	2,935.7	28.4%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 120 GERMANY: HEALTHCARE ANALYTICS MARKET, BY DEPLOYMENT MODEL, 2019–2026 (USD MILLION)

Deployment Model	2019	2020	2021	2026	CAGR (2021–2026)
On-premise Deployment	544.3	733.9	954.1	2,968.5	25.5%
On-demand Deployment	323.3	447.9	598.4	2,143.8	29.1%
Total	867.6	1,181.7	1,552.5	5,112.3	26.9%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 121 GERMANY: HEALTHCARE ANALYTICS MARKET, BY END USER, 2019–2026 (USD MILLION)

End User	2019	2020	2021	2026	CAGR (2021–2026)
Payers	446.8	595.6	764.9	2,200.4	23.5%
Providers	330.2	463.2	626.8	2,392.9	30.7%
ACOs, HIEs, MCOs, and TPAs	90.6	122.9	160.8	519.0	26.4%
Total	867.6	1,181.7	1,552.5	5,112.3	26.9%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 122 GERMANY: HEALTHCARE ANALYTICS MARKET FOR PAYERS, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Private Insurance Companies	257.5	335.3	420.0	1,037.8	19.8%
Government Agencies	143.9	199.5	266.5	932.7	28.5%
Employers & Private Exchanges	45.4	60.8	78.3	229.9	24.0%
Total	446.8	595.6	764.9	2,200.4	23.5%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 123 GERMANY: HEALTHCARE ANALYTICS MARKET FOR PROVIDERS, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Hospitals, Physician Practices, and IDNs	262.4	371.5	507.0	2,003.7	31.6%
Post-acute Care Organizations	40.8	56.5	75.6	271.5	29.2%
Ambulatory Settings	27.0	35.2	44.3	117.7	21.6%
Total	330.2	463.2	626.8	2,392.9	30.7%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Germany, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

11.3.2 UK

11.3.2.1 Availability of venture capital investments to drive market growth in the UK

In 2020, the UK accounted for 28.0% of the European healthcare analytics market. The healthcare analytics market in the UK is expected to reach USD 3,161.3 million by 2026 from USD 1,197.5 million in 2021, at a CAGR of 21.4% during the forecast period.

The UK government expects analytics to become a key growth driver for the British economy in the coming years and has taken various steps to boost its implementation. In 2017, Britain invested around USD 15.8 billion (GBP 12 billion) to improve and enhance its analytics capabilities. Companies in the UK intend to increase their average spending on advanced analytics by 26% to bring it to USD 35 billion (GBP 24 billion) by 2020. In October 2015, the UK government announced an investment of USD 55.5 million (GBP 42 million) in research, where big data is one of the major focus areas.

The growth opportunities in this market have increased investments in the UK's HCIT sector.

- In June 2018, the NHS received USD 36 million (GBP 20 million) from the UK government to develop AI-driven analytics tools.
- The government announced that NHS England's budget would increase by 3.4% every year until 2023–2024.

Such investments by venture capitalists are expected to create a favorable environment for the growth of the healthcare analytics market in the UK.

Several players operating in the healthcare analytics market are adopting inorganic growth strategies to expand their presence in the UK. In February 2016, Apervita collaborated with GE Healthcare to expand its advanced analytics business in the UK, while GE Healthcare focused on making its advanced analytics application—DenialsIQ—available through the Apervita marketplace. In May 2017, IMS Health acquired a UK-based healthcare analytics company, Dataline Software, to expand the capabilities of its IMS platform.

TABLE 124 UK: HEALTHCARE ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Descriptive Analytics	424.3	582.4	709.5	1,572.9	17.3%
Predictive Analytics	149.6	221.9	292.5	999.8	27.9%
Prescriptive Analytics	70.8	102.3	131.5	394.6	24.6%
Cognitive Analytics	34.4	49.7	64.0	194.0	24.8%
Total	679.1	956.3	1,197.5	3,161.3	21.4%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of UK, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 125 UK: HEALTHCARE ANALYTICS MARKET, BY APPLICATION, 2019–2026 (USD MILLION)

Application	2019	2020	2021	2026	CAGR (2021–2026)
Financial Analytics	234.0	321.3	392.3	912.5	18.4%
Clinical Analytics	228.8	337.6	441.2	1,381.2	25.6%
Operational & Administrative Analytics	115.8	154.9	184.3	376.5	15.4%
Population Health Analytics	100.5	142.5	179.7	491.2	22.3%
Total	679.1	956.3	1,197.5	3,161.3	21.4%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of UK, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 126 UK: FINANCIAL ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Claims Processing	96.3	132.4	161.7	371.6	18.1%
Revenue Cycle Management	68.9	92.2	109.8	225.0	15.4%
Payment Integrity & FWA	45.2	64.3	81.2	224.3	22.5%
Risk Adjustment & Assessment	23.7	32.5	39.6	91.6	18.3%
Total	234.0	321.3	392.3	912.5	18.4%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of UK, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 127 UK: CLINICAL ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Quality Improvement & Clinical Benchmarking	77.2	119.8	164.1	619.5	30.4%
Clinical Decision Support	53.5	77.0	98.1	270.5	22.5%
Precision Health	34.0	48.6	61.6	165.7	21.9%
Reporting & Compliance	29.8	42.2	52.9	134.3	20.5%
Medical Imaging Analytics	19.3	27.7	35.2	95.5	22.1%
Comparative Analytics/Effectiveness	15.0	22.3	29.4	95.6	26.6%
Total	228.8	337.6	441.2	1,381.2	25.6%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of UK, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 128 UK: OPERATIONAL & ADMINISTRATIVE ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Supply Chain Analytics	53.8	71.3	83.9	162.2	14.1%
Workforce Analytics	44.1	59.8	72.1	157.0	16.9%
Strategic Analytics	17.9	23.9	28.3	57.3	15.1%
Total	115.8	154.9	184.3	376.5	15.4%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of UK, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 129 UK: HEALTHCARE ANALYTICS MARKET, BY COMPONENT, 2019–2026 (USD MILLION)

Component	2019	2020	2021	2026	CAGR (2021–2026)
Services	366.8	525.0	667.1	1,860.7	22.8%
Software	252.5	354.8	443.4	1,158.8	21.2%
Hardware	59.8	76.5	87.0	141.7	10.3%
Total	679.1	956.3	1,197.5	3,161.3	21.4%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of UK, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 130 UK: HEALTHCARE ANALYTICS MARKET FOR SERVICES, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Support Services	247.3	352.3	445.4	1,210.9	22.1%
Business Analytics Services	119.5	172.7	221.7	649.8	24.0%
Total	366.8	525.0	667.1	1,860.7	22.8%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of UK, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 131 UK: HEALTHCARE ANALYTICS MARKET, BY DEPLOYMENT MODEL, 2019–2026 (USD MILLION)

Deployment Model	2019	2020	2021	2026	CAGR (2021–2026)
On-premise Deployment	431.4	590.1	715.9	1,538.7	16.5%
On-demand Deployment	247.7	366.3	481.6	1,622.6	27.5%
Total	679.1	956.3	1,197.5	3,161.3	21.4%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of UK, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 132 UK: HEALTHCARE ANALYTICS MARKET, BY END USER, 2019–2026 (USD MILLION)

End User	2019	2020	2021	2026	CAGR (2021–2026)
Payers	355.0	488.7	597.4	1,369.9	18.1%
Providers	253.7	367.2	472.7	1,432.7	24.8%
ACOs, HIEs, MCOs, and TPAs	70.4	100.4	127.4	358.7	23.0%
Total	679.1	956.3	1,197.5	3,161.3	21.4%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of UK, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 133 UK: HEALTHCARE ANALYTICS MARKET FOR PAYERS, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Private Insurance Companies	209.3	284.4	343.1	730.5	16.3%
Government Agencies	113.1	159.3	199.2	511.9	20.8%
Employers & Private Exchanges	32.6	45.0	55.1	127.6	18.3%
Total	355.0	488.7	597.4	1,369.9	18.1%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of UK, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 134 UK: HEALTHCARE ANALYTICS MARKET FOR PROVIDERS, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Hospitals, Physician Practices, and IDNs	201.7	293.8	380.3	1,179.5	25.4%
Post-acute Care Organizations	33.8	48.8	62.7	188.3	24.6%
Ambulatory Settings	18.1	24.6	29.7	65.0	17.0%
Total	253.7	367.2	472.7	1,432.7	24.8%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of UK, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

11.3.3 FRANCE

11.3.3.1 Strong financial support for digital health contributes to the growth in the healthcare analytics market in France

France accounted for 24.1% of the European healthcare analytics market in 2020. The healthcare analytics market in France is expected to reach USD 2,948.1 million by 2026 from USD 823.1 million in 2020, at a CAGR of 23.1% during the forecast period.

In France, the adoption of IT in the healthcare industry has been relatively slower than in other European countries due to limited developments in the field and a lack of awareness among healthcare providers. However, in the last decade, IT systems have been increasingly used to improve the efficiency of the healthcare sector in France. In September 2017, the French government announced an investment of EUR 4.9 billion (USD 5.6 billion) to create a digital ecosystem in France for 2018–2022. This investment focused on the development of digital companies in every field, including healthcare.

France has strong data privacy laws, which have created a better framework for EHR systems to connect and operate with advanced capabilities successfully. This has resulted in a growing demand for analytics solutions, and public and private organizations are undertaking several projects to improve the usage of analytics solutions in the French healthcare industry. In 2015, the Assistance Publique-Hôpitaux de Paris, a member of the Healthcare Data Institute, undertook a project aimed at using healthcare analytics to improve patient experience in the emergency department. As of 2017, France is in the process of establishing a legal system, Système national des données de santé. This system will provide access to the

healthcare data collected from national medical insurance, hospitals, and other public organizations. The usage of this system will propel the use of analytics software to carry out data analysis and improve current healthcare practices.

Leading players are also taking initiatives to enter the French market to develop successful partnerships with companies in the healthcare and life sciences industries, based on their growing need for data analytics. For instance, in 2016, Intel collaborated with Healthcare Data Institute (France). Through this collaboration, the company focuses on developing partnerships with players in the French healthcare and life science industry based on their need for data analytics. The company is also working with Unicancer, a federation of 20 French Cancer Centers (FCCs), to deploy analytics solutions.

TABLE 135 FRANCE: HEALTHCARE ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Descriptive Analytics	358.4	481.5	592.2	1,366.5	18.2%
Predictive Analytics	146.0	212.4	283.5	1,031.0	29.5%
Prescriptive Analytics	61.3	87.3	114.0	373.1	26.8%
Cognitive Analytics	29.5	42.0	54.8	177.4	26.5%
Total	595.2	823.1	1,044.6	2,948.1	23.1%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of FRANCE, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 136 FRANCE: HEALTHCARE ANALYTICS MARKET, BY APPLICATION, 2019–2026 (USD MILLION)

Application	2019	2020	2021	2026	CAGR (2021–2026)
Financial Analytics	222.0	301.3	375.0	961.7	20.7%
Clinical Analytics	182.1	263.4	348.2	1,155.4	27.1%
Operational & Administrative Analytics	102.9	135.0	162.6	353.2	16.8%
Population Health Analytics	88.1	123.5	158.7	477.8	24.7%
Total	595.2	823.1	1,044.6	2,948.1	23.1%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of FRANCE, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 137 FRANCE: FINANCIAL ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Revenue Cycle Management	94.41	124.12	149.73	328.00	17.0%
Claims Processing	62.17	86.46	110.03	303.80	22.5%
Payment Integrity & FWA	42.95	60.25	77.56	235.07	24.8%
Risk Adjustment & Assessment	22.52	30.43	37.73	94.82	20.2%
Total	222.05	301.26	375.05	961.69	20.7%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of FRANCE, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 138 FRANCE: CLINICAL ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Quality Improvement & Clinical Benchmarking	58.2	88.8	123.2	494.7	32.1%
Clinical Decision Support	42.4	60.6	79.1	245.7	25.5%
Precision Health	29.5	41.1	52.3	142.9	22.3%
Reporting & Compliance	23.0	31.6	39.8	103.2	21.0%
Medical Imaging Analytics	15.2	21.3	27.4	78.0	23.3%
Comparative Analytics/Effectiveness	13.8	20.0	26.6	91.0	27.9%
Total	182.1	263.4	348.2	1,155.4	27.1%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of FRANCE, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 139 FRANCE: OPERATIONAL & ADMINISTRATIVE ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Supply Chain Analytics	46.6	60.5	72.0	146.9	15.3%
Workforce Analytics	39.8	53.1	64.8	151.8	18.5%
Strategic Analytics	16.4	21.5	25.7	54.5	16.2%
Total	102.9	135.0	162.6	353.2	16.8%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of FRANCE, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 140 FRANCE: HEALTHCARE ANALYTICS MARKET, BY COMPONENT, 2019–2026 (USD MILLION)

Component	2019	2020	2021	2026	CAGR (2021–2026)
Services	305.4	429.7	553.7	1,657.0	24.5%
Software	235.0	324.3	410.7	1,147.7	22.8%
Hardware	54.8	69.1	80.1	143.4	12.4%
Total	595.2	823.1	1,044.6	2,948.1	23.1%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of FRANCE, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 141 FRANCE: HEALTHCARE ANALYTICS MARKET FOR SERVICES, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Support Services	204.7	286.2	366.4	1,059.0	23.6%
Business Analytics Services	100.7	143.5	187.3	598.0	26.1%
Total	305.4	429.7	553.7	1,657.0	24.5%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of FRANCE, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 142 FRANCE: HEALTHCARE ANALYTICS MARKET, BY DEPLOYMENT MODEL, 2019–2026 (USD MILLION)

Deployment Model	2019	2020	2021	2026	CAGR (2021–2026)
On-premise Deployment	386.7	521.0	642.8	1,514.7	18.7%
On-demand Deployment	208.4	302.1	401.8	1,433.4	29.0%
Total	595.2	823.1	1,044.6	2,948.1	23.1%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of FRANCE, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 143 FRANCE: HEALTHCARE ANALYTICS MARKET, BY END USER, 2019–2026 (USD MILLION)

End User	2019	2020	2021	2026	CAGR (2021–2026)
Payers	303.7	409.1	504.9	1,205.1	19.0%
Providers	228.6	326.0	426.5	1,402.1	26.9%
ACOs, HIEs, MCOs, and TPAs	62.9	88.1	113.2	340.9	24.7%
Total	595.2	823.1	1,044.6	2,948.1	23.1%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of FRANCE, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 144 FRANCE: HEALTHCARE ANALYTICS MARKET FOR PAYERS, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Private Insurance Companies	177.1	235.2	286.1	628.8	17.1%
Government Agencies	96.1	132.5	167.5	450.2	21.9%
Employers & Private Exchanges	30.5	41.3	51.3	126.2	19.7%
Total	303.7	409.1	504.9	1,205.1	19.0%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of FRANCE, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 145 FRANCE: HEALTHCARE ANALYTICS MARKET FOR PROVIDERS, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Hospitals, Physician Practices, and IDNs	181.7	260.8	343.3	1,158.9	27.5%
Post-acute Care Organizations	28.5	40.4	52.7	169.7	26.4%
Ambulatory Settings	18.5	24.8	30.5	73.5	19.3%
Total	228.6	326.0	426.5	1,402.1	26.9%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of FRANCE, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

11.3.4 REST OF EUROPE

In 2020, the RoE accounted for 13.3% of the European healthcare analytics market. The healthcare analytics market in this regional segment is expected to reach USD 1,422.6 million by 2026 from USD 454.3 million in 2020, at a CAGR of 20.3% during the forecast period.

The RoE comprises Italy, Spain, Russia, Denmark, Sweden, Finland, and other European countries, which form a smaller portion of the market. Finland, Denmark, and Sweden are the leaders in this regional segment in terms of the adoption of EHRs. General practitioners in Denmark were ranked first in an assessment of the overall implementation of EHRs in 2016. In Denmark, about 98% of primary care physicians have access to the centralized electronic records of the citizens, including all hospital physicians and all pharmacists. All citizens in Denmark have a unique electronic personal identifier, which is used in all public registries, including the health database. Finland has also registered a high rate of EHR adoption. Consequently, the demand for healthcare analytics solutions is expected to increase in their respective healthcare systems in the coming years.

According to a recent report (February 2019) by the Finnish Ministry of Social Affairs, electronic information exchange between organizations has progressed rapidly in the country, and the most frequently used and needed e-Health services for citizens include ePrescriptions, laboratory test results, and electronic booking of services. The Russian government has also taken initiatives to reform its healthcare system. It has been working on establishing international cooperation on healthcare-related issues and strengthening the role of information management in the healthcare system. To improve the overall healthcare system in the country, including establishing an effective hospital infrastructure, Russia and Turkey signed a memorandum of joint investment in March 2017. Both agreed to invest USD 500 million each in developing the healthcare and tourism industries in their respective countries.

Various leading players in the region's healthcare analytics market are adopting organic and inorganic growth strategies to expand their business in some RoE countries. For instance, in November 2018, Wipro established its new office in Porto, Portugal, to expand its European operations and meet growing customer requirements in this region. In April 2018, IBM Watson established an international hub in Italy to advance genomics, big data, aging, and nutrition with an investment of USD 150 million.

TABLE 146 ROE: HEALTHCARE ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Descriptive Analytics	199.6	274.8	331.1	689.6	15.8%
Predictive Analytics	71.0	106.3	139.4	464.2	27.2%
Prescriptive Analytics	33.8	49.1	62.5	179.2	23.5%
Cognitive Analytics	16.5	24.1	30.7	89.5	23.8%
Total	320.9	454.3	563.7	1,422.6	20.3%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of ROE, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 147 ROE: HEALTHCARE ANALYTICS MARKET, BY APPLICATION, 2019–2026 (USD MILLION)

Application	2019	2020	2021	2026	CAGR (2021–2026)
Clinical Analytics	112.6	166.3	214.5	632.4	24.1%
Financial Analytics	113.8	157.2	190.4	425.5	17.5%
Operational & Administrative Analytics	51.0	68.6	80.9	157.9	14.3%
Population Health Analytics	43.5	62.2	78.0	206.9	21.5%
Total	320.9	454.3	563.7	1,422.6	20.3%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of ROE, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 148 ROE: FINANCIAL ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Revenue Cycle Management	47.6	63.7	74.7	142.6	13.8%
Claims Processing	32.9	46.7	57.9	140.2	19.4%
Payment Integrity & FWA	22.0	31.4	39.3	103.4	21.3%
Risk Adjustment & Assessment	11.3	15.4	18.5	39.3	16.3%
Total	113.8	157.2	190.4	425.5	17.5%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of ROE, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 149 ROE: CLINICAL ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Quality Improvement & Clinical Benchmarking	38.7	60.4	81.8	292.6	29.0%
Clinical Decision Support	27.0	38.9	49.0	128.7	21.3%
Precision Health	16.0	22.8	28.4	70.8	20.0%
Reporting & Compliance	15.4	21.6	26.4	59.7	17.7%
Medical Imaging Analytics	9.6	13.8	17.3	44.5	20.8%
Comparative Analytics/Effectiveness	5.9	8.8	11.5	36.0	25.6%
Total	112.6	166.3	214.5	632.4	24.1%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of ROE, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 150 ROE: OPERATIONAL & ADMINISTRATIVE ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Supply Chain Analytics	23.6	31.3	36.4	66.1	12.7%
Workforce Analytics	19.6	26.8	32.0	67.7	16.2%
Strategic Analytics	7.9	10.6	12.4	24.0	14.1%
Total	51.0	68.6	80.9	157.9	14.3%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of ROE, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 151 ROE: HEALTHCARE ANALYTICS MARKET, BY COMPONENT, 2019–2026 (USD MILLION)

Component	2019	2020	2021	2026	CAGR (2021–2026)
Services	172.2	247.6	311.6	829.5	21.6%
Software	119.5	169.0	209.5	526.0	20.2%
Hardware	29.3	37.7	42.6	67.1	9.5%
Total	320.9	454.3	563.7	1,422.6	20.3%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of ROE, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 152 ROE: HEALTHCARE ANALYTICS MARKET FOR SERVICES, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Support Services	116.0	165.9	207.7	537.2	20.9%
Business Analytics Services	56.2	81.7	104.0	292.3	23.0%
Total	172.2	247.6	311.6	829.5	21.6%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of ROE, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 153 ROE: HEALTHCARE ANALYTICS MARKET, BY DEPLOYMENT MODEL, 2019–2026 (USD MILLION)

Deployment Model	2019	2020	2021	2026	CAGR (2021–2026)
On-premise Deployment	206.7	285.7	345.8	747.0	16.7%
On-demand Deployment	114.3	168.5	217.9	675.6	25.4%
Total	320.9	454.3	563.7	1,422.6	20.3%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of ROE, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 154 ROE: HEALTHCARE ANALYTICS MARKET, BY END USER, 2019–2026 (USD MILLION)

End User	2019	2020	2021	2026	CAGR (2021–2026)
Payers	166.7	229.9	277.4	591.4	16.3%
Providers	121.2	177.2	227.1	674.1	24.3%
ACOs, HIEs, MCOs, and TPAs	33.0	47.2	59.2	157.1	21.5%
Total	320.9	454.3	563.7	1,422.6	20.3%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of ROE, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 155 ROE: HEALTHCARE ANALYTICS MARKET FOR PAYERS, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Private Insurance Companies	97.8	132.6	157.4	304.6	14.1%
Government Agencies	53.7	76.1	94.4	231.0	19.6%
Employers & Private Exchanges	15.3	21.1	25.6	55.7	16.8%
Total	166.7	229.9	277.4	591.4	16.3%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of ROE, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 156 ROE: HEALTHCARE ANALYTICS MARKET FOR PROVIDERS, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Hospitals, Physician Practices, and IDNs	96.6	142.3	183.6	560.5	25.0%
Post-acute Care Organizations	15.9	23.0	29.2	82.5	23.1%
Ambulatory Settings	8.6	11.9	14.3	31.2	16.9%
Total	121.2	177.2	227.1	674.1	24.3%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of ROE, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

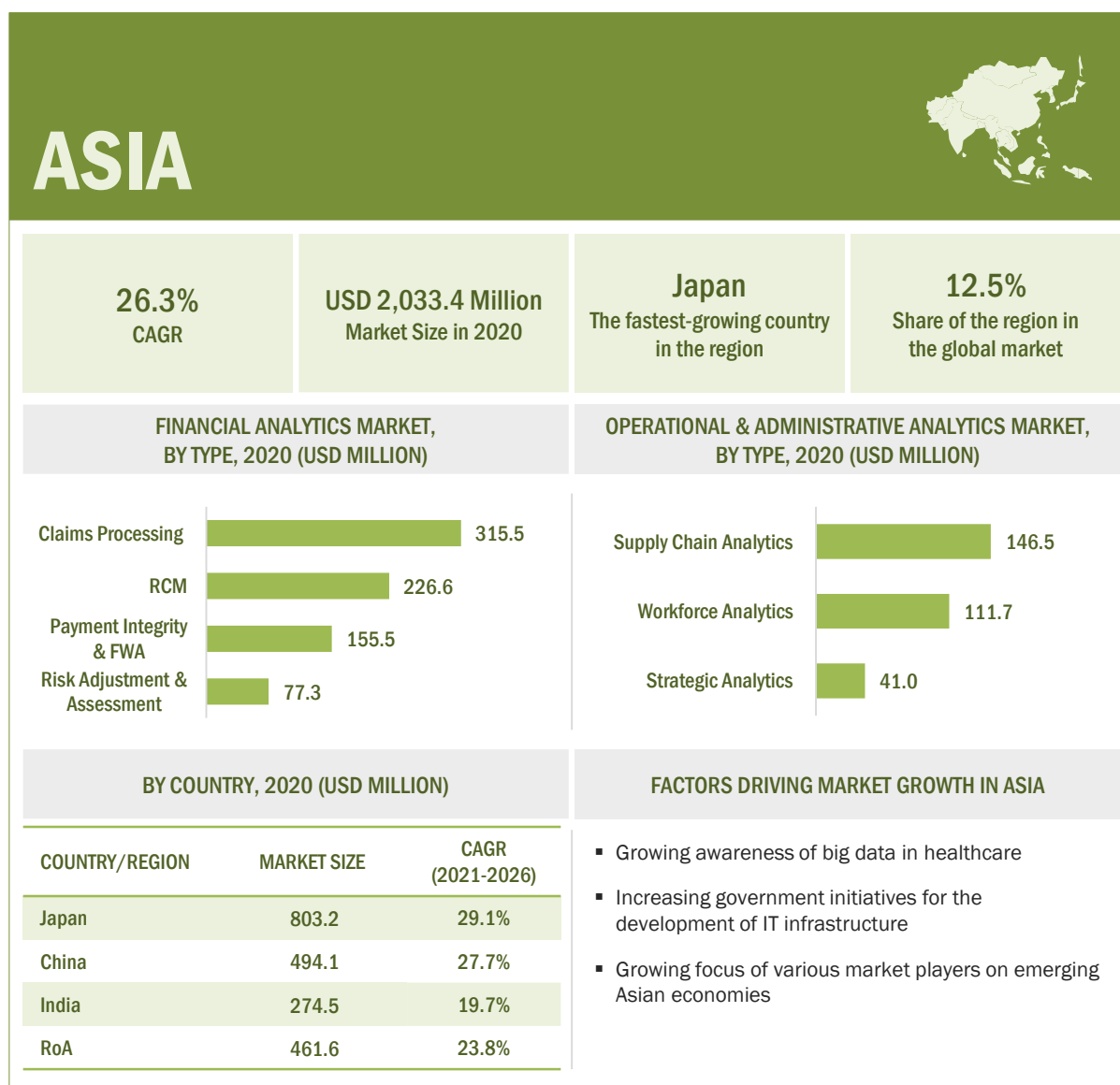
11.4 ASIA

Asia accounted for 12.8% of the global healthcare analytics market in 2020. This segment is expected to reach USD 8,519.1 million by 2026 from USD 2,033.4 million in 2020, at a CAGR of 26.3% during the forecast period.

Since the patient population is diverse and large in Asia, healthcare analytics is expected to play a crucial role in providing quality care at a lower cost. Healthcare organizations in Asia are actively moving towards digitization to ensure patient safety & quality care and streamline workflows in large healthcare enterprises. The healthcare analytics market in Singapore, Malaysia, Japan, China, Taiwan, and South Korea is growing at a higher rate than India, Indonesia, and the Philippines, where market growth is hindered by the slow adoption of EHRs and inadequate HCIT infrastructure. The healthcare sectors in these countries are still in the early stages of digitization.

Asian countries are expected to follow trends in developed countries such as the US and recognize the value that actionable insights can bring in terms of productivity, profitability, and innovation. Factors such as the presence of a larger patient population demanding better healthcare at lower costs, economic growth, the implementation of favorable government policies, rising geriatric population, growing incidence of chronic diseases, the emergence of big data in healthcare, increasing spending on IT infrastructure by healthcare providers, and growing focus of various market players on emerging Asian countries are expected to drive the growth of the healthcare analytics market in Asia.

However, the adoption of healthcare analytics in this region is currently slow. The major factors responsible for this are the high installation costs of systems, traditional mindset of physicians, limited access to and unavailability of structured data, poorly connected IT infrastructure, data security issues, lack of standard protocols for HCIT, lack of suitable software, and low budget allocations of hospitals for implementing HCIT solutions. These restraints can be turned into opportunities by implementing effectively communicated data strategies, better training, and improved access to data, which can significantly improve the ability to adapt and gain value from data analytics.

FIGURE 31 ASIA: HEALTHCARE ANALYTICS MARKET SNAPSHOT (2020)

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 157 ASIA: HEALTHCARE ANALYTICS MARKET, BY COUNTRY, 2019–2026 (USD MILLION)

Country/Region	2019	2020	2021	2026	CAGR (2021–2026)
Japan	565.7	803.2	1,073.0	3,818.2	28.9%
China	384.3	494.1	649.4	2,184.5	27.5%
India	206.5	274.5	338.3	824.3	19.5%
RoA	339.6	461.6	588.0	1,692.2	23.5%
Total	1,496.2	2,033.4	2,648.8	8,519.1	26.3%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 158 ASIA: HEALTHCARE ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Descriptive Analytics	1,001.7	1,331.1	1,692.2	4,636.6	22.3%
Predictive Analytics	253.5	365.3	504.8	2,199.5	34.2%
Prescriptive Analytics	149.6	209.4	281.0	1,053.8	30.3%
Cognitive Analytics	91.3	127.6	170.8	629.2	29.8%
Total	1,496.2	2,033.4	2,648.8	8,519.1	26.3%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 159 ASIA: HEALTHCARE ANALYTICS MARKET, BY APPLICATION, 2019–2026 (USD MILLION)

Application	2019	2020	2021	2026	CAGR (2021–2026)
Financial Analytics	586.7	774.9	981.0	2,723.2	22.7%
Clinical Analytics	498.6	721.0	993.1	3,965.8	31.9%
Operational & Administrative Analytics	235.6	299.2	364.4	833.5	18.0%
Population Health Analytics	175.4	238.2	310.3	996.7	26.3%
Total	1,496.2	2,033.4	2,648.8	8,519.1	26.3%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 160 ASIA: FINANCIAL ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Claims Processing	238.2	315.5	400.0	1,103.5	22.5%
Revenue Cycle Management	176.0	226.6	280.0	686.8	19.7%
Payment Integrity & FWA	114.1	155.5	203.0	657.7	26.5%
Risk Adjustment & Assessment	58.4	77.3	98.1	275.1	22.9%
Total	586.7	774.9	981.0	2,723.2	22.7%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 161 ASIA: CLINICAL ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Quality Improvement & Clinical Benchmarking	170.7	258.3	371.4	1,781.1	36.8%
Clinical Decision Support	115.7	163.2	218.9	760.5	28.3%
Precision Health	71.1	100.2	134.3	465.7	28.2%
Reporting & Compliance	65.0	90.8	120.7	406.7	27.5%
Medical Imaging Analytics	46.3	65.6	88.5	315.5	29.0%
Comparative Analytics/Effectiveness	29.7	43.0	59.2	236.3	31.9%
Total	498.6	721.0	993.1	3,965.8	31.9%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 162 ASIA: OPERATIONAL & ADMINISTRATIVE ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Supply Chain Analytics	114.5	146.5	179.7	424.1	18.7%
Workforce Analytics	87.8	111.7	136.2	314.4	18.2%
Strategic Analytics	33.3	41.0	48.5	95.1	14.4%
Total	235.6	299.2	364.4	833.5	18.0%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 163 ASIA: HEALTHCARE ANALYTICS MARKET, BY COMPONENT, 2019–2026 (USD MILLION)

Component	2019	2020	2021	2026	CAGR (2021–2026)
Services	753.0	1,051.1	1,402.3	4,913.7	28.5%
Software	610.7	824.0	1,065.5	3,304.2	25.4%
Hardware	132.4	158.2	181.0	301.2	10.7%
Total	1,496.2	2,033.4	2,648.8	8,519.1	26.3%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 164 ASIA: HEALTHCARE ANALYTICS MARKET FOR SERVICES, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Support Services	517.9	719.4	954.8	3,253.3	27.8%
Business Analytics Services	235.1	331.7	447.5	1,660.4	30.0%
Total	753.0	1,051.1	1,402.3	4,913.7	28.5%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 165 ASIA: HEALTHCARE ANALYTICS MARKET, BY DEPLOYMENT MODEL, 2019–2026 (USD MILLION)

Deployment Model	2019	2020	2021	2026	CAGR (2021–2026)
On-premise Deployment	1,125.9	1,501.3	1,915.3	5,369.6	22.9%
On-demand Deployment	370.3	532.1	733.5	3,149.5	33.8%
Total	1,496.2	2,033.4	2,648.8	8,519.1	26.3%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 166 ASIA: HEALTHCARE ANALYTICS MARKET, BY END USER, 2019–2026 (USD MILLION)

End User	2019	2020	2021	2026	CAGR (2021–2026)
Payers	880.2	1,183.0	1,524.8	4,623.6	24.8%
Providers	472.5	655.4	870.6	3,087.7	28.8%
ACOs, HIEs, MCOs, and TPAs	143.6	194.9	253.4	807.9	26.1%
Total	1,496.2	2,033.4	2,648.8	8,519.1	26.3%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 167 ASIA: HEALTHCARE ANALYTICS MARKET FOR PAYERS, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Private Insurance Companies	526.6	703.4	900.8	2,641.2	24.0%
Government Agencies	276.1	375.6	490.1	1,579.1	26.4%
Employers & Private Exchanges	77.5	104.0	133.9	403.3	24.7%
Total	880.2	1,183.0	1,524.8	4,623.6	24.8%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 168 ASIA: HEALTHCARE ANALYTICS MARKET FOR PROVIDERS, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Hospitals, Physician Practices, and IDNs	381.6	534.3	715.6	2,625.9	29.7%
Post-acute Care Organizations	57.7	78.8	102.9	336.5	26.7%
Ambulatory Settings	33.1	42.4	52.1	125.2	19.2%
Total	472.5	655.4	870.6	3,087.7	28.8%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

11.4.1 JAPAN

11.4.1.1 Rise in the geriatric population will increase the demand for effective patient management solutions

In 2020, Japan accounted for the largest share of 39.5% of the Asian healthcare analytics market. The Japanese healthcare analytics market is expected to reach USD 3,818.2 million by 2026 from USD 1,073.0 million in 2021, at a CAGR of 28.9% during the forecast period.

Japan strongly focuses on developing and adopting various innovative techniques to provide quality healthcare. Technological developments have helped Japanese healthcare organizations to move from paper-based health record systems to EHRs. This change has created a demand for healthcare analytics solutions to manage health data as and when required. Furthermore, the implementation of government initiatives such as the e-Japan Policy, New IT Reform Strategy, and iJapan Strategy 2015 will prove favorable to market growth by directly driving the adoption of and reliance on HCIT solutions.

The need for streamlining healthcare workflows can be attributed to Japan's rising patient population and the growth of its geriatric population. The percentage of the geriatric population (aged above 65 years) in Japan is expected to increase from 20.1% in 2005 to 40.4% by 2055 (Source: Japanese Journal of Health Economics and Policy). The rapid growth in the geriatric population will burden both payers and providers, which is expected to sustain the demand for healthcare analytics solutions.

Various local and international players are expanding their business in the Japanese analytics market. For instance, in April 2018, Ambra Health partnered with Mitsubishi Corporation to introduce cloud-based medical imaging platforms in Japan. In March 2018, ZoomData, a prominent developer of analytics platforms, partnered with Hitachi INS Software. In May 2017, FUJITSU expanded its alliance with SAS Institute Japan to deliver SAS analytics solutions as a cloud service. Apple and IBM also partnered with the Japan Post Group to provide Japanese senior citizens with iPads equipped with IBM's analytics software/apps to manage health.

TABLE 169 JAPAN: HEALTHCARE ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Descriptive Analytics	375.6	518.1	670.5	1,920.9	23.4%
Predictive Analytics	100.0	152.6	219.2	1,119.6	38.6%
Prescriptive Analytics	57.1	84.3	117.2	507.3	34.1%
Cognitive Analytics	33.0	48.2	66.2	270.4	32.5%
Total	565.7	803.2	1,073.0	3,818.2	28.9%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of JAPAN, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 170 JAPAN: HEALTHCARE ANALYTICS MARKET, BY APPLICATION, 2019–2026 (USD MILLION)

Application	2019	2020	2021	2026	CAGR (2021–2026)
Financial Analytics	226.2	310.8	402.0	1,215.7	24.8%
Clinical Analytics	179.3	273.9	390.4	1,766.5	35.2%
Operational & Administrative Analytics	91.1	120.5	149.8	372.9	20.0%
Population Health Analytics	69.1	98.0	130.8	463.0	28.8%
Total	565.7	803.2	1,073.0	3,818.2	28.9%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of JAPAN, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 171 JAPAN: FINANCIAL ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Claims Processing	91.5	126.2	163.6	492.6	24.7%
Revenue Cycle Management	68.1	91.1	114.6	302.2	21.4%
Payment Integrity & FWA	44.1	62.8	84.0	301.9	29.1%
Risk Adjustment & Assessment	22.4	30.8	39.7	118.9	24.5%
Total	226.2	310.8	402.0	1,215.7	24.8%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of JAPAN, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 172 JAPAN: CLINICAL ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Quality Improvement & Clinical Benchmarking	60.6	98.6	148.7	837.8	41.3%
Clinical Decision Support	42.9	62.4	84.8	301.7	28.9%
Precision Health	25.8	38.1	52.4	198.3	30.5%
Reporting & Compliance	22.6	33.7	47.0	190.1	32.3%
Medical Imaging Analytics	16.4	24.4	33.8	133.4	31.6%
Comparative Analytics/Effectiveness	11.0	16.7	23.7	105.2	34.7%
Total	179.3	273.9	390.4	1,766.5	35.2%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of JAPAN, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 173 JAPAN: OPERATIONAL & ADMINISTRATIVE ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Supply Chain Analytics	43.3	57.6	72.1	184.5	20.7%
Workforce Analytics	34.7	45.9	57.1	142.9	20.1%
Strategic Analytics	13.2	17.0	20.6	45.5	17.1%
Total	91.1	120.5	149.8	372.9	20.0%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of JAPAN, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 174 JAPAN: HEALTHCARE ANALYTICS MARKET, BY COMPONENT, 2019–2026 (USD MILLION)

Component	2019	2020	2021	2026	CAGR (2021–2026)
Services	287.8	420.1	574.7	2,218.9	31.0%
Software	224.6	317.3	421.7	1,463.5	28.3%
Hardware	53.3	65.9	76.5	135.8	12.1%
Total	565.7	803.2	1,073.0	3,818.2	28.9%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of JAPAN, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 175 JAPAN: HEALTHCARE ANALYTICS MARKET FOR SERVICES, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Support Services	199.3	289.0	392.7	1,461.9	30.1%
Business Analytics Services	88.5	131.1	182.0	757.0	33.0%
Total	287.8	420.1	574.7	2,218.9	31.0%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of JAPAN, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 176 JAPAN: HEALTHCARE ANALYTICS MARKET, BY DEPLOYMENT MODEL, 2019–2026 (USD MILLION)

Deployment Model	2019	2020	2021	2026	CAGR (2021–2026)
On-premise Deployment	404.4	559.3	726.1	2,126.7	24.0%
On-demand Deployment	161.3	243.8	346.9	1,691.4	37.3%
Total	565.7	803.2	1,073.0	3,818.2	28.9%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of JAPAN, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 177 JAPAN: HEALTHCARE ANALYTICS MARKET, BY END USER, 2019–2026 (USD MILLION)

End User	2019	2020	2021	2026	CAGR (2021–2026)
Payers	330.0	462.6	609.9	2,015.6	27.0%
Providers	179.7	261.0	356.7	1,422.3	31.9%
ACOs, HIEs, MCOs, and TPAs	55.9	79.5	106.3	380.3	29.0%
Total	565.7	803.2	1,073.0	3,818.2	28.9%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of JAPAN, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 178 JAPAN: HEALTHCARE ANALYTICS MARKET FOR PAYERS, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Private Insurance Companies	198.6	276.7	362.5	1,159.4	26.2%
Government Agencies	102.4	145.3	193.8	679.9	28.5%
Employers & Private Exchanges	29.1	40.7	53.6	176.3	26.9%
Total	330.0	462.6	609.9	2,015.6	27.0%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of JAPAN, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 179 JAPAN: HEALTHCARE ANALYTICS MARKET FOR PROVIDERS, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Hospitals, Physician Practices, and IDNs	142.8	209.4	288.7	1,194.1	32.8%
Post-acute Care Organizations	23.2	33.2	44.6	165.0	29.9%
Ambulatory Settings	13.8	18.5	23.4	63.3	22.0%
Total	179.7	261.0	356.7	1,422.3	31.9%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of JAPAN, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

11.4.2 CHINA

11.4.2.1 Favorable reforms and a growing number of EHRs have contributed to market growth in China

In 2020, China accounted for 24.3% of the Asian healthcare analytics market. The Chinese healthcare analytics market is expected to reach USD 2,184.5 million by 2026 from USD 494.1 million in 2020, at a CAGR of 27.5% during the forecast period.

The adoption of healthcare IT increased significantly after the launch of the National Health Information Development Plan in 2003. Due to the increasing aging population and the growing incidence of chronic diseases, the Chinese government faces a huge challenge in healthcare spending. As a result, the need for new technologies has increased, and the Chinese government is taking steps to create awareness about the use and benefits of HIT. In November 2013, the government issued an overall framework for HIT called '46312,' following a national push for EHR adoption that began back in February 2010. By the end of 2019, 71% of hospitals had implemented an EHR/EMR system to maintain patient health records compared to less than 30% in 2009–2010 (Source: Pulse IT Magazine).

With the growing number of EHRs in China, the amount of patient data collected is expected to increase in the coming years. This will drive the adoption of analytical tools to acquire the necessary insights into patient data and help providers deliver quality care.

In China, the government focuses on implementing several healthcare reforms focused on achieving international care standards. It also emphasizes the development of regional health information networks, which will integrate healthcare services and information across hospitals and other healthcare providers in the country. The number of hospitals in China has increased significantly, from 21,979 in 2011 to 27,587 in 2015. Additionally, 2,000 new private hospitals emerged in China in 2016, which increased the total number of hospitals in the country to 29,000 by the end of 2016. Among them, 160,000 are privately owned, while 130,000 are public hospitals. According to a 2014 Deutsche Bank Report, from 2014, 8,000 public hospitals are expected to be privatized in the next 5–10 years. The demand for advanced analytics solutions is expected to increase in China in the coming years, as more hospitals seek to offer quality patient care while simultaneously reducing healthcare costs.

With the emergence of digital health, the use of big data, analytics, business intelligence, analysis, and reporting has increased. In September 2018, China's National Development and Reform Commission (NDRC) entered into an agreement with the China Development Bank to invest RMB 100 billion (USD 145 billion) in big data and cloud computing for the next five years to enhance its AI capabilities by 2023. In

addition, digital health investment in China has increased from USD 393 million in 2014 to USD 1.1 billion in the first semester of 2016.

The availability of venture capital investments has also grown in the country. For instance, in March 2017, Tencent Holdings led a USD 200-million series D funding round in Haodaifu Online, one of China's largest digital healthcare companies. In January 2016, Yidu Cloud Technology Company Ltd. (China), a solutions provider for big data analytics in healthcare, offered a series B funding to develop a diabetes management platform in China.

TABLE 180 CHINA: HEALTHCARE ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Descriptive Analytics	259.2	327.1	421.5	1,256.7	24.4%
Predictive Analytics	64.0	86.5	119.4	515.2	34.0%
Prescriptive Analytics	38.7	50.9	68.4	257.9	30.4%
Cognitive Analytics	22.4	29.6	40.1	154.7	31.0%
Total	384.3	494.1	649.4	2,184.5	27.5%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of CHINA, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 181 CHINA: HEALTHCARE ANALYTICS MARKET, BY APPLICATION, 2019–2026 (USD MILLION)

Application	2019	2020	2021	2026	CAGR (2021–2026)
Financial Analytics	150.2	186.8	237.4	675.1	23.3%
Clinical Analytics	126.1	174.4	244.6	1,045.1	33.7%
Operational & Administrative Analytics	62.0	74.1	90.6	212.0	18.5%
Population Health Analytics	46.0	58.8	76.9	252.3	26.8%
Total	384.3	494.1	649.4	2,184.5	27.5%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of CHINA, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 182 CHINA: FINANCIAL ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Claims Processing	60.2	75.1	95.7	272.6	23.3%
Revenue Cycle Management	46.7	56.8	70.5	178.5	20.4%
Payment Integrity & FWA	28.4	36.2	47.3	153.7	26.6%
Risk Adjustment & Assessment	14.9	18.7	23.9	70.4	24.1%
Total	150.2	186.8	237.4	675.1	23.3%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of CHINA, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 183 CHINA: CLINICAL ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Quality Improvement & Clinical Benchmarking	43.5	63.1	92.6	474.2	38.6%
Clinical Decision Support	29.1	39.6	54.6	214.0	31.4%
Precision Health	17.8	23.9	32.5	119.3	29.7%
Reporting & Compliance	16.2	21.3	28.3	93.1	26.9%
Medical Imaging Analytics	11.6	15.7	21.5	80.8	30.4%
Comparative Analytics/Effectiveness	7.8	10.8	15.1	63.6	33.3%
Total	126.1	174.4	244.6	1,045.1	33.7%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of CHINA, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 184 CHINA: OPERATIONAL & ADMINISTRATIVE ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Supply Chain Analytics	30.2	36.5	45.0	110.0	19.6%
Workforce Analytics	23.2	27.8	34.0	80.0	18.7%
Strategic Analytics	8.6	9.9	11.6	22.1	13.8%
Total	62.0	74.1	90.6	212.0	18.5%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of CHINA, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 185 CHINA: HEALTHCARE ANALYTICS MARKET, BY COMPONENT, 2019–2026 (USD MILLION)

Component	2019	2020	2021	2026	CAGR (2021–2026)
Services	196.0	259.4	349.6	1,282.7	29.7%
Software	155.7	198.6	259.0	836.8	26.4%
Hardware	32.6	36.1	40.8	65.0	9.7%
Total	384.3	494.1	649.4	2,184.5	27.5%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of CHINA, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 186 CHINA: HEALTHCARE ANALYTICS MARKET FOR SERVICES, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Support Services	133.9	176.7	237.3	856.0	29.2%
Business Analytics Services	62.1	82.8	112.3	426.7	30.6%
Total	196.0	259.4	349.6	1,282.7	29.7%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of CHINA, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 187 CHINA: HEALTHCARE ANALYTICS MARKET, BY DEPLOYMENT MODEL, 2019–2026 (USD MILLION)

Deployment Model	2019	2020	2021	2026	CAGR (2021–2026)
On-premise Deployment	290.7	367.1	473.4	1,410.4	24.4%
On-demand Deployment	93.6	127.0	176.1	774.1	34.5%
Total	384.3	494.1	649.4	2,184.5	27.5%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of CHINA, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 188 CHINA: HEALTHCARE ANALYTICS MARKET, BY END USER, 2019–2026 (USD MILLION)

End User	2019	2020	2021	2026	CAGR (2021–2026)
Payers	234.4	299.4	390.9	1,268.4	26.5%
Providers	115.2	150.2	200.2	721.9	29.2%
ACOs, HIEs, MCOs, and TPAs	34.7	44.5	58.3	194.3	27.2%
Total	384.3	494.1	649.4	2,184.5	27.5%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of CHINA, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 189 CHINA: HEALTHCARE ANALYTICS MARKET FOR PAYERS, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Private Insurance Companies	138.9	176.4	228.9	719.7	25.7%
Government Agencies	75.6	97.6	128.8	441.5	27.9%
Employers & Private Exchanges	19.9	25.5	33.2	107.2	26.4%
Total	234.4	299.4	390.9	1,268.4	26.5%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of CHINA, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 190 CHINA: HEALTHCARE ANALYTICS MARKET FOR PROVIDERS, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Hospitals, Physician Practices, and IDNs	92.3	121.5	163.4	611.9	30.2%
Post-acute Care Organizations	14.0	17.9	23.4	76.6	26.8%
Ambulatory Settings	8.9	10.8	13.4	33.4	20.1%
Total	115.2	150.2	200.2	721.9	29.2%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of CHINA, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

11.4.3 INDIA

11.4.3.1 The HCIT market in India is still in the nascent stage

In 2020, India accounted for a share of 13.5% of the Asian healthcare analytics market. The Indian healthcare analytics market is expected to reach USD 824.3 million by 2026 from USD 274.5 million in 2020, at a CAGR of 19.5% during the forecast period.

India spends around 6% of its GDP on healthcare, which is much lower than the other emerging economies. However, the government of India is expected to increase its healthcare expenditure in its 12th five-year plan. This will improve the healthcare infrastructure in the country.

India is expected to witness slow adoption of healthcare analytics solutions as the HCIT market in the country is in the nascent stage. Factors such as lack of proper infrastructure, low healthcare insurance service coverage, lower expenditure on healthcare, and a reluctance to adopt HCIT solutions among hospitals and other healthcare systems are expected to restrain the growth of the healthcare analytics market in India.

To increase the adoption of HCIT solutions in the country, various initiatives are being undertaken by the Indian government, such as the 2015 launch of Digital India. This initiative focuses on e-healthcare, ensuring access to online medical consultations, records, supplies, and even pan-India patient information. In December 2014, the Telangana government announced the launch of India's first dedicated healthcare app for the Ministry of Health & Family Welfare, powered by Mahindra Comviva. Moreover, an Integrated Health Information Platform is being set up by the Ministry of Health and Family Welfare that provides health IT startups with the opportunity to host their innovative solutions for use by different stakeholders. The states and Union Territories (UTs) in the country are being supported to implement hospital information systems at hospitals and health centers to facilitate the creation of EHRs.

The Indian healthcare system has a unique opportunity to use big data as an integral part of its ecosystem; definite steps are being taken to align the technology infrastructure with business objectives and illustrate quick wins. The consolidation of existing data from various entities, such as healthcare research organizations, public and private hospitals, and public and private payers, can be a good starting point for a big data initiative for the Indian healthcare system.

The potential for growth in the Indian market has served to attract significant funding and investments. In February 2018, Gurugram-based analytics start-up THB received INR 14 crore (USD 2.1 million) funding from Blume Venture and HealthQuad.

With the rising geriatric population and increasing prevalence of chronic diseases in the country, the demand for techniques that can be effectively utilized for maintaining health records is increasing. This, in turn, will increase the use of EHRs in the coming few years. As the importance of digitalization is better understood, the adoption of such techniques will increase and simultaneously drive the demand for analytics solutions.

TABLE 191 INDIA: HEALTHCARE ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Descriptive Analytics	138.3	180.4	217.8	470.2	16.6%
Predictive Analytics	34.3	47.8	61.7	190.0	25.2%
Prescriptive Analytics	20.5	28.0	35.4	98.1	22.6%
Cognitive Analytics	13.4	18.4	23.3	65.9	23.1%
Total	206.5	274.5	338.3	824.3	19.5%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of India, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 192 INDIA: HEALTHCARE ANALYTICS MARKET, BY APPLICATION, 2019–2026 (USD MILLION)

Application	2019	2020	2021	2026	CAGR (2021–2026)
Clinical Analytics	83.0	113.9	144.6	397.9	22.4%
Financial Analytics	69.3	90.9	110.5	252.2	17.9%
Operational & Administrative Analytics	31.5	39.5	46.0	84.0	12.8%
Population Health Analytics	22.7	30.2	37.2	90.1	19.4%
Total	206.5	274.5	338.3	824.3	19.5%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of India, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 193 INDIA: FINANCIAL ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Claims Processing	28.8	37.8	45.9	103.1	17.6%
Revenue Cycle Management	19.8	25.4	30.3	61.8	15.4%
Payment Integrity & FWA	13.8	18.6	23.3	61.6	21.4%
Risk Adjustment & Assessment	6.8	9.0	11.0	25.7	18.5%
Total	69.3	90.9	110.5	252.2	17.9%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of India, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 194 INDIA: CLINICAL ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Quality Improvement & Clinical Benchmarking	28.7	40.3	52.4	160.4	25.1%
Clinical Decision Support	18.7	25.4	31.9	83.5	21.2%
Precision Health	12.0	16.2	20.2	51.0	20.4%
Reporting & Compliance	11.3	15.2	18.8	45.5	19.4%
Medical Imaging Analytics	7.7	10.5	13.2	34.5	21.2%
Comparative Analytics/Effectiveness	4.6	6.4	8.1	23.1	23.2%
Total	83.0	113.9	144.6	397.9	22.4%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of India, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 195 INDIA: OPERATIONAL & ADMINISTRATIVE ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Supply Chain Analytics	15.6	19.8	23.2	43.7	13.6%
Workforce Analytics	11.4	14.3	16.7	30.7	12.9%
Strategic Analytics	4.4	5.4	6.1	9.6	9.4%
Total	31.5	39.5	46.0	84.0	12.8%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of India, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 196 INDIA: HEALTHCARE ANALYTICS MARKET, BY COMPONENT, 2019–2026 (USD MILLION)

Component	2019	2020	2021	2026	CAGR (2021–2026)
Services	102.0	138.1	173.0	449.7	21.1%
Software	86.7	114.7	140.8	336.4	19.0%
Hardware	17.8	21.7	24.5	38.2	9.3%
Total	206.5	274.5	338.3	824.3	19.5%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of India, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 197 INDIA: HEALTHCARE ANALYTICS MARKET FOR SERVICES, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Support Services	70.1	94.4	117.8	299.7	20.5%
Business Analytics Services	32.0	43.6	55.2	150.0	22.1%
Total	102.0	138.1	173.0	449.7	21.1%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of India, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 198 INDIA: HEALTHCARE ANALYTICS MARKET, BY DEPLOYMENT MODEL, 2019–2026 (USD MILLION)

Deployment Model	2019	2020	2021	2026	CAGR (2021–2026)
On-premise Deployment	162.6	213.8	260.5	593.8	17.9%
On-demand Deployment	43.9	60.7	77.7	230.5	24.3%
Total	206.5	274.5	338.3	824.3	19.5%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of India, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 199 INDIA: HEALTHCARE ANALYTICS MARKET, BY END USER, 2019–2026 (USD MILLION)

End User	2019	2020	2021	2026	CAGR (2021–2026)
Payers	119.1	156.5	190.4	433.0	17.9%
Providers	67.3	91.4	115.1	312.8	22.1%
ACOs, HIEs, MCOs, and TPAs	20.1	26.6	32.7	78.5	19.1%
Total	206.5	274.5	338.3	824.3	19.5%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of India, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 200 INDIA: HEALTHCARE ANALYTICS MARKET FOR PAYERS, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Private Insurance Companies	71.1	92.5	111.4	239.6	16.6%
Government Agencies	37.3	49.9	61.9	154.6	20.1%
Employers & Private Exchanges	10.7	14.1	17.1	38.7	17.7%
Total	119.1	156.5	190.4	433.0	17.9%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of India, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 201 INDIA: HEALTHCARE ANALYTICS MARKET FOR PROVIDERS, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Hospitals, Physician Practices, and IDNs	55.5	76.1	96.7	271.9	23.0%
Post-acute Care Organizations	7.8	10.3	12.7	31.3	19.7%
Ambulatory Settings	4.0	4.9	5.7	9.6	11.1%
Total	67.3	91.4	115.1	312.8	22.1%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of India, Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

11.4.4 REST OF ASIA

In 2020, the RoA accounted for 22.7% of the Asian healthcare analytics market. This regional market is expected to reach USD 1,692.2 million by 2026 from USD 461.6 million in 2020, at a CAGR of 23.5% during the forecast period.

The RoA comprises Malaysia, Taiwan, Singapore, Thailand, Indonesia, South Korea, and other Asian countries. The increasing demand for quality healthcare and the rising prevalence of chronic diseases are the key factors driving the growth of the healthcare analytics market in this region.

With the increasing uptake of wearable technology and biosensors, there has been significant growth in the volume of healthcare data generated daily. Various providers in Singapore are focusing on making personalized healthcare data understandable and actionable through cutting-edge technology. Moreover, VC investments in Singapore's healthcare industry have increased. In February 2017, CXA Group received USD 25 million in a Series B investment, co-led by B Capital Group and Singapore-based EDBI. Philips Healthcare and the existing investors, NSI Ventures and Bioveda Capital, also contributed to this investment. This allowed the company to focus on expanding its geographic reach and provide its data analytics tools in markets such as China, India, Japan, South Korea, Taiwan, Indonesia, Malaysia, the Philippines, and Thailand. In August 2017, Biofourmis (Singapore), a data analytics firm, received USD 1 million in angel funding, led by SpesNet and Eden Strategy. Malaysia has also seen such investments—for instance, in May 2017, Fractal Analytics (Malaysia), a data analytics company, raised USD 100 million from

the Malaysian sovereign wealth fund—Khazanah—in one of the largest investments in the technology ecosystem in 2017.

There is a growing demand for risk, fraud, and customer intelligence analytics in these countries, resulting in many companies coming up with innovative solutions to enhance their businesses. In March 2017, Fullerton Health teamed up with KenSci to use predictive analytics to identify and manage fraudulent claims.

TABLE 202 ROA: HEALTHCARE ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Descriptive Analytics	228.7	305.6	382.4	988.8	20.9%
Predictive Analytics	55.3	78.5	104.5	374.6	29.1%
Prescriptive Analytics	33.3	46.2	60.0	190.6	26.0%
Cognitive Analytics	22.4	31.4	41.2	138.2	27.4%
Total	339.6	461.6	588.0	1,692.2	23.5%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 203 ROA: HEALTHCARE ANALYTICS MARKET, BY APPLICATION, 2019–2026 (USD MILLION)

Application	2019	2020	2021	2026	CAGR (2021–2026)
Clinical Analytics	110.1	158.8	213.5	756.3	28.8%
Financial Analytics	141.0	186.5	231.2	580.1	20.2%
Operational & Administrative Analytics	50.9	65.1	77.9	164.6	16.1%
Population Health Analytics	37.6	51.2	65.5	191.2	23.9%
Total	339.6	461.6	588.0	1,692.2	23.5%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 204 ROA: FINANCIAL ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Claims Processing	57.8	76.5	94.8	235.2	19.9%
Revenue Cycle Management	41.3	53.3	64.6	144.3	17.4%
Payment Integrity & FWA	27.8	37.9	48.3	140.6	23.8%
Risk Adjustment & Assessment	14.2	18.8	23.4	60.0	20.7%
Total	141.0	186.5	231.2	580.1	20.2%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 205 ROA: CLINICAL ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Quality Improvement & Clinical Benchmarking	37.9	56.2	77.6	308.7	31.8%
Clinical Decision Support	25.0	35.7	47.6	161.2	27.6%
Precision Health	15.5	22.1	29.3	97.2	27.1%
Reporting & Compliance	14.9	20.6	26.7	77.9	23.9%
Medical Imaging Analytics	10.6	15.1	20.0	66.8	27.2%
Comparative Analytics/Effectiveness	6.2	9.1	12.2	44.4	29.4%
Total	110.1	158.8	213.5	756.3	28.8%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 206 ROA: OPERATIONAL & ADMINISTRATIVE ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Supply Chain Analytics	25.4	32.7	39.4	85.8	16.8%
Workforce Analytics	18.4	23.6	28.4	60.8	16.5%
Strategic Analytics	7.1	8.8	10.2	17.9	12.1%
Total	50.9	65.1	77.9	164.6	16.1%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 207 ROA: HEALTHCARE ANALYTICS MARKET, BY COMPONENT, 2019–2026 (USD MILLION)

Component	2019	2020	2021	2026	CAGR (2021–2026)
Services	167.2	233.6	305.0	962.4	25.8%
Software	143.7	193.4	243.9	667.5	22.3%
Hardware	28.7	34.6	39.2	62.2	9.7%
Total	339.6	461.6	588.0	1,692.2	23.5%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 208 ROA: HEALTHCARE ANALYTICS MARKET FOR SERVICES, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Support Services	114.6	159.3	206.9	635.6	25.2%
Business Analytics Services	52.6	74.3	98.0	326.8	27.2%
Total	167.2	233.6	305.0	962.4	25.8%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 209 ROA: HEALTHCARE ANALYTICS MARKET, BY DEPLOYMENT MODEL, 2019–2026 (USD MILLION)

Deployment Model	2019	2020	2021	2026	CAGR (2021–2026)
On-premise Deployment	268.1	361.0	455.4	1,238.7	22.2%
On-demand Deployment	71.5	100.6	132.7	453.5	27.9%
Total	339.6	461.6	588.0	1,692.2	23.5%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 210 ROA: HEALTHCARE ANALYTICS MARKET, BY END USER, 2019–2026 (USD MILLION)

End User	2019	2020	2021	2026	CAGR (2021–2026)
Payers	196.6	264.5	333.5	906.6	22.1%
Providers	110.2	152.8	198.5	630.8	26.0%
ACOs, HIEs, MCOs, and TPAs	32.9	44.3	56.0	154.8	22.6%
Total	339.6	461.6	588.0	1,692.2	23.5%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 211 ROA: HEALTHCARE ANALYTICS MARKET FOR PAYERS, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Private Insurance Companies	118.0	157.9	198.0	522.5	21.4%
Government Agencies	60.9	82.8	105.5	303.0	23.5%
Employers & Private Exchanges	17.7	23.8	30.0	81.1	22.0%
Total	196.6	264.5	333.5	906.6	22.1%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 212 ROA: HEALTHCARE ANALYTICS MARKET FOR PROVIDERS, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Hospitals, Physician Practices, and IDNs	91.0	127.3	166.8	548.1	26.9%
Post-acute Care Organizations	12.8	17.4	22.2	63.7	23.5%
Ambulatory Settings	6.4	8.1	9.6	19.0	14.7%
Total	110.2	152.8	198.5	630.8	26.0%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

11.5 PACIFIC COUNTRIES

11.5.1 NEED TO REDUCE HEALTHCARE COSTS IN AUSTRALIA AND NEW ZEALAND TO DRIVE MARKET GROWTH IN THIS REGION

The Pacific countries comprise Australia and New Zealand. This segment accounted for 1.6% of the global healthcare analytics market in 2020. The healthcare analytics market in the Pacific countries is expected to reach USD 787.9 million by 2026 from USD 254.2 million in 2020, at a CAGR of 20.1% during the forecast period.

Australia and New Zealand are actively implementing HCIT solutions, especially EHR solutions, in their healthcare systems. In addition, eHealth initiatives taken by governments in this region have also helped boost market growth. The Australian government has been undertaking several initiatives to increase the adoption of IT in healthcare to reduce errors and increase efficiency. According to the Commonwealth Fund International Health Policy Survey of Primary Care Physicians, in 2017, 92% of primary care physicians in Australia and 100% of primary care physicians in New Zealand had EMR systems, which further laid the foundation for the adoption of complementary HCIT solutions such as clinical decision support systems and medical document management systems in these countries.

Australia and New Zealand are lucrative markets for healthcare analytics. These countries have an up-to-date HCIT infrastructure and are focusing on adopting analytics solutions to reduce healthcare costs. The growing incidence of chronic diseases is one of the key factors driving the growth of the healthcare analytics market in Australia. Over seven million people in Australia are estimated to suffer from a chronic condition; this is expected to lead to a growing demand for population health analytics solutions.

TABLE 213 PACIFIC COUNTRIES: HEALTHCARE ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Descriptive Analytics	112.6	159.1	191.2	391.8	15.4%
Predictive Analytics	34.0	52.4	68.8	231.5	27.5%
Prescriptive Analytics	16.5	24.9	32.1	98.3	25.1%
Cognitive Analytics	11.9	17.8	22.7	66.2	23.9%
Total	174.9	254.2	314.9	787.9	20.1%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 214 PACIFIC COUNTRIES: HEALTHCARE ANALYTICS MARKET, BY APPLICATION, 2019–2026 (USD MILLION)

Application	2019	2020	2021	2026	CAGR (2021–2026)
Financial Analytics	70.6	99.6	119.9	258.8	16.6%
Clinical Analytics	61.4	94.3	122.9	373.6	24.9%
Operational & Administrative Analytics	25.3	34.6	40.2	73.4	12.8%
Population Health Analytics	17.6	25.7	32.0	82.0	20.7%
Total	174.9	254.2	314.9	787.9	20.1%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 215 PACIFIC COUNTRIES: FINANCIAL ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Claims Processing	44.4	62.8	75.7	164.3	16.8%
Revenue Cycle Management	12.0	16.6	19.6	38.1	14.2%
Payment Integrity & FWA	8.8	12.9	16.0	41.0	20.7%
Risk Adjustment & Assessment	5.4	7.4	8.6	15.5	12.6%
Total	70.6	99.6	119.9	258.8	16.6%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 216 PACIFIC COUNTRIES: CLINICAL ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Quality Improvement & Clinical Benchmarking	23.9	38.0	51.2	178.8	28.4%
Clinical Decision Support	13.3	20.2	25.9	73.0	23.0%
Precision Health	9.1	13.7	17.4	46.5	21.8%
Medical Imaging Analytics	6.0	9.1	11.8	33.5	23.3%
Reporting & Compliance	5.7	8.3	10.3	24.2	18.7%
Comparative Analytics/Effectiveness	3.3	5.0	6.4	17.5	22.4%
Total	61.4	94.3	122.9	373.6	24.9%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 217 PACIFIC COUNTRIES: OPERATIONAL & ADMINISTRATIVE ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Supply Chain Analytics	11.4	15.3	17.7	30.5	11.5%
Workforce Analytics	10.2	14.1	16.6	32.2	14.2%
Strategic Analytics	3.8	5.1	5.9	10.8	12.7%
Total	25.3	34.6	40.2	73.4	12.8%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 218 PACIFIC COUNTRIES: HEALTHCARE ANALYTICS MARKET, BY COMPONENT, 2019–2026 (USD MILLION)

Component	2019	2020	2021	2026	CAGR (2021–2026)
Services	87.3	129.1	162.6	432.4	21.6%
Software	75.2	108.8	134.2	329.2	19.7%
Hardware	12.5	16.3	18.1	26.3	7.8%
Total	174.9	254.2	314.9	787.9	20.1%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 219 PACIFIC COUNTRIES: HEALTHCARE ANALYTICS MARKET FOR SERVICES, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Support Services	61.5	90.5	113.4	294.3	21.0%
Business Analytics Services	25.8	38.6	49.1	138.1	22.9%
Total	87.3	129.1	162.6	432.4	21.6%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 220 PACIFIC COUNTRIES: HEALTHCARE ANALYTICS MARKET, BY DEPLOYMENT MODEL, 2019–2026 (USD MILLION)

Deployment Model	2019	2020	2021	2026	CAGR (2021–2026)
On-premise Deployment	137.0	195.2	236.8	515.1	16.8%
On-demand Deployment	38.0	59.0	78.1	272.8	28.4%
Total	174.9	254.2	314.9	787.9	20.1%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 221 PACIFIC COUNTRIES: HEALTHCARE ANALYTICS MARKET, BY END USER, 2019–2026 (USD MILLION)

End User	2019	2020	2021	2026	CAGR (2021–2026)
Payers	116.4	166.2	202.3	455.2	17.6%
Providers	46.8	70.7	91.0	277.1	24.9%
ACOs, HIEs, MCOs, and TPAs	11.8	17.3	21.5	55.5	20.9%
Total	174.9	254.2	314.9	787.9	20.1%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 222 PACIFIC COUNTRIES: HEALTHCARE ANALYTICS MARKET FOR PAYERS, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Private Insurance Companies	76.3	107.7	129.4	270.5	15.9%
Government Agencies	31.7	46.5	58.3	151.3	21.0%
Employers & Private Exchanges	8.4	12.0	14.6	33.4	18.0%
Total	116.4	166.2	202.3	455.2	17.6%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 223 PACIFIC COUNTRIES: HEALTHCARE ANALYTICS MARKET FOR PROVIDERS, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Hospitals, Physician Practices, and IDNs	39.4	60.0	77.8	244.1	25.7%
Post-acute Care Organizations	5.0	7.3	9.3	25.2	22.2%
Ambulatory Settings	2.4	3.3	3.9	7.8	14.8%
Total	46.8	70.7	91.0	277.1	24.9%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

11.6 LATIN AMERICA

11.6.1 GROWING EHEALTH INITIATIVES IN LATAM COUNTRIES TO DRIVE MARKET GROWTH IN THIS REGION

In 2020, LATAM accounted for a share of 1.9% of the global healthcare analytics market. This regional market is expected to reach USD 1,049.4 million by 2026 from USD 301.8 million in 2020, at a CAGR of 22.5% during the forecast period.

The growth of the LATAM healthcare analytics market can be attributed to the implementation of favorable government initiatives in several countries in the region. Many countries in LATAM have undertaken eHealth initiatives since 2012, including Mexico, Brazil, Chile, Peru, Colombia, and Argentina that have made significant efforts to standardize the use of EHRs. These countries are exploring alternative ways to manage clinical information within their healthcare systems effectively. Moreover, the rising need to curtail soaring healthcare costs is also contributing to the growth of this market.

Additionally, in February 2014, the HIMSS and the Association of Brazilian Healthcare CIOs (ABCIS) organized the first annual HIMSS Latin American 2014 Conference to introduce the Electronic Medical Record Adoption Model (EMRAM). The model focuses on enhancing clinical and financial health outcomes by using EMRs to provide quality patient care, introduce mobile health for clinicians and patients, introduce applications of predictive analytics for big data in healthcare, and discuss interoperability outlines. Such conferences play an important role in increasing awareness about HCIT and healthcare analytics solutions.

TABLE 224 LATAM: HEALTHCARE ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Descriptive Analytics	146.4	180.5	220.9	496.0	17.6%
Predictive Analytics	52.6	70.6	94.4	344.4	29.6%
Prescriptive Analytics	22.0	28.7	37.2	117.0	25.8%
Cognitive Analytics	16.8	22.0	28.7	92.0	26.2%
Total	237.8	301.8	381.1	1,049.4	22.5%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 225 LATAM: HEALTHCARE ANALYTICS MARKET, BY APPLICATION, 2019–2026 (USD MILLION)

Application	2019	2020	2021	2026	CAGR (2021–2026)
Financial Analytics	93.4	114.7	140.1	326.0	18.4%
Clinical Analytics	75.2	102.3	137.4	476.9	28.3%
Operational & Administrative Analytics	37.1	44.1	52.1	103.0	14.6%
Population Health Analytics	32.0	40.7	51.6	143.4	22.7%
Total	237.8	301.8	381.1	1,049.4	22.5%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 226 LATAM: FINANCIAL ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Claims Processing	37.5	46.5	57.2	135.7	18.9%
Revenue Cycle Management	28.6	34.0	40.3	80.1	14.7%
Payment Integrity & FWA	18.1	22.9	29.0	80.2	22.6%
Risk Adjustment & Assessment	9.2	11.2	13.6	30.1	17.2%
Total	93.4	114.7	140.1	326.0	18.4%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 227 LATAM: CLINICAL ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Quality Improvement & Clinical Benchmarking	24.7	36.0	51.5	225.3	34.3%
Clinical Decision Support	18.5	23.6	29.8	76.0	20.6%
Precision Health	10.7	14.1	18.4	55.5	24.7%
Reporting & Compliance	9.9	13.1	17.1	50.9	24.4%
Medical Imaging Analytics	7.6	10.1	13.4	43.1	26.3%
Comparative Analytics/Effectiveness	3.9	5.3	7.2	26.2	29.4%
Total	75.2	102.3	137.4	476.9	28.3%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 228 LATAM: OPERATIONAL & ADMINISTRATIVE ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Supply Chain Analytics	17.2	20.3	23.8	45.2	13.7%
Workforce Analytics	14.2	17.1	20.4	43.1	16.1%
Strategic Analytics	5.7	6.7	7.8	14.7	13.5%
Total	37.1	44.1	52.1	103.0	14.6%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 229 LATAM: HEALTHCARE ANALYTICS MARKET, BY COMPONENT, 2019–2026 (USD MILLION)

Component	2019	2020	2021	2026	CAGR (2021–2026)
Services	128.0	165.7	212.8	621.5	23.9%
Software	88.5	112.3	141.6	388.0	22.3%
Hardware	21.2	23.8	26.6	39.8	8.4%
Total	237.8	301.8	381.1	1,049.4	22.5%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 230 LATAM: HEALTHCARE ANALYTICS MARKET FOR SERVICES, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Support Services	86.2	111.0	141.8	402.5	23.2%
Business Analytics Services	41.8	54.7	71.0	219.0	25.3%
Total	128.0	165.7	212.8	621.5	23.9%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 231 LATAM: HEALTHCARE ANALYTICS MARKET, BY DEPLOYMENT MODEL, 2019–2026 (USD MILLION)

Deployment Model	2019	2020	2021	2026	CAGR (2021–2026)
On-premise Deployment	151.6	185.9	226.0	477.8	16.2%
On-demand Deployment	86.1	115.9	155.1	571.6	29.8%
Total	237.8	301.8	381.1	1,049.4	22.5%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 232 LATAM: HEALTHCARE ANALYTICS MARKET, BY END USER, 2019–2026 (USD MILLION)

End User	2019	2020	2021	2026	CAGR (2021–2026)
Payers	123.3	152.1	186.4	426.8	18.0%
Providers	89.8	117.7	153.5	497.3	26.5%
ACOs, HIEs, MCOs, and TPAs	24.7	32.0	41.2	125.3	24.9%
Total	237.8	301.8	381.1	1,049.4	22.5%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 233 LATAM: HEALTHCARE ANALYTICS MARKET FOR PAYERS, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Private Insurance Companies	71.7	86.4	103.2	202.6	14.4%
Government Agencies	40.3	51.7	66.0	184.7	22.9%
Employers & Private Exchanges	11.3	14.0	17.2	39.5	18.1%
Total	123.3	152.1	186.4	426.8	18.0%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 234 LATAM: HEALTHCARE ANALYTICS MARKET FOR PROVIDERS, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Hospitals, Physician Practices, and IDNs	71.9	95.5	125.9	426.4	27.6%
Post-acute Care Organizations	11.6	14.7	18.6	51.8	22.7%
Ambulatory Settings	6.3	7.5	9.0	19.0	16.1%
Total	89.8	117.7	153.5	497.3	26.5%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

11.7 MIDDLE EAST & AFRICA

11.7.1 HIGH HEALTHCARE EXPENDITURE AND INFRASTRUCTURAL DEVELOPMENT WILL SUPPORT THE USE OF HCIT SOLUTIONS IN THE MIDDLE EAST

In 2020, the Middle East & Africa accounted for 1.2% of the global healthcare analytics market. This regional segment is expected to reach USD 546.7 million by 2026 from USD 190.6 million in 2021, at a CAGR of 18.6% during the forecast period.

In 2019, healthcare providers in the Middle East and North Africa spent around USD 5.89 billion on HCIT products and services. In 2011, Middle Eastern countries invested an estimated USD 26 billion in healthcare projects; this is estimated to increase to USD 60 billion by 2025 owing to the addition of 120–140 hospitals to the healthcare infrastructure of these countries by 2020 (Source: Ministry of Health and HIMSS). Saudi Arabia is the top spender on healthcare in the Middle East, followed by Egypt, the UAE, Kuwait, and Qatar. The high healthcare expenditure and infrastructural development will prompt the implementation of HCIT solutions to manage costs and handle the growing volume of data.

Middle Eastern Countries are experiencing issues regarding medical insurance fraud and abuse. According to industry experts, an estimated USD 1 billion (ADE 3.67 billion) is lost to identify fraud and over-prescribing. Certain initiatives are being taken to counter such fraudulent activities. For instance, in

September 2018, Netcetera, a Swiss software company, conducted a trial of its AI-powered fraud analytics within a major healthcare insurer in the Middle East. Approximately 37,000 bogus claims made by 4,000 doctors within 2016–2017—totaling Dh21 million (USD 5.72 million)—were identified. These would usually be ignored by insurers, who would end up paying for unnecessary treatments.

Market growth will be aided by a favorable investment scenario, as the potential for growth in the Middle East has attracted significant attention from more developed countries. For instance, in October 2018, DiA Imaging Analysis Ltd (Israel) received USD 5 million from Connecticut Innovations along with Defta Healthcare, Israel's Innovation Authority, and existing investors. This funding will be utilized for developing medical imaging analytics software based on pattern recognition and machine learning algorithms.

In June 2018, Zebra Medical Vision (Israel) raised USD 30 million in a Series C funding round led by aMoon Ventures and healthcare investors Aurum, Johnson & Johnson Innovation JJDC Inc., and Intermountain Healthcare, as well as global AI experts Fei Fei Lee and Richard Socher. This will be utilized to accelerate the development of Zebra's Chest X-ray AI analytics solution. In addition, in May 2016, Zebra received USD 12 million from Intermountain Healthcare and Khosla Ventures to accelerate the development of its AI-driven imaging analytics solutions.

Healthcare fraud is a pervasive and growing problem in Africa, costing high amounts, raising insurance premiums, and depriving patients of the required affordable and quality medical care. This is a significant issue, as African countries do not have strong financial and medical accounting systems. Healthcare fraud is one of the leading crimes in South Africa in terms of monetary value. It is also one of the most complex forms of fraud to detect and prevent, as it tends to occur at all levels of the healthcare delivery chain. An estimated 15–20% of healthcare spending in Africa is lost through fraud annually (Source: National Hospital Insurance Fund). To counter this, initiatives are being taken by insurers. For instance, in September 2015, Kenya National Hospital Insurance Fund (NHIF) implemented a biometric health smart card system for all users of its health facilities. Similar initiatives are expected to support market growth in the coming years.

TABLE 235 MIDDLE EAST & AFRICA: HEALTHCARE ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Descriptive Analytics	90.6	122.8	147.2	306.6	15.8%
Predictive Analytics	24.9	35.8	45.8	133.8	23.9%
Prescriptive Analytics	12.6	17.7	22.1	57.3	21.0%
Cognitive Analytics	10.1	14.3	18.0	49.0	22.1%
Total	138.2	190.6	233.1	546.7	18.6%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 236 MIDDLE EAST & AFRICA: HEALTHCARE ANALYTICS MARKET, BY APPLICATION, 2019–2026 (USD MILLION)

Application	2019	2020	2021	2026	CAGR (2021–2026)
Financial Analytics	51.5	69.6	83.4	176.8	16.2%
Clinical Analytics	42.8	62.0	79.2	222.4	22.9%
Operational & Administrative Analytics	23.6	30.9	35.9	65.1	12.7%
Population Health Analytics	20.4	28.2	34.6	82.4	18.9%
Total	138.2	190.6	233.1	546.7	18.6%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 237 MIDDLE EAST & AFRICA: FINANCIAL ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Claims Processing	21.3	29.0	35.0	76.3	16.8%
Revenue Cycle Management	15.3	20.3	23.8	45.6	13.9%
Payment Integrity & FWA	9.9	13.7	16.9	40.9	19.4%
Risk Adjustment & Assessment	5.0	6.6	7.7	14.0	12.7%
Total	51.5	69.6	83.4	176.8	16.2%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 238 MIDDLE EAST & AFRICA: CLINICAL ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Quality Improvement & Clinical Benchmarking	13.3	20.2	26.9	89.1	27.1%
Clinical Decision Support	10.5	14.6	18.0	41.6	18.3%
Precision Health	6.4	9.2	11.6	30.1	21.1%
Reporting & Compliance	5.4	7.8	9.9	26.5	21.8%
Medical Imaging Analytics	4.3	6.2	7.8	20.9	21.7%
Comparative Analytics/Effectiveness	2.7	4.0	5.1	14.2	22.8%
Total	42.8	62.0	79.2	222.4	22.9%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 239 MIDDLE EAST & AFRICA: OPERATIONAL & ADMINISTRATIVE ANALYTICS MARKET, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Supply Chain Analytics	10.9	14.2	16.3	27.9	11.4%
Workforce Analytics	9.0	12.0	14.1	27.2	14.1%
Strategic Analytics	3.6	4.8	5.5	10.0	12.5%
Total	23.6	30.9	35.9	65.1	12.7%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 240 MIDDLE EAST & AFRICA: HEALTHCARE ANALYTICS MARKET, BY COMPONENT, 2019–2026 (USD MILLION)

Component	2019	2020	2021	2026	CAGR (2021–2026)
Services	74.5	104.3	129.2	319.1	19.8%
Software	51.4	70.7	86.3	200.4	18.3%
Hardware	12.3	15.6	17.6	27.2	9.1%
Total	138.2	190.6	233.1	546.7	18.6%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 241 MIDDLE EAST & AFRICA: HEALTHCARE ANALYTICS MARKET FOR SERVICES, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Support Services	50.1	69.7	85.7	203.3	18.9%
Business Analytics Services	24.4	34.6	43.5	115.8	21.6%
Total	74.5	104.3	129.2	319.1	19.8%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 242 MIDDLE EAST & AFRICA: HEALTHCARE ANALYTICS MARKET, BY DEPLOYMENT MODEL, 2019–2026 (USD MILLION)

Deployment Model	2019	2020	2021	2026	CAGR (2021–2026)
On-premise Deployment	88.6	120.1	144.1	302.3	16.0%
On-demand Deployment	49.5	70.5	89.0	244.4	22.4%
Total	138.2	190.6	233.1	546.7	18.6%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 243 MIDDLE EAST & AFRICA: HEALTHCARE ANALYTICS MARKET, BY END USER, 2019–2026 (USD MILLION)

End User	2019	2020	2021	2026	CAGR (2021–2026)
Payers	72.9	99.3	119.9	261.3	16.9%
Providers	51.3	72.2	90.1	233.3	21.0%
ACOs, HIEs, MCOs, and TPAs	13.9	19.1	23.1	52.1	17.6%
Total	138.2	190.6	233.1	546.7	18.6%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 244 MIDDLE EAST & AFRICA: HEALTHCARE ANALYTICS MARKET FOR PAYERS, BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Private Insurance Companies	43.0	58.0	69.2	141.6	15.4%
Government Agencies	23.2	32.2	39.6	95.3	19.2%
Employers & Private Exchanges	6.7	9.1	11.1	24.3	17.1%
Total	72.9	99.3	119.9	261.3	16.9%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 245 MIDDLE EAST & AFRICA: HEALTHCARE ANALYTICS MARKET FOR PROVIDERS,
BY TYPE, 2019–2026 (USD MILLION)

Type	2019	2020	2021	2026	CAGR (2021–2026)
Hospitals, Physician Practices, and IDNs	40.9	58.1	73.0	195.2	21.7%
Post-acute Care Organizations	6.8	9.5	11.7	28.8	19.7%
Ambulatory Settings	3.6	4.7	5.4	9.3	11.5%
Total	51.3	72.2	90.1	233.3	21.0%

Source: Health Information and Management Systems Society (HIMSS), International Medical Informatics Association (IMIA), American Health Information Management Association (AHIMA), National Health Care Anti-fraud Association (NHCAA), American Medical Informatics Association (AMIA), Certification Commission for Health Information Technology (CCHIT), Healthcare Information Technology Standards Panel (HITSP), Health Level Seven International (HL7), Information Technology Association of Canada (ITAC), Canadian Anti-fraud Centre, European Healthcare Fraud & Corruption Network (EHFCN), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

12 COMPETITIVE LANDSCAPE

12.1 INTRODUCTION

The competitive landscape includes an analysis of the key growth strategies adopted by players between January 2018 and August 2021 to expand their global presence and increase their shares in the healthcare analytics market. The key growth strategies adopted by the top players (profiled in the report) include partnerships, collaborations, agreements, and alliances; product launches, service launches, product upgrades, and product enhancements; expansions; and acquisitions.

The healthcare analytics market is highly fragmented, with several big as well as emerging players competing in the market. The prominent players in this market include IBM, Optum, Cerner, SAS, Allscripts, McKesson, MedeAnalytics, Inovalon, Oracle, Health Catalyst, SCIO Health Analytics, Cotiviti (formerly Verscend Technologies), CitiusTech, Wipro, and VitreosHealth. Some of the other market players include Infosys Limited, PluralSoft, Informatica, Microsoft, iVantage Health Analytics, SAP SE, Philips Healthcare, LexisNexis Risk Solutions, Information Builders, EMC Corporation, Precision Software, Predixion Software, MedAssets, Caradigm, Emdeon, Accenture, Wellcentive, Atena/Active Health, Availity, Lightbeam, Xerox, Interpreta, Inc., CPSI, Premium, Inc., FICO, Predilytics, Saama Technologies, Tableau Software, Avanade, Kronos, hCentive, Rapid Insight, Vantage Point Healthcare Information Systems, and Vital Spring Technologies.

12.2 KEY PLAYER STRATEGIES/RIGHT TO WIN

- The healthcare analytics market is fragmented, with consolidation expected to occur at a faster pace in the forecast period.
- High-level strategic choices made by players in this market include product/service launches, partnerships, agreements, collaborations, acquisitions, and expansions. Companies in this market are focused on product and service launches for offering technology-powered solutions that utilize artificial intelligence, data analytics, and wearable technologies.
- Market players are also moving toward offering more comprehensive solutions to end-user organizations through strategic partnerships, agreements, collaborations, and acquisitions.
- The top players in this market have stayed relevant by offering suitable services and products for organizations during the COVID-19 pandemic, where much of the executive workforce has been engaged in remote work. This has resulted in the rising adoption of healthcare analytics solutions and services due to limited in-person interactions.
- The healthcare analytics market is well-established in North America and most countries in Europe, with a growing focus of market players on the emerging markets across the Asia Pacific and Latin America.

12.2.1 OVERVIEW OF STRATEGIES ADOPTED BY PLAYERS IN THE HEALTHCARE ANALYTICS MARKET

KEY PLAYERS	KEY PRODUCTS/SERVICES OFFERED	KEY PRODUCT/SERVICE LAUNCHES/APPROVALS	MAJOR DEALS/DEVELOPMENTS	REGIONS
	- IBM Watson Health - Population Health Management	- Watson Data Kits - Analytics System with Machine Learning	Opened four new IBM Cloud data centers—two in London, one in California, and one in Sydney	Predominantly serves North America, with market presence in Europe, the Asia Pacific, Latin America, the Middle East, and Africa
	- SAS Viya - SAS Enterprise Miner - SAS Visual Data Mining and Machine Learning	SAS Viya	- Entered into an agreement with Prime Therapeutics - Partnered with Teradata	Operates in countries across North America, Europe, the Asia Pacific, Latin America, the Middle East, and Africa
	- LifeWorks Wellbeing Platform - AbilitiCBT	Launched AvaFinance, an automated personal assistant, in Australia	- Partnered with Availity - Partnered with Quest Diagnostics	North America accounted for more than 90% of the company's revenue
	HealthIntent		- Collaborated with The Unified Government of Wyandotte County and Kansas City - Collaborated with XG Health Solutions	15 global offices and service centers across North America, Europe, the Asia Pacific, Latin America, the Middle East, and Africa
	- VP Transform - Virgin Pulse Navigate - VP Passport	Established Veradigm, a new division of the Allscripts Payer and Life Science segment	- Acquired Care Port - Partnered with Liverpool Heart and Chest Hospital NHS Foundation Trust - Partnered with Optum Care	Has a business presence in North America, Europe, the Asia Pacific, and Latin America

Source: Company Website and Annual Reports

In the following write-ups, the strategies of various healthcare analytics market players have been assessed, along with the choices they made:

- **Product Portfolio Focus:** The product portfolios of major healthcare analytics market players are comprehensive and cover a wide range of programs and services, including health risk assessment, population health management, patient risk assessment, report writing, clinical analytics, claims, and financial analytics, and other employee assistance programs. Smaller players in the market may be limited in their offerings to certain specific products or segments.
- **Regional Focus:** Market players are mainly located in developed countries across North America and Europe. However, there is a trend of expansion toward emerging markets. The regional focus has been assessed based on where companies source their revenues and the location of their major client base. However, as the market is fragmented, many local players are starting to offer these services and solutions in developing countries such as India, China, and Brazil.

- **Organic/Inorganic Growth Strategies:** Players in the healthcare analytics market have been investing in both organic strategies (such as product/service launches) and inorganic growth strategies (such as partnerships, collaborations, agreements, and acquisitions) to maintain their market share.

Companies such as IBM, Cerner, Allscripts, and Optum focus on the North American market. In general, the companies in this market are all strong in their home region (mostly North America and Europe) and are currently exploring geographic diversification alternatives to grow their businesses.

12.3 MARKET RANKING ANALYSIS

As of 2020, this market is dominated by IBM Corporation (US), SAS Institute (US), Optum (a part of United Health Group, US), Allscripts (US), and Cerner Corporation (US).

The above-mentioned leading market players are based in the US, but the market also shows the presence of numerous small players operating at the global as well as local levels. These small and medium-sized players operating in both developed and emerging regions are making this a highly competitive market. However, dominant market players have numerous advantages, such as stronger marketing & distribution networks, larger budgets for R&D, and better brand recognition.

Companies across the market are increasing their focus on R&D to add innovative products to their portfolios, which helps them to meet consumer demands and strengthen their market presence. The lucrative growth opportunities offered by Pacific and Middle Eastern countries are also encouraging market players to strengthen their presence in these regions.

FIGURE 32 IBM WAS THE LEADING PLAYER IN THE HEALTHCARE ANALYTICS MARKET IN 2020



Note: The market ranking is based on the product portfolio and geographic presence of companies, as well as insights from demand- and supply-side primaries.

Source: MarketsandMarkets Analysis

IBM is the leading player in the healthcare analytics market. The company's strong position in this market can be attributed to its robust healthcare analytics product portfolio, including IBM Content, Predictive Analytics for Healthcare, IBM Cognos, and IBM Smarter Analytics Signature. The company also offers quality services such as business analytics, optimization services, and analytics consulting services. IBM has a strong presence in the commercial and government healthcare payer markets. It also has a robust sales and distribution network across more than 175 countries. IBM invests a significant amount of its revenue in research activities to continuously improve its product portfolio. Since 2005, the company has invested USD 24 billion in the development of its big data and analytics software and service capabilities and has 6,809 patents to its name. One of the major steps taken by IBM to improve its position in the market is the acquisition of Truven Health Analytics in April 2016. This acquisition helped the company to acquire a major share of the healthcare analytics market.

SAS Institute held the second position in the healthcare analytics market in 2020. The company is known for its innovative analytics, business intelligence, and data management software and services. With a wide product portfolio and large geographic presence, SAS Institute has managed to create a unique identity in the healthcare analytics market. The company offers a variety of analytics solutions for healthcare payers and providers, such as Advanced Analytics, SAS Visual Analytics, SAS Claims Analytics, Data Management Software, SAS Fraud Framework for Health Care, and SAS Episode Analytics. The company's presence in around 148 countries plays an important role in increasing its market share. SAS focuses on organic and inorganic strategies such as collaborations, partnerships, and agreements to enhance its market presence.

Optum held the third position in the healthcare analytics market. With around 40 years of experience, the company offers high-quality solutions for analytics in the healthcare industry. Its products include Optum One, Optum Risk Reporting, population management tools, RCM solutions for hospitals, RCM solutions for physicians, impact intelligence, and health management and population analytics. The company offers these products to healthcare payers, care providers, employers, and government agencies across 150 locations worldwide. The company has a strong presence in North America, South America, Europe, the Asia Pacific, and the Middle East. Optum focuses on agreements, partnerships, and collaborations to maintain and grow its share in the healthcare analytics market. For instance, in November 2019, the company collaborated with the Advisory Board Company to combine the latter's research and advisory services with its own data analytics and technology capabilities.

12.4 COMPANY EVALUATION MATRIX

12.4.1 COMPANY EVALUATION MATRIX: HEALTHCARE ANALYTICS MARKET

The competitive leadership mapping is based on parameters such as the strength of the product portfolio and business strategy excellence. These parameters have a significant impact on the growth and development of companies in the healthcare analytics market. In this evaluation, vendors have been placed into four categories—Stars, Pervasive Players, Emerging Leaders, and Participants—based on their performance in each criterion.

FIGURE 33 GLOBAL HEALTHCARE ANALYTICS MARKET: COMPANY EVALUATION MATRIX, 2020



Source: Press Releases, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

12.4.1.1 Stars

These are leading market players in terms of new developments and the adoption of growth strategies and received high scores for most evaluation criteria. They have a strong product portfolio, a very strong market presence, and robust business strategies.

12.4.1.2 Pervasive players

Pervasive players are established vendors with very strong business strategies and market presence. However, their product portfolio may not be as vast as the star players. They generally offer specialized healthcare analytics solutions (pertaining to specialized applications or delivery models).

12.4.1.3 Emerging leaders

Emerging leaders demonstrate substantial product innovations compared to their competitors. These players focus on their product portfolio but do not have very strong strategies for overall business growth (and often have a limited geographical presence). They also show considerable investments in research and development activities compared to their competitors.

12.4.1.4 Participants

Participants are companies that have niche product offerings and are starting to gain traction in the market. They do not have strong business strategies compared to other established vendors. They might be new entrants in the market and require some more time before getting significant traction. Emerging companies usually do not have a wide market presence.

12.4.1.5 Competitive benchmarking

12.4.1.5.1 Overall footprint of companies (20 companies)

COMPANY	SERVICE FOOTPRINT	END-USER FOOTPRINT	REGIONAL FOOTPRINT
IBM	●●●●	●●●●	●●●●
SAS Institute	●●●●	●●●●	●●●●
Cerner	●●●●	●●●●	●●●●
Optum	●●●●	●●●●	●●●●
Allscripts	●●●●	●●●●	●●●●
Health Catalyst	●●●●	●●●●	●●●●
Oracle	●●●●	●●●●	●●●●
Cotiviti	●●●●	●●●●	●●●●
Inovalon	●●●●	●●●●	●●●●
CitiusTech	●●●●	●●●●	●●●●
HealthEC	●●●●	●●●●	●●●●
CVS Health	●●●●	●●●●	●●●●
SCIO Health Analytics	●●●●	●●●●	●●●●
Wipro	●●●●	●●●●	●●●●
Linguamatics	●●●●	●●●●	●●●●
Flatiron	●●●●	●●●●	●●●●
Komodo Health	●●●●	●●●●	●●●●
Roam Analytics	●●●●	●●●●	●●●●
Enlitic, Inc.	●●●●	●●●●	●●●●
Evidation Health	●●●●	●●●●	●●●●

RATING : ●●●● EXCELLENT ●●●● GOOD ●●●● AVERAGE ●●●● BELOW AVERAGE ●●●● NOT APPLICABLE

Note: The product/service footprint of companies is based on the analysis of the service offerings, end users catered to, and regional penetration. The breadth and depth of product/service offerings, product/service features and functionalities, product/service innovations, product/service launches/developments, and company branding have also been considered.

Source: Press Releases, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 246 SERVICE FOOTPRINT OF COMPANIES (20 COMPANIES)

COMPANY	CLINICAL ANALYTICS	FINANCIAL ANALYTICS	OPERATIONAL ANALYTICS	POPULATION HEALTH MANAGEMENT	OTHER SERVICE OFFERINGS	OVERALL SERVICE FOOTPRINT
IBM	Y	Y	Y	Y	Y	4.00
UnitedHealth Group (Optum)	Y	Y	Y	Y	Y	4.00
Cerner	Y	Y	Y	Y	Y	4.00
Allscripts	Y	Y	Y	Y	Y	4.00
SAS Institute	Y	Y	Y	Y	Y	4.00
Health Catalyst	Y	Y	Y	Y	N	3.43
Cerner Corporation	Y	N	Y	Y	Y	3.43
Oracle	Y	Y	Y	Y	Y	4.00
Cotiviti	Y	Y	Y	N	N	3.43
Inovalon	Y	N	N	Y	Y	2.29
CitiusTech	Y	N	N	N	Y	2.29
HealthEC	Y	N	N	N	N	1.71
CVS Health	Y	Y	N	N	Y	2.29
SCIO Health Analytics	Y	Y	N	N	Y	2.29
Wipro	Y	Y	N	N	Y	2.29
Linguamatics	Y	N	N	N	N	1.71
Flatiron	Y	Y	Y	N	Y	3.43
Komodo Health	Y	Y	Y	N	N	2.86
Roam Analytics	Y	N	N	N	N	1.71
Evidation Health	Y	N	N	N	N	1.71
Enlitic	Y	N	N	N	N	1.71

Source: Company Websites and MarketsandMarkets Analysis

TABLE 247 **END-USER FOOTPRINT OF COMPANIES (20 COMPANIES)**

COMPANY NAME	SMALL ORGANIZATIONS	MEDIUM-SIZED ORGANIZATIONS	LARGE ORGANIZATIONS	OVERALL END-USER FOOTPRINT
IBM	Y	Y	Y	4.00
UnitedHealth Group (Optum)	Y	Y	Y	4.00
Cerner	Y	Y	Y	4.00
Allscripts	Y	Y	Y	4.00
SAS Institute	Y	Y	Y	4.00
Health Catalyst	Y	Y	Y	4.00
Cerner Corporation	Y	Y	Y	4.00
Oracle	Y	Y	Y	4.00
Cotiviti	Y	Y	Y	4.00
Inovalon	Y	Y	Y	4.00
CitiusTech	Y	Y	Y	4.00
HealthEC	Y	Y	Y	4.00
CVS Health	Y	Y	Y	4.00
SCIO Health Analytics	Y	Y	Y	4.00
Wipro	Y	Y	Y	4.00
Linguamatics	Y	Y	Y	4.00
Flatiron	Y	Y	Y	4.00
Komodo Health	Y	Y	Y	4.00
Roam Analytics	Y	Y	Y	4.00
Evidation Health	Y	Y	Y	4.00

Source: Company Websites and MarketsandMarkets Analysis

TABLE 248 REGIONAL FOOTPRINT OF COMPANIES (20 COMPANIES)

COMPANY NAME	NORTH AMERICA	EUROPE	ASIA PACIFIC	MIDDLE EAST & AFRICA	LATIN AMERICA	OVERALL REGIONAL FOOTPRINT
IBM	Y	Y	Y	Y	Y	4.00
UnitedHealth Group (Optum)	Y	Y	Y	Y	Y	4.00
Cerner	Y	N	N	N	N	1.60
Allscripts	Y	Y	Y	Y	Y	4.00
SAS Institute	Y	Y	Y	Y	Y	4.00
Health Catalyst	Y	Y	N	N	N	2.40
Cerner Corporation	Y	Y	Y	Y	Y	4.00
Oracle	Y	Y	Y	Y	Y	4.00
Cotiviti	Y	Y	Y	Y	Y	4.00
Inovalon	Y	N	N	N	N	1.60
CitiusTech	Y	Y	N	N	N	2.40
HealthEC	Y	Y	Y	Y	Y	4.00
CVS Health	Y	Y	Y	Y	Y	4.00
SCIO Health Analytics	Y	Y	N	N	N	2.40
Wipro	Y	Y	Y	Y	Y	4.00
Linguamatics	Y	N	N	N	N	1.60
Flatiron	Y	N	N	N	N	1.60
Komodo Health	Y	Y	Y	Y	Y	4.00
Roam Analytics	Y	N	N	N	N	1.60
Evidation Health	Y	N	Y	N	N	1.60

Source: Company Websites and MarketsandMarkets Analysis

12.5 COMPETITIVE SITUATIONS AND TRENDS

Players operating in the healthcare analytics market adopted various strategies to expand their global presence and increase their market shares. Product/service launches & approvals; agreements, partnerships, and collaborations; acquisitions; and expansions are some of the major growth strategies adopted by market players.

12.5.1 PRODUCT/SERVICE LAUNCHES & APPROVALS

Product/service launches and approvals accounted for a large share of the overall growth strategies adopted by market players between January 2018 and August 2021. Companies adopted this strategy to strengthen their product portfolios and cater to the evolving needs of customers.

TABLE 249 PRODUCT/SERVICE LAUNCHES (JANUARY 2018 TO AUGUST 2021)

MONTH & YEAR	COMPANY	TYPE	SEGMENT	PRODUCT/SERVICE
June 2021	APIXIO	Infostream	Infostream-AI	Launched additional functionalities in the Infostream healthcare interoperability platform
May 2020	APIXIO	Accurate Risk Adjustment	HCC Complete	Launched HCC Complete, which uses AI to support a comprehensive chart review experience for Medicare Advantage and Affordable Care Act organizations
May 2019	Enlitic	Enlitic Curie	Cognitive Analytics	Enlitic Curie comprises patient encounters data and algorithms to provide clinical expertise
October 2018	Cotiviti (Verscend Technology)	Payer Management	Financial Analytics	Introduced an upgraded version of its risk adjustment solution suite with in-home assessment capabilities
August 2018	Cotiviti (Verscend Technology)	Performance Analytics	Clinical Analytics	Launched Medical Intelligence v7.8
February 2018	Health Catalyst	Touchstone	Clinical Analytics	Launched Health Catalyst Touchstone
March 2018	IBM	IBM Analytics	Clinical Analytics	Launched Watson Data Kits (used to accelerate the adoption of AI technologies)
January 2018	SAS	Healthcare Analytics	Clinical Analytics	Launched SAS Viya

Source: Company Websites and Secondary Sources

12.5.2 DEALS

Market players focused on strategic partnerships, acquisitions, and collaborations to expand their business operations, diversify their product offerings, enhance their technology portfolio, and enhance their overall performance.

TABLE 250 DEALS (JANUARY 2018 TO AUGUST 2021)

MONTH & YEAR	DEAL TYPE	COMPANY 1	COMPANY 2	DESCRIPTION
June 2021	Partnership	Flatiron	Foundation Medicine, Inc.	Partnered with Foundation Medicine, Inc. to integrate its solutions with OncoEMR to offer integrated genomic profiling to customers
July 2021	Partnership	Flatiron	NICE	This research partnership will leverage Flatiron's OncoEMR data to explore RWE to help evaluate the cost-effectiveness and clinical efficacy of oncology treatments
July 2021	Partnership	Enlitic	NMC Healthcare	This is a multi-faceted partnership with private healthcare companies to incorporate Enlitic's solutions in GCC countries
May 2021	Partnership	Flatiron	Genentech and Foundation Medicine	Partnered with Foundation Medicine and Genentech to launch a novel prospective lung cancer clinical study
November 2020	Acquisition	Centene	Apixio	Centene announced its plans to acquire Apixio. Post the acquisition, Apixio will continue to operate independently.
December 2019	Partnership	Enlitic	MLP Care	Partnered to bring clinical AI solutions to Turkey and neighboring countries in East Europe and Central Asia
August 2018	Acquisition	Verscend Technologies, Inc.	Cotiviti Holdings, Inc.	Acquired Cotiviti Holdings, Inc., a leading provider of payment accuracy and analytics-driven solutions focused primarily on the healthcare industry.
March 2018	Acquisition	Verscend Technologies, Inc.	General Dynamics (health insurance products)	This acquisition helped Verscend gain one of the most robust sets of solutions for tracking healthcare fraud, waste, and abuse (FWA). This acquisition also enabled Verscend to strengthen its payer quality analytics and reporting solutions.
July 2018	Acquisition	Health Catalyst	Medicity	Health Catalyst acquired Medicity, a population health management company offering solutions for HIEs, business intelligence, and provider and patient engagement. This acquisition enabled the company to offer improved health outcomes and advanced population health management solutions to its existing customer base.

Source: Press Releases

13 COMPANY PROFILES

13.1 KEY PLAYERS

13.1.1 ALLSCRIPTS HEALTHCARE SOLUTIONS

13.1.1.1 Business overview

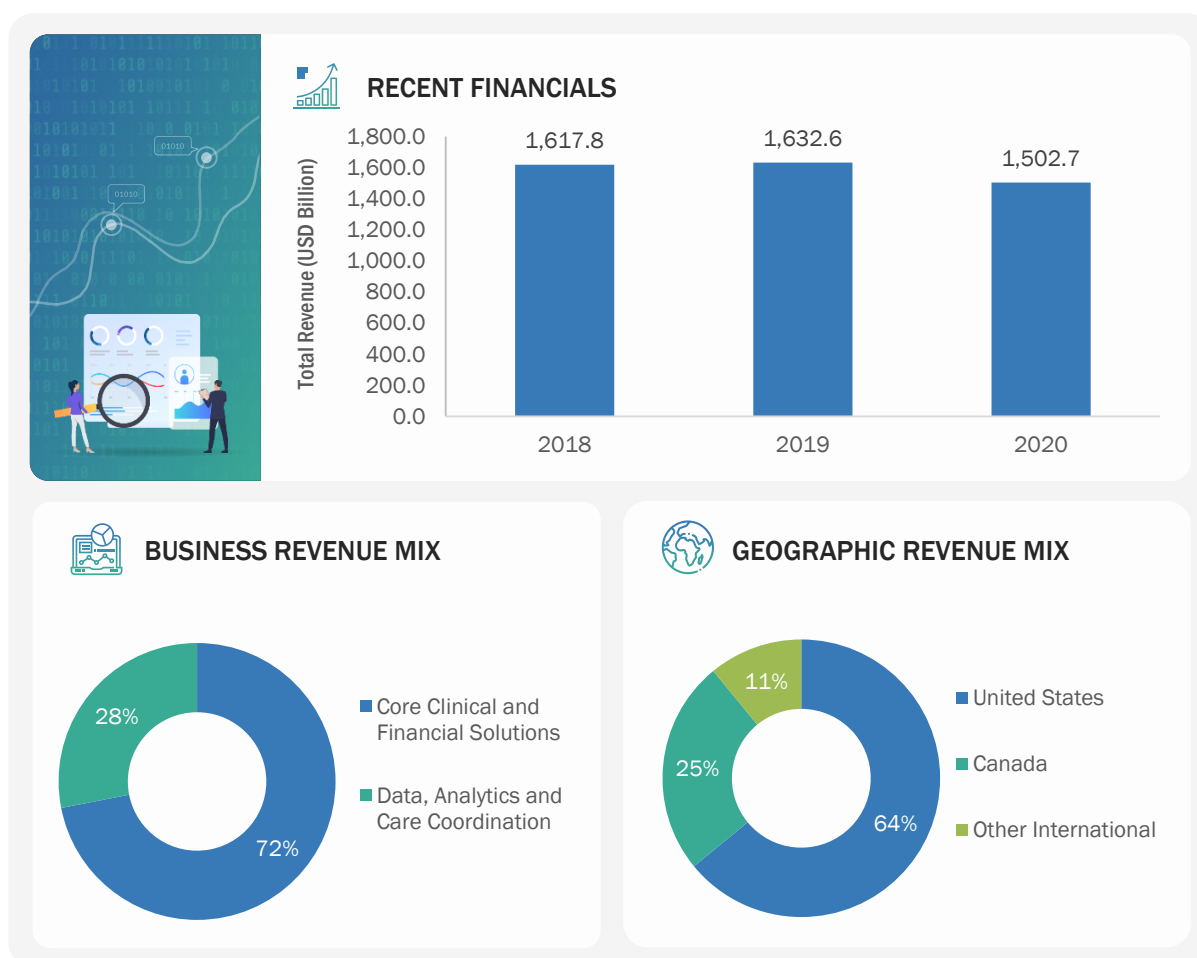
Allscripts is a leading provider of HCIT solutions and services. It offers solutions such as electronic healthcare records (EHRs), connectivity, hosting, outsourcing, analytics, patient engagement, and population health management (PHM). The company provides these solutions to physicians, hospitals, governments, health systems, retail clinics and pharmacies, pharmacy benefit managers, and post-acute healthcare providers (such as home health and hospice agencies) to help them achieve better clinical, financial, and operational results.

The company operates in the US, Mauritius, Malaysia, Singapore, Israel, Australia, Canada, India, and the UK. Some of its notable subsidiaries are Allscripts Holdings US, Allscripts Healthcare US, Allscripts Canada, Allscripts Mauritius Limited, Allscripts India, Allscripts Malaysia, Allscripts Healthcare IT Singapore, Allscripts Healthcare IT Australia, Allscripts Healthcare IT UK, and dbMotion Israel.

TABLE 251 ALLSCRIPTS HEALTHCARE SOLUTIONS: BUSINESS OVERVIEW

Year Founded	1986
Country	US
City/State	Chicago, Illinois
Ownership	Public

Source: Company Website

FIGURE 34 ALLSCRIPTS HEALTHCARE SOLUTIONS: COMPANY SNAPSHOT (2020)

Source: Company Website, SEC Filings, and Annual Reports

13.1.1.2 Products & services offered

CATEGORY	PRODUCT/SERVICE	DESCRIPTION	APPLICATION
Physician Practices	Analytics and Insights Solutions	Using these solutions, administrators can review organizational, clinical, and financial key performance indicators (KPIs); investigate provider productivity, service utilization, outcomes, and satisfaction; and make decisions about expanding services, capital investments, and resource allocation.	Clinical practice analytics
	Allscripts Practice Performance		
	Clinical Quality Solution		
PACOs	Analytics and Performance Solutions	These products allow end users to compare against benchmarks to provide quality patient care.	Financial analytics
	Allscripts Patient Flow		
	Allscripts EPSi		
	Allscripts Clinical Performance Management		
Hospitals and Healthcare Systems	Allscripts Sunrise	This EHR leverages cloud-based technology to offer patient-centric	Hospital management

		solutions.	and EHR
Acute Care Solutions	▪ Sunrise Financial Manager	This solution helps identify opportunities for cost savings so that end users can drive accountability and accelerate transformation	Financial analytics and supply chain management
PHM	▪ Population Health Analytics ▪ dbMotion Clinical Analytics Gateway ▪ Allscripts dbMotion ▪ Sunrise Acute Care ▪ CareInMotion	These solutions provide safer, more informed patient care, better measure performance with data management, and robust insights for informed decisions	Population health management
Services	▪ Managed IT Services ▪ Allscripts Hosting Solutions ▪ Allscripts Professional Services	These are various analytics and IT services provided to hospitals and clinics	Operational and administrative management

13.1.1.3 Recent developments

DEALS

MONTH & YEAR	DEAL TYPE	COMPANY 1	COMPANY 2	DESCRIPTION
September 2018	Partnership	EPSi, Allscripts	MPA Healthcare Solutions	The partnership with MPA Healthcare Solutions offered risk adjustment and clinical analytics solutions, which helped in developing a deeper understanding of performance and informed decision-making.
September 2018	Partnership	EPSi, Allscripts	Ilerasoft	The partnership with Ilerasoft offered additional capital budgeting capabilities and provided analytical solutions to improve operational efficiency around medical equipment use.
September 2017	Partnership	Allscripts	Liverpool Heart and Chest Hospital NHS Foundation Trust (LHCH)	This partnership was aimed at utilizing Allscripts' Sunrise solution suite.
August 2017	Collaboration	Allscripts	Catholic Medical Centre (CMC)	This collaboration allowed CMC to utilize Allscripts' Sunrise solution suite. This suite includes Sunrise Ambulatory Care, Sunrise Emergency Care, Sunrise Surgical Care, Sunrise Radiology, Sunrise Financial Manager, Sunrise Enterprise Registration and

				Scheduling, Sunrise Critical Care, Sunrise Mobile MD, and Sunrise Mobile Care.
August 2017	Acquisition	Allscripts	McKesson Corporation	Allscripts acquired McKesson Corporation's hospital and health system IT business—Enterprise Information Solutions (EIS). This acquisition added McKesson's clinical and financial solutions to Allscripts' product offerings, thus helping the company offer more comprehensive, high-performing HCIT solutions.
October 2016	Acquisition	Allscripts	CarePort	The acquisition of CarePort was made to extend Allscripts' comprehensive PHM platform.
August 2016	Partnership	Allscripts	Hospital IQ	This strategic partnership was aimed to make Hospital IQ's predictive analytics platform available to Allscripts Sunrise and Allscripts EPSi clients. The partnership helped the company in increasing its product offerings.
August 2016	Partnership	Allscripts	OptumCare	Allscripts integrated its EHR and practice management software with the technology that OptumCare currently provides to its physicians. This partnership helped provide its physicians a connected health IT platform.
March 2016	Collaboration	Allscripts	University Hospitals (UH)	The company extended its collaboration to promote the utilization of the Allscripts Sunrise Clinical Manager platform in nine hospitals in the US. UH enhanced the use of Allscripts' dbMotion across the Northeast Ohio healthcare community.

Source: Press Releases

EXPANSIONS

MONTH & YEAR	LOCATION	DESCRIPTION
November 2018	US	Veradigm is the new division of the Allscripts Payer and Life Sciences business segment. It provides next-generation analytics solutions that are designed to increase efficiency and manage healthcare costs.

Source: Press Releases

PRODUCT LAUNCHES

MONTH & YEAR	SEGMENT	APPROACH	PRODUCT
April 2017	Population Health	Product Launch	Allscripts made the dbMotion Solution available through the cloud via Microsoft Azure.

Source: Press Releases

13.1.1.4 MnM view

Allscripts offers software solutions and services that help healthcare organizations and other stakeholders analyze dependencies, trends, and patterns. The company holds a prominent position in the healthcare analytics market due to its wide product portfolio and brand positioning. Allscripts offers analytical solutions for various applications, including clinical, operative, administration, and PHM. The company has subsidiaries in strategic locations across North America, Europe, Asia, and Oceania. Allscripts pursues geographical expansion as a key strategy to enhance its market share. The other growth strategies adopted by the company to maintain its leading position include partnerships, agreements, and collaborations.

13.1.1.4.1 Right to win

The company provides healthcare analytics services to various organizations covering more than 190 countries. The company has been focusing on organic as well as inorganic growth strategies to establish its brand image among end users. For instance, in September 2018, EPSi (Allscripts) partnered with MPA Healthcare Solutions to offer risk adjustment and clinical analytics solutions that will help in developing a deeper understanding of performance-driven and informed decision making. Also, in the same month, EPSi (Allscripts) partnered with Ilerasoft to offer additional capital budgeting capabilities and provide analytics solutions to help improve operational efficiency around medical equipment use powered by Ilerasoft's software. Such strategic developments will enable Allscripts to strengthen its presence in the healthcare analytics market.

13.1.1.4.2 Strategic choices made

During the COVID-19 pandemic, the company has been engaged in conducting employee surveys to determine its effect on employees, especially with regard to disruptions in the work environment, challenges faced by human resource departments, and changes in employer absence programs. The company also conducts regular, relevant polls among employees regarding their concerns for the return-to-home work process, overall mental health and well-being, and other stress factors.

Allscripts is a pioneer in the field of corporate wellness solutions, assisting organizations in overcoming challenges by adapting to their business needs and increasing attraction and retention of employees with improved productivity and performance.

13.1.2 CERNER CORPORATION

13.1.2.1 Business overview

Cerner is a leading supplier of HCIT solutions, medical devices, and related services. It operates through four business units, namely, System Sales, Support and Maintenance, Services, and Reimbursed Travel.

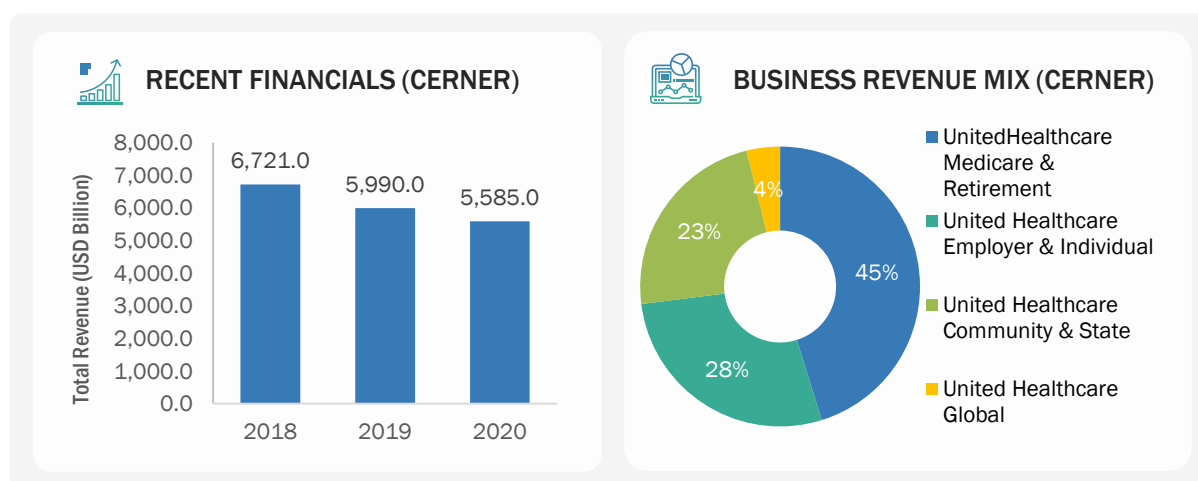
The company offers its HCIT products and services to hospitals and physicians as well as ambulatory facilities, home facilities, retail pharmacies, and local, regional, and national government agencies.

Cerner has a wide geographical presence, spanning North America, Europe, the Asia Pacific, Latin America, and the Middle East. Some of its major subsidiaries are Cerner Australia, Cerner Limited (UK), Cerner Healthcare Solutions Private Limited (India), Cerner Canada, Cerner France SAS (France), Cerner México, S. de R. L. de C.V. (Mexico), and Cerner Middle East FZ-LLC (Dubai).

TABLE 252 CERNER CORPORATION: BUSINESS OVERVIEW

Year Founded	1979
Country	US
City/State	North Kansas City, Missouri
Ownership	Public

Source: Company Website

FIGURE 35 CERNER CORPORATION: COMPANY SNAPSHOT (2020)

Note: Revenues include premiums, products, services, investments, and other income.

Source: Company Website, SEC Filings, and Annual Reports

13.1.2.2 Products & services offered

CATEGORY	PRODUCT/SERVICE	DESCRIPTION	APPLICATION
Hospital and Health Systems	- Reporting and Outcomes - Discern nCode - Infection Control - Rapid Response Team Dashboard - Millennium Lighthouse - Workforce Management - Outcomes-driven Acuity - CareAware Tracking - Supply Chain - CareAware Critical Care - Inpatient Summary - Revenue Cycle - Claims Editing - Millennium Revenue Cycle - Clinically Driven Revenue Cycle - Business Office Services	This is a suite of hospital management and clinical analytics solutions, which includes functionalities such as supply chain management, claims management, reporting, and patient data management	Clinical analytics and hospital management
PHM	HealthIntent	This population health management software provides data for patient care in large hospitals	Population health management

13.1.2.3 Recent developments

DEALS

MONTH & YEAR	DEAL TYPE	COMPANY 1	COMPANY 2	DESCRIPTION
December 2016	Collaboration	Cerner	The Unified Government of Wyandotte County and Kanas City	This collaboration was aimed to open the Road to Wellness Health Center to improve the health of Wyandotte's county and city employees and to provide convenient and affordable care.
February 2016	Collaboration	Cerner	xG Health Solutions	This collaboration was aimed at providing the Cerner HealthIntent PHM platform, which helps in managing patient outcomes and risks. xG's clinical content was integrated into Cerner's Healthcare Analytics solutions. This enabled the company to expand its PHM turnkey catalog, which helped in reducing costs and improving the quality of care and patient outcomes.

Source: Press Releases

EXPANSIONS

MONTH & YEAR	LOCATION	DESCRIPTION
September 2016	India	Opened Healthe Clinic and Healthe Fitness Center at its Manyata campus in Bangalore, India

Source: Press Releases

13.1.2.4 MnM view

Cerner Corporation is one of the leading suppliers of software, hardware, and HCIT services. The company's leading position in the healthcare analytics market can be attributed to its expertise in HCIT, strong brand recognition, and robust portfolio of healthcare analytics solutions. Cerner provides its solutions and services to customers through more than 27,000 facilities located worldwide.

13.1.2.4.1 Right to win

Its wide presence across the globe enables the company to maintain its position in this market. To gain a competitive edge, Cerner focuses on adopting organic and inorganic growth strategies. For instance, in February 2016, xG Health Solutions collaborated with Cerner to provide the Cerner HealthIntent PHM platform, which helps in managing patient outcomes and risks. xG's clinical content was further integrated into Cerner's Healthcare Analytics solutions. Also, a robust distribution network and an efficient sales team help the company in capturing most of the global market.

13.1.2.4.2 Strategic choices made

The company has leveraged the lucrative opportunities given by the COVID-19 pandemic. Cerner has taken strategic steps to innovate and improve digital health and healthcare analytics after the outbreak of the pandemic. As of July 2021, several health systems are working directly with Cerner to harness the power of AI to help identify and engage patients at risk for severe COVID-19 complications before they get sick. In Philadelphia, teams at Einstein Healthcare Network teamed up with Cerner to run an open-source model that predicts patients' COVID-19 complication risk.

13.1.3 COTIVITI, INC. (VERSCEND TECHNOLOGIES)

13.1.3.1 Business overview

Verscend Technologies, formerly known as Verisk Health, was founded in 1996 and is headquartered in Massachusetts, US. Verscend is an HCIT company that provides data-driven healthcare solutions to payers, providers, employers, life insurance carriers, and legal providers. Verscend provides solutions for payment accuracy, revenue integrity, risk assessment and stratification, and quality improvement, which help organizations utilize their data efficiently and cost-effectively.

Verscend operates through its offices in the US and Nepal. The company has received accreditation and is associated with various bodies such as the National Health Care Anti-Fraud Association (NHCAA), HIPAA, National Committee for Quality Assurance (NCQA), and Resource Initiative and Society for Education (RISE). In August 2018, Verscend technologies acquired Cotiviti Holdings, Inc., a leading provider of payment accuracy and analytics-driven solution focused primarily on the healthcare industry. Post-merger, the company will operate under Cotiviti's name.

Founded in 1979 and headquartered in the US, Cotiviti (formerly known as Connolly Super Holdings, Inc.) is one of the leading players that offer analytics-driven payment accuracy and spend management. Cotiviti operates through two business segments, namely, Healthcare and Global Retail and Others. The company offers its solutions primarily for the healthcare sector in the US, Canada, the UK, and India. Cotiviti offers a wide range of solutions from prospective claims accuracy to retrospective claims audit and recovery, clinical chart validation solutions, and optimization analytics to its healthcare customer base. Apart from these offerings, the company provides analytics and support services, including anti-fraud, waste, and abuse analytics to identify abnormal patterns in coding and billing practices, surveillance and longitudinal analytics, and claims history to identify areas for direct interaction, which enable the enhancement of future payment accuracies.

Note: Cotiviti is a privately held company, and no financial information is available in the public domain. Hence, financial details are not added in this section.

TABLE 253 COTIVITI, INC.: BUSINESS OVERVIEW

Year Founded	1979
Country	US
City/State	South Jordan, Utah
Ownership	Private

Source: Company Website

13.1.3.2 Products & services offered

CATEGORY	PRODUCT/SERVICE	DESCRIPTION	APPLICATION
Payment Accuracy	- Claims Accuracy - Fraud Detection - Inpatient Accuracy - Dental Accuracy	The products provide health plans with the resources, analysis, and strategies to pay claims more accurately.	Financial analytics
Quality Improvement	- Quality Measurement and Reporting - Medical Record Retrieval - Medical Record Abstraction - Star Navigator	These quality and performance solutions allow health plans to move from reporting quality measures to driving the purchase, delivery, and utilization of higher value healthcare.	Medical reporting and quality measurement
Risk Adjustment	- Medicare Risk Adjustment - Commercial Risk Adjustment	This risk adjustment suite includes complete prospective and retrospective risk adjustment solutions supported by technology, analytics, artificial intelligence, and deep subject-matter expertise	Claims management and business intelligence

13.1.3.3 Recent developments

PRODUCT LAUNCHES AND UPGRADES

MONTH & YEAR	SEGMENT	APPROACH	PRODUCT
October 2018	Payer Management	Product Upgrade	Introduced an upgraded version of its risk adjustment solution suite with in-home assessment capabilities
August 2018	Performance Analytics	Product Launch	Launched Medical Intelligence v7.8
July 2018	Quality Improvement	Product Launch	Launched Star Navigator

Source: Press Releases

DEALS

MONTH & YEAR	DEAL TYPE	COMPANY 1	COMPANY 2	DESCRIPTION
August 2018	Acquisition	Verscend Technologies, Inc.	Cotiviti Holdings, Inc. (US)	Acquired Cotiviti Holdings, Inc., a leading provider of payment accuracy and analytics-driven solutions focused primarily on the healthcare industry.
March 2018	Acquisition	Verscend Technologies, Inc. (US)	General Dynamics (health insurance products) (US)	This acquisition helped Verscend gain one of the most robust sets of solutions for tracking healthcare fraud, waste, and abuse (FWA). This acquisition also enabled Verscend to strengthen its payer quality analytics and reporting solutions.

Source: Press Releases

13.1.3.4 MnM view

Cotiviti is one of the prominent players in the healthcare analytics market. Fraud detection, claims accuracy, risk adjustment and assessment, and performance analytics are the company's areas of expertise for which it provides innovative solutions. To maintain its leading position in the healthcare analytics market, Cotiviti aims to pursue product launches as its key growth strategy. Some of the recent launches by the company include Risk Adjustment Suite with In-house Assessment capabilities (October 2018), Medical Intelligence v7.8 (August 2018), and Star Navigator (July 2018).

13.1.3.4.1 Right to win

Cotiviti works with over 180 healthcare payers globally. Over 125 members have been processed through the solution provided by the company. Cotiviti increased its focus on acquisitions over the last five years to improve product offerings and expand customer presence. Furthermore, the company participates in various conferences, symposia, and workshops to display its analytics solutions. This enables the firm to enhance its customer base by improving its visibility in the healthcare analytics industry.

13.1.3.4.2 Strategic choices made

Cotiviti has joined forces with other companies in the COVID-19 Healthcare Coalition to preserve the healthcare delivery system and help protect US populations. Such developments will help the company in future strategic collaborations and partnerships.

13.1.4 CITIUSTECH INC.

13.1.4.1 Business overview

Founded in 2005 and headquartered in New Jersey, US, CitiusTech is a provider of healthcare technology solutions and services to medical technology companies, payers, providers, and life science organizations. CitiusTech helps customers accelerate innovation in healthcare by providing solutions for clinical quality reporting, big data, cloud computing, mobile health, and predictive analytics. The company also offers solutions and services for healthcare software development, healthcare interoperability, regulatory compliance, consumer engagement, care coordination, and PHM.

CitiusTech has a presence in North America, Europe, Asia, and the Middle East.

Note: CitiusTech is a privately held company, and no financial information is available in the public domain. Hence, financial details are not added in this section.

TABLE 254 CITIUSTECH INC.: BUSINESS OVERVIEW

Year Founded	2005
Country	US
City/State	Princeton, New Jersey
Ownership	Private

Source: Company Website

13.1.4.2 Products & services offered

CATEGORY	PRODUCT/SERVICE	DESCRIPTION	APPLICATION
Healthcare Analytics	- BI- Clinical - H-Scale - CQ-IQ - Medictiv - Integrated AnalytiX - SCORE+	- BI-Clinical: An analytics platform with an extensive range of configurable apps that cover over 750+ KPIs. It is a modular, configurable, and scalable platform. - H-Scale: A healthcare data integration and management platform that helps make data the backbone of organizational strategy.	Data integration, transformation, quality, security, and analytics
Services	- EDW, BI/Analytics, and Reporting - Consulting	The company offers consulting services for health plans, providers, and life sciences organizations. The company has a team of consultants with expertise in US government programs, digital innovation, business operations, and enterprise software platforms.	Business operations, healthcare management, and analytics

13.1.4.3 Recent developments

PRODUCT LAUNCHES

MONTH & YEAR	SEGMENT	APPROACH	PRODUCT
April 2017	Clinical Analytics	Product Launch	Launched SCORE+ for HEDIS to address key payer challenges around HEDIS compliance
April 2016	Clinical Analytics	Product Launch	Launched BI-Clinical, a clinical analytics product for healthcare providers

Source: Press Releases

OTHER DEVELOPMENTS

MONTH & YEAR	STRATEGY	PRODUCT	DESCRIPTION
February 2019	Showcase	BI-Clinical and H-scale	The company showcased its BI-Clinical and H-scale product offerings on Microsoft Azure at the Healthcare Information Management Systems Society Conference in 2019.

Source: Press Releases

13.1.5 HEALTH CATALYST

13.1.5.1 Business overview

Founded in 2008 and headquartered in Salt Lake City, Utah (US), Health Catalyst (formerly known as Healthcare Quality Catalyst) designs, develops, and delivers healthcare data warehousing and process improvement solutions for healthcare organizations and health systems.

The company's solutions help hospitals manage operational, financial, clinical, and research data. It provides automated reporting software, which helps users discover, design, and implement automated reporting for internal and external needs and clinical improvement solutions (such as PHM, care transformation, and personalized wellness management). The company also offers services necessary to identify, prioritize, execute, and measure clinical improvement efforts. Health Catalyst primarily operates in the US.

Note: Health Catalyst is a privately held company, and no financial information is available in the public domain. Hence, financial details are not added in this section.

TABLE 255 HEALTH CATALYST: BUSINESS OVERVIEW

Year Founded	2008
Country	US
City/State	South Jordan, Utah
Ownership	Private

Source: Company Website

13.1.5.2 Products & services offered

CATEGORY	PRODUCT/SERVICE	DESCRIPTION	APPLICATION
Healthcare Analytics Applications	- Accountable Care/Shared Risk - ACO Explorer - ACO Measures - Attribution Modeler - Bundled Payments - Leakage and Referrals Explorer - PMPM Analyzer - Risk Model Analyzer - Financial Management - Clinical Operations Resource Utilization System (CORUS) - Financial Management Explorer - General Ledger Explorer - Revenue Cycle Explorer: Hospital - Operation and Performance - ACO Measures - Labor Management Explorer - Patient Experience Explorer - Patient Flow Explorer - Practice Management: Patient Access	- These solutions help proactively identify: - Where an organization is performing well and where it is underperforming (solutions recommend and prioritize opportunities for improvement) - Identify—and address—gaps in care, especially for high-risk patients. - Quality improvement leadership - Population health management and patient injury	Clinical and operational analytics

	- Practice Management: Professional Billing - Population Health and Patient Injury Prevention - Catheter-associated Urinary Tract Infection (CAUTI) Prevention Advanced Application - Central Line-associated Bloodstream Infection (CLABSI) Prevention Advanced Application - Cohort Builder - Community Care Advanced Application - Exclusive Breast Milk Feeding Advanced Application - Heart Failure Advanced Application - Labor and Delivery Application - Pediatric Appendectomy Advanced Application - Pediatric Asthma Advanced Application - Population Explorer - Readmission Explorer - Sepsis Advanced Application - Surgical-site Infection Prevention Advanced Application - Translation Research - De-identified Cohort Builder - De-identified Population Explorer - Shared Applications - Attribution Modeler - Executive Dashboard Integration Tool - Key Process Analysis - Risk Model Analyzer - Health Catalyst Touchstone	- prevention - Financial management and accounting of organizations	
- Services - Installation Services - Improvement Services		These are infrastructure and information technology services provided for healthcare providers and payers	Infrastructure and IT

13.1.5.3 Recent developments

PRODUCT LAUNCHES

MONTH & YEAR	SEGMENT	APPROACH	PRODUCT
February 2018	Healthcare Analytics	Product Launch	Launched Health Catalyst Touchstone
June 2017	Financial Management	Product Launch	Launched CORUS Suite
March 2016	Healthcare Analytics Applications	Product Launch	Keeping the Health Catalyst Enterprise Data Warehouse (EDW) and analytics platform as its foundation, the company announced product development in its nine product lines. These updates mainly revolved around accountable care, PHM, and personalized medicine.

Source: Press Releases

DEALS

MONTH & YEAR	DEAL TYPE	COMPANY 1	COMPANY 2	DESCRIPTION
July 2018	Acquisition	Health Catalyst	Medicity	Health Catalyst acquired Medicity, a population health management company offering solutions for HIEs, business intelligence, and provider and patient engagement. This acquisition enabled the company to offer improved health outcomes and advanced population health management solutions to its existing customer base
February 2017	Partnership	Health Catalyst	Gunnison Valley Hospital	This partnership was aimed to utilize the analytical solutions and expertise of the Health Catalyst team
February 2017	Agreement	Health Catalyst	Regenstrief Institute	This agreement was signed to commercialize Regenstrief's nDepth AI-powered text analytics technology. Health Catalyst agreed to incorporate the nDepth solution into the Health Catalyst data analytics platform to serve patients across the US
March 2016	Partnership	Health Catalyst	UPMC	This partnership was aimed to develop a cost-improvement process infrastructure and reduce healthcare costs jointly. This helped the firm develop innovative technologies and provide high-value care.

Source: Press Releases

OTHER DEVELOPMENTS

MONTH & YEAR	COMPANY	DEVELOPMENT	DESCRIPTION
February 2016	Health Catalyst	Venture Capital Funding	Health Catalyst raised USD 70 million in Series E funding from Norwest Venture Partners, UPMC, MultiCare Health System, and OSF Healthcare. The company announced to use the raised amount for the development of analytics solutions for healthcare organizations.

Source: Press Releases

13.1.6 CVS HEALTH CORPORATION

13.1.6.1 Business overview

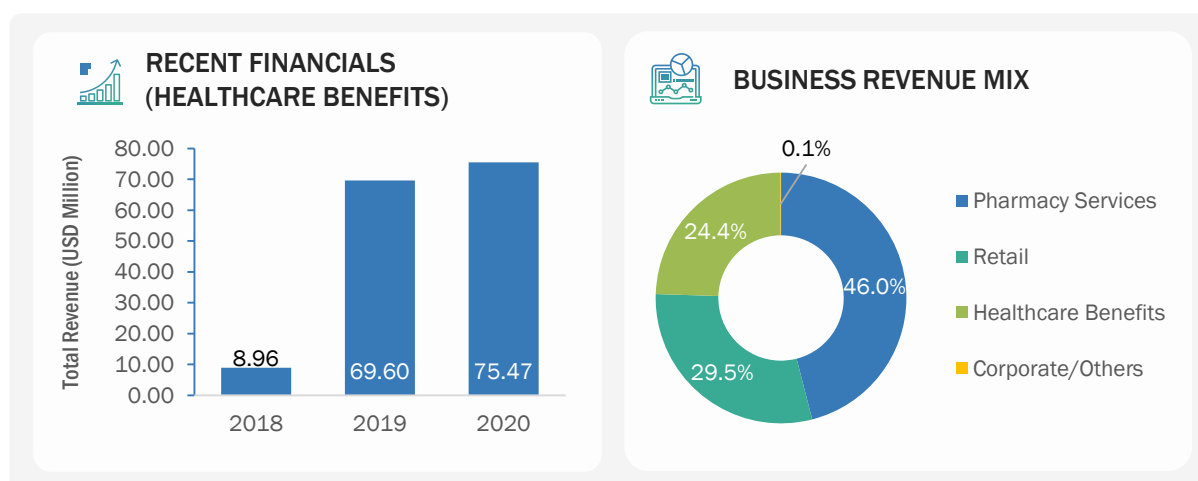
CVS Health Corporation is a diversified health services company with four reportable segments, namely, Pharmacy Services, Retail/LTC, Health Care Benefits, and Corporate/Other. The company added the Health Care Benefits segment because of its acquisition of Aetna in 2018. The company provides employee assistance products through its Health Care Benefits segment. CVS Health, through Aetna, offers employee wellness solutions such as well-being tools, on-site well-being support, lifestyle coaching, and digital tools for personalized care and for tracking health programs.

The company has a business presence either directly or through its subsidiaries across North America, Europe, the Asia Pacific, Latin America, and the Middle East & Africa. CVS Health's major subsidiaries include CVS International, LLC (US), Aetna Global Benefits (UK) Limited (England and Wales), Aetna (Shanghai) Enterprise Services Co. Ltd. (China), Aetna Global Benefits (Asia) Limited (Hong Kong), Aetna Global Benefits (Middle East) LLC (UAE), Aetna Global Benefits (Bahamas) Limited (Bahamas), and Drogaria Onofre Ltda (Brazil). As of December 31, 2020, the company has approximately 300,000 employees.

TABLE 256 CVS HEALTH CORPORATION: BUSINESS OVERVIEW

Year Founded	1963
Country	US
City/State	Woonsocket, Rhode Island
Ownership	Public

Source: Company Website

FIGURE 36 CVS HEALTH CORPORATION: COMPANY SNAPSHOT (2020)

Note: The geographic revenue mix was not available for the company. Hence, this section has not been provided.

Source: Company Website and Annual Reports

13.1.6.2 Products & services offered

CATEGORY	PRODUCT/SERVICE	DESCRIPTION	APPLICATION
Administrative Offerings	Healthagen Analytical and Advisory Services	Healthagen, part of the CVS Health family of companies, offers analytical and advisory services for the clinical challenges faced by pharmaceutical, biotech, and medical device companies.	Administrative analytical services for healthcare payers and healthcare providers
Administrative Offerings	ActiveHealth	The tools review patient information to find gaps in care, as well as safety issues (such as potentially dangerous drug interactions).	Operational Analytics and administrative analytics for healthcare providers

13.1.6.3 Recent developments

DEALS

MONTH & YEAR	APPROACH	COMPANY 1	COMPANY 2	DESCRIPTION
November 2018	Acquisition	CVS Health (US)	Aetna (US)	CVS completed the acquisition of Aetna (US) to establish an innovative healthcare model

Source: Press Releases

13.1.7 IBM

13.1.7.1 Business overview

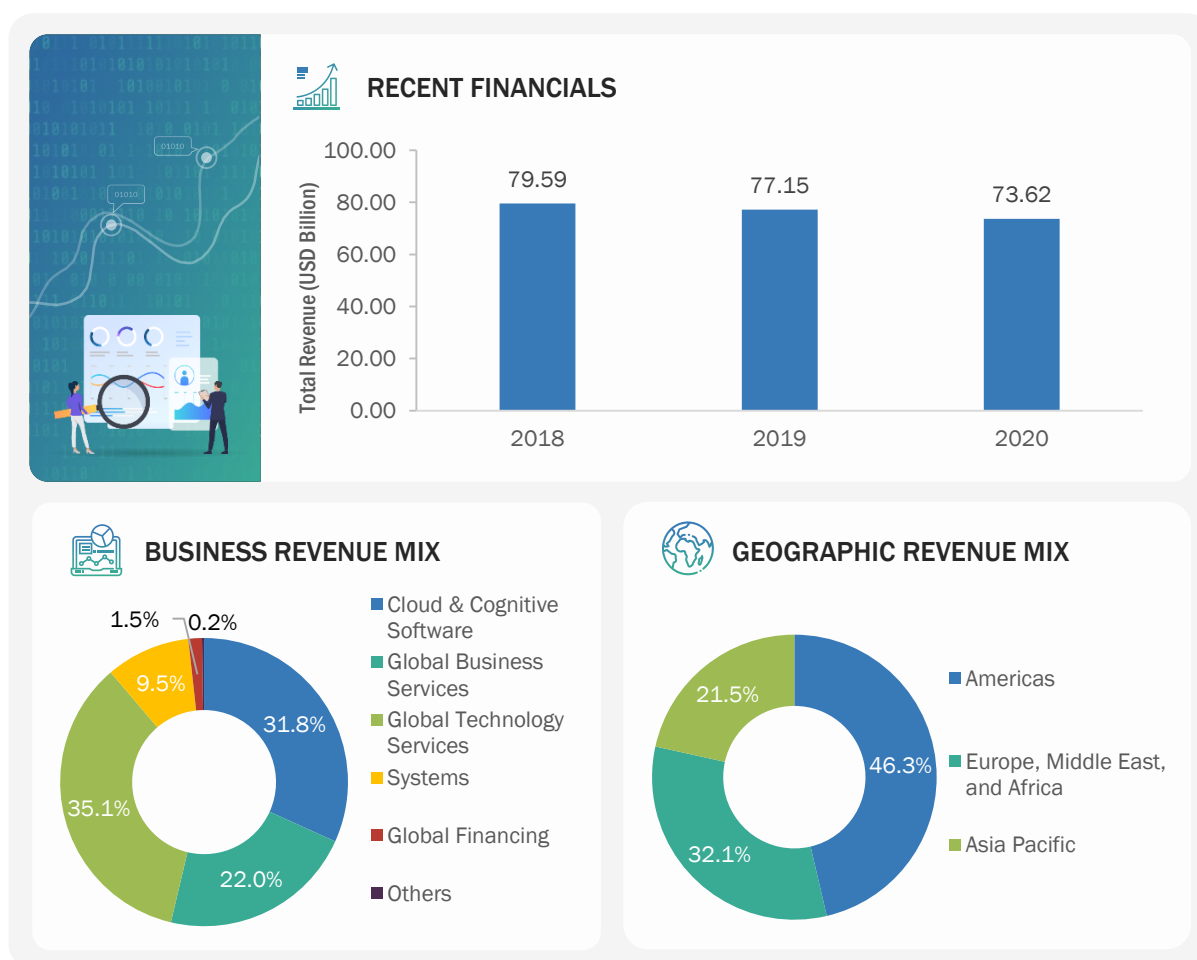
IBM is an IT company that provides business, technology, and consulting services. The company operates through five business segments, namely, Global Technology Services (GTS), Software, Global Business Services (GBS), Systems Hardware, and Global Financing. The Global Business Services segment offers analytics, consulting, and systems integration services for strategy and transformation, enterprise applications, and application management maintenance. It also provides Watson solutions based on a cognitive computing platform, which analyzes big data to provide accurate insights to its customers. In April 2015, IBM established a new business unit—Watson Health—which provides cloud-based access to the Watson supercomputer for the purpose of analyzing healthcare data. In April 2016, IBM acquired Truven Health Analytics, which helped the company expand its healthcare analytics segment offerings.

IBM has a presence in more than 175 countries in North America, Europe, APAC, MEA, and Latin America. Some of its subsidiaries are IBM Argentina Sociedad de Responsabilidad Limitada (Argentine), IBM Global Financing Australia Limited, IBM Bangladesh Private Limited, WTC Insurance Corporation, Ltd. (Bermuda), and IBM de Chile S.A.C.

TABLE 257 IBM: BUSINESS OVERVIEW

Year Founded	1911
Country	US
City/State	Armonk, New York
Ownership	Public

Source: Company Website

FIGURE 37 IBM: COMPANY SNAPSHOT (2020)

Source: Company Website and Annual Reports

13.1.7.2 Products & services offered

CATEGORY	PRODUCT/SERVICE	DESCRIPTION	APPLICATION
Content Analytics	- IBM Content and Predictive Analytics for Healthcare - IBM Integrated Analytics System	These products collect and analyze structured and unstructured content in documents, e-mails, databases, websites, and other enterprise repositories	Operational analytics
IBM Business Analytics Software for Healthcare	- IBM Cognos Business Intelligence - IBM Cognos TM1 - IBM Cognos Analytic Application - IBM SPSS Software	These are planning solutions that help customers to transcend the limits of manual planning to create more accurate plans for FP&A, sales, and supply chain, among other areas.	Business and administrative analytics
IBM Business Analytics Software for Healthcare	- IBM Infosphere Clinical Analytics - IBM Infosphere Data Architect - Mainline Enterprise	These planning solutions help customers to transcend the limits of manual planning to create more accurate plans for FP&A, sales, and supply chain, among other areas.	Supply chain analytics

Healthcare Analytics V2.0			
Fraud and Abuse Management for Payers and Providers	▪ IBM Research ▪ IBM DB2 ▪ IBM WebSphere ▪ IBM Cognos	These solutions help payers identify providers that may be submitting claims and potentially engaging in fraudulent, abusive, or wasteful practices.	Claims management and fraud detection analytics
Provider Performance Management Solutions	▪ Action OI ▪ Care Discovery ▪ Market Expert	These products help identify drivers of clinical quality, detect unseen inefficiencies, benchmark against the best care for patients, and understand variations in care based on benchmarking	Clinical analytics
IBM Watson Health	▪ ActionOI ▪ ActionOI Practice Insight ▪ Advantage Suite – Employer ▪ Advantage Suite – Government ▪ Advantage Suite – Health Plan ▪ Complementary & Alternative Medicine (formerly AltCareDex (System)) ▪ AltMed-REAX for the Patient ▪ AltMedDex System ▪ IBM Micromedex Patient Connect Suite	IBM offers analytics tools, benchmarks, and services to a variety of healthcare institutions, including hospitals, government agencies, employers, payers, clinicians, and pharmaceutical, biotech, and medical device companies.	Healthcare analytics
Services	▪ Business Analytics and Optimization Services ▪ Analytics and Consulting Services ▪ Strategic Consulting and Advisory Services ▪ Research Services ▪ Data Management and Systems Integrations ▪ Other Services	These are business analytics and consultation services for healthcare providers and payers, including database management services and operational management services	Operational and administrative consultation

13.1.7.3 Recent developments

PRODUCT LAUNCHES

MONTH & YEAR	SEGMENT	APPROACH	PRODUCT
March 2018	IBM Analytics	Product Launch	Launched Watson Data Kits (used to accelerate the adoption of AI technologies)
November 2017	IBM Analytics	Product Launch	Launched Analytics Systems with Machine Learning
February 2017	Watson Health	Product Launch	Launched expanded capabilities for the Watson Platform for Health Cloud and a specialized Watson Health Consulting Services Unit
March 2016	Healthcare Providers	Product Launch	Launched the Community Hospital Solution Suite
March 2016	Hospital and Health Systems	Product Launch	Launched Truven Health Custom Solutions

Source: Press Releases

DEALS

MONTH & YEAR	DEAL TYPE	COMPANY 1	COMPANY 2	DESCRIPTION
June 2017	Contract	IBM and SAP	Helse Vest (the Western Norway Regional Health Authority)	Helse entered a USD 42 million contract with IBM to design and implement a wide-ranging finance and logistics support project.
July 2016	Collaboration	IBM	VU University Medical Center Amsterdam	This collaboration was meant to implement analytics-driven IBM MobileFirst for an iOS app. This app is designed to coordinate and communicate data across healthcare professionals with the help of analytics.
June 2016	Agreement	Truven Health Analytics (an IBM Company)	SabiaMed	Truven entered into a multi-year agreement with SabiaMed to resell Truven Health Micromedex evidence-based clinical decision support to customers in Latin America. Along with this development, SabiaMed resold Micromedex CareNotes for patient and consumer education. Such market developments will assist IBM in expanding its customer base in Latin American markets.
February 2016	Acquisition	IBM (US)	Truven Health Analytics (US)	IBM acquired Truven Health Analytics, a provider of cloud-based

healthcare data, analytics, and insights. This acquisition strengthened IBM's product portfolio and customer base.

Source: Press Releases

EXPANSIONS

MONTH & YEAR	LOCATION	DESCRIPTION
April 2017	England, US, and Australia	The company opened four new IBM Cloud data centers—two in London, one in California, and one in Sydney.
April 2016	US	IBM opened four new cloud data centers in the US. The opening of two centers in Dallas, Texas, and two in Washington, D.C., has helped the company expand its footprint in the US.
March 2016	South Africa	IBM Research opened its second research center at the University of the Witwatersrand as part of its 10-year investment program through South Africa's Department of Trade and Industry.
March 2016	Italy	The company established the Watson Health European Center of Excellence that supports the Italian government's initiative to establish an international hub for the advancement of genomics, big data, aging, and nutrition. With this establishment, IBM expanded its operations in Europe.

Source: Press Releases

13.1.7.4 MnM view

IBM is one of the key players in the healthcare BI market and provides a comprehensive range of solutions for this market. IBM's R&D expenditure increased substantially in 2020 as compared to 2019. In 2019, it invested 8.4% of its annual revenue in R&D activities. This significant investment allows the company to offer innovative solutions, such as the Cognos Analytics platform that enables companies to access, analyze, and share business performance data across the organization.

13.1.7.4.1 Right to win

IBM is a prominent player in the healthcare analytics market. The company has a strong presence in over 175 countries across North America, Europe, the Asia Pacific, Latin America, the Middle East, and Africa. The company focuses on increasing its presence in various information technology sectors. IBM is an established player in information technology, which gives the company an upper hand in establishing partnerships, strategic acquisitions, and innovating products in the healthcare domain.

13.1.7.4.2 Strategic choices made

With an aim to increase its geographical footprint in various sectors, IBM has adopted inorganic strategies, such as acquisitions, collaborations, and partnerships. For instance, in June 2017, Helse entered a USD 42 million contract with IBM and SAP to design and implement a wide-ranging finance and logistics support project. Also, IBM launches and upgrades its products regularly to have a competitive edge in this market.

13.1.8 INOVALON

13.1.8.1 Business overview

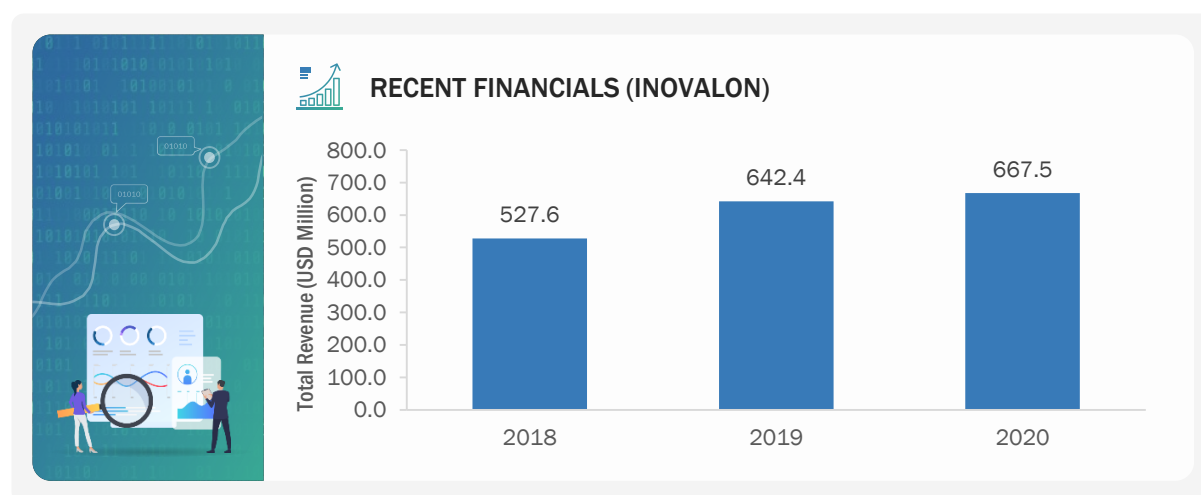
Inovalon (formerly known as MedAssurant, Inc.) is a healthcare technology company that provides advanced cloud-based data analytics and data-driven intervention platforms to the healthcare industry. The company's data analytics and intervention platforms consist of four components—data integration, advanced analytics, intervention platforms, and business processing. The company utilizes superior data analytics with targeted intervention to enhance clinical outcomes and financial performance in the healthcare market. It also offers solutions for clinical and quality outcomes, claims and payment integrity, care coordination and enhancement, and healthcare data insights to physicians, patients, and researchers. Inovalon provides proprietary datasets, sophisticated analytics, and expertise in technology and operations, and it also helps identify gaps in care quality, data integrity, and financial performance. Inovalon primarily operates in the US.

TABLE 258 INOVALON: BUSINESS OVERVIEW

Year Founded	1998
Country	US
City/State	Bowie, Maryland
Ownership	Public

Source: Company Website and Annual Reports

FIGURE 38 INOVALON: COMPANY SNAPSHOT (2020)



Note: The geographic revenue mix and business revenue mix were not mentioned in the annual report of the company. Hence, they have not been included in this section.

Source: Annual Reports, SEC Filings, and Company Website

13.1.8.2 Products & services offered

CATEGORY	PRODUCT/SERVICE	DESCRIPTION	APPLICATION
Clinical and Quality Outcomes	- Natural Language Processing Solutions - Clinical Data Extraction - Cloud-based Healthcare Data Lake - Population Health Insights - Quality Measurement, Reporting & Improvement - ScriptMed Cloud Clinical	- Population Health Management: Integrates and enriches clinical, social determinants of health (SDOH), and national and regional benchmark data to provide health plans with a holistic view of members within a population. - Cloud-Based Healthcare Data Lake solution begins with the integration, aggregation, and normalization of complex, disparate, structured, and unstructured data	Improving clinical and quality outcomes

13.1.8.3 Recent developments

SERVICE LAUNCHES

MONTH & YEAR	SEGMENT	APPROACH	SERVICE
April 2018	Services	Service Launch	Launched Clinical Data Extraction-as-a-Service (CDEaaS) and Natural Language Processing-as-a-Service (NLPaaS) to the Inovalon ONE Platform

Source: Press Releases

Deals

MONTH & YEAR	DEAL TYPE	COMPANY 1	COMPANY 2	DESCRIPTION
November 2018	Engagement/ Agreement	Virginia Premier	Inovalon	Virginia Premier entered into an agreement to implement Inovalon's CDEaaS and NLPaaS within the Inovalon ONE Platform.
April 2018	Acquisition	Inovalon	ABILITY Network	Inovalon acquired ABILITY Network, one of the leading cloud-based providers of real-time connectivity and analytics solutions to payers and providers.
November 2017	Collaboration	Inovalon	University of Maryland Center of Health Information & Decision	As per the agreement, CHIDS will work with Inovalon to accelerate its development of Natural Language Processing

			Systems (CHIDS)	(NLP), Machine Learning (ML), and Deep Learning (DL) solutions for its Inovalon ONE Platform.
January 2017	Partnership	Inovalon	Anthem	Inovalon extended its partnership with Anthem for commercial health insurance and Medicare and Medicaid business.
September 2016	Partnership	Inovalon	MDLIVE	The companies signed an agreement to provide on-demand, real-time patient insights, and analytics.
July 2016	Agreement	Inovalon	athenahealth	The companies partnered to establish interoperability between Inovalon's advanced healthcare data aggregation and analytics platform, athenaNet.

Source: Press Releases

13.1.9 MCKESSON CORPORATION

13.1.9.1 Business overview

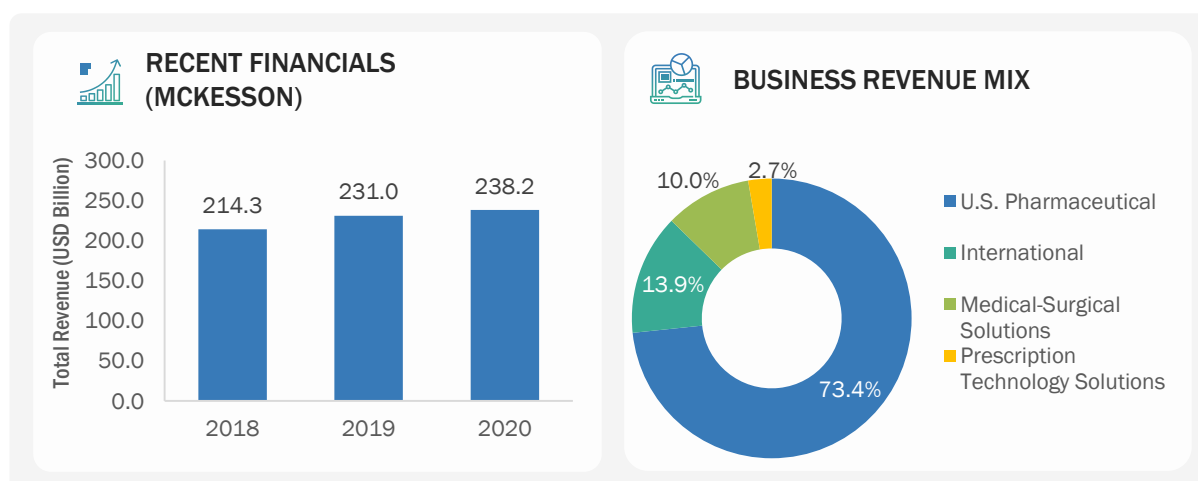
McKesson Corporation offers pharmaceuticals, medical supplies, and HCIT solutions. The company operates through two business segments, namely, Distribution Solutions and Technology Solutions. The Distribution Solutions segment distributes proprietary drugs, medical-surgical supplies and equipment, and beauty products throughout North America and internationally. The Technology Solutions segment offers McKesson health solutions, connected care and analytics, imaging and workflow solutions, business performance services, and enterprise information solutions. It also offers solutions to hospitals, physicians, home care providers, retail pharmacies, and payers, primarily from North America.

The company operates in the US, the UK, Canada, Australia, New Zealand, and Israel. Some of its subsidiaries are US Oncology, McKesson Medical-Surgical US, Northstar Rx US, McKesson Canada, and McKesson Water Products Company US.

TABLE 259 MCKESSON CORPORATION: BUSINESS OVERVIEW

Year Founded	1833
Country	US
City/State	San Francisco, California
Ownership	Public

Source: Company Website

FIGURE 39 MCKESSON CORPORATION: COMPANY SNAPSHOT (2020)

Note: The geographic revenue mix was not provided by McKesson in its annual reports. Hence, this section has not been provided.

Source: Company Website and Annual Reports

13.1.9.2 Products & services offered

CATEGORY	PRODUCT/SERVICE	DESCRIPTION	APPLICATION
Healthcare Analytics	- McKesson Analytics Explorer - McKesson Performance Analytics - McKesson Practitioner Evaluation - McKesson Applied Analytics - McKesson Performance Budgeting - Capacity Benchmark Collaborative - RelayAnalytics Pulse - McKesson Intelligence Hub	These solutions provide actionable insights into clinical performance, improve financial decision support, and help in practitioner evaluation and patient care and safety	Clinical and operational analytics

13.1.9.3 Recent developments

DEALS

MONTH & YEAR	DEAL TYPE	COMPANY 1	COMPANY 2	DESCRIPTION
December 2016	Partnership	McKesson	Infor	Partnered to improve customer choice and aid healthcare organizations in better management by reducing supply chain costs with the integration of McKesson Strategic Supply Sourcing and Infor CloudSuite Healthcare
June 2016	Partnership	McKesson	Change Healthcare	Partnered to form a new HCIT company named Change Healthcare, which focuses on delivering a broad portfolio of solutions that help lower healthcare costs, improve patient access and outcomes, and make it simpler for payers, providers, and consumers to manage the transition to value-based care
February 2016	Agreement	McKesson	HealthQX	This agreement was focused on assisting payers with quick design, scaling complex bundled payment models, and ensuring that these programs provide timely and accurate payment to providers

Source: Press Releases

EXPANSIONS

MONTH & YEAR	LOCATION	DESCRIPTION
March 2017	US	Created a new healthcare information technology company named Change Healthcare
December 2016	US	Opened a new office in Evansville, Indiana, to expand its revenue cycle services operations

Source: Press Releases

PRODUCT LAUNCHES

MONTH & YEAR	SEGMENT	APPROACH	PRODUCT
September 2016	Healthcare Analytics	Product Launch	Launched McKesson Intelligence Hub

Source: Press Releases

13.1.10 MEDEANALYTICS, INC.

13.1.10.1 Business overview

Founded in 1994 and headquartered in California, US, MedeAnalytics (formerly known as Mede Finance) provides cloud-based healthcare analytics solutions to healthcare providers. The company offers a wide range of solutions for executive dashboards, patient access, revenue integrity, ICD-10 compliance, revenue cycle management, performance management, provider engagement, Medicare and Medicaid management, employer reporting, enterprise master patient index, HEDIS quality management, and PHM. It also provides consulting services such as business planning, organizational alignment, PHM, RCM, and healthcare economics for healthcare providers and health plans. The company primarily operates in the US and the UK.

Note: MedeAnalytics, Inc. is a privately held company, and no financial information is available in the public domain. Hence, financial details are not included in this section.

TABLE 260 MEDEANALYTICS, INC.: BUSINESS OVERVIEW

Year Founded	1993
Country	US
City/State	Richardson, Texas
Ownership	Private

Source: Company Website

13.1.10.2 Products & services offered

CATEGORY	PRODUCT/SERVICE	DESCRIPTION	APPLICATION
For Healthcare Providers	- Executive Dashboards - Patient Access - Revenue Integrity - ICD-10 - Compliance - RCM - PHM - Performance Management	These solutions enable the analysis of readmissions, gaps in care, patient safety indicators, patient mortality, hospital-acquired conditions, and potentially preventable procedures—so that providers can easily determine patient compliance	Clinical and operational analytics
For Health Plans	- Provider Engagement - PHM - Medical and Medicaid Management - Employer Reporting - HEDIS Quality Management - Performance Management		
Services	- Consulting Services for Healthcare Providers - Business Planning - Organizational Alignment	These business consulting and planning services for population management are	Operational and business analytics

- Population Health
 - RCM
 - Consulting Services for Health Plans
 - Business Planning
- offered to healthcare providers

13.1.10.3 Recent developments

DEALS

MONTH & YEAR	DEAL TYPE	COMPANY 1	COMPANY 2	DESCRIPTION
June 2018	Partnership	MedeAnalytics	HealthEdge	The partnership aims for integrated quality, advanced analytics, insights, and reporting for population health to HealthRule customers. HealthRule is a product offered by HealthEdge. This partnership helps provide advanced analytics insights to customers who have purchased HealthRule.
November 2016	Partnership	MedeAnalytics	West Tennessee Healthcare	West Tennessee Healthcare extended its partnership with MedeAnalytics to utilize patient access and denials management capabilities.
May 2016		MedeAnalytics	Presbyterian Healthcare Services	This partnership was used to deliver MedeAnalytics' advanced analytics tools to strengthen Presbyterian's integrated care management.

Source: Press Releases

PRODUCT LAUNCHES

MONTH & YEAR	SEGMENT	APPROACH	PRODUCT
June 2016	Revenue Cycle	Product Launch	Launched the Business Office Solution

Source: Press Releases

13.1.11 OPTUM, INC.

13.1.11.1 Business overview

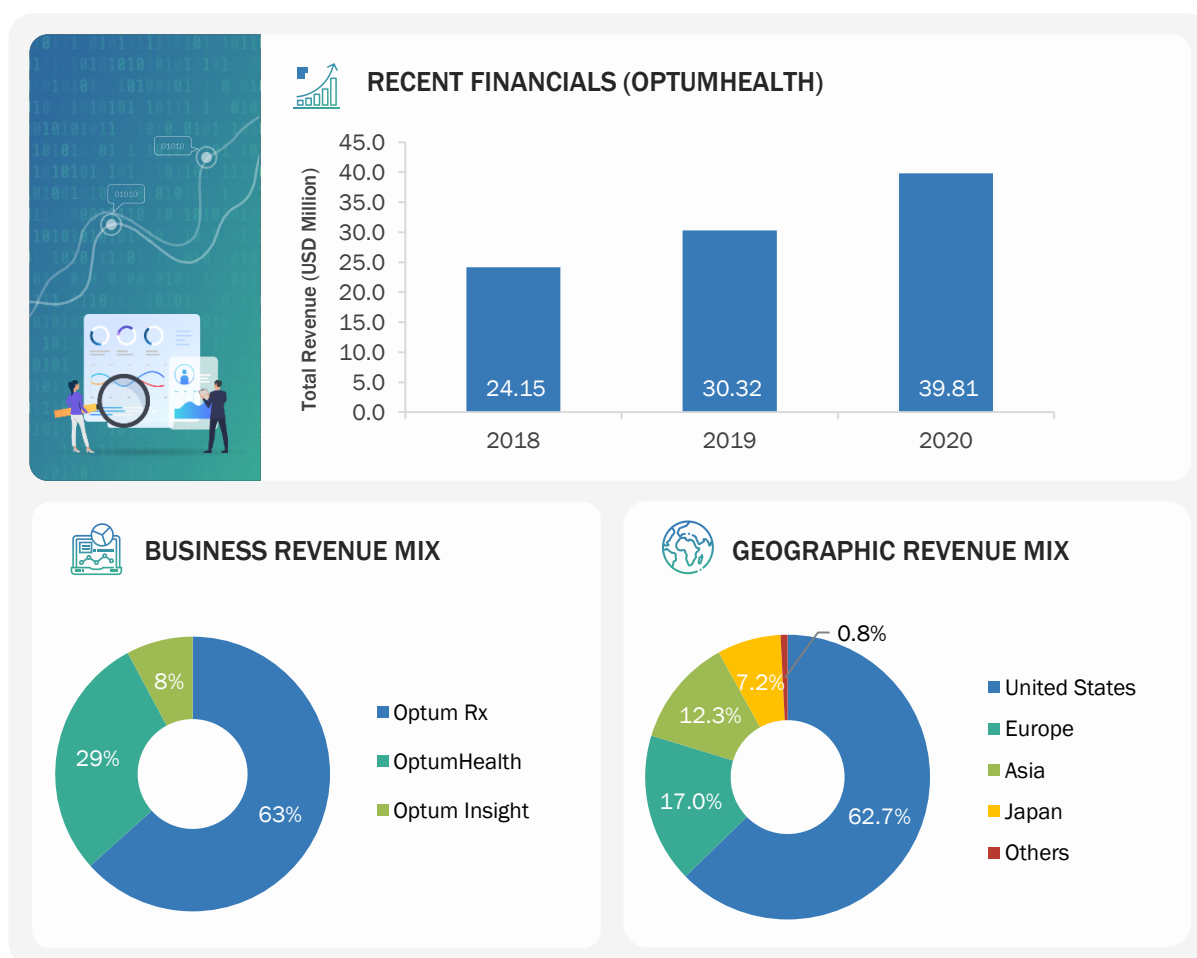
Incorporated in 1977 and headquartered in Minnesota, US, Optum (part of UnitedHealth Group) is a healthcare solutions and services provider. It provides solutions to payers, care providers, employers, governments, and life sciences companies. Optum operates through three reportable segments, namely, Optum Health, OptumInsight, and OptumRx. These segments offer products and solutions for data and analytics, pharmacy care services, population health, health care delivery, and health care operations. The OptumInsight segment provides data and analytics and other HCIT services, software and information products, advisory consulting services, and business process outsourcing to organizations in the healthcare industry. These solutions help healthcare organizations to reduce costs, meet compliance mandates, improve clinical performance, and adapt to the changing healthcare landscape.

Optum has a wide geographical presence across the US, Australia, the UK, Canada, Brazil, Singapore, and India.

TABLE 261 OPTUM, INC.: BUSINESS OVERVIEW

Year Incorporated	1978
Country	US
City	Eden Prairie, Minnesota
Ownership	Public

Source: Company Website

FIGURE 40 OPTUM, INC.: COMPANY SNAPSHOT (2020)

Source: Company Website and Annual Reports

13.1.11.2 Products & services offered

CATEGORY	PRODUCT/SERVICE	DESCRIPTION	APPLICATION
Healthcare Analytics	- Clinical Analytics - Optum One - Population Risk and Reporting - Optum Risk Reporting and Population Management Tools - Actuarial Services - Provider-sponsored Health Plan Analytics - Optum Impact Suite - Provider-sponsored Healthcare Analytics - Optum Symmetry	These clinical analytics solutions enable risk reporting and population health management	Clinical healthcare analytics
	- Optum Claims Connectivity - Physician Claims - Health Systems and Hospitals Claims		
RCM		These are revenue cycle management solutions for claim processing and	Operational and financial analytics

	Electronic Payments and Statements Enrolment	electronic payment	
	Optum 360		
	- RCM Solutions for Hospitals - RCM Solutions for Physicians		
Operations and Administration	- Payment Integrity - Pre-payment Solutions - Post-payment Solutions - Administrative Cost Optimization	These are payment management solutions for improving administrative effectiveness	Operational and financial analytics
Federal and State	- Analytics and Health Information Technology - Business Intelligence - Health Management and Population Analytics - Provider Network Optimization - Health Insurance Marketplace Solutions - Analytics Services - Provider Information Management - Operations and Support - Medicaid Management - Optum Medicaid Business Services - Optum Medicaid and Health Services - Optum Triple Aim Analytic Services	These are business intelligence, Medicaid management, population health management, and healthcare insurance solutions	Clinical and financial analytics

13.1.11.3 Recent developments

DEALS

MONTH & YEAR	DEAL TYPE	COMPANY 1	COMPANY 2	DESCRIPTION
November 2017	Collaboration	Optum	The Advisory Board Company	The Advisory Board Company's research and advisory services came together with Optum's data analytics and technology capabilities to assist healthcare clients in adapting to the changing healthcare landscape.
November 2017	Acquisition	Optum	Advisory Board	Optum acquired Advisory Board's Healthcare Business for USD 1.3 billion. The company's healthcare business unit will complement Optum's products and services for the benefit of healthcare providers, thus enabling it to expand its presence in new markets.

September 2016	Collaboration	UnitedHealth Group	The University of California (UC)	UC Health's five academic medical centers expanded the use of Optum's clinically integrated network services and advanced data analytics services.
September 2016	Partnership	Optum	Quest Diagnostics	This partnership increased the use of diagnostic information services, such as data analytics, population health insights, and connectivity solutions to help improve healthcare effectiveness and manage costs for health plans and care providers.
January 2016	Partnership	Optum	Availity	This partnership helped improve the completeness and accuracy of health insurance claims and reduce claims-processing errors and appeals.

Source: Press Releases

13.1.11.4 MnM view

13.1.11.4.1 Right to win

The sales for Optum Health increased to USD 39.81 million in 2020 from USD 30.32 million in 2019. The increase in revenues was primarily driven by the increase in the number of individuals served, pricing trends, and organic and acquisition growth across the Optum business. Increases were partially offset by decreased individuals served through certain voluntary customer assistance programs and rebate requirements. Revenues were also negatively impacted by decreases in fee-for-service care delivery and other volume-based businesses, primarily due to care deferral and the economic impact of the COVID-19 pandemic.

13.1.11.4.2 Strategic choices made

The company mainly focuses on organic growth strategies, such as product launches, to maintain its presence in the healthcare analytics market. In July 2021, the company increased its focus on predictive analytics solutions to help improve well-being, lower costs, and drive engagement in clinical intervention programs by addressing social determinants of health for people in some employer-sponsored benefit plans. The company also adopts inorganic growth strategies such as partnerships and collaborations. In December 2020, UnitedHealthcare, a part of Optum, partnered with Apple, Inc. (US) to provide its members with access to Apple Fitness + for six months. Such initiatives help the company create a strong customer base and improve its brand visibility.

13.1.12 ORACLE CORPORATION

13.1.12.1 Business overview

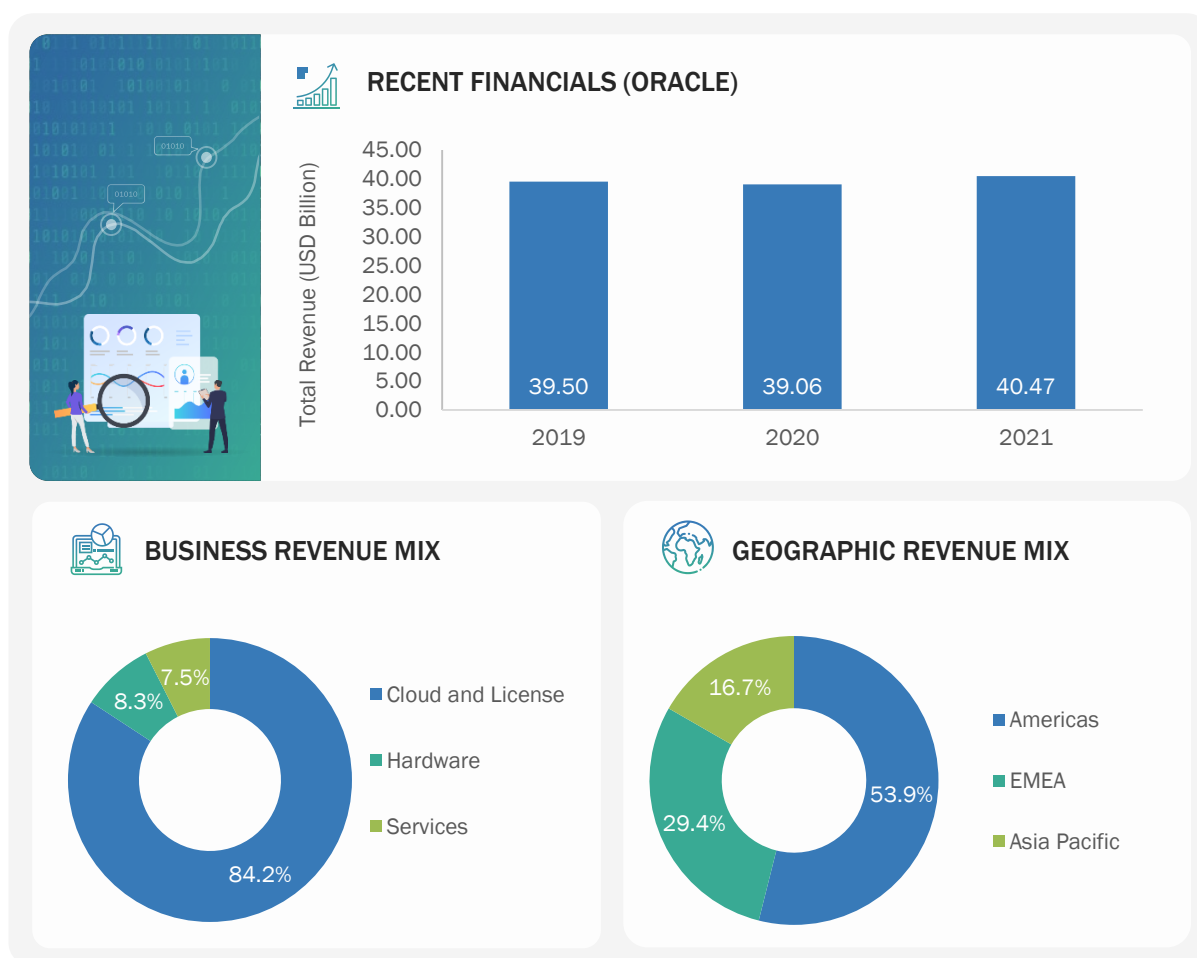
Oracle is a software and hardware systems company that operates through three business segments, namely, Cloud and On-premise Software Business, Hardware Business, and Service Business. The company develops, manufactures, markets, and maintains Oracle database and middleware software, application software, cloud infrastructure, and hardware systems. Its healthcare product portfolio comprises data discovery software, enterprise performance management software, analytic applications software, supply chain software, business intelligence software, and predictive analytics and self-learning decision optimization software.

Oracle's geographical operations are spread across Europe, the Middle East, Africa, the Asia Pacific, North America, and South America. Some of its major subsidiaries are Logical Applications US, Oracle Argentina, G-Log, Oracle Canada, BEA Technology Beijing, Oracle India, Oracle Financial Services Software India, I-flex Solutions Singapore, Siebel Systems US, and Virtual Iron Software US.

TABLE 262 ORACLE CORPORATION: BUSINESS OVERVIEW

Year Founded	1977
Country	US
City	Santa Clara, California
Ownership	Public

Source: Company Website

FIGURE 41 ORACLE CORPORATION: COMPANY SNAPSHOT (2020)

Note: EMEA refers to Europe, Middle East, and Africa

Source: Company Website and Annual Reports

13.1.12.2 Products & services offered

CATEGORY	PRODUCT/SERVICE	DESCRIPTION	APPLICATION
Analytics Applications	Healthcare	These completely integrated solutions encompass data governance and pre-built analytics	Clinical and operational analytics
	Oracle Enterprise Healthcare Analytics		
	Oracle Clinical Development Analytics		
	Oracle Operating Room Analytics		
	Health Sciences		
	Oracle Health Sciences Clinical Development Analytics		
	Oracle Argus Analytics		
Advanced Analytics	Oracle Advanced Analytics	These are advanced analytics solutions for decision support and workflow management	Clinical and operational analytics
	Oracle Real-time Decisions		

Oracle Exalytics	- Oracle Exalytics In-memory Machine - Oracle WebCenter Portal - Oracle's Siebel CRM - Oracle Hyperion Performance Management - Oracle Enterprise Performance Management and Business Intelligence	These products have an open, integrated software infrastructure that is flexible enough to enable interoperability with various other analytics sources	Business intelligence analytics
Clinical Analytics	- Oracle Healthcare Analytics Data Integration - Oracle Enterprise Healthcare Analytics Solution - Oracle Healthcare Data Warehouse Foundation	These clinical analytics solutions offer data integration and analysis for healthcare providers	Clinical analytics
Human Resources Management	HR Analytics	This healthcare HR analytics solution streamlines operational efforts	Operational analytics
Financial Management	- Oracle Financial Analytics - Financial Control and Reporting - Oracle Financial Services Revenue Management and Billing Analytics	These financial analytics solutions offer billing and financial reporting and control	Financial analytics

13.1.12.3 Recent developments

PRODUCT LAUNCHES AND UPGRADES

MONTH & YEAR	SEGMENT	APPROACH	PRODUCT
February 2017	Cloud	Product Launch	Launched the Oracle Cloud Platform
February 2017	Cloud	Product Launch	Introduced a breadth of services through Oracle Cloud
March 2016	Healthcare Analytics	Product Upgrade	Introduced an upgraded version of Oracle Enterprise healthcare analytics

Source: Press Releases

13.1.13 SAS INSTITUTE INC.

13.1.13.1 Business overview

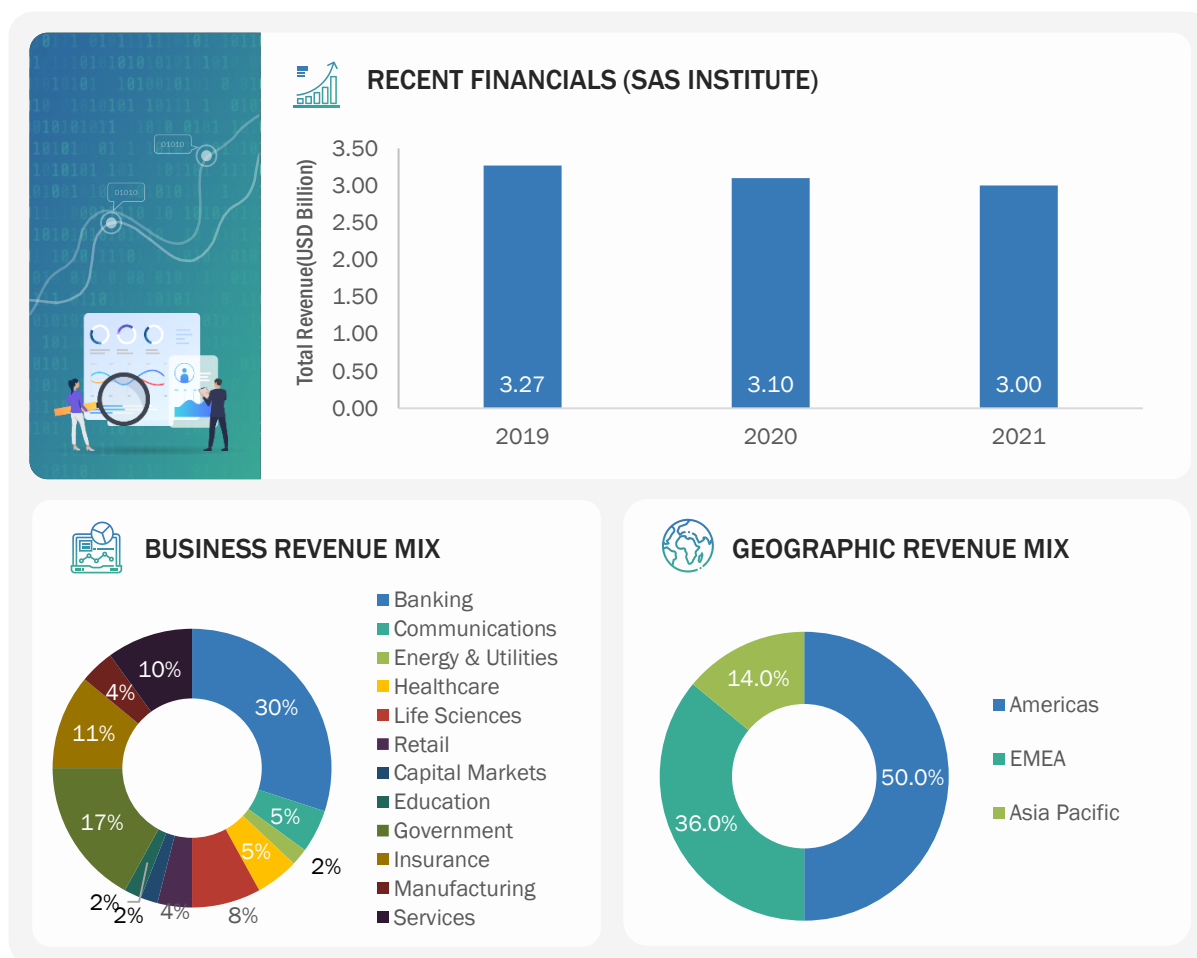
SAS Institute is a business analytics software and services company. It offers solutions such as advanced analytics, business intelligence and analytics, cloud analytics, customer intelligence, data management, decision management, fraud and security intelligence, in-memory analytics, performance management, risk management, and supply chain management. These solutions help access, manage, analyze, and report on data that support the decision-making process. SAS Institute offers its products to various industries, including communications, education, financial services, government, health insurance, healthcare, hotels, insurance, life sciences, manufacturing, media, oil and gas, retail, travel and transportation, and utilities.

SAS Institute has a wide geographical presence in the Americas, the Asia Pacific, Europe, the Middle East, and Africa. The company has its customers in 145 countries at more than 80,000 business, government, and university sites. Accenture, Capgemini, EMC, Oracle, IBM, and Teradata are some of the featured partners of SAS.

TABLE 263 SAS INSTITUTE INC.: BUSINESS OVERVIEW

Year Founded	1976
Country	US
City/State	Cary, North Carolina
Ownership	Public

Source: Company Website

FIGURE 42 SAS INSTITUTE INC.: COMPANY SNAPSHOT (2020)

Note: EMEA refers to Europe, Middle East, and Africa

Source: Company Website and Annual Reports

13.1.13.2 Products & services offered

CATEGORY	PRODUCT/SERVICE	DESCRIPTION	APPLICATION
Healthcare Insurance	- Advanced Analytics - SAS Enterprise Miner - SAS Visual Data Mining and Machine Learning - SAS Customer Intelligence - Data Management Software - SAS Detection and Investigation for Healthcare - SAS Enterprise Case Management - SAS Episode Analytics - SAS Visual Analytics	These financial analytics solutions provide visual analytics, data management, machine learning tools, and customer intelligence	Financial analytics
Healthcare Analytics	- AI Solutions for Healthcare - SAS/OR Software - SAS Enterprise Miner	These clinical analytics solutions provide visual analytics, data management,	Clinical analytics

CATEGORY	PRODUCT/SERVICE	DESCRIPTION	APPLICATION
	- SAS Episode Analytics - SAS Optimization - SAS Real-world Evidence - SAS Visual Data Mining and Machine Learning - SAS Visual Forecasting - SAS Visual Text Analytics	machine learning tools, and customer intelligence	
	- Advanced Analytics - SAS Visual Analytics - SAS Contextual Analytics - SAS Data Quality - SAS Enterprise Case Management - SAS Enterprise Guide - SAS Real-time decision manager - Data Management Software - SAS Detection and Investigation for Health Care - SAS Episode Analytics - SAS for Procurement Integrity		
Services	- Consulting - SAS Assessment - SAS Enablement - SAS Solution Implementation - SAS Value Management	These business and software implementation consulting services are provided to healthcare providers and payers	Operational and financial analytics

13.1.13.3 Recent developments

PRODUCT LAUNCHES

MONTH & YEAR	SEGMENT	APPROACH	PRODUCT
January 2018	Healthcare Analytics	Product Launch	Launched SAS Viya
October 2017	Healthcare Analytics	Product Launch	Introduced AI capabilities in its existing SAS Viya and SAS Platform

Source: Press Releases

DEALS

MONTH & YEAR	DEAL TYPE	COMPANY 1	COMPANY 2	DESCRIPTION
June 2018	Agreement	Prime Therapeutics LLC	SAS	Prime entered into an agreement with SAS to deploy SAS analytics software to identify FWA and recover overpayments due to fraudulent claims.
2017	Partnership	SAS	International City/Country Management Association (ICMA)	SAS and ICMA renewed their partnership to promote the use of analytics by local governments in areas such as public health, utility management, fleet maintenance, and law enforcement.
2016	Agreement	SAS	Teradata	As per this agreement, Teradata is the global authorized reseller of SAS data management and analytics software

Source: Press Releases

EXPANSIONS

MONTH & YEAR	LOCATION	DESCRIPTION
April 2018	US	SAS created a new division to fight fraud globally. The newly established division will focus and improve the SAS fraud and cybersecurity portfolio with the help of AI and advanced analytics solutions.

Source: Press Releases

13.1.14 SCIO HEALTH ANALYTICS (AN EXL COMPANY)

13.1.14.1 Business overview

Founded in 2007 and headquartered in Connecticut, US, SCIO Health is a provider of analytics solutions to payers, providers, healthcare service companies, and life science companies. SCIO specializes in the areas of population health, payment integrity, risk adjustment, and care management, opportunity analysis and incentive design, consumer segmentation and engagement, network performance, and commercial effectiveness. SCIO also provides commercial analytics services, including patient-centric analytics and insights, target setting, forecasting, and resource deployment.

SCIO has a total of six delivery centers in the US, India, and the UK. The company serves its clients in 30 countries, which include over 80 healthcare organizations (including 20 provider groups) and 8 of the top 15 global pharmaceutical companies.

Note: SCIO Health is a privately held company, and no financial information is available in the public domain. Hence, financial details are not added in this section.

TABLE 264 SCIO HEALTH ANALYTICS: BUSINESS OVERVIEW

Year Founded	2007
Country	US
City/State	West Hartford, Connecticut
Ownership	Private

Source: Company Website

13.1.14.2 Products & services offered

CATEGORY	PRODUCT/SERVICE	DESCRIPTION	APPLICATION
Care & Network Optimizations	- SCIOVantage - SCIOClarity Analytics - SCIOClarity	These are clinical analytics solutions for healthcare providers	Clinical analytics
Reimbursement Optimizations	- MY SOCRATES - SCIOMine - SCIOClarity RAPS Error Correction - SCIOClarity Inpatient Data Pursuit	These reimbursement claims management solutions for healthcare providers and payers help access inpatient and outpatient data	Financial analytics
Services	- SCIOXpert Consulting Services - Data Integration & Management - Medical Pharmacy & Claims Auditing - Prepayment Auditing - Chart Validation - SCIOClarity Inpatient Data Pursuit - SCIOClarity Member Outreach & Provider Collaboration	These are consulting, payment, auditing, and claims management solutions for healthcare payers and providers	Business and operational analytics

13.1.14.3 Recent developments

DEALS

MONTH & YEAR	DEAL TYPE	COMPANY 1	COMPANY 2	DESCRIPTION
May 2018	Acquisition	EXL	SCIOInspire, Corp.	EXL acquired SCIOInspire, Corp., a leading healthcare analytics solutions and services company. This acquisition will help EXL to strengthen its presence in the healthcare analytics market.
January 2016	Acquisition	SCIOInspire, Corp.	Clear Vision Information Systems	SCIOInspire, Corp. acquired Clear Vision Information Systems, a provider of analytics solutions and services for health plans and providers.

Source: Press Releases

PRODUCT/SERVICE LAUNCHES AND UPGRADES

MONTH & YEAR	SEGMENT	APPROACH	PRODUCT/SERVICE
February 2018	Care Network & Optimization	Product Upgrade	Upgraded SCIOVantage
October 2017	Care Network & Optimization	Product Upgrade	Upgraded SCIOVantage
June 2017	Services	Service Launch	Launched SCIOClarity Chart Validation
June 2017	Services	Service Upgrade	Expanded prepayment auditing services for a diagnosis-related group (DRG) and DRG Short Stay audit concepts

Source: Press Releases

13.1.15 VITREOSHEALTH, INC.

13.1.15.1 Business overview

Founded in 2009 and headquartered in Texas, US, VitreosHealth is a global provider of big data and healthcare analytics solutions and services. The company offers the first IaaS delivery model for advanced population health analytics. VitreosHealth leverages big data and predictive and prescriptive health insights for population risk models that provide a complete view of the healthcare industry.

VitreosHealth has offices in the US and India. Through these offices, it serves Fortune 500 clients in the US, the UK, Australia, the UAE, Malaysia, and India.

Note: VitreosHealth, Inc. is a privately held company, and no financial information is available in the public domain. Hence, financial details are not added in this section.

TABLE 265 VITROSHEALTH, INC.: BUSINESS OVERVIEW

Year Founded	1992
Country	US
Province/City	Drive Plano, Texas
Ownership	Private

Source: Company Website

13.1.15.2 Products & services offered

CATEGORY	PRODUCT/SERVICE	DESCRIPTION	APPLICATION
Population Health	- Care Management Insights Application - Vitreous Digital Outreach Engagement Solution - Performance Application - HCC Predictive Risk Adjustment Solution - ETL-as-a-Service	These AI-driven healthcare data enrichment services help providers and payers who are struggling with reconciling data from multiple sources to create a sole source of for population management and engagement solutions	Population health management

13.1.15.3 Recent developments

DEALS

MONTH & YEAR	DEAL TYPE	COMPANY 1	COMPANY 2	DESCRIPTION
December 2016	Partnership	VitreosHealth	CitiusTech	VitreosHealth leveraged CitiusTech's cloud-hosted clinical rules engine.
October 2016	Partnership	VitreosHealth	PrecisionBI	PrecisionBI partnered with VitreosHealth to leverage the latter's Advanced Population Health Analytics solutions.

Source: Press Releases

PRODUCT/SERVICE LAUNCHES

MONTH & YEAR	SEGMENT	APPROACH	PRODUCT/SERVICE
November 2017	Service	Service Launch	Launched the AI-driven ETL-as-a-Service for the healthcare market
August 2017	Service	Service Launch	Launched the AI-driven Digital Member Engagement Service
July 2016	Service	Product Launch	Launched new predictive risk models

Source: Press Releases

13.1.16 WIPRO

13.1.16.1 Business overview

Wipro is a global information technology, consulting, and outsourcing company. The company offers its products to various industries through different segments such as banking, financial services and insurance (BFSI), healthcare and life sciences, retail, consumer, transport and government, energy, natural resources and utilities, manufacturing, and global media and telecom. Through its healthcare and life sciences segment, the company offers manufacturing & supply chain; R&D; risk, safety, compliance; and sales & marketing solutions and services. The solutions offered by Wipro include social, mobility, analytics, and cloud-powered offerings.

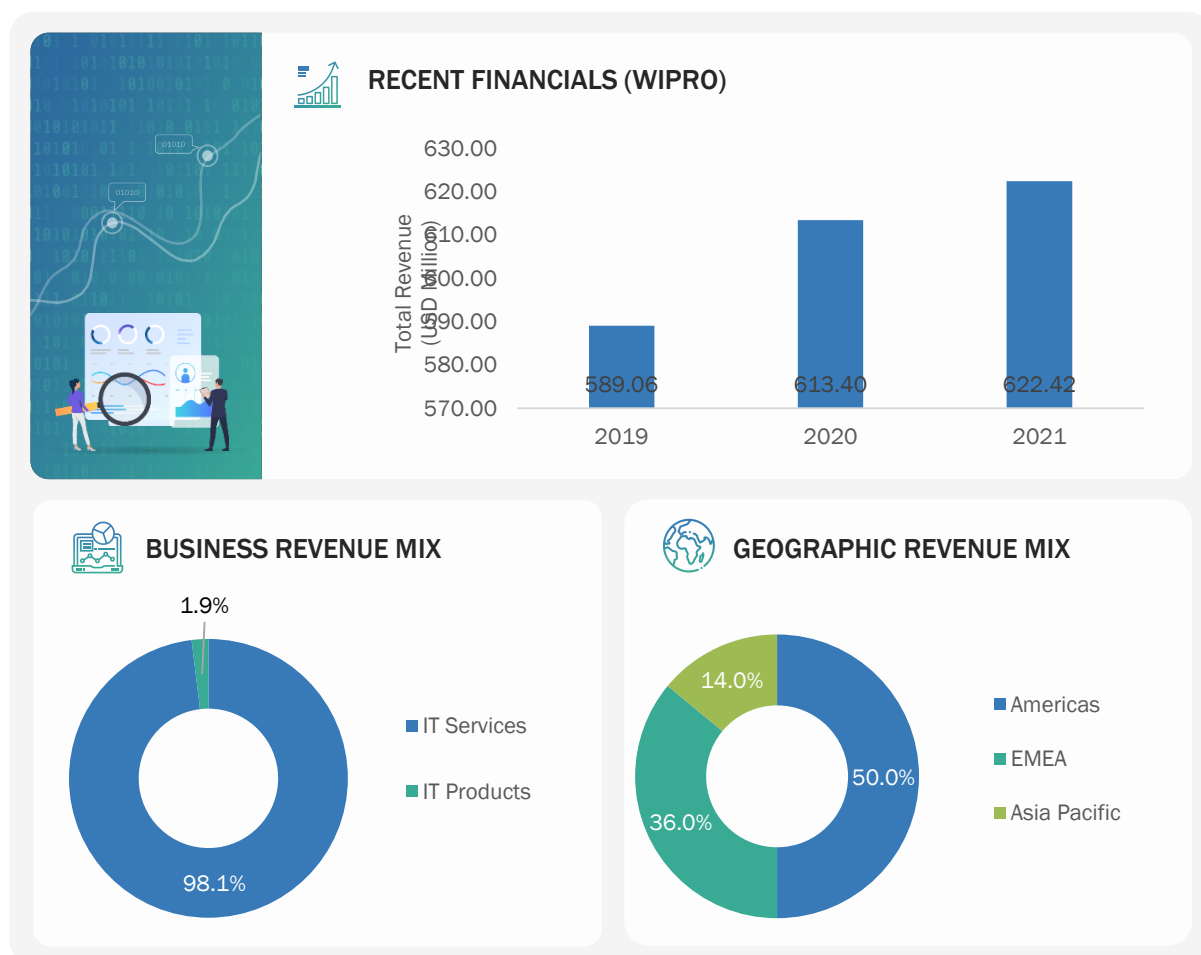
The company has a wide geographical presence spanning the Americas, Europe, the Asia Pacific, and the Middle East. Wipro serves its clients in over 175 cities across six continents.

TABLE 266 WIPRO: BUSINESS OVERVIEW

Year Founded	1945
Country	India
Province/City	Bangalore, Karnataka
Ownership	Public

Source: Company Website

FIGURE 43 WIPRO: COMPANY SNAPSHOT (2020)



13.1.16.2 Products & services offered

CATEGORY	PRODUCT/SERVICE	DESCRIPTION	APPLICATION
Analytics	Insurance Analytics	This is a suite of analytics solutions that offer clinical analytics, finance analytics, and fraud detection	Financial and clinical analytics
	Analytics for Healthcare & Life Sciences		
	Fraud Detection		
Services	Big Data Analytics	Supply chain and operating analytics are provided as a service offering	Supply chain analytics and operational analytics
	Supply Chain Analytics		
	Operational Analytics		

13.1.16.3 Recent developments

DEALS

MONTH & YEAR	DEAL TYPE	COMPANY 1	COMPANY 2	DESCRIPTION
September 2017	Partnership	Wipro	McLaren Technology Group	Wipro became the official technology provider to McLaren Technology Group to help drive digitalization across its businesses.
September 2017	Acquisition	Wipro	Appirio	This acquisition will help Wipro become the world's largest cloud transformation practices company.
September 2016	Partnership	Wipro	Reltio	This partnership was focused on developing faster and cost-effective Master Data Management (MDM) solutions.
May 2016	Partnership	Wipro	Qlik	Wipro partnered with Qlik to offer advanced enterprise analytics solutions to joint customers across industry sectors.
February 2016	Acquisition	Wipro	HealthPlan Services	Wipro acquired HealthPlan Services, an independent technology and Business Process-as-a-Service (BPaaS) provider in the US health insurance market.

Source: Press Releases

PRODUCT LAUNCHES

MONTH & YEAR	SEGMENT	APPROACH	PRODUCT
July 2016	Analytics	Product Launch	Launched a data discovery platform

Source: Press Releases

EXPANSIONS

MONTH & YEAR	LOCATION	DESCRIPTION
November 2018	Portugal	Established its new office in Porto to expand its European Operations and meet the growing customer requirements in Portugal
March 2018	US	Established the Texas Technology Center in Plano
August 2017	California	Launched the Silicon Valley Innovation Center

Source: Press Releases

13.2 OTHER PLAYERS

13.2.1 FLATIRON

13.2.1.1 Business overview

Flatiron is one of the key emerging players in the healthcare analytics market. The company was established in 2012 and is focused on using RWD to improve treatment opportunities for cancer patients. The company offers its services and products to life science researchers and hospitals. Looking at the potential of the company, Roche acquired Flatiron for USD 1.9 billion in 2018. However, the company continues to operate as a subsidiary of Roche.

Flatiron currently operates in the US and has partnered with over 270 community cancer centers and 14 oncology pharma companies to use healthcare analytics to improve cancer research.

Note: Flatiron is a privately held company, and no financial information is available in the public domain. Hence, financial details are not added in this section.

TABLE 267 FLATIRON: BUSINESS OVERVIEW

Year Founded	2012
Country	US
City/State	New York
Ownership	Private

Source: Company Website

13.2.1.2 Products & services offered

CATEGORY	PRODUCT/SERVICE	DESCRIPTION	APPLICATION
Life Science	Flatiron HC	Flatiron HC is a platform that connects patient data and with RWE for advanced research	Comparison of real-world data
Academia and Hospitals	- Flatiron Assist - OncoEMR - OncoAir - OncoBilling - OncoAnalytics - OncoTrials	- Flatiron Assist offers a pool of data that physicians can refer to and select evidence-based treatment options. - OncoEMR is an oncology-specific workflow management system - OncoAir is a mobile application that helps access patient data from OncoEMR - OncoBilling is an integrated system of OncoEMR - OncoAnalytics offers a dashboard design that provides clinicians with actionable insights for oncology treatment - OncoTrials is a tool that is built for clinical research and the management of clinical trial data	Oncology

13.2.1.3 Recent developments

DEALS

MONTH & YEAR	DEAL TYPE	COMPANY 1	COMPANY 2	DESCRIPTION
June 2021	Partnership	Flatiron	Foundation Medicine, Inc.	Partnered with Foundation Medicine, Inc. to integrate its solutions with OncoEMR to offer integrated genomic profiling to customers
July 2021	Partnership	Flatiron	NICE	This research partnership will leverage Flatiron's OncoEMR data to explore RWE to help evaluate the cost-effectiveness and clinical efficacy of oncology treatments
May 2021	Partnership	Flatiron	Genentech and Foundation Medicine	Partnered with Foundation Medicine and Genentech to launch a novel prospective lung cancer clinical study
September 2019	Partnership	Flatiron	Pfizer, Inc.	Expansion of this strategic partnership will help Pfizer evaluate real-world data sets and source expertise from Flatiron for several clinical development projects
February 2019	Partnership	Flatiron	US FDA	Expansion of this collaboration would enable the FDA to better understand RWE derived from de-identified patient datasets curated from the electronic health records of Flatiron
January 2019	Acquisition	Flatiron	American Family Archives & Chronicles, Inc.	The acquisition of AFAC will help improve Flatiron's real-world mortality data and would benefit its product offerings
May 2018	Partnership	Flatiron	Bristol-Myers Squibb	This three-year partnership agreement was signed to enable Bristol-Myers Squibb to use Flatiron's RWD to accelerate its research and development efforts
February 2018	Acquisition	Roche	Flatiron	Roche acquired Flatiron to improve its research and development efforts in treating cancer patients

Source: Press Releases

PRODUCT LAUNCHES

MONTH & YEAR	SEGMENT	APPROACH	PRODUCT
November 2019	Analytics	Product Launch	Launched Flatiron Assist

Source: Press Releases

13.2.2 APIXIO

13.2.2.1 Business overview

Apixio was founded in 2009 when the federal government was pushing the universal adoption of digital health records through its Meaningful Use initiative. The company's first product was created to make use of the unstructured text being recorded in new medical records by enabling a physician to search and find information hidden away in these systems easily. Headquartered in California, Apixio is a privately held company. The company offers products in financial and clinical analytics segments.

Note: Apixio is a privately held company, and no financial information is available in the public domain. Hence, financial details are not added in this section.

TABLE 268 APIXIO: BUSINESS OVERVIEW

Year Founded	2009
Country	US
City/State	San Mateo, California
Ownership	Private

Source: Company Website

13.2.2.2 Products & services offered

CATEGORY	PRODUCT/SERVICE	DESCRIPTION	APPLICATION
Risk Adjustment Solutions	HCC Identifier	HCC Identifier creates an accurate risk score for payment adjustment and care management with the least amount of time and effort	Financial analytics
	HCC Auditor	HCC Auditor ensures that reported risk adjustment diagnoses have the required supporting evidence in linked clinical encounters.	Financial analytics
Infostream	Infostream Epic App Orchard	Using these products, organizations can gain direct access to patient and administrative data from EHR platforms	Electronic health management
Quality Identifier		The Quality Identifier solution uses proprietary AI algorithms to locate relevant clinical facts in medical charts. These facts are presented in an application where abstractors can easily confirm the evidence to support predicted facts, significantly reducing the amount of time needed to gather clinical evidence for quality reporting	Clinical analytics

13.2.2.3 Recent developments

PRODUCT/SERVICE LAUNCHES

MONTH & YEAR	SEGMENT	PRODUCT/SERVICE	APPROACH	DESCRIPTION
June 2021	Infostream	Infostream-AI	Product Launch	Launched additional functionalities in the Infostream healthcare interoperability platform
May 2020	Accurate Risk Adjustment	HCC Complete	Product Launch	Launched HCC Complete, which uses AI to support a comprehensive chart review experience for Medicare Advantage and Affordable Care Act organizations
September 2019	Quality Identifier	Quality Identifier	Product Launch	Launched Quality Identifier, a solution that allows health plans and providers to unlock critical patient insights for better quality measurement and care delivery
July 2018	Audit Solution	HCC Auditor	Product Launch	Launched HCC Auditor, which helps health plans and provider organizations to undertake internal audits of their risk adjustment payment data with increased accuracy and a shorter time frame than traditional methods

Source: Press Releases

DEALS

MONTH & YEAR	DEAL TYPE	COMPANY 1	COMPANY 2	DESCRIPTION
November 2020	Acquisition	Centene	Apixio	Centene announced its plans to acquire Apixio. Post the acquisition, Apixio will continue to operate independently.

Source: Press Releases

13.2.3 ENLITIC, INC.

13.2.3.1 Business overview

Enlitic, Inc. is a healthcare AI-based company that develops non-clinical and clinical AI-driven workflow solutions for radiologists for the improvement of disease diagnosis. Enlitic provides high-quality software-as-a-medical-device to radiologists, with a commitment to meet or exceed applicable regulatory, statutory, and quality requirements. The company's Enlitic Curie is catered to analyze chest radiograph images for physiological anomalies. The company is headquartered in California and operates predominantly in North America and Europe.

Note: Enlitic is a privately held company, and no financial information is available in the public domain. Hence, financial details are not added in this section.

TABLE 269 ENLITIC, INC.: BUSINESS OVERVIEW

Year Founded	2014
Country	US
City/State	San Francisco, California
Ownership	Private

Source: Company Website

13.2.3.2 Products & services offered

CATEGORY	PRODUCT/SERVICE	DESCRIPTION	APPLICATION
Cognitive Analytics	Enlitic Curie	This platform automatically standardizes data and provides a host of AI-powered solutions for reading, reporting, QA, and research in radiology workflow	Analysis of chest X-ray images

13.2.3.3 Recent developments

DEALS

MONTH & YEAR	DEAL TYPE	COMPANY 1	COMPANY 2	DESCRIPTION
December 2019	Partnership	Enlitic	MLP Care	Partnered to bring clinical AI solutions to Turkey and neighboring countries in East Europe and Central Asia
July 2021	Partnership	Enlitic	NMC Healthcare	This is a multi-faceted partnership with private healthcare companies to incorporate Enlitic's solutions in GCC countries

Source: Press Releases

13.2.4 KOMODO HEALTH, INC.

13.2.4.1 Business overview

Komodo Health, Inc. is an upcoming healthcare analytics company focused on delivering patient-level insights through a wide array of collected data. The Komodo Health software platform provides access to a broader array of healthcare stakeholders in over fifty therapeutic areas. The platform has information about patient encounters and software solutions for various professionals in the healthcare field, such as those in pharmaceutical marketing, research and development, and account directors. Komodo Health, Inc. has a strong focus on securing investor funding to develop innovative solutions. For instance, in March 2021, the company secured USD 220 million in Series E funding led by Tiger Global.

Note: Komodo Health, Inc. is a privately held company, and no financial information is available in the public domain. Hence, financial details are not added in this section.

TABLE 270 KOMODO HEALTH, INC.: BUSINESS OVERVIEW

Founded	2014
Country	US
City/State	San Francisco, California
Ownership	Private

Source: Company Website

13.2.4.2 Products & services offered

CATEGORY	PRODUCT/SERVICE	DESCRIPTION	APPLICATION
Healthcare Analytics	Healthcare Map	This product comprises data on patient encounters and algorithms to provide clinical expertise	Healthcare analytics

Note: No significant developments were reported by the company in the healthcare analytics market between January 2016 and August 2021.

13.2.5 HEALTHEC, LLC

13.2.5.1 Business overview

HealthEC, LLC is a population health technology company. The product offerings of the company aggregate claims, clinical, and SDoH data from multiple sources and deliver key insights into the cost drivers and quality measures that impact performance-based reimbursements. The company provides solutions and services to hospitals, payers, as well as government agencies such as State and Country Health Departments.

Note: HealthEC, LLC is a privately held company, and no financial information is available in the public domain. Hence, financial details are not added in this section.

TABLE 271 HEALTHEC, LLC: BUSINESS OVERVIEW

Year Founded	2012
Country	US
City	Edison, New Jersey
Ownership	Private

Source: Company Website

13.2.5.2 Products & services offered

CATEGORY	PRODUCT/SERVICE	DESCRIPTION	APPLICATION
Healthcare Analytics	3D Analytics	This analytics platform accesses and visualizes data with point-and-click reporting and customizable, drag-and-drop dashboards	Healthcare analytics

13.2.5.3 Recent developments

PRODUCT LAUNCHES

MONTH & YEAR	SEGMENT	PRODUCT/SERVICE	APPROACH	DESCRIPTION
August 2018	Healthcare Analytics	HealthEC Opioid Analytics	Product Launch	HealthEC Opioid Analytics integrates with MMIS systems and combines powerful data analytics and care management tools to provide a data-driven analysis of opioid consumption and use.

Source: Press Releases

DEALS

MONTH & YEAR	DEAL TYPE	COMPANY 1	COMPANY 2	DESCRIPTION
September 2018	Acquisition	Limeade	Sitrion	Acquired Sitrion and its employee experience solution, Sitrion ONE
May 2018	Partnership	Limeade	Optisom	Partnered with Optisom to allow Limeade's users access to Optisom's sleep solution—ProjectZ platform
May 2018	Collaboration	Limeade	Microsoft Corporation	Collaborated with Microsoft Corporation to drive future initiatives relating to enhancing employee experiences by leveraging MicrosoftGraph to integrate data insights from Microsoft365

Source: Press Releases

13.2.6 SEMA4

Founded	2017
Headquarters	Connecticut, US
Business Overview	Sema4 is a patient-centered health intelligence company that focuses on advancing healthcare through data-driven insights. The company develops and markets a unique advanced analytics platform to build predictive models of human health and deliver personalized insights for patients in the areas of women's health and precision oncology, as well as more generalized genomic health screening and modeling across the spectrum of diseases.
Products & Services Offered	Centrelils
Geographic Presence	North America

Source: Company Website

13.2.7 EVIDATION HEALTH, INC.

Founded	2012
Headquarters	California, US
Business Overview	Evidation Health is a technology and services company that helps individuals and healthcare companies understand and influence everyday behaviors that create better health outcomes. Evidation pairs its discovery engine designed for better, faster, and more efficient studies with a delivery platform, Achievement, that connects individuals and the healthcare industry, including life sciences organizations, providers, payers, and digital health companies.
Products Offered	Evidation Platform
Geographic Presence	Global

Source: Company Website

13.2.8 INSILICO MEDICINE

Founded	2014
Headquarters	Hong Kong, China
Business Overview	Insilico Medicine pioneered the applications of generative adversarial networks (GANs) and reinforcement learning for the generation of novel molecular structures for diseases with a known target and with no known targets. In addition to working collaborations with large pharmaceutical companies, the company is pursuing internal drug discovery programs in cancer, dermatological diseases, fibrosis, Parkinson's Disease, Alzheimer's Disease, ALS, diabetes, sarcopenia, and aging. Through a partnership with LifeExtension.com, the company launched a range of nutraceutical products compounded using advanced bioinformatics techniques and deep learning approaches. It also provides a range of consumer-facing applications, including Young.AI.
Products & Services Offered	▪ PHARMA.AI • PandaOmics • Chemistry42 • InClinico
Geographic Presence	North America, Europe, and Asia

Source: Company Website

13.2.9 EXSCIENTIA

Founded	2012
Headquarters	Oxford, UK
Business Overview	Exscientia is an AI-based company that specializes in automated drug design. The company focuses on implementing AI design as well as precision data generation. The company's knowledge-driven systems design millions of novel, project-specific compounds and pre-assesses them for predicted potency, selectivity, ADME, and other key criteria.
Products & Services Offered	▪ CentaurAI • CentaurBiologist • CentuarChemist ▪ ExperimentAI
Geographic Presence	North America and Europe

Source: Company Website

13.2.10 AETION, INC.

Founded	2012
Headquarters	New York, US
Business Overview	The Aetion Evidence Platform delivers real-world evidence for life sciences companies, payers, and at-risk providers. Aetion is based in New York City and backed by several investors, including New Enterprise Associates (NEA), Flare Capital Partners, Lakestar, and Town Hall Ventures.
Products & Services Offered	Aetion Evidence Platform
Geographic Presence	North America

Source: Company Website

14 APPENDIX

14.1 DISCUSSION GUIDE

- Q. 1.** What are your views on the current scenario in the healthcare analytics market? At what rate do you think this market will grow in the next five years (2021-2026)?
- Q. 2.** Kindly provide the market shares and projected growth rates of the following type segments in the healthcare analytics market. What are the key factors that will drive the growth of the fastest-growing segment in the next five years? Kindly validate the market segmentation and let us know if we are missing anything or if there is any overlap between these segments.

SR. NO.	TYPE	MARKET SHARE (2020)	CAGR (2021-2026)
1.	Descriptive Analytics		
2	Predictive Analytics		
3	Prescriptive Analytics		
4	Cognitive Analytics		

- Q. 3.** Kindly provide the market shares and projected growth rates of the following component segments in the healthcare analytics market. What are the key factors that will drive the growth of the fastest-growing segment in the next five years? Kindly validate the market segmentation and let us know if we are missing anything or if there is any overlap between these segments.

SR. NO.	COMPONENT	SUBSEGMENT	MARKET SHARE (2020)	CAGR (2021-2026)
1.	Services	Support Services		
		Business Analytics Services		
2	Software			
3	Hardware			

- Q. 4.** Kindly provide the market shares and projected growth rates of the following application segments in the healthcare analytics market. What are the key factors that will drive the growth of the fastest-growing segment in the next five years? Kindly validate the market segmentation and let us know if we are missing anything or if there is any overlap between these segments.

SR. NO.	APPLICATION	SUBSEGMENT	MARKET SHARE (2020)	CAGR (2021-2026)
1.	Financial Analysis	Claims Processing		
		Revenue Cycle Management		
		Payment Integrity & FWA		
		Risk Adjustment & Assessment		
2.	Clinical Analytics	Quality Improvement & Clinical Benchmarking		
		Clinical Decision Support		
		Reporting & Compliance		
		Comparative Analytics/Effectiveness		
		Precision Health		
3.	Operational & Administrative Analytics	Medical Imaging Analytics		
		Supply Chain Analytics		
		Workforce Analytics		
4	Population Health Analytics	Strategic Analytics		

- Q. 5.** Kindly provide the market shares and projected growth rates of the following deployment model segments in the healthcare analytics market. What are the key factors that will drive the growth of the fastest-growing segment in the next five years? Kindly validate the market segmentation and let us know if we are missing anything or if there is any overlap between these segments.

SR. NO.	DEPLOYMENT MODEL	MARKET SHARE (2020)	CAGR (2021-2026)
1.	On-premise Deployment		
2	On-demand Deployment		

- Q. 6.** Kindly provide the market shares and projected growth rates of the following end-user segments in the healthcare analytics market. What are the key factors that will drive the growth of the fastest-growing segment in the next five years? Kindly validate the market segmentation and let us know if we are missing anything or if there is any overlap between these segments.

SR. NO.	END USER	SUBSEGMENT	MARKET SHARE (2020)	CAGR (2021-2026)
1.	Payers	Private Insurance Companies		
		Government Agencies		
		Employers and Private Exchanges		
2.	Providers	Hospitals, Physician Practices, and IDNs		
		Post-acute Care Organizations		
		Ambulatory Settings		
3.	ACOs, HIEs, MCOs, and TPAs			

- Q. 7.** What are the major drivers, restraints, opportunities, and challenges that will impact the growth of the healthcare analytics market in the next five years?

PARAMETER	DESCRIPTION
Drivers	
Restraints	
Opportunities	
Challenges	

Q. 8. Kindly provide the market shares of the following players in the healthcare analytics market for 2020.

COMPANY NAME	MARKET SHARE (2020)
Allscripts Healthcare Solutions	
Cerner Corporation	
Health Catalyst	
IBM	
Inovalon	
McKesson Corporation	
MedeAnalytics	
Optum	
Oracle	
SAS Institute	
SCIO Health Analytics	
Verscend Technologies	
Wipro	
VitreosHealth	

Q. 9. Kindly provide the market shares and projected growth rates of the following countries/regions in the healthcare analytics market.

REGION	MARKET SHARE (2020)	CAGR (2021-2026)
North America		
▪ US		
▪ Canada		
Europe		
▪ Germany		
▪ France		
▪ UK		
▪ Rest of Europe (RoE)		
Asia		
▪ Japan		
▪ China		

- India

- Rest of Asia (RoA)

Pacific Countries

Latin America

Middle East & Africa

Q. 10. Kindly provide additional comments/suggestions, if any.

14.2 KNOWLEDGE STORE: MARKETSandMARKETS' SUBSCRIPTION PORTAL

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- Banking & Financial Services
- Chemicals & Material
- Energy & Power
- Food & Beverage
- Healthcare**
- Information & Communications Technology
- Packaging, Construction, Mining & Gases
- Retail & eCommerce
- Semiconductor & Electronics

Advanced Genetic Sequencing

Analytical & Scientific Instrumentation

Animal Health

Bio-Production

Cardiovascular Devices

Dental Devices

Diagnostic Imaging

Drug Delivery Devices

Healthcare Analytics

Healthcare IT

In-vitro Diagnostics

Medical Supplies

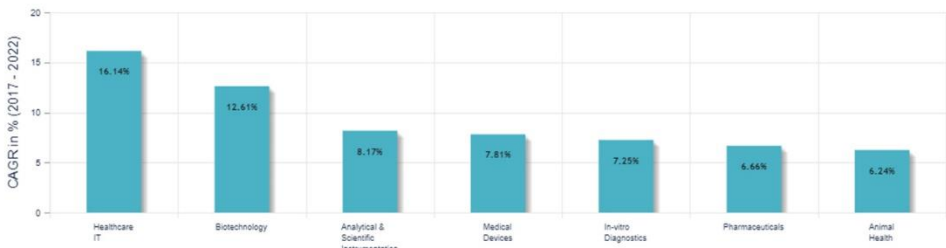
Pharmaceuticals

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- Agriculture (198)
- Automotive & Transportation (253)
- Chemicals & Material (1078)
- Energy & Power (354)
- Food & Beverage (298)
- Healthcare (881)**
 - Analytical & Scientific Instrumentation (59)
 - Animal Health (14)
 - Biotechnology (168)
 - Healthcare IT (132)
 - In-vitro Diagnostics (33)
 - Medical Devices (432)
 - Pharmaceuticals (43)
- Information & Communications Technology (809)
- Packaging, Construction, Mining & Gases (152)
- Semiconductor & Electronics (885)

HIGH GROWTH MARKETS



Market	CAGR in % (2017 - 2022)
Healthcare IT	16.14%
Biotechnology	12.61%
Analytical & Scientific Instrumentation	8.17%
Medical Devices	7.81%
In-vitro Diagnostics	7.25%
Pharmaceuticals	6.66%
Animal Health	6.24%

* CAGR (Compound Annual Growth Rate) represents Market Value (USD) growth forecast for a period of 5 years. It is an average CAGR based on the high growth niche markets covered in MarketsandMarkets syndicated reports and should not be interpreted as Industry CAGR.
Source: MarketsandMarkets Reports published since Jan 2015

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Report Title	Domain	Publish Date	Market Size in 2017 (USD Bn)	CAGR %	Action
Blockchain Technology in Healthcare Market by Application (Supply Chain Management, Clinical Data Exchange & Interoperability, Claims Adjudication and Billing), End User (Pharmaceutical Companies, Healthcare Payers, Providers) - Global Forecast to 2023	Healthcare	Jul 2018	0.03	72.8	 
Artificial Intelligence in Healthcare Market by Offering (Hardware, Software and Services), Technology (Deep Learning, Querying Method, NLP, and Context Aware Processing), Application, End-User Industry, and Geography - Global Forecast to 2022	Healthcare	May 2017	0.96	52.68	 

14.3 AVAILABLE CUSTOMIZATIONS

With the given market data, MarketsandMarkets offers customizations as per the company's specific needs.

The following customization options are available for the report:

- Company profiles for five additional companies.
- Healthcare analytics market size and growth rate estimates for Pacific countries, Latin America, and the Middle East & Africa

14.4 RELATED REPORTS

SR. NO.	REPORT TITLE	PUBLISHING DATE
1	HEALTHCARE IT MARKET - GLOBAL FORECAST TO 2024 By Product (EHR, RIS, PACS, VNA, CPOE, HIE, Telehealth, Healthcare Analytics, Population Health Management, Supply Chain Management, CRM, Fraud Management, Claims Management) End User (Provider, Payer) https://www.marketsandmarkets.com/Market-Reports/healthcare-it-252.html	April 2019
2	POPULATION HEALTH MANAGEMENT MARKET - GLOBAL FORECAST TO 2026 By End User (Healthcare Providers, Healthcare Payers, Employer Groups, Government Bodies), Component (Software, Services), Mode of Delivery (Web-based, Cloud-based, On-premise), Region https://www.marketsandmarkets.com/Market-Reports/population-health-management-market-263411936.html	June 2021
3	REVENUE CYCLE MANAGEMENT MARKET - GLOBAL FORECAST TO 2022 By Product (Integrated Solutions, Standalone), Function (Claim & Denial, Medical Coding, Insurance Verification), Deployment (Web-based, On-premise, Cloud-based), End User (Hospitals, Physicians, Lab) https://www.marketsandmarkets.com/Market-Reports/population-health-management-market-263411936.html	January 2018

14.5 AUTHOR DETAILS

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*Associate Vice President
Healthcare*

Rajiv has over 13+ years of experience in market research and consulting in the healthcare vertical. He has worked across diverse areas in the pharma/biotech, medical device, and healthcare IT industries.

He has been instrumental in tracking high-growth and niche markets in the healthcare vertical and has worked on a number of projects requiring expertise in market estimations, primary & secondary research techniques, strategic & competitive assessment of companies, investment opportunity analysis, and product opportunity evaluations. His current role involves tracking industry developments to come up with research areas in collaboration with clients.

Rajiv has diligently worked on addressing the problems of clients regarding market and product strategy, and formulated detailed solutions in terms of research and consulting outputs. He has completed his Master's degree in Pharmacy with a major in Pharmacology from Manipal University and also has a Postgraduate Diploma in Financial Management.

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